Digital Banking and Financial Technology

The Malaysian Perspective ››› A Textbook for Diploma and Undergraduate Students

First Edition ››› November 2026

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Digital Banking and Financial Technology

The Malaysian Perspective

A textbook for diploma and undergraduate students of banking, finance, business and information technology

First Edition

Preface

Financial services are being rebuilt in real time. A hawker in Kota Bharu now accepts QR payments through DuitNow; a fresh graduate in Johor Bahru invests RM300 a month through a licensed robo-advisor; a micro-enterprise in Kuching raises working capital from hundreds of small investors on a peer-to-peer platform; and five newly licensed digital banks are opening accounts for Malaysians whom conventional banking never reached. Understanding these changes is no longer optional for students of business, banking or technology — it is foundational.

This textbook was written specifically for **Malaysian diploma and undergraduate students**. Global FinTech theory is presented rigorously and cited to the scholarly literature (APA 7th edition), but every concept is anchored to the Malaysian market: Bank Negara Malaysia's (BNM) digital bank licensing framework, the Securities Commission Malaysia's (SC) regulation of P2P financing, equity crowdfunding, digital asset exchanges and robo-advisors, PayNet's national payment rails, and Malaysia's world-leading Islamic finance ecosystem.

How to Use This Book

Each chapter follows a consistent pedagogical structure designed around Bloom's Taxonomy:

- **Chapter Outline** — the sections you will cover.
- **Learning Outcomes** — six outcomes tagged to Bloom's cognitive levels (Remember, Understand, Apply, Analyse, Evaluate, Create).
- **Opening Vignette** — a realistic Malaysian scenario that motivates the chapter.
- **Core sections** — with definitions cited to the academic literature and Malaysian regulatory documents.
- **Pedagogical boxes** throughout the text:

  - **Quick Note** — clarifies a tricky point in one or two sentences.
  - **Did You Know? (Trivia)** — surprising, verified facts to keep you curious.
  - **Brainstormer** — open-ended questions to discuss in class or reflect on.
  - **Malaysian Spotlight** — a local institution, initiative or statistic.
- **Case Study** — a factual Malaysian case with discussion questions.
- **Chapter Summary** — key takeaways by section.
- **Key Terms** — definitions that feed into the end-of-book Glossary.
- **Test Your Knowledge** — 10 multiple-choice questions and 5 short-answer questions with an answer guide.
- **Problem-Solving Corner** — worked numerical and scenario problems.
- **References** — APA 7th edition.

For lecturers: each chapter maps to roughly one to two weeks of a 14-week semester. Chapters 1–3 suit the first half (foundations, banking and payments, digital financing); Chapters 4–7 suit the technology core (blockchain, trading and robo-advisory, crypto-assets, AI); Chapters 8–9 close with risk, regulation, ethics and Islamic/sustainable finance — a distinctive Malaysian strength.

For practitioners: the Case Studies, Malaysian Spotlights and Problem-Solving Corners are self-contained and can be read independently as professional briefings.

Recommended Textbooks

Main Recommended Textbook

Pompella, M., & Matousek, R. (Eds.). (2021). *The Palgrave handbook of FinTech and blockchain*. Palgrave Macmillan.

Recommended as the main companion text. Published within the last five years, this peer-reviewed handbook covers virtually every topic in this book — digital payments, crowdfunding, blockchain, cryptocurrencies, robo-advisory, RegTech and financial inclusion — with contributions from leading scholars. Its academic depth makes it ideal for students progressing to degree level, while individual chapters are readable enough for diploma students. Pair its conceptual chapters with the Malaysian applications presented in this book.

Supplementary Textbooks

1. **Liaw, K. T. (Ed.). (2021). *The Routledge handbook of fintech*. Routledge.** — A comprehensive survey of FinTech development, funding and applications across banking, blockchain and capital markets; strong on the economics of FinTech business models (supports Chapters 1, 3 and 5).
2. **Gupta, P., & Tham, T. M. (2019). *Fintech: The new DNA of financial services*. De Gruyter.** — A structured, teaching-oriented treatment of FinTech segments with an Asian perspective; particularly useful for payments, P2P lending and robo-advisory (supports Chapters 1–3 and 5).
3. **Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). *DeFi and the future of finance*. Wiley.** — A Duke University scholar's accessible yet rigorous introduction to decentralised finance; the best single short book for Chapters 4 and 6.

4. **Arslanian, H. (2022).** ***The book of crypto: The complete guide to understanding Bitcoin, cryptocurrencies and digital assets*****. Palgrave Macmillan.** — A practitioner-scholar's guide to crypto-assets, tokenisation and institutional adoption (supports Chapter 6).
5. **Oseni, U. A., & Ali, S. N. (Eds.). (2019).** ***Fintech in Islamic finance: Theory and practice*****. Routledge.** — The leading academic treatment of Islamic FinTech, essential for Chapter 9 and for understanding Malaysia's distinctive advantage.
6. **Chishti, S., Craddock, T., Courtneidge, R., & Zachariadis, M. (Eds.). (2020).** ***The PayTech book: The payment technology handbook for investors, entrepreneurs and FinTech visionaries*****. Wiley.** — Practitioner chapters on payments infrastructure, QR and real-time payments (supports Chapter 2).
7. **King, B. (2021).** ***The rise of technosocialism: How inequality, AI and climate will usher in a new world order*****. Marshall Cavendish Business.** — A big-picture, provocative read on where finance, AI and climate intersect; excellent for tutorial debates (supports Chapters 7–9).

In addition to these textbooks, students should treat the primary sources of **Bank Negara Malaysia** (e.g., the *Financial Sector Blueprint 2022–2026* and the *Licensing Framework for Digital Banks*), the **Securities Commission Malaysia** (e.g., the *Guidelines on Recognized Markets* and *Guidelines on Digital Assets*), and the **World Bank Global Findex Database** as required reading — they are cited throughout and are freely available online.

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End matter: Master References · Glossary · Index

Edition notice: First edition, compiled November 2026. Statistics reflect the latest available official data (Bank Negara Malaysia Annual Report 2025, Securities Commission Malaysia Annual Report 2025, PayNet 2026 press release, World Bank Global Findex 2025). Students should verify fast-moving figures against current regulator publications.

Chapter 1: Introduction to Digital Banking and Financial Technology

Chapter Outline

- 1.1 Overview of Digital Banking and FinTech
- 1.2 Evolution of Financial Services: From FinTech 1.0 to 4.0
- 1.3 The Digital Financial Services Ecosystem and Landscape in Malaysia
- 1.4 Key Players and Business Models
- 1.5 Benefits and Challenges: Financial Inclusion and Operational Viability

Learning Outcomes

After studying this chapter, you should be able to:

1. **Define** digital banking and financial technology (FinTech), and state the differences between digital banking, online banking and neobanks. *(Bloom's level: Remember)*
2. **Explain** the evolution of financial services from FinTech 1.0 through FinTech 3.0/3.5 to the emerging FinTech 4.0 era. *(Bloom's level: Understand)*
3. **Describe** the regulators, enablers and market segments that make up Malaysia's digital financial services ecosystem. *(Bloom's level: Understand)*
4. **Apply** knowledge of FinTech business models to classify the revenue strategies of real Malaysian providers. *(Bloom's level: Apply)*
5. **Analyse** the benefits of digital financial services against their challenges for Malaysian consumers and providers. *(Bloom's level: Analyse)*
6. **Evaluate** whether Malaysia's digital bank licensing approach can balance financial inclusion with operational viability. *(Bloom's level: Evaluate)*

Opening Vignette

At 7.30 a.m. in Kota Bharu, Kelantan, twenty-year-old diploma student Aina Sofia unlocks her phone before she unlocks her hostel room door. She pays for her nasi lemak breakfast by scanning the stallholder's DuitNow QR code with her Touch 'n Go eWallet, then sends RM50 to her mother in Bachok through her GXBank savings account. The money arrives in seconds, and neither of them has visited a bank branch in months. On the bus to campus, Aina checks a

robo-advisory app, where the RM200 she saved last semester has been invested automatically according to her risk profile. Her uncle, who runs a small batik workshop, recently raised RM40,000 through a peer-to-peer financing platform after being turned away by two banks.

None of these services existed in Malaysia when Aina was born. This chapter explains what they are, where they came from, who provides and regulates them in Malaysia, how they make money, and whether they truly make finance fairer for everyone.

1.1 Overview of Digital Banking and FinTech

Few pairs of words are used as loosely in business today as "digital banking" and "FinTech". News articles, bank advertisements and even regulators sometimes use them differently. Before we can study the Malaysian landscape, we need precise, citable definitions. This section defines financial technology, separates its two common meanings, defines digital banking, and distinguishes digital banking from online banking and neobanks.

1.1.1 Defining Financial Technology (FinTech)

Financial technology (FinTech) is defined as digital innovations and technology-enabled business model innovations in the financial sector (Puschmann, 2017). Two parts of this definition deserve attention. First, FinTech involves *digital innovation* — the application of computing, mobile communications and data analytics to financial problems. Second, and equally important, FinTech involves *business model innovation* — new ways of producing, distributing and charging for financial services, not merely faster versions of old processes. A bank that scans paper forms into a computer is digitising; a platform that replaces the bank's lending process entirely is a FinTech business model innovation.

Gomber, Kauffman, Parker and Weber (2018) interpret the phenomenon as three forces acting on financial services: *innovation* (new products and processes), *disruption* (new entrants displacing incumbents) and *transformation* (deep structural change in how finance operates). Thakor (2020) adds that FinTech matters economically because it can reduce the cost of producing financial services, improve the matching of providers and customers, and allow better use of data in decisions such as credit assessment. In short, FinTech is not a single product or app; it is an ongoing technological re-organisation of the financial sector.

> **QUICK NOTE:** Do not confuse "FinTech" (the field or industry) with "a fintech" (a single company). In this book, *FinTech* refers to the field, while a firm operating in it is called a *fintech company* or *fintech start-up*.

1.1.2 FinTech-as-Industry versus FinTech-as-Technology

The word "FinTech" is used in two distinct ways, and recognising both will help you read reports and job advertisements critically.

- **FinTech-as-industry.** In this sense, FinTech is a *sector of the economy* made up of firms — usually young, technology-led start-ups — that deliver financial services such as payments, lending, fund-raising, investment or insurance. When a report says "Malaysia's fintech industry is growing", it means this population of firms. Examples in Malaysia include Funding Societies in peer-to-peer financing, MoneyMatch in cross-border remittance, and HelloGold in Shariah-compliant digital gold savings.

- **FinTech-as-technology.** In this sense, FinTech is a *toolbox of technologies* applied to finance: mobile applications, cloud computing, big data analytics, artificial intelligence (AI), application programming interfaces (APIs), biometrics and distributed ledger technology (DLT). When an incumbent bank announces it is "investing in fintech", it usually means adopting these technologies, not buying start-ups.

The two meanings interact. Start-ups in the FinTech *industry* are typically built entirely on FinTech *technologies*, while incumbent banks increasingly deploy the same technologies to defend their markets (Puschmann, 2017; Lee & Shin, 2018). Throughout this book, context will indicate which meaning is intended, but as a student you should always ask: is the speaker talking about the firms, or the technology?

1.1.3 Defining Digital Banking

Bank Negara Malaysia (BNM), the central bank, provides the authoritative local definition. Under its licensing framework, a **digital bank** is a bank licensed under the Financial Services Act 2013 or the Islamic Financial Services Act 2013 that conducts banking business *wholly or almost wholly through digital or electronic means* (Bank Negara Malaysia [BNM], 2020). Building on this, **digital banking** can be defined as the delivery of regulated banking services — accepting deposits, providing financing or lending, and facilitating payments — through digital channels end-to-end, with customer onboarding, verification and servicing conducted remotely rather than at physical branches (BNM, 2020).

Three implications follow. First, digital banking in Malaysia is a *licensed, regulated* activity: a digital bank is still a bank, subject to capital, liquidity and conduct requirements. Second, the digital channel is the primary — often the only — channel; branch networks are absent or minimal. Third, remote identity verification is essential, which is why BNM's electronic know-your-customer (e-KYC) policy, first issued in 2020 and revised in April 2024, is a critical enabler of the model (BNM, 2024a).

1.1.4 Digital Banking versus Online Banking versus Neobanks

Students frequently use these three terms interchangeably, but they describe different things.

Online banking is the provision of banking services over the internet as an *additional channel* of a traditional, branch-based bank. The bank existed first; the website or app came later. Malaysia's first internet banking platform, Maybank2u, launched in 2000, and services such as CIMB Clicks remain examples of incumbent banks offering digital channels alongside their branches.

Digital banking, as defined in Section 1.1.3, describes a banking operation designed from the ground up to be digital: account opening via e-KYC, no reliance on branches, and products built for a smartphone-first customer (BNM, 2020). A licensed digital bank such as GXBank or Boost Bank is a regulated bank in its own right, not a channel of another bank.

A **neobank** is a term from global industry usage for a digital-only bank that operates without physical branches and typically targets customers through a mobile app. Internationally, some neobanks hold full banking licences (for example, UK challenger banks), while others operate by partnering with licensed banks. In Malaysia, however, the five digital banks all hold full BNM licences, so the closest local equivalent to the global "neobank" concept is the licensed digital bank itself (BNM, 2020).

Table 1.1 summarises the distinctions.

Table 1.1: Online banking, digital banking and neobanks compared

Feature	Online banking	Licensed digital bank	Neobank (global usage)
What it is	A digital channel of an incumbent bank	A fully licensed bank operating almost wholly digitally	A digital-only banking provider
Licence	Held by the parent bank	Digital bank licence under FSA 2013 / IFSA 2013	Varies: own licence or partner's licence
Physical branches	Yes, often extensive	None (or minimal)	None
Customer onboarding	May still require a branch visit for some products	Fully remote via e-KYC	Fully remote
Malaysian examples	Maybank2u, CIMB Clicks	GXBank, Boost Bank, AEON Bank, KAF Digital Bank, Ryt Bank	The five licensed digital banks are the local equivalent
Global examples	Internet portals of large banks	—	Chime (US), Revolut (UK/EU)

1.2 Evolution of Financial Services: From FinTech 1.0 to 4.0

FinTech did not begin with the smartphone. In an influential framework, Arner, Barberis and Buckley (2016) divide the technological evolution of finance into three historical eras — FinTech 1.0, 2.0 and 3.0 (with a 3.5 variant for parts of Asia) — and subsequent literature, notably Gupta and Tham (2019), has extended the story with an emerging FinTech 4.0. Understanding this evolution matters because each era left behind institutions, regulations and habits that still shape Malaysia's financial system today.

1.2.1 FinTech 1.0: The Analogue Era (1866–1967)

Arner et al. (2016) date FinTech 1.0 from 1866, when the first successful transatlantic telegraph cable connected North America and Europe. For the first time, financial information — prices, orders, settlement instructions — could cross an ocean in minutes rather than weeks. Finance became *globalised* through telecommunications even though the underlying processes remained analogue and paper-based.

Other milestones of the era include the establishment of Fedwire in 1918, through which the United States Federal Reserve transferred funds electronically using telegraph lines and Morse code, and the launch of the Diners Club card in 1950, often described as the first widely used payment card. Arner et al. (2016) mark the end of FinTech 1.0 in 1967, when Barclays Bank installed the world's first automated teller machine (ATM) in London — a fitting symbol of the shift from analogue to digital technology in finance.

> **DID YOU KNOW? (TRIVIA):** The first "electronic" money transfers were sent over telegraph wires more than a century ago. Fedwire, launched in 1918, moved money

using Morse code — proof that the idea of fintech is far older than the internet (Arner et al., 2016).

1.2.2 FinTech 2.0: Digitalisation of Traditional Finance (1967–2008)

FinTech 2.0, from 1967 to 2008, is the era in which *traditional financial institutions* digitalised their own processes and delivery channels (Arner et al., 2016). The technology was new, but the providers were familiar: banks, insurers and exchanges remained firmly in control.

Global milestones included the spread of ATMs from 1967 onwards; the creation of SWIFT (the Society for Worldwide Interbank Financial Telecommunication) in 1973, which standardised secure interbank messaging across borders; the rise of electronic securities trading, exemplified by the NASDAQ electronic exchange from 1971; the introduction of mainframe computing into back offices; and, from the mid-1990s, the arrival of internet banking, with Security First Network Bank in the United States (1995) widely cited as the first internet-only bank.

Malaysia participated in this era. The central bank launched RENTAS (Real Time Electronic Transfer of Funds and Securities), the country's real-time gross settlement system, in 1999, allowing large interbank payments to be settled instantly and irrevocably. One year later, Maybank2u (2000) became Malaysia's first internet banking platform, bringing the FinTech 2.0 model — an incumbent bank adding a digital channel — to Malaysian retail customers.

1.2.3 FinTech 3.0 and 3.5: New Entrants Post-2008 and Asia's Rise

Arner et al. (2016) identify 2008 as a turning point for two converging reasons. First, the global financial crisis damaged public trust in incumbent banks and triggered heavier regulation of them, opening space for new entrants. Second, the smartphone — popularised by the iPhone's launch in 2007 — put a powerful, always-connected computer in consumers' pockets. FinTech 3.0 is therefore the era of *new entrants*: start-ups and technology firms delivering financial services directly to consumers, often bypassing banks altogether.

The hallmarks of the era appeared quickly: online payment services such as PayPal (founded 1998) and Alipay (launched 2004 in China); mobile money such as M-Pesa (2007, Kenya), which brought basic financial services to millions without bank accounts; peer-to-peer lending, pioneered by Zopa in the United Kingdom (2005); and, in 2008 itself, the publication of the Bitcoin whitepaper proposing a peer-to-peer electronic cash system outside the banking system entirely (Nakamoto, 2008).

Arner et al. (2016) add a special label, **FinTech 3.5**, for economies — above all China and India — that skipped the card-and-branch stages of development and leapfrogged directly to mobile-first financial services. In China, Alipay and WeChat Pay evolved into "super-apps" combining payments, lending, investment and insurance inside a single platform. Southeast Asia followed a similar path: in Malaysia, bank-led and non-bank e-wallets proliferated, the DuitNow instant payment service launched in 2018, and the DuitNow QR national QR standard followed in 2019, allowing any participating wallet or bank app to pay any participating merchant.

> **BRAINSTORMER:** Arner et al. (2016) argue that the 2008 global financial crisis helped launch the FinTech 3.0 era. Do you think FinTech would have grown just as quickly without the crisis? What role did falling trust in banks play, compared with the simple availability of smartphones?

1.2.4 FinTech 4.0: Platformisation, BigTech, AI and Embedded Finance

Where does the story go next? Building on the FinTech 1.0–3.0 framework, subsequent literature — notably Gupta and Tham (2019) — frames an emerging **FinTech 4.0** era characterised by four mutually reinforcing forces.

1. **Platformisation.** Financial services are increasingly distributed through large digital *platforms* that host many providers and services in one ecosystem, rather than through stand-alone products sold by individual institutions. The platform — not the branch or even the app of a single bank — becomes the main point of contact with the customer.
2. **BigTech.** Large technology companies with vast user bases and data advantage — regionally, firms such as Grab and Sea — have moved into payments, lending and banking. Zetzsche, Buckley, Arner and Barberis (2017) describe this data-driven entry of technology firms into finance as "TechFin", the mirror image of FinTech.
3. **Artificial intelligence.** AI now supports credit scoring, fraud detection, customer service and personalised financial advice. In Malaysia, BNM issued a discussion paper on artificial intelligence in the financial sector in August 2025, signalling how central AI has become to regulatory thinking (BNM, 2025a).
4. **Embedded finance and tokenisation. Embedded finance** is the integration of financial services — payments, lending, insurance — directly into non-financial apps and platforms, so that, for example, a consumer can obtain financing at the checkout of an e-commerce app without ever visiting a bank. Alongside it, BNM's October 2025 discussion paper on asset tokenisation explores representing financial and real-world assets on programmable ledgers (BNM, 2025b).

A note of caution: eras overlap and labels are debated. FinTech 4.0 is an *emerging* framing from subsequent literature, not a settled historical fact, and elements of FinTech 2.0 and 3.0 remain firmly in place (Gupta & Tham, 2019). Table 1.2 summarises the evolution.

Table 1.2: The evolution of FinTech eras

Era	Period	Character	Defining features	Malaysian milestones
FinTech 1.0	1866–1967	Analogue, globalising	Transatlantic cable (1866), Fedwire (1918), Diners Club (1950)	—
FinTech 2.0	1967–2008	Incumbents digitalise	ATMs, SWIFT (1973), electronic trading, internet banking	RENTAS (1999); Maybank2u (2000)
FinTech 3.0	2008–present	New entrants	Smartphones, start-ups, mobile money, P2P, crypto	E-wallet growth; DuitNow (2018); DuitNow QR (2019)
FinTech 3.5	2010s	Asia leapfrogs	Super-apps, mobile-first models in China, India, SEA	QR-based merchant payments

Era	Period	Character	Defining features	Malaysian milestones
FinTech 4.0	Emerging	Platforms and intelligence	Platformisation, BigTech, AI, embedded finance, tokenisation	Five digital banks; BNM AI and tokenisation discussion papers (2025)

Sources: Arner et al. (2016); Gupta and Tham (2019); BNM (2025a, 2025b).

1.3 The Digital Financial Services Ecosystem and Landscape in Malaysia

A FinTech *ecosystem* is the network of regulators, market operators, industry bodies, infrastructure providers and firms whose interactions produce digital financial services (Lee & Shin, 2018). Malaysia's ecosystem is distinctive in two ways: it is shared between two powerful regulators, and it contains an unusually strong Islamic finance dimension. This section maps the key institutions and market segments.

1.3.1 Regulators and Ecosystem Enablers

Bank Negara Malaysia (BNM) is the central bank, established under the Central Bank of Malaysia Act 2009, and regulates banks, payment systems, electronic money (e-money) and money services under the Financial Services Act 2013 (FSA 2013) and the Islamic Financial Services Act 2013 (IFSA 2013). Its major fintech-relevant instruments include:

- the **Licensing Framework for Digital Banks** (December 2020), which created the digital bank licence (BNM, 2020);
- the **Financial Sector Blueprint 2022–2026**, which sets five desired outcomes for the financial sector and targets roughly 15% compound annual growth in e-payments (BNM, 2022);
- the **Financial Technology Regulatory Sandbox Framework** (revised February 2024), which allows firms to test innovative services with real customers under relaxed but supervised conditions — a **regulatory sandbox** (BNM, 2024b);
- the **e-KYC policy** (2020, revised April 2024), enabling non-face-to-face customer onboarding (BNM, 2024a); and
- the **Risk Management in Technology (RMiT)** policy (2020; revised 2023; updated November 2025), which imposes technology and cyber-risk requirements on financial institutions (BNM, 2025c).

Securities Commission Malaysia (SC) regulates capital market fintech under the Capital Markets and Services Act 2007: peer-to-peer financing, equity crowdfunding, digital asset exchanges, digital investment management and initial exchange offering (IEO) operators, principally through the Guidelines on Recognized Markets and the Guidelines on Digital Assets (Securities Commission Malaysia [SC], 2024a, 2024b). Since the Capital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019 took effect on 15 January 2019, digital currencies and digital tokens have been prescribed as securities in Malaysia — meaning they fall under the SC's jurisdiction and are *not* legal tender.

Around the two regulators sit important ecosystem enablers:

- **Payments Network Malaysia (PayNet)**, owned by BNM, operates the national shared payments infrastructure described in Section 1.3.3.
- **Malaysia Digital Economy Corporation (MDEC)** champions the growth of the digital economy, including fintech.
- The **Fintech Association of Malaysia** is the industry's main representative body.
- The **Malaysia International Islamic Financial Centre (MIFC)** promotes Malaysia as a global hub for Islamic finance, including Islamic fintech.

Table 1.3: Key institutions in Malaysia's digital financial services ecosystem

Institution	Type	Role in the ecosystem
Bank Negara Malaysia (BNM)	Regulator	Banks, payment systems, e-money, money services; digital bank licensing; sandbox; e-KYC; RMiT
Securities Commission Malaysia (SC)	Regulator	P2P financing, equity crowdfunding, digital asset exchanges, digital investment management, IEOs
PayNet	Infrastructure	Operates national payment rails (RENTAS, IBG, FPX, JomPAY, MyDebit, DuitNow, DuitNow QR)
MDEC	Government agency	Digital economy development and fintech promotion
Fintech Association of Malaysia	Industry body	Represents and connects fintech firms
MIFC	Promotion agency	Develops Malaysia as an international Islamic finance centre

1.3.2 Key Segments of Malaysian FinTech

Payments is the largest and most visible segment. Malaysia recorded 18.4 billion e-payment transactions in 2025, up 25% from 14.7 billion in 2024 — an average of 538 transactions per Malaysian — with retail e-payment value reaching RM831 billion (BNM, 2026). **Electronic money (e-money)** is a digitally stored monetary value issued under BNM licence that can be used to pay for goods and services; the average e-money transaction was RM43 in 2025, up from RM33 in 2024 (BNM, 2026).

E-wallets — smartphone apps holding e-money — are the everyday face of this segment. Major Malaysian e-wallets include Touch 'n Go eWallet, MAE by Maybank, GrabPay, Boost, ShopeePay and BigPay.

Digital banks. Malaysia licensed five digital banks in 2022; their journey is told in this chapter's case study.

Peer-to-peer (P2P) financing is the raising of debt funding by businesses from many investors through a SC-registered online platform. Since 2016, Malaysian P2P platforms have raised a cumulative RM11.32 billion across 154,283 campaigns; 98% of campaigns fund working capital, 62% raise RM50,000 or less, and 26% of funds were Shariah-compliant in 2025 (SC, 2026). Example operators include Funding Societies, Fundaztic, B2B FinPAL, microLEAP, Alixco, CapBay and MoneySave.

Equity crowdfunding (ECF) is the raising of equity capital by companies from many investors through a SC-registered online platform. Cumulatively, RM917.51 million has been raised across 445 campaigns since 2016, mostly at seed and Series A stages; example platforms include pitchIN, Ata Plus, MyStartr, Ethis and Leet Capital (SC, 2026).

> **DID YOU KNOW? (TRIVIA):** Malaysia was the first ASEAN country to regulate equity crowdfunding (2015) and among the first in the region to regulate P2P financing (2016). Its digital investment management framework of May 2017 was also the first robo-advisory licensing framework in the region (SC, 2017, 2024b).

Digital investment management (DIM) — commonly called robo-advisory — provides automated risk profiling, suitability assessment, asset allocation and portfolio rebalancing with minimal human supervision, licensed by the SC since May 2017 (SC, 2017). Nine DIM firms are registered, including StashAway (first licensed, 2018), Wahed Invest, MYTHEO, Raiz, Akru, Versa, Kenanga Digital Investing and BEST Invest. Although DIM assets remain below 1% of the RM1.07 trillion fund management industry (2024), the segment is growing — 15% growth in 2023 — with over one million accounts, 51% of them held by investors in their twenties (SC, 2026).

Digital asset exchanges (DAX) are SC-registered platforms for trading digital assets; registered operators include Luno Malaysia, SINEGY Technologies, Tokenize Technology and Hata Digital (SC, 2024a). Because digital assets are prescribed as securities under the 2019 Order, trading on registered DAXs is regulated — but crypto remains *not* legal tender in Malaysia.

Insurtech — the application of FinTech technologies to insurance and takaful — is a smaller but growing segment, with digital distribution and embedded insurance appearing inside e-wallets, e-commerce platforms and digital bank apps.

Islamic fintech deserves special emphasis. Malaysia is consistently ranked among the world's top Islamic fintech ecosystems, alongside Saudi Arabia, the UAE, Indonesia and Bahrain. The global Islamic fintech market reached approximately US$198 billion in transaction volume in 2024/25 and is projected to reach US$341 billion by 2029, with 484 Islamic fintech firms operating globally (DinarStandard & Elipses, 2026). Malaysian examples include HelloGold (Shariah-compliant digital gold savings), Ethis (Islamic P2P/ECF), Global Sadaqah (waqf and sadaqah platform), microLEAP, Wahed Invest, BEST Invest, and the Islamic digital banks AEON Bank and KAF Digital Bank. BNM's Value-based Intermediation initiative (2018) and Shariah standards underpin this segment, anchored by the MIFC.

> **MALAYSIAN SPOTLIGHT:** DuitNow QR, Malaysia's national QR standard operated by PayNet, doubled its volume to roughly 3 billion transactions in 2025 and reached about 3 million merchant touchpoints — many of them micro, small and medium enterprises that previously accepted only cash. A single interoperable QR code now lets a customer pay a night-market stall from almost any bank app or e-wallet in the country (PayNet, 2026).

1.3.3 Payments Infrastructure: PayNet and the National Rails

Behind every tap, scan and transfer lies shared infrastructure operated mainly by PayNet. The principal national rails are:

Rail	Launched	Function
RENTAS	1999	Real-time gross settlement of large-value interbank payments
Interbank GIRO (IBG)	—	Batch-based interbank fund transfers
FPX	—	Online bank-to-bank payments for e-commerce
JomPAY	2015	National bill payment service
MyDebit	—	Domestic debit card scheme
DuitNow	2018	Instant account-to-account transfers using proxies such as mobile numbers
DuitNow QR	2019	National interoperable QR payment standard

Usage has grown explosively. PayNet processed 8.44 billion digital payment transactions in 2025 (PayNet, 2026). DuitNow QR has also gone regional: cross-border QR payment linkages are live with Thailand, Indonesia, Singapore and China, with Cambodia added in 2025 and India expected in 2026; cross-border QR transactions grew 2.5 times in 2025 (PayNet, 2026). A Malaysian traveller in Bangkok can therefore pay a Thai merchant with the same app used at home — a quiet but remarkable piece of financial diplomacy built on shared rails.

1.4 Key Players and Business Models

Technology alone does not explain the transformation of finance; equally important is *who* provides the services and *how* they earn revenue. This section identifies five groups of players and examines the business models that sustain them, drawing on Lee and Shin's (2018) analysis of fintech ecosystems and business models.

1.4.1 Key Players in the Digital Financial Services Arena

Five groups of players compete and cooperate in Malaysia's digital financial services arena.

1. **Incumbent banks digitising.** Established banks have responded to fintech competition by rebuilding their own services around mobile apps — for example, MAE by Maybank and CIMB Octo/CIMB Clicks. Their advantages are large customer bases, trusted brands and deposits; their weaknesses are legacy systems and branch cost structures (Thakor, 2020).
2. **Digital challenger banks.** The five licensed digital banks — GXBank, Boost Bank, AEON Bank, KAF Digital Bank and Ryt Bank — are branchless, app-native institutions built on modern technology stacks, targeting underserved and unserved segments (BNM, 2020).

3. **BigTech.** Large technology platforms embed financial services into ecosystems customers already use daily. In Malaysia this is no longer theoretical: Grab (with the Kuok Group) backs GXBank, and Sea (with YTL Digital Capital) backs Ryt Bank. Zetzsche et al. (2017) warn that BigTech's data advantage — knowing users' locations, purchases and social graphs — can shift competitive power decisively.
4. **Fintech start-ups.** Focused firms attack single pain points: remittances (MoneyMatch, a graduate of BNM's regulatory sandbox), SME financing (Funding Societies), gold savings (HelloGold) and crowdfunding (pitchIN). Their advantages are speed and specialisation; their challenge is achieving scale and profitability (Lee & Shin, 2018).
5. **Infrastructure providers.** Beneath the visible players sit firms and utilities that everyone else depends on: PayNet's national payment rails, e-KYC and credit-data services, cloud computing and core-banking technology vendors, and cybersecurity providers.

Table 1.4: Five groups of players and their strategic goals

Player group	Malaysian examples	Main strengths	Strategic goal
Incumbent banks digitising	Maybank (MAE), CIMB (Octo/Clicks)	Customers, trust, deposits	Defend relationships by matching fintech convenience
Digital challenger banks	GXBank, Boost Bank, AEON Bank, KAF Digital Bank, Ryt Bank	Modern technology, low cost base	Win underserved segments profitably
BigTech	Grab (GXBank), Sea (Ryt Bank)	Data, daily-use platforms	Embed finance into existing ecosystems
Fintech start-ups	MoneyMatch, Funding Societies, HelloGold, pitchIN	Focus, speed	Solve one pain point better than banks
Infrastructure providers	PayNet, e-KYC and cloud/core-banking vendors	Scale, neutrality	Provide shared rails and services

1.4.2 Business Models: How FinTech Firms Make Money

A **business model** describes how a firm creates, delivers and captures value. Lee and Shin (2018) show that fintech business models differ fundamentally from the traditional bank model of earning the spread between deposit and lending rates. Six revenue models dominate the Malaysian landscape.

1. **Payments fees.** Charging merchants or users per transaction — for example, the merchant discount rate (MDR) a merchant pays to accept card or QR payments.
2. **Interchange.** The **interchange fee** is the portion of a card transaction fee paid by the merchant's bank to the cardholder's bank (the issuer). Card-issuing digital banks and wallet providers earn interchange on every card spend.
3. **Subscription.** Recurring monthly or annual fees for premium features, used by some robo-advisors, business tools and premium account tiers.
4. **Marketplace/platform fees.** Platforms that match fund-raisers with investors charge listing, success or servicing fees. P2P and ECF operators in Malaysia earn revenue this way (SC, 2026).

5. **Lending margins.** Lending at a higher rate than the cost of funds — the classic banking model, also used by digital banks and some P2P-funded lenders.
6. **Freemium.** A **freemium** model offers basic services free to build a large user base, then charges for premium features or cross-sells other products. Many e-wallets and digital banks use free accounts as their entry point.

   QUICK NOTE: Keep MDR and interchange distinct. The *merchant discount rate* is the total fee a merchant pays to accept a cashless payment; *interchange* is only the slice of that fee passed to the customer's bank. The remainder goes to the merchant's bank and the scheme or switch operator.

Table 1.5: Major fintech business models with Malaysian illustrations

Model	How revenue is earned	Malaysian illustration	Key limitation
Payments fees (MDR)	Fee per merchant transaction	QR and card acceptance services	Thin margins; price competition
Interchange	Issuer's share of card transaction fees	Debit cards issued by digital banks	Depends on card usage volumes
Subscription	Recurring user fees	Premium tiers of investment/finance apps	Users resist paying for finance apps
Marketplace/platform	Listing, success, servicing fees	P2P and ECF platforms (SC-registered RMOs)	Needs steady deal flow both sides
Lending margins	Spread between lending and funding costs	Digital bank financing products	Constrained for digital banks by the RM3 billion asset cap
Freemium	Free basic service; paid upgrades/cross-sell	E-wallets and digital banks' entry accounts	Monetisation is uncertain and slow

In practice, successful firms combine several models. A digital bank, for instance, may offer free accounts (freemium) to acquire customers, earn interchange on card spend, collect lending margins on financing, and later add subscription or platform income. Diversification matters because no single fintech revenue line has proven sufficient on its own at small scale (Lee & Shin, 2018).

BRAINSTORMER: If customers expect payments, accounts and transfers to be free, where should a digital bank's profits come from? Would you rather pay visible fees, a subscription, or allow the provider to earn from your data and spending behaviour? What are the trade-offs?

1.5 Benefits and Challenges: Financial Inclusion and Operational Viability

Is the fintech transformation actually good for Malaysians? The honest answer is: mostly yes, with important qualifications. This section weighs the documented benefits — cost ef-

ficiency, convenience and financial inclusion — against the challenges of the digital divide, trust, regulation and, especially for new providers, operational viability.

1.5.1 Benefits: Cost Efficiency, Convenience and Inclusion

Cost efficiency. Digital delivery removes branches and paper from the cost base, so services can be produced at a fraction of the traditional cost (Thakor, 2020). These savings can be passed on as cheaper remittances, lower account fees and micro-sized products — such as robo-advisory accounts with minimal balances or P2P notes of a few hundred ringgit — that would be uneconomical to serve at a branch.

Convenience and speed. e-KYC allows an account to be opened from a phone in minutes (BNM, 2024a); DuitNow moves money between accounts in seconds; bill payment, investment and insurance can be done at any hour. Convenience is not a trivial benefit: for a hawker or small trader, a day spent travelling to a branch is a day of lost income.

Financial inclusion. Financial inclusion is defined as the access of individuals and businesses to useful and affordable financial products and services — transactions, payments, savings, credit and insurance — delivered in a responsible and sustainable way (World Bank, 2022). The World Bank's Global Findex database shows Malaysia's remarkable progress in adult (15+) account ownership:

Table 1.6: Account ownership in Malaysia, adults aged 15+ (%)

2011	2014	2017	2021	2024
66.2%	80.7%	85.3%	88.4%	88.7%

Source: World Bank Global Findex Database (World Bank, 2022, 2025).

By 2024 the gender gap in Malaysian account ownership had narrowed to less than one percentage point, and even among the poorest 40% of households, ownership reached 82.0% (World Bank, 2025). Digital services reinforce these gains: DuitNow QR has brought cashless acceptance to around 3 million merchant touchpoints, many of them formerly cash-only micro, small and medium enterprises (PayNet, 2026), while P2P financing has channelled RM11.32 billion largely to working-capital needs of smaller businesses — 99% of them non-tech firms — that banks often overlook (SC, 2026). Thakor (2020) notes that alternative data can extend credit to "thin-file" borrowers with no conventional credit history, which is precisely the segment Malaysia's digital banks were licensed to serve (BNM, 2020).

1.5.2 Challenges: Digital Divide, Trust, Regulation and Profitability

The digital divide. Inclusion statistics can mask exclusion: elderly Malaysians, rural communities with weak connectivity, and those with low digital literacy may find app-only finance harder, not easier, to use. When branches close in favour of apps, the customers least able to adapt can be left behind — a risk that regulators expect providers to manage (BNM, 2022).

Trust and fraud. Digital finance moves money instantly, which criminals exploit. Malaysia has responded with the National Scam Response Centre (hotline 997, established 2022) to rapidly freeze scam-linked funds, and BNM has directed banks to tighten controls such as kill-switches to freeze accounts, blocking SMS one-time passwords for high-risk transactions,

and binding accounts to one secured device. Trust remains fragile: a single widely publicised scam can push cautious consumers back to cash.

Regulatory burden. Compliance is expensive. Financial institutions must meet RMiT technology and cyber-resilience requirements (BNM, 2025c), anti-money-laundering obligations, and personal data protection duties under the Personal Data Protection Act 2010 (amended 2024). For start-ups, these fixed costs can be the difference between viability and failure — even though the same rules protect consumers.

Operational viability of digital banks. The sharpest challenge concerns the five digital banks themselves. During a **foundational phase** of three to five years, each digital bank's total assets are capped at RM3 billion (BNM, 2020). The cap protects financial stability while the banks prove themselves, but it also limits the size of their financing books — and therefore the lending-margin income that funds a traditional bank. With heavy upfront technology investment, free customer acquisition and constrained balance sheets, profitability is a genuine challenge. Global experience is sobering: many digital-only banks elsewhere have taken years to break even (Thakor, 2020). BNM's Governor noted in 2025 that the digital banks remain in their foundational phase, subject to asset size limitations, with deposits growing from underserved and unserved segments and credit products rolling out gradually (BNM, 2026).

> **QUICK NOTE:** An asset cap limits the *balance sheet*, not the number of customers. A digital bank can still grow deposits, accounts and fee income beyond the cap's shadow — which is why fee-based business models (Section 1.4.2) matter so much to its survival.

The central policy tension of Malaysian digital banking is now clear: the same licence conditions that push digital banks towards *financial inclusion* (serving small, remote, thin-file customers) make *operational viability* harder, because such customers are costly to serve and generate small balances. Whether Malaysia's digital banks can resolve this tension — through fee income, ecosystem partnerships and disciplined cost control — is one of the most important questions this book will follow.

Case Study 1.1: Malaysia's Five Digital Banks — The Licensing Journey

In December 2020, in the middle of the COVID-19 pandemic, Bank Negara Malaysia issued its Licensing Framework for Digital Banks, inviting applications for up to five licences. The policy objective was explicit: digital banks should serve the *unserved and underserved* — micro-enterprises, gig workers, low-income and rural Malaysians — using technology rather than branches (BNM, 2020). To manage risk, licensees would operate under a foundational phase of three to five years with total assets capped at RM3 billion.

On 29 April 2022, BNM announced the five winners, each a consortium combining financial experience with technology and distribution muscle:

- **GXBank**, led by Grab together with the Kuok Group;
- **Boost Bank**, uniting Boost/Axiata with RHB Banking Group;
- **AEON Bank**, formed by AEON Financial Service, AEON Credit and MoneyLion;
- **KAF Digital Bank**, a consortium anchored by KAF Investment Bank with Carsome, MoneyMatch, Jirnexu and StoreHub; and
- **Ryt Bank**, pairing Sea Ltd with YTL Digital Capital.

The licensees then had to build their banks — systems, compliance, products — and satisfy BNM's readiness requirements before opening to the public. **GXBank** was the first to begin operations, on 1 September 2023. **Boost Bank** commenced on 15 January 2024. **AEON Bank** launched on 26 May 2024 as Malaysia's first Islamic digital bank, carrying the inclusion mandate into Shariah-compliant banking. Finally, **KAF Digital Bank** — the second Islamic digital bank — and **Ryt Bank** were both approved to commence operations on 20 December 2024 (BNM, 2026).

Early progress, while deliberately measured, fits the policy design. BNM reported that digital banks' deposits are growing from underserved and unserved segments, with credit products being rolled out gradually under the foundational phase's asset limitations (BNM, 2026). The journey illustrates three themes of this chapter: the shift from FinTech 3.0 start-ups to FinTech 4.0 platform-backed banking (BigTech names appear throughout the shareholder lists); the centrality of regulation in shaping market structure (only five licences, tightly conditioned); and the unresolved question of whether inclusion-oriented digital banking can also be commercially viable under a RM3 billion asset cap.

Discussion Questions

1. BNM chose to licence consortia that pair BigTech firms (Grab, Sea) with established financial or corporate groups (Kuok, RHB, YTL, AEON, KAF). What advantages and risks does this pairing strategy create, compared with licensing stand-alone start-ups?
2. The foundational phase caps total assets at RM3 billion for three to five years. Explain how this cap protects the financial system, and analyse how it simultaneously limits the digital banks' ability to earn lending margins.
3. Two of the five digital banks are Islamic (AEON Bank and KAF Digital Bank). Why might Shariah-compliant digital banking be strategically important for Malaysia's inclusion goals and for its ambition as a global Islamic finance hub under MIFC?
4. GXBank took roughly 16 months from licence announcement to launch. Identify the major tasks a new digital bank must complete in this period, and discuss which you think is the hardest.

Chapter Summary

- **1.1** FinTech is defined as digital innovations and technology-enabled business model innovations in the financial sector (Puschmann, 2017); the term is used both for an *industry* of firms and for a *toolbox of technologies*.
- **1.1** A digital bank conducts banking business wholly or almost wholly through digital or electronic means under an FSA 2013 or IFSA 2013 licence (BNM, 2020); digital banking differs from online banking (a channel of an incumbent) and from the global neobank concept.
- **1.2** Arner et al. (2016) trace FinTech 1.0 (1866–1967, analogue globalisation), FinTech 2.0 (1967–2008, incumbents digitalising) and FinTech 3.0/3.5 (post-2008 new entrants and Asia's mobile-first leapfrog).
- **1.2** Subsequent literature frames an emerging FinTech 4.0 era of platformisation, BigTech, AI and embedded finance (Gupta & Tham, 2019).
- **1.3** Malaysia's ecosystem is co-regulated by BNM (banks, payments, e-money) and the SC (P2P, ECF, DAX, DIM, IEOs), supported by PayNet, MDEC, the Fintech Association of Malaysia and MIFC.

- **1.3** Key segments — payments, digital banks, e-wallets, P2P/ECF, DIM, DAX, insurtech and Islamic fintech — show rapid, well-documented growth, including 18.4 billion e-payment transactions in 2025 (BNM, 2026).
- **1.4** Five player groups compete and cooperate: incumbent banks digitising, digital challenger banks, BigTech, fintech start-ups and infrastructure providers.
- **1.4** Fintech business models include payments fees, interchange, subscription, marketplace/platform fees, lending margins and freemium; sustainable firms usually combine several (Lee & Shin, 2018).
- **1.5** Benefits include cost efficiency, convenience and financial inclusion — Malaysian adult account ownership rose from 66.2% (2011) to 88.7% (2024) (World Bank, 2022, 2025).
- **1.5** Challenges include the digital divide, trust and fraud, regulatory compliance costs, and the operational viability of digital banks constrained by the RM3 billion asset cap during their foundational phase (BNM, 2020).

Key Terms

- **BigTech**: Large technology companies with vast user bases and data advantages that expand into financial services such as payments, lending and banking.
- **Business model**: The plan describing how a firm creates, delivers and captures value, including its sources of revenue and cost structure.
- **Digital asset exchange (DAX)**: A platform registered with the Securities Commission Malaysia on which digital assets, prescribed as securities under Malaysian law, may be traded.
- **Digital banking**: The delivery of licensed banking services wholly or almost wholly through digital or electronic means, with remote onboarding and no reliance on branches.
- **Digital investment management (DIM)**: SC-licensed automated investment services — robo-advisory — covering risk profiling, suitability assessment, asset allocation and rebalancing with minimal human supervision.
- **Electronic know-your-customer (e-KYC)**: Regulator-approved digital processes for verifying a customer's identity remotely, enabling non-face-to-face account opening.
- **Electronic money (e-money)**: Digitally stored monetary value issued under a BNM licence that holders can use to pay merchants or transfer to other users.
- **Embedded finance**: The integration of financial services such as payments, lending or insurance directly into non-financial apps and platforms at the point of need.
- **Equity crowdfunding (ECF)**: The raising of equity capital by companies from many investors through an online platform registered with the Securities Commission Malaysia.
- **E-wallet**: A smartphone application that stores e-money and enables payments, transfers and often additional services such as investments or insurance.
- **Financial inclusion**: Access of individuals and businesses to useful, affordable financial products and services — payments, savings, credit and insurance — delivered responsibly and sustainably.
- **Financial technology (FinTech)**: Digital innovations and technology-enabled business model innovations in the financial sector; used for both the industry of firms and the underlying technologies.
- **Freemium**: A business model offering basic services free to build a large user base, then charging for premium features or cross-selling other products.
- **Interchange fee**: The portion of a card transaction fee paid by the merchant's bank to

the cardholder's issuing bank, forming a key revenue source for card issuers.
- **Neobank**: Global industry term for a digital-only, branchless banking provider; in Malaysia the nearest equivalents are the five fully licensed digital banks.
- **Online banking**: The provision of banking services over the internet as an additional channel of a traditional branch-based bank, such as Maybank2u or CIMB Clicks.
- **Peer-to-peer (P2P) financing**: The raising of debt funding by businesses from many investors through an online platform registered with the Securities Commission Malaysia.
- **Regulatory sandbox**: A supervised framework allowing firms to test innovative financial services with real customers under relaxed conditions and safeguards before full licensing.

Test Your Knowledge

Part A: Multiple Choice

1. According to Puschmann (2017), FinTech is best defined as: A. any company that uses computers B. digital innovations and technology-enabled business model innovations in the financial sector C. cryptocurrencies and blockchain only D. software sold exclusively to banks

2. According to Arner et al. (2016), FinTech 1.0 began in 1866 with the: A. installation of the first ATM B. founding of SWIFT C. completion of the first successful transatlantic telegraph cable D. publication of the Bitcoin whitepaper

3. In the framework of Arner et al. (2016), "FinTech 3.5" refers to: A. the era of mainframe computers in banks B. post-crisis regulation of American banks C. markets such as China and India that leapfrogged directly to mobile-first financial services D. central bank digital currency experiments

4. Which set of forces characterises the emerging FinTech 4.0 era as framed by Gupta and Tham (2019)? A. Telegraphs, payment cards and branch networks B. Platformisation, BigTech, artificial intelligence and embedded finance C. Cheques, passbooks and telephone banking D. Gold standard, paper currency and clearing houses

5. In Malaysia, peer-to-peer financing and equity crowdfunding are regulated by the: A. Bank Negara Malaysia B. Securities Commission Malaysia C. Payments Network Malaysia D. Malaysia Digital Economy Corporation

6. Which of the following is NOT one of Malaysia's five licensed digital banks? A. GXBank B. Boost Bank C. AEON Bank D. Maybank

7. During the foundational phase, a Malaysian digital bank's total assets must not exceed: A. RM300 million B. RM3 billion C. RM30 billion D. There is no limit

8. An e-wallet offers a free basic account but charges users for premium features. This is an example of the: A. marketplace model B. lending-margin model C. freemium model D. interchange model

9. According to the Global Findex Database, adult account ownership in Malaysia in 2024 was approximately: A. 66.2% B. 80.7% C. 85.3% D. 88.7%

10. A licensed digital bank attracts large deposits from underserved customers but cannot expand its financing book because of the asset cap. Its MOST realistic additional revenue strategy is to: A. ignore the cap and keep lending B. build fee-based income such as

payments, interchange and subscriptions C. close its smallest deposit accounts D. open physical branches in city centres

Part B: Short Answer

1. Distinguish between online banking, digital banking and a neobank, giving a Malaysian example of each where possible.
2. Explain two reasons why Arner et al. (2016) treat 2008 as the turning point between FinTech 2.0 and FinTech 3.0.
3. Describe the respective roles of Bank Negara Malaysia and the Securities Commission Malaysia in the fintech ecosystem, identifying two market segments supervised by each.
4. Identify two challenges that Malaysia's digital banks face in achieving operational viability during the foundational phase, and explain how fee-based business models could help.
5. Using evidence from the Global Findex Database, discuss how digital financial services have contributed to financial inclusion in Malaysia.

Answer Guide

Part A

1. **B** — Puschmann's (2017) definition combines digital innovation with business model innovation in finance.
2. **C** — The 1866 transatlantic telegraph cable opened FinTech 1.0; the ATM (1967) closed it.
3. **C** — FinTech 3.5 labels Asian markets that skipped card-and-branch stages and went mobile-first.
4. **B** — Platformisation, BigTech, AI and embedded finance define the 4.0 framing (Gupta & Tham, 2019).
5. **B** — The SC regulates capital market fintech; BNM regulates banks, payments and e-money.
6. **D** — Maybank is an incumbent conventional bank, not one of the five digital bank licensees.
7. **B** — The Licensing Framework for Digital Banks imposes a RM3 billion asset limit in the foundational phase (BNM, 2020).
8. **C** — Free basic service with paid premium features is the freemium model.
9. **D** — Account ownership reached 88.7% in 2024 (World Bank, 2025).
10. **B** — The cap limits the balance sheet, not fee income, making payments, interchange and subscriptions the viable path.

Part B (model answers)

1. Online banking is a digital channel added to an incumbent branch-based bank (e.g., Maybank2u, launched 2000). Digital banking is banking conducted wholly or almost wholly through digital or electronic means by an institution holding a digital bank licence (e.g., GXBank, Boost Bank), with remote e-KYC onboarding and no branches. A neobank is the global term for a digital-only bank; in Malaysia the closest equivalents are the five licensed digital banks, since all hold full BNM licences rather than partnering with other banks.
2. First, the 2008 global financial crisis damaged trust in incumbent banks and increased regulatory pressure on them, creating space for new entrants. Second, smartphones

(from 2007) matured into an always-connected delivery channel, letting start-ups reach consumers directly with payments, lending and investment apps — shifting innovation from incumbents (FinTech 2.0) to new entrants (FinTech 3.0).

3. BNM, under the FSA 2013, IFSA 2013 and Central Bank of Malaysia Act 2009, regulates banks (including the five digital banks) and payment systems/e-money; segments include payments/e-wallets and digital banking. The SC, under the Capital Markets and Services Act 2007, regulates capital market fintech; segments include P2P financing and equity crowdfunding (also DAXs and DIM). Accept any two valid segments per regulator.
4. Challenges: (i) the RM3 billion asset cap limits the financing book and hence lending-margin income; (ii) heavy upfront technology and compliance costs (e.g., RMiT) combined with free customer acquisition delay profitability. Fee-based models — payments fees, interchange on card spend, subscriptions and marketplace income — generate revenue that does not enlarge the balance sheet, allowing growth in income even while assets are capped.
5. Account ownership rose from 66.2% (2011) to 88.7% (2024), the gender gap fell below one percentage point, and ownership among the poorest 40% reached 82.0% (World Bank, 2022, 2025). Digital services support this trend: e-KYC enables remote onboarding, DuitNow QR reached about 3 million merchant touchpoints including formerly cash-only MSMEs, and P2P financing serves small businesses banks overlook. However, digital divide risks (elderly, rural connectivity) mean inclusion gains are not automatic.

Problem-Solving Corner

Problem 1: Interchange Revenue of a Digital Bank

A licensed digital bank issues a debit card linked to its savings account. In one month, cardholders make 2,000,000 purchase transactions with an average value of RM50. Assume, purely for illustration, that the bank earns an average interchange fee of 0.5% of transaction value. The bank's monthly processing and servicing cost for the card programme is RM450,000.

(a) Calculate the monthly card spending volume, monthly interchange revenue, and annual interchange revenue. **(b)** Calculate the net monthly contribution of the card programme. **(c)** The bank's total operating cost is RM30 million per year. Comment on whether interchange alone can sustain the bank.

Solution

(a) Monthly spending volume = 2,000,000 × RM50 = **RM100,000,000**. Monthly interchange revenue = RM100,000,000 × 0.5% = **RM500,000**. Annual interchange revenue = RM500,000 × 12 = **RM6,000,000 (RM6 million)**.

(b) Net monthly contribution = RM500,000 − RM450,000 = **RM50,000** (RM600,000 per year).

(c) Annual net contribution from cards (RM0.6 million) covers only 2% of the RM30 million operating cost. Interchange alone cannot sustain the bank. The bank must add other revenue lines — lending margins (within the RM3 billion asset cap), subscriptions, marketplace or partnership fees — illustrating why successful fintech firms combine several business models (Lee & Shin, 2018).

Problem 2: Measuring Financial Inclusion Progress

The Global Findex reports Malaysian adult account ownership of 66.2% in 2011 and 88.7% in 2024 (World Bank, 2022, 2025). For parts (b)–(c), *assume* an adult population of 24.0 million in both years (an illustrative simplification).

(a) Calculate the change in account ownership in percentage points and as a percentage increase relative to 2011. **(b)** Estimate the number of adults *without* an account in 2011 and in 2024. **(c)** By how much did the estimated unbanked adult population fall?

Solution

(a) Change in percentage points = 88.7% − 66.2% = **22.5 percentage points**. Relative increase = 22.5 ÷ 66.2 × 100% ≈ **34.0%**.

(b) Unbanked share in 2011 = 100% − 66.2% = 33.8% → 33.8% × 24.0 million ≈ **8.11 million adults**. Unbanked share in 2024 = 100% − 88.7% = 11.3% → 11.3% × 24.0 million ≈ **2.71 million adults**.

(c) Reduction ≈ 8.11 − 2.71 = **5.40 million fewer unbanked adults** (under the constant-population assumption; in reality population grew, so interpret this as an illustration of method).

Problem 3: The RM3 Billion Asset Cap and Growth Planning

Digital bank *Bank Maya* (a fictional example) is in its foundational phase with total assets of RM2.7 billion, of which RM2.2 billion is its financing (loan) book. Its business plan calls for growing the financing book by 25% next year. BNM's asset limit is RM3 billion (BNM, 2020).

(a) If the plan is executed and all growth appears as new financing assets, what would total assets become? Does the bank breach the cap, and by how much? **(b)** What is the maximum financing growth (in RM and %) the bank can book without breaching the cap, assuming other assets stay constant? **(c)** Suggest two strategies that would let Bank Maya keep growing income without breaching the cap.

Solution

(a) New financing = 25% × RM2.2 billion = RM0.55 billion → total assets = RM2.7 + RM0.55 = **RM3.25 billion**. This **breaches the cap by RM250 million** (RM3.25b − RM3.0b).

(b) Available headroom = RM3.0b − RM2.7b = **RM300 million**. Maximum financing growth = RM300 million ÷ RM2.2 billion ≈ **13.6%**, not 25%.

(c) Strategies: (i) pivot towards *fee-based income* — payments, interchange, subscriptions, marketplace or referral fees — which adds revenue without adding assets; (ii) *distribute rather than hold* financing, e.g., originate financing and sell or refer it to partners, or focus on smaller, higher-yield Shariah-compliant or unsecured products within the headroom. Both keep growth inside the foundational-phase constraint while building a path to operational viability.

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Chapter 2: Digital Banking, Open Banking and Payment Systems

Chapter Outline

- **2.1** Mobile and Digital Banking Services
- **2.2** Digital Customer Journey and e-KYC
- **2.3** Open Banking and APIs
- **2.4** Digital Payment Systems (QR, Instant Transfer, International)
- **2.5** Stablecoins and Central Bank Digital Currencies (CBDCs)

Learning Outcomes

After studying this chapter, you should be able to:

1. **Describe** the evolution of digital banking in Malaysia, from internet banking to mobile-first applications and app-only digital banks. *(Bloom's level: Remember)*
2. **Explain** how electronic know-your-customer (e-KYC) procedures and tiered customer due diligence enable secure non-face-to-face onboarding. *(Bloom's level: Understand)*
3. **Apply** cost-comparison methods to digital payment and remittance decisions faced by consumers and small businesses. *(Bloom's level: Apply)*
4. **Differentiate** between mandated and market-led open banking models, and between wholesale and retail central bank digital currencies. *(Bloom's level: Analyse)*
5. **Evaluate** the benefits and risks of data sharing, instant payments and digital currencies for Malaysian consumers and merchants. *(Bloom's level: Evaluate)*
6. **Develop** a practical cashless-adoption plan for a Malaysian micro, small or medium enterprise (MSME). *(Bloom's level: Create)*

Opening Vignette

Aina, 19, is a foundation student at a polytechnic in Johor Bahru. On a Monday morning she downloads a banking app, photographs her MyKad, records a short selfie video, and — fifteen minutes later — her first savings account is approved. A virtual debit card appears in the app instantly; the plastic card arrives by post days later. At the campus canteen she pays for nasi lemak by scanning a DuitNow QR sticker. At lunch she splits the bill with three friends using DuitNow transfers addressed to their mobile numbers — no bank account numbers needed. In the evening she sends RM100 to her grandmother in Mersing, checks the app's spending summary, and moves RM50 into a savings goal. During the semester break, she pays for boat

noodles in Bangkok by scanning a Thai QR code with the same Malaysian app. Not once did she visit a branch, fill in a paper form, or handle cash she could not immediately replace. This chapter explains the services, regulations and payment rails that make Aina's ordinary day possible — and the risks that regulators and banks must manage behind the scenes.

2.1 Mobile and Digital Banking Services

2.1.1 From internet banking to mobile-first

Digital banking is the delivery of banking products and services — accepting deposits, moving money, granting credit — through digital channels such as websites and mobile applications rather than physical branches (Thakor, 2020). It is one visible part of the broader fintech transformation, in which technology changes how financial services are produced, distributed and consumed (Arner et al., 2016; Gomber et al., 2018).

Malaysia's digital banking journey began on the desktop computer. **Maybank2u**, launched in 2000, is recognised as Malaysia's first internet banking platform, allowing customers to check balances and transfer funds from a web browser at a time when most banking still required a branch visit. Other banks followed, and by the 2010s internet banking had become a standard expectation rather than a novelty.

The smartphone shifted the centre of gravity again. Banks moved from designing websites that were later squeezed onto phone screens to a **mobile-first** design philosophy, in which the mobile application is the primary channel and other channels are built around it (Gupta & Tham, 2019). Today the flagship applications of the largest banks — **MAE by Maybank**, which combines banking with wallet-style features, and CIMB's pairing of **CIMB Clicks** (web) with **CIMB Octo** (mobile) — serve as the main daily touchpoint between millions of Malaysians and their banks. Features that once required a counter visit, such as opening an account or disputing a transaction, are increasingly completed inside the app.

Era	Dominant channel	Typical tasks	Access pattern
1990s	Branch counter	Deposits, withdrawals, loan applications	Physical, office hours
2000s	Internet banking (web)	Balance checks, transfers, bill payment	Desktop browser
2010s–2020s	Mobile-first apps	Full service: onboarding, payments, cards, savings tools	Smartphone, 24/7
2020s	App-only digital banks	Entire banking relationship inside one app	Smartphone only

Table 2.1: The shifting primary channel of Malaysian retail banking.

2.1.2 What modern banking apps can do

Although each bank's app differs, four clusters of features now define the Malaysian mobile banking experience.

1. **Remote account opening.** New customers can open savings or e-wallet accounts entirely within the app, using the electronic identity checks described in Section 2.2. This removes geography as a barrier to joining the formal financial system.
2. **Payments and transfers.** Apps connect directly to national payment rails — DuitNow for instant transfers, Interbank GIRO for ordinary transfers, FPX for online checkout payments and JomPAY for bills (see Section 2.4). To the customer these appear as simple menu options; underneath, each routes through different infrastructure.
3. **Personal financial management (PFM).** Many apps now categorise spending automatically, display monthly summaries, and let users set budgets and savings goals. These tools turn raw transaction lists into guidance, helping customers — especially students and first-jobbers — understand where their money goes.
4. **Virtual cards.** A **virtual card** is a payment card credential that exists only in digital form, issued instantly inside an app and usable for online purchases or wallet-based payments before any plastic card is printed (Chishti et al., 2020). Virtual cards can be frozen, deleted or limited to single merchants, giving users finer control over security.

Modern apps also embed security controls that have become standard across Malaysian banking, such as biometric login, device binding (linking an account to one secured device) and a kill-switch that lets customers freeze their own accounts quickly if they suspect fraud. These controls reflect the technology-risk obligations that Bank Negara Malaysia places on financial institutions (Bank Negara Malaysia [BNM], 2025b).

2.1.3 Digital banks: the app-only model

A **digital bank** is a licensed bank that delivers all of its services through digital channels, with no physical branch network at all (BNM, 2020). In December 2020 BNM issued its *Licensing Framework for Digital Banks*, and on 29 April 2022 it announced five successful applicants. To protect stability while these new institutions prove themselves, each digital bank operates under an asset limit of RM3 billion during a foundational phase of three to five years (BNM, 2020).

The five licensees entered the market in stages:

- **GXBank** (led by Grab with the Kuok Group) — the first to begin operations, on 1 September 2023.
- **Boost Bank** (Boost/Axiata with RHB Banking Group) — commenced 15 January 2024.
- **AEON Bank** (AEON Financial Service, AEON Credit and MoneyLion) — launched 26 May 2024 as Malaysia's first Islamic digital bank.
- **KAF Digital Bank** (a consortium led by KAF Investment Bank) — approved to commence operations on 20 December 2024, as the second Islamic digital bank.
- **Ryt Bank** (Sea Ltd with YTL Digital Capital) — approved to commence operations on 20 December 2024.

The app-only model matters for three reasons. First, without branches, digital banks have far lower fixed costs, which makes it economical to serve customers with small balances. Second, their products are deliberately simple — savings, payments and, gradually, small-ticket credit — designed for unserved and underserved segments rather than to replicate everything a large bank does. BNM has observed that the digital banks remain in their foundational phase

under asset-size limitations, with deposits growing from underserved segments and credit products rolling out gradually (BNM, 2026). Third, their presence pressures incumbent banks to improve their own apps and pricing — competition as a policy tool.

The model also carries risks: new institutions must build trust without a physical presence, achieve profitability under an asset cap, and manage cyber risk with no room for operational failure. Students should therefore view digital banks as an important experiment in inclusion, not as a guaranteed replacement for traditional banks.

> **QUICK NOTE:** A mobile banking app and a digital bank are not the same thing. MAE by Maybank is a *channel* of a branch-based bank; GXBank or AEON Bank is a separately *licensed institution* that exists only as an app.
>
> **DID YOU KNOW? (TRIVIA):** When Maybank2u launched in 2000, checking your balance without visiting a branch felt revolutionary. By 2025, Malaysians were making an average of 538 electronic payment transactions each per year (BNM, 2026) — a behavioural revolution in one generation.

2.2 Digital Customer Journey and e-KYC

2.2.1 The digital onboarding funnel

Onboarding is the process by which a person becomes a customer of a financial institution. In the digital era it is often described as a funnel, because some potential customers drop out at every stage.

Stage	What happens	Typical friction
1. Discovery and download	Customer finds and installs the app	Data cost, phone storage, trust
2. Registration	Personal details, mobile number verification	Form fatigue, typing errors
3. Identity verification	Document capture, selfie, database checks	Poor lighting, damaged identity card
4. Approval	Risk checks, account tier assigned	Manual review delays
5. Activation	First deposit, first transaction	Funding hurdle, habit formation

Table 2.2: The digital onboarding funnel.

Each abandoned application represents not just a lost customer but potentially a person remaining outside the formal financial system. This matters in Malaysia: adult account ownership rose from 85.3% in 2017 to 88.4% in 2021 and 88.7% in 2024, with a gender gap below one percentage point — yet about one in nine adults, and 18% of the poorest 40%, still lacked an account in 2024 (World Bank, 2022, 2025). Smooth digital onboarding is one of the most practical tools for closing that gap.

2.2.2 e-KYC under Bank Negara Malaysia's policy

Electronic know-your-customer (e-KYC) is the electronic verification of a customer's identity, enabling an account to be opened without any face-to-face meeting at a branch (BNM,

2024). BNM first issued its e-KYC policy document in 2020 and revised it in April 2024, setting out how banks, insurers and e-money issuers may conduct non-face-to-face verification safely (BNM, 2024).

A typical e-KYC flow combines four building blocks:

- **Document capture.** The customer photographs an identity document, usually the MyKad. Software checks security features and reads the printed data.
- **Liveness detection. Liveness detection** is a set of techniques that confirm a live human being is present during verification — for example, asking the user to blink, turn their head or record a short video — so that a photograph, video replay or mask cannot fool the system (BNM, 2024).
- **Database checks.** The captured details are checked against authoritative databases to confirm the document is genuine and the identity exists.
- **Risk assessment and audit trail.** The institution scores the application's risk and records evidence, so that every approved account can be justified later to supervisors.

The policy significance is large. Before e-KYC, regulation effectively required a branch visit for full account opening, which excluded rural customers, shift workers and anyone far from a bank. Non-face-to-face onboarding removes that barrier while keeping identity assurance high. It is no coincidence that Malaysia's five digital banks — which have no branches at all — could only exist after the e-KYC framework was in place.

2.2.3 AML/CFT and tiered customer due diligence

Opening accounts remotely creates an obvious tension: the same convenience that helps a student in Johor Bahru could also help a criminal open accounts for laundering money. Malaysian law addresses this through the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (Act 613), known as AMLA, which obliges reporting institutions — banks, e-money issuers and others — to know who their customers are and to report suspicious transactions (Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001).

The operational tool is **customer due diligence (CDD)**: the process by which an institution identifies and verifies a customer, understands the purpose of the relationship, and monitors transactions against that understanding (Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001). Crucially, Malaysian regulation applies CDD on a risk-sensitive, tiered basis:

Tier	When applied	Practical meaning
Simplified due diligence	Lower-risk products, e.g., basic accounts with balance or transaction limits	Fast onboarding; fewer checks; tighter usage caps
Standard due diligence	Ordinary customers and products	Full e-KYC verification as in Section 2.2.2
Enhanced due diligence	Higher-risk customers (e.g., politically exposed persons) or unusual patterns	Extra verification, senior approval, closer monitoring

Table 2.3: Tiered customer due diligence.

Tiering is the compromise that makes both inclusion and integrity possible. A basic account with small limits can be opened in minutes with simplified checks; a customer who later wants higher limits or complex products undergoes progressively deeper verification. The design principle — proportionality between risk and control — recurs throughout financial regulation, and you will meet it again when we discuss fraud controls in later chapters.

Money laundering itself — the process of disguising criminal proceeds so they appear legitimate — and terrorism financing harm not only institutions but society: mule accounts opened through weak checks are a key enabler of the scam networks that Malaysian authorities, through bodies such as the National Scam Response Centre (hotline 997), work to disrupt (BNM, 2026).

> **BRAINSTORMER:** Tighter identity checks stop criminals, but they also slow down honest customers and can exclude people with damaged documents or old phones. Where should the balance lie — and who should bear the cost when the balance is wrong?

2.3 Open Banking and APIs

2.3.1 APIs and the idea of open banking

An **application programming interface (API)** is a standardised set of rules and protocols that allows one software system to request data or services from another in a controlled, machine-readable way (Chishti et al., 2020). When a weather app on your phone shows tomorrow's forecast, it has almost certainly pulled that data from a meteorological service through an API. The same technique can move *financial* data — with far higher stakes for privacy and security.

Open banking is the practice of banks sharing customer data with authorised third-party providers through secure APIs, but only with the customer's explicit consent (Zetzsche et al., 2017). A typical use case is account aggregation: a budgeting app that, with your permission, reads balances and transactions from several banks so you can see your whole financial life in one place. Other use cases include applying for a loan using verified income data instead of paper payslips, or initiating a payment from a merchant's app directly from your bank account.

The concept is now widening into **open finance**, which extends consented data sharing beyond bank accounts to a broader range of financial data — such as investments, insurance and pensions — under the same principles of consent and standardised access (BNM, 2022). Open finance recognises that a customer's financial life is spread across many institutions, and that portability of that data can increase competition and better products.

> **QUICK NOTE:** APIs are the *safe* way to share financial data. The unsafe alternative is "screen scraping", where you hand your banking username and password to a third party that logs in as you. APIs give granular, revocable permission for specific data; screen scraping hands over the keys to everything.

2.3.2 Two models: mandated and market-led

Around the world, open banking has followed two broad models.

In the **mandated model**, law compels banks to open their data. The European Union's revised Payment Services Directive (Directive (EU) 2015/2366, known as PSD2) requires banks

to provide licensed third parties with API access to account data and payment initiation, with customer consent (European Parliament and Council, 2015). The United Kingdom went further: after a competition inquiry, the Competition and Markets Authority ordered the nine largest banks to build common, standardised APIs, giving rise to the UK's Open Banking framework from 2018. Mandates produce fast, uniform adoption, but impose heavy compliance costs and can freeze standards into law before technology settles.

In the **market-led model**, regulators encourage and shape data sharing but leave the pace and specific arrangements to the industry. The United States evolved this way through bilateral commercial agreements; Singapore and Malaysia have adopted similar philosophies, with the central bank acting as catalyst rather than lawgiver. Market-led approaches are flexible and let standards follow real use cases, but adoption can be slower and patchier, and large players may prefer closed ecosystems.

Dimension	Mandated model (EU, UK)	Market-led model (Malaysia)
Legal basis	PSD2; UK CMA order	Blueprint encouragement, industry standards
Who must share	Covered banks, by law	Voluntary, use-case driven
Pace	Fast and uniform	Gradual and flexible
Main benefit	Level playing field, rapid entry	Innovation fitted to local needs
Main risk	Compliance cost, rigid standards	Fragmentation, slow coverage

Table 2.4: Mandated versus market-led open banking.

2.3.3 Malaysia's open finance direction

Malaysia has firmly chosen the market-led path. The BNM *Financial Sector Blueprint 2022–2026* sets five desired outcomes for the financial system and commits the central bank to advancing open finance, alongside a target of roughly 15% compound annual growth in e-payments over the blueprint period (BNM, 2022). Rather than ordering banks to open their data, BNM works with the industry on common standards and use cases where sharing creates clear value — for example, helping SMEs obtain credit on the strength of their transaction histories rather than collateral alone.

The potential benefits are substantial:

- **Competition.** New providers can build services on customer data that incumbents previously locked away.
- **Better products.** Consolidated data enables more accurate budgeting tools, credit scoring and product recommendations.
- **Financial inclusion.** Thin-file customers — gig workers, micro-entrepreneurs — can prove creditworthiness through cash-flow data instead of formal documents.

The risks are equally real, and they centre on **consent**. Did the customer genuinely understand what they agreed to share, for how long, and with whom? Poorly designed consent screens create "consent fatigue", in which users tap Agree without reading — a risk noted in the broader literature on data-driven finance (Zetzsche et al., 2017). Shared data must also be protected: the Personal Data Protection Act 2010 (Act 709), amended in 2024, governs how commercial entities collect, use and safeguard personal data (Personal Data Protection Act

2010), while BNM's Risk Management in Technology (RMiT) policy imposes cyber-resilience duties on financial institutions that build or consume APIs (BNM, 2025b). Open banking succeeds only if customers trust it; a single high-profile data breach could set adoption back years.

> **BRAINSTORMER:** Would you allow a budgeting app to read all your bank transactions? What would you demand in return — and what would make you withdraw consent later?

2.4 Digital Payment Systems (QR, Instant Transfer, International)

2.4.1 The rails beneath the apps

Every tap, scan or click ultimately travels on a payment rail — shared infrastructure that moves money between institutions. In Malaysia these rails are operated by **Payments Network Malaysia (PayNet)**, the national payments operator owned by Bank Negara Malaysia. The table below summarises the principal rails you should know.

Rail	Launched	Function	Character
RENTAS	1999	High-value interbank settlement	Real-time, final settlement
IBG	—	Ordinary account-to-account transfers	Cleared in batches, not instant
FPX	—	Online bank-to-bank payments at checkout	Near-instant authorisation
JomPAY	2015	National bill-payment scheme	Pay any registered biller via any bank
MyDebit	—	Domestic debit card payments	Card-present and online
DuitNow	2018	Instant transfers using proxies	Real-time, 24/7
DuitNow QR	2019	National interoperable QR standard	Real-time, scan-based

Table 2.5: Malaysia's principal payment rails (Payments Network Malaysia [PayNet], 2026).

Three ideas in this table deserve precise definitions. First, RENTAS is a **real-time gross settlement (RTGS)** system: one in which high-value payments between banks are settled individually, immediately and with finality, rather than being netted in batches — the backbone on which all other rails ultimately settle (Chishti et al., 2020). Second, **DuitNow** introduced proxy-based instant transfer: instead of asking for a bank account number, the sender addresses money to a **proxy** such as the recipient's mobile number or MyKad number, and PayNet's addressing service routes it to the registered account. Third, **DuitNow QR** is a single national QR standard (QR stands for quick-response code): one QR code at a merchant can

be scanned by any participating banking app or e-wallet, ending the confusion of competing, incompatible codes (PayNet, 2026).

2.4.2 Malaysia's e-payment boom: the numbers

The scale of adoption is easiest to grasp in numbers. PayNet processed 8.44 billion digital payment transactions in 2025 (PayNet, 2026). Across all e-payment channels, BNM counted 18.4 billion transactions in 2025, up 25% from 14.7 billion in 2024 — an average of 538 transactions for every Malaysian (BNM, 2026). The value of retail e-payments reached RM831 billion in 2025, up from RM698.2 billion in 2024, while the average e-money transaction was RM43, up from RM33 a year earlier (BNM, 2026).

DuitNow QR is the standout story. Its transaction volume roughly doubled to about 3 billion transactions in 2025, from 1.5 billion in 2024, and merchant acceptance grew to approximately 3 million touchpoints, from 2.6 million in 2024 — many of the new acceptors being previously cash-only MSMEs such as hawker stalls and pasar malam traders (BNM, 2026; PayNet, 2026).

What explains the boom? Network effects play a central role: each new merchant QR makes consumer apps more useful, and each new user makes acceptance more attractive to merchants. Habit persistence matters too — payment habits formed during the pandemic years endured. And crucially, DuitNow QR settles directly into the merchant's bank or wallet account through rails they already trust, with simple pricing that a micro-business can absorb. Critics of the cashless shift, however, warn that digital payment systems can exclude those without smartphones or connectivity, and concentrate power in the platforms that run them (Scott, 2022). A balanced view keeps both the efficiency gains and the exclusion risks in sight.

> **MALAYSIAN SPOTLIGHT:** Payments Network Malaysia (PayNet), owned by Bank Negara Malaysia, is the quiet giant of Malaysian fintech. Its DuitNow QR standard reached roughly 3 million merchant touchpoints by 2025 — many of them MSMEs that were cash-only until recently — and processed about 3 billion QR transactions in 2025 alone (PayNet, 2026). When you buy roti canai with your phone, you are using national infrastructure, not just an app.

2.4.3 E-wallets and electronic money

Electronic money (e-money) is a prepaid, stored-value payment instrument: the user loads funds in advance and spends from that balance, with the issuer obliged to hold equivalent backing funds. In Malaysia, e-money issuers are licensed by BNM under section 11 of the Financial Services Act 2013 or, for Islamic issuers, the Islamic Financial Services Act 2013 (Financial Services Act 2013; Islamic Financial Services Act 2013).

The major wallets are household names: Touch 'n Go eWallet (operated by TNG Digital), MAE by Maybank, GrabPay, Boost, ShopeePay and BigPay. Early wallets were closed gardens — a balance in one could not pay a merchant signed to another. DuitNow QR changed that: because the national QR standard is interoperable, a merchant's single QR can now accept payment from banking apps and multiple wallets alike, and wallets increasingly connect to the same rails as banks. For consumers, e-money sits alongside deposits rather than replacing them; for merchants, the practical question is no longer *which* app a customer uses, but simply whether the business displays its QR.

2.4.4 Cross-border QR payments and remittances

Malaysia has extended DuitNow QR beyond its borders through bilateral linkages that let travellers pay overseas with their home banking app or wallet, in ringgit, at the foreign merchant's local QR code. Cross-border QR linkages are live with Thailand, Indonesia, Singapore and China; Cambodia was added in 2025, and India is expected in 2026. Usage is growing rapidly — cross-border QR transactions grew about 2.5 times in 2025 (BNM, 2026). For a Malaysian student in Bangkok, this means scanning a Thai merchant's QR and paying from a Malaysian account, with the currency conversion shown in the app; for Thai tourists in Kuala Lumpur, the mirror image applies.

A different cross-border flow is the **remittance**: a cross-border transfer of funds, typically sent by migrant workers to family members in their home country (World Bank, 2022). Malaysia is a major remittance-sending country, hosting large communities of workers from Indonesia, Bangladesh, Nepal and elsewhere. Remittances have historically been expensive, especially through cash-based agents. The World Bank's Remittance Prices Worldwide database tracks these costs and reports that the global average cost of sending US$200 remains several per cent of the amount sent (World Bank, n.d.). Sustainable Development Goal target 10.c commits countries to reducing the cost of migrant remittances to below 3% by 2030.

Fintech providers attack this cost directly. **MoneyMatch**, a Malaysian cross-border remittance firm that graduated from BNM's regulatory sandbox, illustrates the model: digital onboarding via e-KYC, transparent fees and exchange rates shown up front, and delivery to overseas bank accounts — typically at a fraction of traditional costs. When cheaper formal channels exist, fewer senders resort to informal ones, improving both consumer welfare and regulatory visibility.

> **DID YOU KNOW? (TRIVIA):** Malaysia's real-time gross settlement system, RENTAS, has been running since 1999 — older than most of this book's readers. The QR code a hawker displays today ultimately settles on infrastructure that has moved interbank billions for more than two decades (PayNet, 2026).

2.5 Stablecoins and Central Bank Digital Currencies (CBDCs)

2.5.1 Stablecoins: private digital money

A **stablecoin** is a crypto-asset designed to maintain a stable value relative to a reference asset, most commonly a national currency such as the US dollar (Arslanian, 2022). The most familiar design is the **fiat-collateralised** stablecoin: the issuer holds reserves — bank deposits, treasury bills and similar assets — and issues tokens one-for-one against them. Tether (USDT) and USD Coin (USDC), both pegged to the US dollar, are the largest examples.

Stablecoins matter because they combine the programmability and 24/7 transferability of crypto-assets with (in principle) the price stability of fiat money. They are widely used for settlement between crypto trades and, increasingly, for cross-border transfers. The risks, however, are serious: if reserves prove inadequate or opaque, a stablecoin can "de-peg" and collapse in value; because holders can redeem quickly, stablecoins are exposed to run dynamics similar to unregulated bank deposits.

Malaysia's stance is cautious. Digital assets are not legal tender in Malaysia; under the Cap-

ital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019, digital currencies and digital tokens are prescribed as securities, placing their issuance and trading within the securities regulatory perimeter (Capital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019). At the same time, BNM is studying the underlying concepts seriously: work on tokenised deposits and ringgit-denominated stablecoins forms part of its 2026 agenda (BNM, 2025a).

2.5.2 Central bank digital currencies: wholesale and retail

A **central bank digital currency (CBDC)** is a digital form of central-bank money — a direct liability of the central bank, denominated in the national unit of account (Auer et al., 2020; Bank for International Settlements [BIS], 2021). Two forms are distinguished:

- A **wholesale CBDC** is restricted to banks and other licensed financial institutions, used for interbank settlement and securities transactions. Think of it as a modernised version of the settlement balances banks already hold at the central bank.
- A **retail CBDC** would be available to the general public — households and businesses — as a digital equivalent of cash.

Central banks explore CBDCs for several reasons: more efficient and resilient payments, cheaper and faster cross-border settlement, financial inclusion, and preserving the role of public money in an increasingly digital economy (Auer et al., 2020; BIS, 2021). The motivations differ by country — small islands worry about the cost of physical cash; trading nations worry about cross-border frictions.

It is essential to understand why a CBDC is *not* a cryptocurrency. A CBDC is issued by, and is a liability of, the central bank; it is denominated in ringgit (or dollars, or euro) and does not fluctuate against them; and it runs on permissioned ledgers in which identified, authorised participants validate transactions — not open networks where anonymous miners compete for rewards. BNM has stated precisely this position: a Malaysian CBDC would be a central-bank liability, not a crypto-asset, built on permissioned infrastructure (BNM, 2022). Bitcoin was designed to bypass central banks; a CBDC *is* the central bank, in digital form.

> **QUICK NOTE:** If a government digital currency can be mined, traded at a floating price, or exists outside the central bank's balance sheet, it is not a CBDC. Three quick tests: Who issues it? Is it a central-bank liability? Is one unit always worth one ringgit?

2.5.3 Malaysia's measured exploration

Malaysia has chosen experimentation over haste. The *Financial Sector Blueprint 2022–2026* commits BNM to a multi-year CBDC exploration through proof-of-concept (PoC) work in three phases: first, wholesale cross-border applications; second, a full-scale wholesale model; and only third, consideration of retail use (BNM, 2022).

Two flagship international projects anchor the first phase:

- **Project Dunbar (2021)** brought together BNM, the BIS Innovation Hub, the Reserve Bank of Australia, the Monetary Authority of Singapore and the South African Reserve Bank to build prototypes of a multi-CBDC (mCBDC) platform — showing that several national CBDCs could coexist on a shared platform to make cross-border settlement cheaper, faster and safer (BIS Innovation Hub, 2021).

- **Project mBridge**, in which BNM participates, reached the minimum viable product (MVP) stage in 2024, testing multi-CBDC settlement for real commercial transactions.

Domestically, BNM has run wholesale digital ringgit trials and published a digital ringgit PoC report in 2025. Crucially, BNM has made **no decision to issue a retail CBDC**; the research programme exists precisely so that any future decision rests on evidence rather than fashion (BNM, 2022, 2026).

The frontier has also widened beyond CBDCs. BNM's work programme for 2026 includes **tokenised deposits** — ordinary commercial-bank deposits represented as programmable digital tokens that can move on modern ledgers — and ringgit-denominated stablecoins, explored through its 2025 discussion paper on asset tokenisation (BNM, 2025a). The distinction is worth memorising: a CBDC is *central-bank* money, a tokenised deposit is *commercial-bank* money, and a stablecoin is *private* money — three different issuers, three different risk profiles, one digital future.

Feature	Wholesale CBDC	Retail CBDC
Users	Banks and licensed institutions	General public
Purpose	Interbank and cross-border settlement	Everyday payments; digital cash
Closest existing analogy	Settlement balances at the central bank	Banknotes and coins
Malaysia's status	Active PoCs: Dunbar, mBridge, digital ringgit trials	Under study; no issuance decision

Table 2.6: Wholesale versus retail CBDC.

> **BRAINSTORMER:** Suppose BNM offered every Malaysian a retail digital ringgit tomorrow. Who would benefit most — and who might be harmed? What would happen to commercial-bank deposits if people could hold central-bank money directly?

Case Study 2.1: A Hawker's Journey to Cashless

Pak Rashid, 54, runs a char kway teow stall in George Town, Penang. For more than twenty years his till was a biscuit tin: customers paid in notes and coins, his wife counted takings every night, and twice a week he queued at the bank to deposit damp, sooty cash. He kept no proper records, which later hurt him — when he applied for a small loan to replace his stove, he had little documentation of his income.

In 2024 his daughter Mei Lin helped him register as a merchant through his bank's app. The e-KYC check — MyKad photo plus a selfie with liveness detection — took minutes, and he received a DuitNow QR sticker to display at the stall. Because DuitNow QR is a single national standard, that one sticker accepts payment from virtually any Malaysian banking app or e-wallet, with settlement going straight into his account.

Pak Rashid is one face behind the national statistics. Merchant touchpoints for DuitNow QR grew from 2.6 million in 2024 to about 3 million in 2025, many of them formerly cash-only MSMEs like his stall; QR transaction volume roughly doubled over the same period, from 1.5 billion to about 3 billion transactions (BNM, 2026). His average sale is well below the national

average e-money transaction of RM43 — char kway teow is cheap — but the behavioural shift is the same: younger customers now reach for phones, not wallets.

The benefits surprised him. Queues move faster; he no longer worries about fake RM50 notes or keeping change; and his monthly statement is now a business record he can show a lender. The challenges are real too: festival crowds strain the mobile signal, and he has learned to confirm payment in his *own* app rather than trusting a customer's screen, after a neighbouring stall was shown a fake screenshot. When in doubt, he knows the National Scam Response Centre hotline (997) exists to freeze scam-linked funds quickly.

The family's cashless life even crosses borders. On a 2025 holiday in Bangkok, Mei Lin paid a Thai street vendor by scanning the local QR with her Malaysian banking app — one of the cross-border QR linkages now live between Malaysia and Thailand, Indonesia, Singapore, China and (from 2025) Cambodia, a corridor whose transactions grew about 2.5 times in 2025 (BNM, 2026). One sticker in George Town, one scan in Bangkok: the same rails underneath.

Discussion Questions

1. Beyond speed, identify two benefits DuitNow QR brought to Pak Rashid's business. How did cashless acceptance change his *relationship with the banking system*? *(Understand/Analyse)*
2. What fraud risks does QR acceptance create for micro-merchants, and what practical and institutional mitigations (such as verifying credits in one's own app, or the NSRC hotline) exist? *(Evaluate)*
3. Explain how cross-border QR linkages change the spending experience of travellers and the market reach of small merchants in both countries. *(Analyse)*
4. About 3 million merchant touchpoints existed by 2025, yet some MSMEs remain cash-only. Propose three concrete actions — by banks, PayNet or government — to reach the remaining holdouts. *(Create)*

Chapter Summary

- **2.1** Digital banking in Malaysia evolved from branch counters to internet banking — Maybank2u (2000) was the first — and then to mobile-first apps such as MAE by Maybank and CIMB Clicks/Octo, which now offer account opening, payments, personal financial management and virtual cards.
- **2.1** Five app-only digital banks (GXBank, Boost Bank, AEON Bank, KAF Digital Bank, Ryt Bank) were licensed under BNM's 2020 framework and operate under an RM3 billion asset limit during a three-to-five-year foundational phase, targeting underserved segments.
- **2.2** The digital onboarding funnel runs from app download to activation; e-KYC under BNM's 2020 policy (revised April 2024) enables non-face-to-face verification through document capture, liveness detection and database checks.
- **2.2** AMLA 2001 requires customer due diligence, applied in risk-based tiers — simplified, standard and enhanced — balancing financial inclusion against money-laundering and terrorism-financing risks.
- **2.3** Open banking is consent-based sharing of bank data with third parties through APIs; models range from mandated (EU's PSD2, the UK) to market-led, with Malaysia pursuing a market-led open-finance direction under the Financial Sector Blueprint 2022–2026.

- **2.3** Data sharing promises competition, better products and inclusion, but raises consent and privacy risks governed by the Personal Data Protection Act 2010 and BNM's technology-risk rules.
- **2.4** PayNet operates Malaysia's payment rails: RENTAS (RTGS, 1999), IBG, FPX, JomPAY (2015), MyDebit, DuitNow (2018, proxy-based instant transfer) and DuitNow QR (2019, national QR standard).
- **2.4** Adoption is booming: PayNet processed 8.44 billion transactions in 2025; BNM counted 18.4 billion e-payments (average 538 per Malaysian); DuitNow QR doubled to about 3 billion transactions across roughly 3 million merchant touchpoints.
- **2.4** Cross-border QR linkages are live with Thailand, Indonesia, Singapore, China and Cambodia (2025), with India expected in 2026; fintech remittance providers such as MoneyMatch support the SDG 10.c goal of cutting remittance costs below 3% by 2030.
- **2.5** Stablecoins are crypto-assets pegged to fiat currencies (e.g., USDT, USDC); CBDCs are digital central-bank liabilities, divided into wholesale and retail forms, and are not cryptocurrencies.
- **2.5** BNM is exploring CBDCs in three phases — wholesale cross-border, full-scale wholesale, then retail consideration — via Project Dunbar (2021), Project mBridge (MVP 2024) and wholesale digital ringgit trials, with no retail decision; tokenised deposits and ringgit stablecoins are on its 2026 agenda.

Key Terms

- **Application programming interface (API)**: a standardised set of rules allowing one software system to request data or services from another in a controlled, machine-readable way.
- **Central bank digital currency (CBDC)**: a digital form of central-bank money; a direct liability of the central bank, denominated in the national unit of account, in wholesale or retail form.
- **Customer due diligence (CDD)**: the process by which a financial institution identifies and verifies a customer, understands the purpose of the relationship, and monitors transactions accordingly.
- **Digital bank**: a licensed bank that delivers all services through digital channels, operating without any physical branch network.
- **Digital banking**: the delivery of banking products and services — deposits, payments, credit — through digital channels such as websites and mobile applications rather than branches.
- **DuitNow**: Malaysia's instant account-to-account transfer service (2018), addressed to proxies such as a mobile or MyKad number instead of a bank account number.
- **DuitNow QR**: Malaysia's national interoperable QR payment standard (2019); one merchant QR code can be scanned by any participating banking app or e-wallet.
- **Electronic know-your-customer (e-KYC)**: the electronic verification of a customer's identity, enabling non-face-to-face account opening under BNM's policy framework.
- **Electronic money (e-money)**: a prepaid, stored-value payment instrument; in Malaysia, issuers are licensed by BNM under the Financial Services Act 2013 or Islamic Financial Services Act 2013.
- **Liveness detection**: techniques used during e-KYC to confirm a live person is present — such as blinking or head movements — defeating photos, videos and masks.
- **Mobile-first**: a design philosophy in which the smartphone application is the primary banking channel and all other channels are built around it.

- **Open banking**: the consent-based sharing of customer banking data with authorised third-party providers through secure APIs.
- **Open finance**: the extension of consented, API-based data sharing beyond bank accounts to wider financial data such as investments and insurance.
- **Real-time gross settlement (RTGS)**: the settlement of high-value interbank payments individually, immediately and with finality, rather than in netted batches.
- **Remittance**: a cross-border transfer of funds, typically sent by migrant workers to family members in their home country.
- **Stablecoin**: a crypto-asset designed to maintain stable value against a reference asset, usually a fiat currency; fiat-collateralised examples include USDT and USDC.
- **Tokenised deposit**: an ordinary commercial-bank deposit represented as a programmable digital token transferable on modern ledgers.
- **Virtual card**: a payment card credential that exists only in digital form, issued instantly inside an app and usable for online or wallet-based payments.

Test Your Knowledge

Part A: Multiple Choice

Q1. Malaysia's first internet banking platform, launched in 2000, was: A. CIMB Clicks B. Maybank2u C. DuitNow D. FPX

Q2. Under BNM's Licensing Framework for Digital Banks, a digital bank's assets are capped at what level during its foundational phase? A. RM300 million B. RM1 billion C. RM3 billion D. RM30 billion

Q3. The main purpose of liveness detection during e-KYC is to: A. speed up the banking app B. confirm that a live person, not a photo or video, is being verified C. translate identity documents automatically D. assign the customer's account tier

Q4. In Malaysia, anti-money laundering and counter-terrorism financing obligations for banks arise primarily under: A. the Personal Data Protection Act 2010 B. the Financial Services Act 2013 C. the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 D. the Central Bank of Malaysia Act 2009

Q5. Which pairing correctly matches the jurisdiction to its open banking model? A. European Union — market-led B. United Kingdom — market-led C. Malaysia — mandated D. European Union — mandated

Q6. DuitNow QR, Malaysia's national QR payment standard, was launched in: A. 2015 B. 2018 C. 2019 D. 2023

Q7. PayNet's total processed digital payment transactions in 2025 were approximately: A. 844 million B. 3 billion C. 8.44 billion D. 18.4 billion

Q8. Sustainable Development Goal target 10.c aims to reduce the cost of migrant remittances to: A. below 1% by 2025 B. below 3% by 2030 C. below 5% by 2030 D. zero by 2035

Q9. Which statement best explains why a CBDC is not a cryptocurrency? A. A CBDC uses no digital technology B. A CBDC is a central-bank liability on a permissioned ledger, not a decentralised asset with a floating price C. A CBDC can be mined by the public D. A CBDC is pegged to the US dollar

Q10. The key difference between wholesale and retail CBDCs is that: A. wholesale CBDCs are

for banks' settlement among themselves; retail CBDCs are for the general public B. wholesale CBDCs use blockchain; retail CBDCs do not C. wholesale CBDCs are issued by commercial banks; retail CBDCs by central banks D. there is no difference; the terms are interchangeable

Part B: Short Answer

Q1. Describe the five stages of the digital onboarding funnel and explain why drop-offs at each stage matter for financial inclusion in Malaysia.

Q2. Distinguish between simplified, standard and enhanced customer due diligence, giving one example of a customer or product suited to each tier.

Q3. Compare the mandated and market-led models of open banking, state which model Malaysia follows, and give one advantage and one risk of that choice.

Q4. Trace the path of a RM10 DuitNow QR payment from the moment a customer scans a hawker's QR code to settlement, naming the rail used and the institution that operates it.

Q5. Summarise BNM's three-phase approach to CBDC exploration and state Malaysia's current position on issuing a retail digital ringgit.

Answer Guide

Part A

- **Q1 — B.** Maybank2u (2000) is recognised as Malaysia's first internet banking platform.
- **Q2 — C.** Digital banks face an RM3 billion asset limit during the 3–5-year foundational phase (BNM, 2020).
- **Q3 — B.** Liveness detection resists spoofing by photos, videos or masks during non-face-to-face verification (BNM, 2024).
- **Q4 — C.** AMLA 2001 (Act 613) is Malaysia's principal AML/CFT statute.
- **Q5 — D.** The EU mandates data sharing under PSD2; the UK also mandates; Malaysia is market-led.
- **Q6 — C.** DuitNow QR launched in 2019 (DuitNow itself, 2018; JomPAY, 2015).
- **Q7 — C.** PayNet processed 8.44 billion transactions in 2025 (PayNet, 2026); 18.4 billion is BNM's wider e-payment count.
- **Q8 — B.** SDG 10.c targets remittance costs below 3% by 2030.
- **Q9 — B.** A CBDC is central-bank money on permissioned infrastructure — stable in value, centrally issued.
- **Q10 — A.** The defining distinction is the user base: institutions (wholesale) versus the public (retail).

Part B (model answers)

- **Q1.** The stages are discovery/download, registration, identity verification (e-KYC), approval, and activation. Drop-offs matter because each abandoned application can leave a person outside the formal financial system — significant when about one in nine Malaysian adults still lacked an account in 2024 (World Bank, 2025). Reducing friction at every stage is an inclusion policy, not merely a commercial one.
- **Q2.** Simplified CDD applies to lower-risk products such as basic accounts with balance limits, enabling fast onboarding; standard CDD covers ordinary customers with full e-KYC verification; enhanced CDD applies to higher-risk customers, such as politically exposed persons, requiring extra checks and closer monitoring. Tiering matches the intensity of control to the level of risk.

- **Q3.** Mandated models (EU's PSD2; the UK) legally compel banks to open APIs; market-led models rely on regulator encouragement and industry initiative. Malaysia follows the market-led model under the Financial Sector Blueprint 2022–2026. Advantage: flexibility to fit standards to local use cases. Risk: slower, uneven adoption and possible dominance by closed ecosystems.
- **Q4.** The customer scans the merchant's DuitNow QR with a banking app or e-wallet and confirms RM10; the instruction travels over the DuitNow QR rail operated by PayNet; funds move from the customer's account to the merchant's account in real time; both parties receive confirmation. Settlement between the institutions is completed through the national infrastructure, ultimately anchored on RENTAS.
- **Q5.** Phase one explores wholesale cross-border CBDC (e.g., Project Dunbar, Project mBridge); phase two would build a full-scale wholesale model; phase three would consider retail use. Malaysia has conducted wholesale digital ringgit trials but has made no decision to issue a retail CBDC.

Problem-Solving Corner

Problem 2.1: Comparing remittance costs

Karim, a migrant worker in Kuala Lumpur, sends RM3,000 home to Dhaka every month. He compares two options: his bank quotes a *total* cost (transfer fee plus exchange-rate margin) of 6.0% of the amount sent, while a licensed remittance fintech app quotes 1.5%.

(a) Calculate the monthly cost of each option and the amount that reaches his family (in ringgit terms). **(b)** How much more does his family receive in a full year if he uses the fintech? **(c)** Which option satisfies SDG target 10.c?

Solution. (a) Bank: 6.0% × RM3,000 = **RM180 cost**, so the family receives RM3,000 − RM180 = **RM2,820**. Fintech: 1.5% × RM3,000 = **RM45 cost**, so the family receives RM3,000 − RM45 = **RM2,955**. (b) Monthly difference: RM2,955 − RM2,820 = RM135. Over 12 months: RM135 × 12 = **RM1,620** more reaches the family each year. (c) SDG 10.c targets costs below 3% by 2030. The fintech (1.5%) **meets** the target; the bank (6.0%) does not. For perspective, the World Bank's benchmark tracks the cost of sending US$200, for which 3% equals US$6 (World Bank, n.d.).

Problem 2.2: Reading the DuitNow QR statistics

DuitNow QR handled about 1.5 billion transactions across 2.6 million merchant touchpoints in 2024, and about 3.0 billion transactions across 3.0 million touchpoints in 2025 (BNM, 2026).

(a) Compute transactions per touchpoint per year for 2024 and 2025, and the daily equivalents. **(b)** Compute the percentage growth in (i) transaction volume, (ii) touchpoints, and (iii) transactions per touchpoint. **(c)** What does your answer to (b)(iii) tell a merchant such as Pak Rashid?

Solution. (a) 2024: 1.5 billion ÷ 2.6 million ≈ **577 transactions per touchpoint per year**, or 577 ÷ 365 ≈ **1.6 per day**. 2025: 3.0 billion ÷ 3.0 million = **1,000 per year**, or about **2.7 per day**. (b) (i) Volume: (3.0 ÷ 1.5) − 1 = **+100%**. (ii) Touchpoints: (3.0 ÷ 2.6) − 1 ≈ **+15.4%**. (iii) Intensity: (1,000 ÷ 577) − 1 ≈ **+73%**. (c) Usage per touchpoint grew much faster than the number of touchpoints. QR payments are not merely spreading wider — they are being *used more deeply* at each merchant. For Pak Rashid, the trend means QR is becoming a mainstream habit among his customers, not a novelty.

Problem 2.3: E-payment growth and the 15% target

BNM counted 14.7 billion e-payment transactions in 2024 and 18.4 billion in 2025, an average of 538 per Malaysian (BNM, 2026). The Financial Sector Blueprint 2022–2026 targets roughly 15% compound annual growth in e-payments (BNM, 2022).

(a) Verify the reported 2025 growth rate. **(b)** Estimate Malaysia's population implied by the 538-per-person figure. **(c)** Project 2026 transactions if the 15% target growth rate holds from the 2025 actual level, and comment on whether Malaysia is ahead of or behind the target pace.

Solution. (a) Growth = $(18.4 \div 14.7) - 1 = 0.2517 \approx$ **25%**, matching the reported figure. (b) Population ≈ 18.4 billion $\div 538 \approx$ **34.2 million** — consistent with Malaysia's population. (c) Projection: $18.4 \text{ billion} \times 1.15 =$ **21.16 billion transactions in 2026**. Because actual 2025 growth (25%) exceeded the 15% target pace, Malaysia is currently running **ahead** of the Blueprint trajectory — evidence of how quickly payment behaviour is digitising.

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Chapter 3: Digital Financing, Peer-to-Peer Lending and Crowdfunding

Chapter Outline

- 3.1 Capital Raising and Digital Financing
- 3.2 Peer-to-Peer (P2P) Financing
- 3.3 Types of Crowdfunding and Equity Crowdfunding
- 3.4 Platform Fees and Financing Costs
- 3.5 Risks, Financial Inclusion and Regulatory Considerations

Learning Outcomes

After studying this chapter, you should be able to:

1. **Define** key terms in digital financing, including peer-to-peer financing, investment notes, crowdfunding and equity crowdfunding. *(Bloom's level: Remember)*
2. **Explain** why traditional banks underserve micro-SMEs and how digital financing platforms help to close this gap in Malaysia. *(Bloom's level: Understand)*
3. **Apply** financing-cost formulas to calculate the effective annual cost of a P2P investment note and the ownership dilution from an equity crowdfunding campaign. *(Bloom's level: Apply)*
4. **Analyse** the risks faced by issuers and investors on digital financing platforms, including default risk, platform failure and information asymmetry. *(Bloom's level: Analyse)*
5. **Evaluate** the Securities Commission Malaysia's regulatory approach to recognized markets, weighing financial innovation against investor protection. *(Bloom's level: Evaluate)*
6. **Develop** a suitable digital financing plan for a Malaysian SME, justifying the choice among bank, P2P and equity crowdfunding options. *(Bloom's level: Create)*

Opening Vignette

Aina Damia runs *Dapur Aina*, a small frozen curry-puff business in Klang, Selangor. For three years she has supplied school cooperatives and neighbourhood grocers, buying ingredients with her own savings. One morning, a supermarket chain offers her a contract to supply 40

outlets — but she must first buy a second freezer, a packaging machine and larger stocks of flour and potatoes. She needs RM40,000 within a month.

Her bank asks for audited accounts, collateral and a guarantor. Aina has none of these; her accounts are simple spreadsheets, and the shop lot is rented. A friend mentions a peer-to-peer financing platform registered with the Securities Commission Malaysia. Aina applies online, uploads six months of bank statements and her purchase order, and receives a decision in days. Investors on the platform fund her campaign, and she repays the financing over six months as the supermarket pays her invoices.

Aina's story shows why digital financing exists — and why it matters to thousands of Malaysian micro, small and medium enterprises. This chapter explains how it works, what it costs, and what can go wrong.

3.1 Capital Raising and Digital Financing

Every business needs capital. Money is needed to start a business (seed capital), to buy equipment (capital expenditure) and to pay for daily running costs such as wages, rent and raw materials (working capital). Traditionally, Malaysian businesses raised this money from their own savings, from family and friends, from bank loans or — for larger companies — from issuing shares and bonds. This section explains why that traditional system leaves many small businesses behind, and how digital channels are filling the gap.

3.1.1 The Traditional SME Financing Gap

Digital financing refers to the provision of financial services — particularly credit and capital raising — through digital technologies, platforms and channels rather than through traditional branch-based processes (Arner et al., 2016; Puschmann, 2017). It matters because the traditional system has a persistent gap.

Micro, small and medium enterprises (MSMEs) form the backbone of the Malaysian economy. They employ millions of Malaysians and include everything from roadside food stalls and online sellers to small manufacturers and professional firms. Yet many of them cannot borrow as much as they need, when they need it. The difference between the financing that businesses demand and the financing that traditional institutions are willing to supply is known as the **financing gap**.

The financing gap exists on both sides of the market. On the demand side, micro-SMEs typically want small loans — sometimes only RM10,000 to RM50,000 — for short periods, often urgently. On the supply side, banks find such loans unattractive: the administrative cost of assessing a RM30,000 loan is almost the same as assessing a RM3 million loan, but the profit is a tiny fraction. Thakor (2020) notes that this mismatch between what small borrowers need and what banks find profitable is one of the main reasons technology-based lenders emerged around the world.

> **QUICK NOTE:** Financing comes in two basic forms. *Debt* financing is borrowed money that must be repaid with interest or profit; the lender does not own any part of the business. *Equity* financing means selling a share of ownership; there is no repayment, but the investor shares in future profits and losses. P2P financing is debt; equity crowdfunding is equity.

3.1.2 Why Banks Underserve Micro-SMEs

Understanding the financing gap requires understanding how banks make lending decisions. Four factors explain why banks underserve micro-SMEs.

1. Information asymmetry. Information asymmetry exists when one party in a transaction has more or better information than the other party (Thakor, 2020). A small business owner knows far more about the real condition of the business than the bank does. Banks protect themselves by demanding audited financial statements, several years of track record and credit histories. Many micro-SMEs cannot produce these documents, not because they are dishonest, but because they are small and informal. Faced with uncertainty, the bank either declines the application or charges a high price.

2. Collateral requirements. Banks typically secure their loans against assets such as property or fixed deposits. Micro-entrepreneurs often rent their premises and have few assets to pledge, so they fail this test even when their businesses are profitable.

3. High cost-to-serve. Manual underwriting — visiting the business, verifying documents, assessing character — costs time and money. For small loans, these fixed costs make the business uneconomic for banks.

4. Speed and rigidity. Bank approval processes can take weeks. A small trader who must pay a supplier within days to secure a festive-season order cannot wait. The result is that many micro-SMEs rely on personal savings, borrow from family, use supplier credit or — in the worst cases — turn to unlicensed moneylenders.

3.1.3 Digital Channels and Malaysia's Alternative Financing Market

Digital financing platforms attack each of these four problems. Online onboarding replaces branch visits; electronic know-your-customer (e-KYC) processes verify identity remotely; automated credit assessment lowers the cost of evaluating small loans; and platform marketplaces allow many small funders to combine their money to meet one borrower's needs (Gomber et al., 2018).

Malaysia's alternative financing market is supervised by the Securities Commission Malaysia (SC) and consists mainly of two segments: peer-to-peer (P2P) financing and equity crowdfunding (ECF). The scale is now significant. By the end of 2025, P2P financing platforms had channelled a cumulative RM11.32 billion to Malaysian businesses through 154,283 campaigns, while ECF platforms had raised a cumulative RM917.51 million across 445 campaigns (Securities Commission Malaysia [SC], 2026). These flows complement, rather than replace, the banking system, and they align with the national agenda in the *Financial Sector Blueprint 2022–2026*, which champions a more digital and inclusive financial system (Bank Negara Malaysia [BNM], 2022).

Table 3.1 summarises how the two digital channels compare with traditional bank financing.

Table 3.1: Traditional bank financing versus P2P financing versus equity crowdfunding

Dimension	Bank financing	P2P financing	Equity crowdfunding
Instrument	Loan or overdraft	Investment note (debt)	Shares (equity)

Dimension	Bank financing	P2P financing	Equity crowdfunding
Source of funds	Bank's balance sheet	Many individual/institutional investors	Many individual/institutional investors
Repayment obligation	Yes, with interest/profit	Yes, with agreed returns	No; investors share profits and losses
Collateral	Commonly required	Usually unsecured	Not applicable
Typical applicant	Established firms with track record	MSMEs, including thin-file firms	Early-stage, high-growth firms
Regulator (Malaysia)	Bank Negara Malaysia	Securities Commission Malaysia	Securities Commission Malaysia

3.1.4 The Regulatory Perimeter: The SC and the CMSA 2007

In Malaysia, responsibility for financial regulation is divided. Bank Negara Malaysia regulates banks, payment systems, e-money and money services, while the Securities Commission Malaysia regulates capital-market fintech — including P2P financing, equity crowdfunding, digital asset exchanges and digital investment management (SC, 2026).

The legal foundation is the Capital Markets and Services Act 2007 (Act 671), which governs securities and capital-market activities. Because investment notes and ECF shares are capital-market products, the platforms that offer them must be registered with the SC as **Recognized Market Operators (RMOs)** — that is, operators of electronic platforms approved to bring together issuers and investors under the *Guidelines on Recognized Markets* (SC, 2024). The Guidelines set out obligations on governance, due diligence over issuers, disclosure to investors, handling of client monies and fair treatment.

Malaysia moved early. It was the first country in ASEAN to regulate equity crowdfunding, with the SC's framework announced in 2015, and it was among the first in the region to regulate P2P financing in 2016 (SC, 2026). This early, rules-based approach gave legitimate operators certainty and gave investors a regulated environment in which to participate.

> **MALAYSIAN SPOTLIGHT:** The Securities Commission Malaysia, established in 1993, has become one of Asia's most active regulators of alternative finance. Its decision to license P2P and ECF platforms in 2015–2016 — years before many neighbours — helped build a market that by 2025 had supplied more than RM12 billion in cumulative alternative financing to Malaysian MSMEs, much of it to businesses too small for conventional bank credit (SC, 2026).

3.2 Peer-to-Peer (P2P) Financing

P2P financing is the larger of Malaysia's two alternative-financing segments and the one most relevant to ordinary small businesses. This section defines the concept, explains how investment notes work, shows how platforms assess credit using alternative data, and presents the Malaysian market in numbers.

3.2.1 What Is P2P Financing?

Peer-to-peer (P2P) financing is the practice of raising debt funding from a large number of investors through an online platform that matches funders directly with borrowers, without the funding passing through a traditional financial intermediary such as a bank (Thakor, 2020; Lee & Shin, 2018). In Malaysia the borrowers are businesses, so the SC uses the term "P2P financing" rather than "P2P lending", and the products are structured as securities called investment notes (SC, 2024).

Three parties take part in every transaction:

- The **issuer** — the business that raises funds by issuing an investment note and promises to repay it with returns.
- The **investors** — individuals or institutions who subscribe to the note in amounts of their choosing and receive the repayments.
- The **platform operator** — the RMO that vets issuers, hosts the campaign, handles the money and administers repayments.

The platform is a marketplace, not a bank. It does not lend its own money; it earns fees for arranging the transaction. This distinction is important for understanding both the costs (Section 3.4) and the risks (Section 3.5) of P2P financing.

> **DID YOU KNOW? (TRIVIA):** Malaysia was the first country in ASEAN to put equity crowdfunding under a formal regulatory framework (2015) and among the first in the region to regulate P2P financing (2016). Within a decade, those early frameworks had channelled RM11.32 billion in P2P funding to Malaysian businesses (SC, 2026).

3.2.2 How Investment Notes Work: Issuers and Investors

An **investment note** is a security issued through a registered P2P platform that evidences the issuer's obligation to repay investors the principal amount together with agreed returns over a fixed period, known as the tenor (SC, 2024). Tenors in the market commonly run from a few months to a few years, and returns may be structured on conventional (interest-based) or Shariah-compliant (profit-based) terms.

The issuer's journey typically follows six steps:

1. **Application.** The business applies online, submitting its registration documents (for example, Suruhanjaya Syarikat Malaysia or SSM records), bank statements, financial information and details of the funding purpose.
2. **Credit assessment.** The platform verifies the information, analyses the business's ability to repay and assigns an internal risk grade, which determines the rate of return offered to investors.
3. **Listing.** The approved note is listed on the platform with its amount, tenor, rate and purpose. Investors browse listings and decide how much to subscribe.
4. **Funding.** When subscriptions reach the target, the campaign closes. If the target is not met within the listing period, the campaign is cancelled and monies are returned to investors.
5. **Disbursement.** The platform deducts its fees and releases the net proceeds to the issuer.
6. **Repayment.** The issuer repays according to the schedule — either in regular instalments of principal and returns, or as a single "bullet" repayment at maturity. The platform collects and distributes the money to investors.

For investors, the attraction is access to a fixed-income style product with disclosed rates and the ability to spread money across many notes. A investor with RM10,000 can place RM500 into twenty different issuers instead of betting everything on one. This practice of spreading exposure across many borrowers is called **diversification**, and it is the single most important risk-management habit for P2P investors.

> **QUICK NOTE:** An investment note is a *security*, not a bank deposit. It is not protected by deposit insurance, and the advertised rate of return is a promise, not a guarantee. If the issuer fails to pay, investors can lose part or all of their money.

3.2.3 Credit Assessment with Alternative Data

How can a platform lend to a business with no audited accounts and no collateral? The answer is technology plus data. **Credit scoring** is the use of statistical models to estimate the probability that a borrower will repay a debt, usually expressed as a score or grade (Thakor, 2020). Traditional scoring depends heavily on formal credit-bureau records and financial statements. Digital platforms supplement these with **alternative data** — non-traditional digital footprints such as bank-account transaction flows, e-payment and e-wallet records, online marketplace sales histories, invoices and utility payment records (Gomber et al., 2018).

Alternative data helps in three ways. First, it is *timely*: a platform can observe the business's actual cash receipts last month, not last year's audited figures. Second, it is *granular*: daily inflows and outflows reveal seasonality and stress that annual accounts hide. Third, it is *cheap to process*: automated analysis reduces the cost of underwriting small loans, directly attacking the cost-to-serve problem described in Section 3.1.2.

For a food manufacturer, for instance, the platform might examine six months of bank statements to confirm steady sales deposits, review a confirmed purchase order from a supermarket to verify the purpose of the financing, and check public records for litigation or winding-up petitions. The result is a risk grade that prices the note: stronger businesses pay lower returns; weaker ones pay more, or are declined. Section 3.5 returns to the limitations and risks of data-driven credit assessment.

3.2.4 Malaysia's P2P Market in Numbers

A decade after regulation, Malaysia's P2P market has clear contours. Table 3.2 summarises the headline statistics reported by the SC (2026).

Table 3.2: Malaysia's P2P financing market at a glance (SC Annual Report 2025)

Indicator	Value
Cumulative funds raised since 2016	RM11.32 billion
Cumulative campaigns since 2016	154,283
Funds raised in 2025	RM2.83 billion (up 13% year-on-year)
Campaigns in 2025	33,913
Campaigns raising RM50,000 or less (2025)	62%
Campaigns for working-capital purposes	98%
Largest sector in 2025 (wholesale and retail trade)	51% of funds (RM1.44 billion)
Total issuers since inception	24,683
Cumulative investors	More than 53,000

Indicator	Value
Investors active in 2025	15,257 (of whom 20% were new)
Share of funds that are Shariah-compliant (2025)	26%
Issuers that are non-technology businesses	99%
Issuers operating for six years or more	49%
Issuers located in Selangor and Kuala Lumpur	54%

Three features stand out. First, this is overwhelmingly a *small-ticket, working-capital* market: 62% of campaigns raised RM50,000 or less, and 98% funded working capital rather than long-term investment (SC, 2026). Second, issuers are ordinary businesses, not tech startups — 99% are non-technology firms, and nearly half have operated for six years or more. Third, the market is still growing: 2025 fundraising rose 13% to RM2.83 billion, and one in five investors was new to the market (SC, 2026).

Registered operators active in Malaysia include Funding Societies, Fundaztic, B2B FinPAL, microLEAP (which offers Shariah-compliant notes), Alixco, CapBay and MoneySave (SC, 2026). Students should always check the SC's current list of registered operators before engaging with any platform, because registration status can change.

> **BRAINSTORMER:** If 98% of P2P campaigns finance working capital, what does that suggest about the *types* of needs banks are still not meeting for Malaysian MSMEs? What might happen to these businesses if P2P platforms suddenly disappeared?

3.3 Types of Crowdfunding and Equity Crowdfunding

P2P financing is, in fact, one member of a larger family: crowdfunding. This section defines the four main types of crowdfunding, then focuses on equity crowdfunding (ECF), the segment in which investors become part-owners of the businesses they back.

3.3.1 Four Types of Crowdfunding

Crowdfunding is the practice of raising small amounts of money from a large number of people — "the crowd" — through an online platform, typically to fund a project, cause or business (Chishti & Barberis, 2016; Lee & Shin, 2018). What the contributor receives in return defines the type of crowdfunding, as summarised in Table 3.3.

Table 3.3: The four main types of crowdfunding

Type	What the backer receives	Typical use	Example
Donation-based	Nothing material; the satisfaction of giving	Charity, community, disaster relief	Funding a mosque's flood-relief appeal
Reward-based	A product, service or token of appreciation	Creative projects, product pre-orders	Pre-ordering a gadget before it is manufactured

Type	What the backer receives	Typical use	Example
Debt-based (P2P)	Repayment of principal with returns	Business working capital	An investment note for a food manufacturer
Equity-based (ECF)	Shares in the company	Early-stage business growth	Buying shares in a tech startup

Donation and reward crowdfunding are generally outside securities regulation because backers do not receive financial instruments. Debt and equity crowdfunding involve securities and therefore fall within the SC's regulatory perimeter in Malaysia (SC, 2024). Islamic variants exist across the family: Malaysian platforms operate Shariah-compliant debt and equity crowdfunding, and donation-based crowdfunding aligns naturally with *sadaqah* and *waqf* giving (Oseni & Ali, 2019).

3.3.2 Equity Crowdfunding: Concept and Investor Returns

Equity crowdfunding (ECF) is a form of crowdfunding in which investors subscribe for shares in a company through an online platform, becoming part-owners who share in the company's future profits and losses (SC, 2024). Unlike a P2P investor, an ECF investor is *not* promised repayment. Returns come only if the company prospers — through dividends, a trade sale of the company, a buy-back of shares or, rarely, a stock-exchange listing.

Two valuation concepts drive every ECF deal. The **pre-money valuation** is the value agreed for the company *before* the new money arrives; the **post-money valuation** is the pre-money valuation *plus* the amount raised. The percentage of the company that new investors receive equals the amount raised divided by the post-money valuation. The corresponding reduction in existing shareholders' ownership percentage is called **dilution** — founders still hold the same number of shares, but those shares represent a smaller slice of a larger company. The worked example in the Problem-Solving Corner shows the arithmetic in detail.

ECF suits businesses that are young, fast-growing and hungry for cash that debt cannot safely carry. A startup with no stable revenue cannot promise monthly repayments, but it can offer investors a share of a potentially valuable future. The trade-off is risk: many early-stage companies fail, and ECF shares are *illiquid* — there is usually no ready market on which to sell them, so investors must be prepared to hold for years or lose the entire investment.

3.3.3 Malaysia's ECF Framework and Market

Malaysia was the first country in ASEAN to regulate equity crowdfunding, announcing the SC framework in 2015, with campaigns beginning in 2016 (SC, 2026). A decade later, the market remains much smaller than P2P financing — reflecting the smaller population of businesses suited to equity — but it is growing quickly. Table 3.4 presents the headline figures.

Table 3.4: Malaysia's equity crowdfunding market at a glance (SC Annual Report 2025)

Indicator	Value
Cumulative funds raised since 2016	RM917.51 million

Indicator	Value
Cumulative campaigns since 2016	445
Share of campaigns that are Shariah-compliant	4.1%
Funds raised in 2025	RM139.34 million (up 42.4% year-on-year)
Successful campaigns in 2025	42
2025 campaigns raising more than RM3 million	45%
Dominant funding stages	Seed and Series A
Issuers located in Selangor and Kuala Lumpur	82%

Three observations follow. First, growth is rapid from a small base: 2025 fundraising jumped 42.4% to RM139.34 million across 42 successful campaigns (SC, 2026). Second, campaigns are concentrated at the earliest stages of company life — seed and Series A rounds dominate — confirming ECF's role as startup finance. Third, the market is geographically concentrated: 82% of issuers are in Selangor and Kuala Lumpur, the country's business heartland. Registered ECF operators include pitchIN, Ata Plus, MyStartr, Ethis and Leet Capital (SC, 2026).

> **DID YOU KNOW? (TRIVIA):** Nearly half of all successful Malaysian ECF campaigns in 2025 — 45% — raised more than RM3 million each, showing that equity crowdfunding has matured from tiny community raises into a serious funding channel for growth-stage companies (SC, 2026).
>
> **BRAINSTORMER:** A social enterprise wants to build recycling centres but will earn only modest profits for its first five years. Which type of crowdfunding — donation, reward, debt or equity — would you recommend, and why? Would your answer change if the founders refused to give up any ownership?

3.4 Platform Fees and Financing Costs

Digital financing is convenient, but it is not free. Platforms charge fees to run their marketplaces, and investors demand returns for bearing risk. This section maps the typical fee structures, shows how to compute the *effective* cost of P2P financing, and explains the different cost logic of equity crowdfunding.

3.4.1 Typical Fee Structures on Digital Financing Platforms

Platforms earn revenue by charging issuers, investors or both. The exact amounts vary between operators and are disclosed on each platform, as required by the disclosure obligations in the *Guidelines on Recognized Markets* (SC, 2024). Table 3.5 lists the fee types a student is likely to encounter.

Table 3.5: Common fee types on digital financing platforms

Fee type	Who pays	When charged	Purpose
Listing / **origination fee**	Issuer	On successful funding (often deducted from proceeds)	Covers application processing, credit assessment and documentation

Fee type	Who pays	When charged	Purpose
Success fee	Issuer	Only if the campaign succeeds	Platform's main revenue for arranging the deal
Servicing / administration fee	Issuer or investor	During the tenor (e.g., monthly)	Covers collection and distribution of repayments
Investor fee	Investor	On returns received	Platform's share for providing the marketplace
Late fee / penalty	Issuer	On overdue payments	Compensates for delay; may be shared with investors
Early-settlement fee	Issuer (some platforms)	If the note is repaid early	Compensates investors/platform for lost returns

An **origination fee** is a charge levied by a platform for processing, assessing and arranging a financing facility, usually expressed as a percentage of the amount raised and deducted from the proceeds (SC, 2024). Two practical points follow. First, because the fee is deducted upfront, the issuer receives *less* cash than the face amount of the note — but must repay the *full* face amount. Second, fees stack: a note may carry an origination fee, servicing fees *and* late fees, so the true cost exceeds the headline rate of return.

> **QUICK NOTE:** The advertised rate on an investment note (say, 10% per annum) is the *investor's* return, not the *issuer's* cost. The issuer's true cost is higher once origination and servicing fees are included, and it must be measured against the net cash actually received.

3.4.2 The Effective Cost of P2P Financing versus Bank Loans

To compare financing options fairly, businesses should compute the **effective annual cost** — the total cost of financing (returns plus all fees) expressed as an annual rate on the net funds actually received by the borrower (Thakor, 2020). A simple illustration makes the idea concrete (a fuller version appears in the Problem-Solving Corner):

- An SME issues a six-month investment note for RM150,000 at 10% per annum (simple), with a 2% origination fee.
- Returns payable at maturity: RM150,000 × 10% × 6/12 = RM7,500.
- Origination fee: RM150,000 × 2% = RM3,000, deducted upfront, so the SME receives RM147,000.
- Total six-month cost: RM7,500 + RM3,000 = RM10,500, on RM147,000 actually received — a six-month cost of about 7.14%, or roughly 14.3% annualised.

How does this compare with a bank loan? Secured bank financing is usually *cheaper in rate terms*, because banks fund themselves with low-cost deposits. P2P financing is usually *more expensive*, because investors bear unsecured credit risk and platforms must cover their costs. But the comparison is not only about price. P2P financing typically offers faster approval, no collateral requirement, smaller ticket sizes and shorter tenors — precisely the features micro-SMEs need. For a business facing a time-sensitive opportunity, a more expensive facility that

arrives in a week can be worth more than a cheaper one that arrives in three months, or never. The rational borrower compares *availability, speed and cost* together (Thakor, 2020).

3.4.3 The Cost of Equity Crowdfunding

ECF has a different cost logic, because there is no debt to repay. Instead, an ECF issuer faces three kinds of cost:

1. **Cash fees.** Platforms typically charge a success fee on the amount raised, payable only if the campaign completes; some also charge listing or administrative fees. Legal, due-diligence and marketing expenses add to the bill, because a public campaign requires a polished pitch, financial disclosures and often a video and media outreach.
2. **Time and disclosure costs.** Founders must prepare detailed information for public scrutiny and manage relations with dozens or hundreds of new shareholders after the campaign.
3. **Dilution.** The largest economic "cost" is the ownership given up. Selling 20% of the company means the founders' claim on future profits falls by one-fifth — permanently, unless shares are later bought back.

The absence of repayment obligations, however, is a genuine benefit: an ECF-funded company keeps all its cash for growth and owes investors nothing if the plan fails. For high-risk, high-growth ventures, equity is often the *only* sustainable financing — no prudent lender will fund a firm that cannot service debt. Students should therefore resist the question "which is cheaper?" and ask instead "which instrument matches the firm's cash-flow profile and risk?".

3.5 Risks, Financial Inclusion and Regulatory Considerations

Digital financing creates real opportunities, but it also creates new ways to lose money. This final section analyses the main risks, explains how the SC's framework protects investors, introduces Shariah-compliant options, and assesses what digital financing has — and has not — achieved for financial inclusion in Malaysia.

3.5.1 Default Risk, Platform Failure and Information Asymmetry

Default risk (credit risk) is the risk that a borrower will fail to make required payments, causing lenders or investors to lose interest, profit or principal (Thakor, 2020). Default risk is the central risk of P2P financing: notes are usually unsecured, issuers are small businesses with fragile cash flows, and there is no deposit insurance. Mitigants include the platform's credit assessment, diversification across many notes, and — on some platforms — provision funds or security arrangements. None of these eliminates the risk.

Platform failure is a second, distinct risk. Investors and issuers depend on the platform to hold monies safely, collect repayments and enforce the notes. If an operator becomes insolvent or exits the market, servicing of outstanding notes could be disrupted even if the underlying borrowers remain healthy. The *Guidelines on Recognized Markets* address this by imposing obligations on governance, segregation of client monies and orderly conduct of the platform's business (SC, 2024), but investors should still prefer registered operators and read the platform's arrangements for business continuity.

Third, **information asymmetry** does not disappear online — it changes shape. Platforms know more about their credit models than investors do; issuers know more about their true condition than the platform does; and polished campaign pages can flatter weak businesses (Zetzsche et al., 2017). Related dangers include issuers misstating their finances, campaigns raising money for fictitious purposes, and outright **fraud** by fake platforms. The first line of defence is simple: deal only with operators on the SC's register of recognized market operators, and treat any unsolicited "guaranteed high return" offer as a red flag.

3.5.2 Investor Protection under the Recognized Market Framework

The SC's approach combines gatekeeping, disclosure and investment limits:

- **Gatekeeping.** Platforms must be registered as RMOs and satisfy fit-and-proper, governance and systems requirements. Operators are obliged to conduct due diligence on issuers before listing their campaigns (SC, 2024).
- **Disclosure.** Issuers must provide prescribed information about their business, financial condition and the purpose of the fundraising, enabling investors to make informed decisions (SC, 2024).
- **Investment limits.** To contain the damage any one failure can do, the Guidelines prescribe investment limits for retail investors — caps on how much an individual may commit to a single issuer and in aggregate over a period — while sophisticated investors, who are deemed better able to assess and absorb risk, enjoy wider access (SC, 2024).

This architecture illustrates a permanent regulatory trade-off. Rules that are too lax invite scandals that destroy public trust; rules that are too strict smother a young industry that serves borrowers the banks ignore. Malaysia's answer has been to allow experimentation *inside* a regulated perimeter — licensing real operators, mandating disclosure and capping retail exposure — rather than banning the activity or leaving it unregulated (Zetzsche et al., 2017).

> **BRAINSTORMER:** Should ordinary Malaysians be free to invest as much as they like in P2P notes and ECF shares, or are legal caps on retail investment amounts justified? How would you balance personal freedom against the risk of life-changing losses?

3.5.3 Shariah-Compliant Options

Malaysia's position as a global leader in Islamic finance extends into alternative financing. **Shariah-compliant** financing is financing structured in accordance with Islamic law — notably the prohibition of *riba* (interest) and excessive uncertainty — using contracts such as cost-plus sale (*murabahah*/*tawarruq*) or agency (*wakalah*) instead of conventional interest-bearing loans (Oseni & Ali, 2019).

The supply of Shariah-compliant digital financing is substantial and growing. In 2025, 26% of all funds raised on Malaysian P2P platforms were Shariah-compliant, and operators such as microLEAP specialise in Shariah-compliant notes, while Ethis operates Islamic P2P and ECF offerings (SC, 2026; DinarStandard & Elipses, 2026). Shariah-compliant campaigns remain a smaller share of ECF — 4.1% of campaigns — suggesting room for growth (SC, 2026). Globally, Malaysia ranks among the top Islamic-fintech ecosystems, within a market estimated at about US$198 billion in transaction volume and projected to reach US$341 billion by 2029 (DinarStandard & Elipses, 2026). For Muslim investors and issuers, these options make it

possible to participate in digital financing without compromising religious principles; for platforms, they widen the investor base.

3.5.4 Financial Inclusion: MSMEs and B40 Entrepreneurs

Financial inclusion means that individuals and businesses have access to, and actually use, affordable and suitable financial products and services (World Bank, 2025). Malaysia performs well on basic access — 88.7% of adults owned a financial account in 2024, with a gender gap below one percentage point — but access to *credit* is weaker for poorer households: account ownership among the poorest 40% stood at 82.0% in 2024, and owning an account does not guarantee the ability to borrow (World Bank, 2025).

Digital financing helps close this credit gap in several ways. Small ticket sizes (62% of 2025 P2P campaigns raised RM50,000 or less) match the needs of micro-businesses; unsecured notes remove the collateral barrier; alternative-data credit scoring serves thin-file applicants; and fully digital journeys reduce travel and paperwork for rural entrepreneurs (SC, 2026). The issuer base confirms the inclusive character of the market: 99% of issuers are non-technology businesses, and 54% come from Selangor and Kuala Lumpur, meaning the other 46% are spread across the rest of the country (SC, 2026). For entrepreneurs in the B40 (bottom 40% income) group — a food-stall operator formalising her business, a village craftsman buying equipment — a successful small campaign can be the first rung on the ladder of formal credit, building a repayment record that unlocks larger facilities later.

Two cautions complete the picture. First, digital financing presupposes digital access and literacy; entrepreneurs without smartphones, connectivity or the skills to prepare an online application may be left further behind. Second, inclusion must not become *over-indebtedness*: easy credit that funds consumption rather than productive investment can harm the very households it is meant to help. Sustainable inclusion pairs access with financial education — which is precisely why you are studying this chapter.

Case Study 3.1: Sri Cahaya Food Products Sdn. Bhd. — Raising RM150,000 Working Capital through a P2P Investment Note

Note: the company and figures in this case are fictional and for illustration only; fee and rate assumptions are stated explicitly.

Sri Cahaya Food Products Sdn. Bhd. is a halal food manufacturer in Shah Alam, Selangor, producing bottled sambal and cooking sauces. The company, owned by Encik Rahman Hassan, has operated for eight years, employs 22 workers and sells through grocers and online marketplaces. In March, a national supermarket chain offers Sri Cahaya a RM480,000 contract to supply a festive-season promotion, with delivery in five months and payment within 60 days of delivery.

The problem. Fulfilling the order requires RM150,000 upfront for chillies, glass bottles, labels and seasonal workers. Sri Cahaya's bank declines an overdraft: the company rents its factory, has no property to pledge, and its thin credit file produces a weak bureau score. The contract will be lost without alternative funding.

The application. Encik Rahman applies to an SC-registered P2P financing platform. He uploads the company's SSM registration, six months of business bank statements, the latest

management accounts, and the supermarket's signed purchase order.

Credit scoring. The platform verifies the documents and analyses alternative data: steady monthly sales deposits in the bank statements, growing online-marketplace sales, and the creditworthiness of the supermarket behind the purchase order. Sri Cahaya receives a risk grade consistent with a sound, established business, and the platform approves an investment note of RM150,000 for a six-month tenor at an illustrative 10% per annum (simple), repayable in a single bullet payment at maturity when the supermarket's invoice is settled.

Funding. The note is listed, and over several days more than 90 investors subscribe amounts of their choosing until the RM150,000 target is filled. The platform deducts its 2% origination fee (RM3,000) and disburses RM147,000 to Sri Cahaya.

Repayment. Production runs through Ramadan; delivery is made on schedule, and the supermarket pays in month five. At month six, Sri Cahaya repays RM157,500 — the RM150,000 principal plus RM7,500 in returns — which the platform distributes to investors. The contract yields a healthy gross margin even after the total financing cost of RM10,500, and Sri Cahaya gains two assets beyond cash: a national retail customer and a successful repayment record that improves its standing for future financing.

Discussion Questions

1. Why was conventional bank financing unsuitable for Sri Cahaya, and which specific features of P2P financing addressed each barrier?
2. Identify two items of alternative data the platform used and explain what each reveals about repayment capacity that a credit-bureau score would miss.
3. Calculate Sri Cahaya's total financing cost and its effective annual cost on the net funds received. Compare your answer with the note's headline 10% rate and explain the difference.
4. From an investor's perspective, evaluate the risks of subscribing to Sri Cahaya's note and explain how diversification across many notes changes the risk–return trade-off.

Chapter Summary

- **3.1:** Digital financing provides credit and capital raising through digital platforms rather than branch-based processes; it exists because a financing gap separates what MSMEs need from what banks profitably supply.
- **3.1:** Banks underserve micro-SMEs because of information asymmetry, collateral requirements, high cost-to-serve and slow approval processes.
- **3.1:** In Malaysia, the Securities Commission Malaysia regulates P2P financing and equity crowdfunding under the Capital Markets and Services Act 2007, with platforms registered as Recognized Market Operators under the *Guidelines on Recognized Markets*.
- **3.2:** P2P financing raises debt from many investors via an online platform; businesses issue investment notes, which are securities promising repayment of principal with agreed returns.
- **3.2:** Platforms assess credit using traditional records plus alternative data such as bank transaction flows, e-payment records and marketplace sales, enabling them to serve thin-file SMEs.
- **3.2:** Malaysia's P2P market has raised RM11.32 billion cumulatively through 154,283 campaigns; in 2025 it raised RM2.83 billion, with 62% of campaigns at RM50,000 or less and 98% funding working capital (SC, 2026).

- **3.3:** Crowdfunding comes in four types — donation, reward, debt and equity; equity crowdfunding gives investors shares, with returns depending entirely on the company's success.
- **3.3:** Malaysia was the first ASEAN country to regulate ECF (2015); the market has raised RM917.51 million cumulatively across 445 campaigns, dominated by seed and Series A stage companies (SC, 2026).
- **3.4:** Platforms charge listing/origination, success, servicing, investor and late fees; because fees are deducted upfront, the issuer's effective annual cost exceeds the headline rate on the note.
- **3.5:** Key risks are default, platform failure, information asymmetry and fraud; the SC mitigates them through RMO registration, disclosure duties and retail investment limits.
- **3.5:** Digital financing advances financial inclusion for MSMEs and B40 entrepreneurs — 26% of P2P funds in 2025 were Shariah-compliant — but digital literacy gaps and over-indebtedness remain concerns.

Key Terms

- **Digital financing**: the provision of financial services, especially credit and capital raising, through digital technologies and platforms rather than traditional branch-based processes.
- **Peer-to-peer (P2P) financing**: raising debt funding from many investors through an online platform that matches funders directly with business borrowers, bypassing banks as intermediaries.
- **Investment note**: a security issued through a registered P2P platform evidencing the issuer's obligation to repay investors principal plus agreed returns over a fixed tenor.
- **Issuer**: the business that raises funds on a P2P or ECF platform by issuing investment notes or shares, and bears the obligation to repay or to share ownership.
- **Recognized Market Operator (RMO)**: an electronic platform operator registered with the Securities Commission Malaysia to run a market — such as P2P, ECF or digital asset trading — under the Guidelines on Recognized Markets.
- **Crowdfunding**: raising small amounts of money from a large number of people through an online platform to fund a project, cause or business.
- **Equity crowdfunding (ECF)**: a form of crowdfunding in which investors subscribe for shares in a company through an online platform, becoming part-owners who share its profits and losses.
- **Origination fee**: a charge levied by a platform for processing, assessing and arranging a financing facility, usually a percentage of the amount raised, deducted from the proceeds.
- **Effective annual cost**: the total cost of financing, including returns and all fees, expressed as an annual rate on the net funds actually received by the borrower.
- **Credit scoring**: the use of statistical models to estimate the probability that a borrower will repay a debt, typically summarised as a score or risk grade.
- **Alternative data**: non-traditional digital information — such as bank transaction flows, e-payment records and online sales — used to assess creditworthiness when formal records are thin.
- **Information asymmetry**: a situation in which one party to a transaction holds more or better information than the other, complicating lending and investment decisions.
- **Default risk**: the risk that a borrower fails to make required payments, causing lenders or investors to lose returns, principal or both.
- **Pre-money valuation**: the agreed value of a company immediately before new invest-

ment money is added in a funding round.

- **Post-money valuation**: a company's value immediately after a funding round, equal to the pre-money valuation plus the amount of new capital raised.
- **Dilution**: the reduction in existing shareholders' ownership percentage when new shares are issued to new investors, even though their share count is unchanged.
- **Financial inclusion**: the condition in which individuals and businesses can access and use affordable, suitable financial products and services.
- **Shariah-compliant**: structured in accordance with Islamic law, avoiding prohibited elements such as riba (interest) through contracts like murabahah, tawarruq or wakalah.

Test Your Knowledge

Part A: Multiple Choice

1. In Malaysia, peer-to-peer financing and equity crowdfunding platforms are regulated by: A. Bank Negara Malaysia B. Securities Commission Malaysia C. Payments Network Malaysia D. The Ministry of Finance

2. According to the SC Annual Report 2025, the vast majority of P2P campaigns in Malaysia — 98% — were conducted to finance: A. Property purchases B. Working capital C. Research and development D. Share buy-backs

3. An investment note is best described as: A. A bank deposit insured against loss B. A share in the issuing company C. A security evidencing the issuer's obligation to repay principal with agreed returns D. A licence to operate a digital platform

4. Which of the following best explains why banks underserve micro-SMEs? A. Micro-SMEs never repay their loans B. The fixed cost of assessing small loans is high relative to the profit they generate C. Banks are legally forbidden from lending to small businesses D. Micro-SMEs prefer to borrow from unlicensed moneylenders

5. Which type of crowdfunding gives backers shares in the company? A. Donation-based B. Reward-based C. Debt-based D. Equity-based

6. A business issues a six-month, RM100,000 investment note at 8% per annum (simple) with a 2% origination fee deducted upfront. Ignoring other fees, the approximate effective annual cost on the net funds received is: A. 8.0% B. 10.0% C. 12.2% D. 16.0%

7. A company valued at RM2 million pre-money raises RM500,000 through ECF. The new investors' ownership stake after the round is: A. 15% B. 20% C. 25% D. 33%

8. The main regulatory purpose of investment limits for retail investors under the Guidelines on Recognized Markets is to: A. Guarantee positive returns for small investors B. Limit the losses any single individual can suffer from platform-based investments C. Reserve the best investments for institutions D. Increase the fees platforms can charge

9. Which of the following is an example of alternative data used in digital credit assessment? A. A company's memorandum of association B. Six months of business bank-account transaction flows C. The founder's university degree D. A property title deed pledged as collateral

10. The share of P2P funds in Malaysia that was Shariah-compliant in 2025 was approximately: A. 4.1% B. 26% C. 51% D. 98%

Part B: Short Answer

1. Explain two reasons why traditional banks underserve Malaysian micro-SMEs, and describe how P2P financing platforms address each reason.

2. Distinguish between donation, reward, debt and equity crowdfunding, giving one suitable Malaysian example for each type.

3. Describe the six steps a Malaysian SME typically passes through to raise working capital via a P2P investment note, from application to repayment.

4. A note carries a headline rate of 10% per annum. Explain, with a simple numerical illustration, why the issuer's effective annual cost is higher than 10%.

5. Discuss the trade-off between financial innovation and investor protection in the regulation of P2P financing and ECF in Malaysia, referring to the role of the Securities Commission Malaysia.

Answer Guide

Part A

1. **B** — The SC regulates capital-market fintech, including P2P and ECF, under the CMSA 2007; BNM regulates banks and payments.
2. **B** — 98% of P2P campaigns finance working capital (SC, 2026).
3. **C** — An investment note is a debt security issued through an RMO; it is neither a deposit nor a share.
4. **B** — Small loans cost almost as much to underwrite as large ones but generate far less revenue, making them uneconomic for banks.
5. **D** — Only equity crowdfunding transfers ownership; donation and reward types provide no financial return, and debt provides repayment.
6. **C** — Returns = RM4,000; fee = RM2,000; total cost RM6,000 on RM98,000 net received = 6.12% for six months ≈ 12.2% annualised.
7. **B** — Post-money = RM2.5 million; RM500,000 ÷ RM2,500,000 = 20%.
8. **B** — Caps per issuer and per period contain how much a retail investor can lose (SC, 2024).
9. **B** — Transaction-flow data is digital, behavioural and non-traditional; the other options are formal documents or collateral.
10. **B** — 26% of P2P funds were Shariah-compliant in 2025 (SC, 2026); 4.1% is the ECF figure.

Part B (model answers)

1. Two reasons are (i) *information asymmetry*: banks demand audited accounts and credit histories that informal micro-SMEs cannot produce, so applications are declined; and (ii) *high cost-to-serve/collateral requirements*: small loans are unprofitable to underwrite manually, and entrepreneurs without property cannot meet collateral demands. P2P platforms address these by using alternative data (bank transaction flows, e-payment records, marketplace sales) for automated credit scoring, which lowers assessment cost and serves thin-file firms, and by issuing unsecured notes sized to small needs, removing the collateral barrier.

2. *Donation-based*: backers receive nothing material — for example, funding a community flood-relief appeal. *Reward-based*: backers receive a product or token — for example, preordering a Malaysian board game before production. *Debt-based (P2P)*: backers receive repayment with returns — for example, an investment note financing a Selangor food manufac-

turer's working capital. *Equity-based (ECF)*: backers receive shares — for example, buying equity in a Kuala Lumpur tech startup on a registered platform. Debt and equity types are regulated by the SC because they involve securities.

3. (1) *Application*: the SME applies online with SSM records, bank statements, financials and funding purpose. (2) *Credit assessment*: the platform verifies documents, analyses traditional and alternative data, and assigns a risk grade. (3) *Listing*: the note is published with amount, tenor, rate and purpose. (4) *Funding*: investors subscribe until the target is met; otherwise the campaign lapses. (5) *Disbursement*: the platform deducts fees and releases net proceeds. (6) *Repayment*: the issuer pays instalments or a bullet payment at maturity, which the platform distributes to investors.

4. The headline rate is paid on the face amount, but fees reduce the cash the issuer actually receives. For example, on a RM100,000 one-year note at 10% with a 2% origination fee, the issuer pays RM10,000 in returns plus a RM2,000 fee (RM12,000 total) while receiving only RM98,000. The effective annual cost is RM12,000 ÷ RM98,000 ≈ 12.2%, which is higher than the headline 10%. Servicing or late fees would raise the effective cost further.

5. Digital financing widens access to capital for MSMEs excluded from bank credit, but exposes retail investors to default, platform failure and fraud. Overly lax rules invite scandals that destroy trust; overly strict rules would strangle a channel that serves borrowers banks ignore. Malaysia balances the two by allowing the activity inside a regulated perimeter: the SC registers platforms as Recognized Market Operators, requires due diligence and disclosure, and caps retail investment amounts, while sophisticated investors enjoy wider access (SC, 2024; Zetzsche et al., 2017). This permits innovation while containing the harm any single failure can inflict.

Problem-Solving Corner

Problem 1: Effective Annual Cost of a Six-Month Investment Note

Sri Cahaya Food Products Sdn. Bhd. issues a **RM150,000** investment note with a **six-month tenor** at **10% per annum (simple)**, repayable as a single bullet payment at maturity. The platform charges a **2% origination fee**, deducted from the proceeds. There are no other fees.

Assumptions: simple (not compounded) returns; bullet repayment at month six; fee deducted upfront.

Solution

Step 1 — Returns payable to investors. Returns = RM150,000 × 10% × (6/12) = **RM7,500**

Step 2 — Origination fee. Fee = RM150,000 × 2% = **RM3,000**

Step 3 — Net cash received by the issuer. Net proceeds = RM150,000 − RM3,000 = **RM147,000**

Step 4 — Amount repaid at maturity. Repayment = RM150,000 + RM7,500 = **RM157,500**

Step 5 — Total financing cost and six-month cost rate. Total cost = RM7,500 + RM3,000 = **RM10,500** Six-month cost = RM10,500 ÷ RM147,000 = **7.14%** (per six months)

Step 6 — Annualise. Simple annualisation: 7.14% × 2 = **14.29% per annum** Effective annual rate (compounded): $(1 + 0.0714)^2 - 1$ = **14.80% per annum**

Interpretation. Although the note's headline rate is 10% per annum, the issuer's effective annual cost is about **14.3%–14.8%**, because the fee raises the cost and the fee deduction shrinks the usable funds. If the platform also charged a monthly servicing fee, the effective cost would be higher still.

Problem 2: Investor Dilution in an Equity Crowdfunding Campaign

TechSihat Sdn. Bhd., a health-tech startup in Kuala Lumpur, has **1,000,000 ordinary shares** outstanding, all held by its two founders. The company raises **RM500,000** through an ECF campaign at an agreed **pre-money valuation of RM2,000,000**.

(a) Compute the price per share, the number of new shares issued, and the founders' ownership after the campaign.

(b) Suppose the founders had accepted a pre-money valuation of RM1,500,000 instead. How does their ownership change?

(c) Does the dilution in part (a) necessarily make the founders worse off? Explain.

Solution

(a) Price per share = pre-money valuation ÷ existing shares = RM2,000,000 ÷ 1,000,000 = **RM2.00**. New shares issued = amount raised ÷ price per share = RM500,000 ÷ RM2.00 = **250,000 shares**. Post-money valuation = RM2,000,000 + RM500,000 = **RM2,500,000**; total shares = 1,250,000. Investors' stake = 250,000 ÷ 1,250,000 = **20%**; founders' stake = **80%** — dilution of 20 percentage points.

(b) Price per share = RM1,500,000 ÷ 1,000,000 = RM1.50. New shares = RM500,000 ÷ RM1.50 ≈ **333,333 shares**; total ≈ 1,333,333. Investors' stake = RM500,000 ÷ RM2,000,000 post-money = **25%**; founders' stake falls to **75%**. A lower pre-money valuation means the founders give up more ownership for the same money.

(c) Not necessarily. Before the round, the founders owned 100% of a company worth RM2,000,000 (RM2,000,000 in value). After the round, they own 80% of a company worth RM2,500,000 — also RM2,000,000 on paper, *plus* the company now holds RM500,000 in cash to fund growth. If the money raises the company's value over time, the founders' smaller percentage of a larger company can be worth far more than 100% of a stagnant one. Dilution reduces *percentage ownership*, not automatically *value*.

Problem 3: Diversification and Default Risk for a P2P Investor

Pn. Lim, a retail investor, places **RM10,000** into P2P notes. Consider two strategies over six months. Every note pays 10% per annum (simple) with bullet repayment; one issuer out of every twenty defaults, paying nothing during the tenor, and investors eventually recover **40% of principal** from the defaulted issuer's assets. Ignore platform fees and taxes.

Strategy A: all RM10,000 in one note. *Strategy B:* RM500 into each of 20 notes.

Solution

Strategy A (concentrated). If the single note is the one that defaults: recovery = 40% × RM10,000 = RM4,000. Net outcome = **−RM6,000 (−60%)** in six months. If it does not default, the gain is RM10,000 × 5% = RM500 (+5%). The outcome is binary — large loss or modest gain.

Strategy B (diversified). Nineteen notes perform; one defaults. - Principal returned by performing notes: 19 × RM500 = RM9,500 - Returns from performing notes: 19 × (RM500 × 10% × 6/12) = 19 × RM25 = **RM475** - Recovery from the defaulted note: 40% × RM500 = **RM200** - Total received = RM9,500 + RM475 + RM200 = **RM10,175** - Net gain = **RM175**, i.e. about **1.75% over six months (≈3.5% annualised)**

Interpretation. With diversification, a single default is an expected cost that returns from the other nineteen notes largely absorb; the portfolio still earns a profit. Without diversification, the same default event destroys 60% of the capital. This is why spreading small amounts across many notes — and respecting the retail investment limits in the *Guidelines on Recognized Markets* — is fundamental risk management for P2P investors. It also shows that a portfolio's *expected* return is always lower than the headline note rate once realistic defaults are priced in.

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Chapter 4: Introduction to Blockchain Technology and Applications

Chapter Outline

- **4.1 Blockchain and Distributed Ledger Technology (DLT)**
- **4.2 Key Components of Blockchain (Blocks, Transactions, Nodes)**
- **4.3 Types of Blockchain Networks and Applications**
- **4.4 Benefits, Limitations and Real-World Uses of Blockchain**

Learning Outcomes

After completing this chapter, you should be able to:

1. **Define** blockchain and distributed ledger technology (DLT), and state the relationship between them. *(Bloom's level: Remember)*
2. **Explain** the anatomy of a block, the transaction lifecycle, and the roles of full nodes and light nodes in a blockchain network. *(Bloom's level: Understand)*
3. **Apply** the logic of hash-linkage and public-key cryptography to reason about tamper-evidence and transaction security. *(Bloom's level: Apply)*
4. **Differentiate** public, private and consortium blockchain networks, and compare proof-of-work and proof-of-stake consensus mechanisms. *(Bloom's level: Analyse)*
5. **Evaluate** the benefits and limitations of blockchain technology for financial services, including the scalability trilemma, energy use and regulatory uncertainty. *(Bloom's level: Evaluate)*
6. **Recommend** an appropriate record-keeping design (blockchain or conventional database) for a given Malaysian business scenario, justifying your choice. *(Bloom's level: Create)*

Opening Vignette

Mei Lin is a second-year diploma student in Penang. Her aunt runs a small food business that exports halal-certified frozen *kuih* to buyers in Singapore and Indonesia. Every month, her aunt complains about the same problems: cross-border payments take days to arrive and

the bank charges are hard to predict; buyers keep asking for "proof" that the supply chain is truly halal; and paperwork for trade financing seems to go around in circles.

At a college fintech talk, Mei Lin hears that a technology called **blockchain** is being tested by Bank Negara Malaysia and other central banks to settle cross-border payments in minutes, and that some halal producers use it to record every step of a product's journey. She wonders: Is this the same "blockchain" behind Bitcoin? Can it really help a small business like her aunt's? And if it is so wonderful, why is everyone not already using it?

This chapter answers Mei Lin's questions. By the end, you will understand what blockchain is, how it works, where it is genuinely useful, and — just as importantly — where it is not.

4.1 Blockchain and Distributed Ledger Technology (DLT)

4.1.1 Defining Blockchain and DLT

When people hear the word "blockchain", many immediately think of Bitcoin. The two are related, but they are not the same thing. Bitcoin is one application; blockchain is the underlying record-keeping technology that makes it possible.

A **distributed ledger technology (DLT)** is a digital system for recording transactions in which the ledger — the record of who owns what and who paid whom — is shared, replicated and synchronised across multiple computers (nodes) in a network, rather than being kept by a single central authority (Pompella & Matousek, 2021). In a conventional system, a bank keeps *the* master ledger on its own servers, and everyone else must trust the bank's copy. In a DLT system, every participating node can hold a copy of the same ledger, and the copies are kept in agreement through automated rules.

A **blockchain** is a specific type of DLT in which transactions are grouped into *blocks*, and each new block is cryptographically linked to the block before it, forming a chronological chain (Nakamoto, 2008). The key idea is captured in the original Bitcoin white paper: a "peer-to-peer electronic cash system" in which transactions are recorded in a chain of blocks secured by cryptographic proof instead of trust in a central intermediary (Nakamoto, 2008).

This leads to an important logical relationship that examiners love to test:

- **All blockchains are DLT** — a blockchain is always a shared, replicated ledger.
- **Not all DLT is blockchain** — some distributed ledgers organise records in other structures (for example, directed graphs or transaction-by-transaction validation) rather than chaining blocks.

4.1.2 Three Defining Properties: Append-Only, Cryptographic Linking, Consensus

Three properties give a blockchain its distinctive character.

First, the ledger is append-only. New transactions can be added, but past entries are not edited or deleted. If a mistake is made, it is corrected by adding a *new* reversing entry — similar to how accountants correct errors with adjustment journals rather than erasing figures. Arslanian (2022) describes this as creating an immutable, time-stamped audit trail that any participant can inspect.

Second, records are cryptographically linked. Each block contains a digital fingerprint (a *hash*) of the previous block. Because of this linking, altering an old record would break the chain and be immediately visible to every node. We will work through exactly how this functions in Section 4.2 and again in the Problem-Solving Corner.

Third, the network reaches agreement by consensus. Since there is no central administrator, the nodes follow a shared set of rules — a **consensus mechanism** — to decide which new transactions are valid and which version of the ledger is the official one (Nakamoto, 2008). Consensus replaces the role traditionally played by a bank's back office or a clearing house.

> **QUICK NOTE:** "Distributed" does not mean "no rules". A blockchain network still has strict rules — they are simply enforced by software and mathematics across many computers, instead of by one institution's internal procedures.

4.1.3 A Brief History: From Nakamoto (2008) to Enterprise DLT

In October 2008, in the middle of the global financial crisis, a person or group using the pseudonym Satoshi Nakamoto published a nine-page white paper, *Bitcoin: A Peer-to-Peer Electronic Cash System* (Nakamoto, 2008). The Bitcoin network itself launched in January 2009. Its design solved a long-standing problem in digital money — the *double-spending problem*, in which the same digital coin could be copied and spent twice — without needing a central bank or payment company to police it.

The Bitcoin design also introduced an unusual monetary policy: the supply of bitcoin is capped at 21 million coins, and the reward paid to the computers that maintain the network is cut in half roughly every four years in an event known as the *halving*. The April 2024 halving reduced the block reward to 3.125 BTC.

The second great leap came from Vitalik Buterin, who proposed Ethereum in a 2014 white paper and launched the network in 2015 (Buterin, 2014). Ethereum generalised Nakamoto's idea: instead of recording only payments, its blockchain could run small programs called *smart contracts* (Section 4.3.2). This opened the door to decentralised applications, token issuance and much of today's digital-asset ecosystem. The years 2017–2018 saw a global boom in initial coin offerings (ICOs), followed by sharp corrections and, later, high-profile failures such as the collapse of the FTX exchange in November 2022 — events that pushed regulators worldwide, including in Malaysia, to build clearer rules (Securities Commission Malaysia [SC], 2024a).

From about 2015 onwards, banks and technology firms began asking whether the *ledger-sharing* idea could be separated from public cryptocurrencies. This produced "enterprise DLT" platforms — private or consortium ledgers designed for institutions that know each other. Central banks joined in: the Bank for International Settlements (BIS) has documented a worldwide wave of central bank digital currency (CBDC) research that builds on DLT concepts (Auer et al., 2020; Bank for International Settlements [BIS], 2021). Malaysia's own Project Dunbar, the case study at the end of this chapter, is part of this story.

> **DID YOU KNOW? (TRIVIA):** Nobody knows who Satoshi Nakamoto is. The white paper appeared on a cryptography mailing list in 2008, the author corresponded with early developers until about 2010, and then disappeared — leaving behind an invention that has been copied thousands of times. The identity remains one of technology's great unsolved mysteries.

For Malaysia, this history matters because local regulation evolved in response to it. Digital currencies and digital tokens were formally prescribed as securities under Malaysian law in January 2019, bringing them within the SC's jurisdiction (Capital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019). In other words, Malaysia did not ban the technology; it brought it inside the regulatory perimeter.

4.2 Key Components of Blockchain (Blocks, Transactions, Nodes)

To understand why a blockchain is hard to tamper with, you need to look inside it. This section examines three building blocks of the technology: the blocks themselves, the transactions they contain, and the nodes that maintain the network.

4.2.1 Anatomy of a Block

A **block** is a bundle of validated transactions plus a header of metadata. Think of a page in an accounting ledger: the page lists entries, and the page header carries a page number, a date and a reference to the previous page. Table 4.1 summarises the main fields of a typical block.

Component	What it contains	Why it matters
Block header	Metadata about the block	The part that is hashed and linked
Hash (block ID)	The block's own digital fingerprint	Uniquely identifies the block
Previous-hash	The hash of the block before it	Creates the "chain"; detects tampering
Timestamp	Approximate creation time	Orders the chain chronologically
Nonce	A number miners adjust (in PoW)	Used to satisfy the consensus puzzle
Merkle root	Fingerprint of all transactions in the block	Allows efficient verification
Transaction list	The validated transactions	The actual ledger content

Table 4.1: Simplified anatomy of a block.

Two ideas deserve closer attention: hashing and the Merkle tree.

A **hash** is the output of a hash function — a mathematical algorithm that converts any input (a document, a transaction, an entire block) into a fixed-length string of characters, such as a 256-bit number. Hash functions used in blockchains have three vital properties (Arslanian, 2022):

1. **Deterministic** — the same input always gives the same output.
2. **One-way** — you cannot reconstruct the input from the output.
3. **Avalanche effect** — changing even one character of the input completely changes the output.

The previous-hash field is what turns isolated blocks into a *chain*. Block 101 stores the hash of Block 100; Block 102 stores the hash of Block 101; and so on. If anyone alters the data in Block 101, its hash changes (avalanche effect), so the previous-hash stored in Block 102 no longer matches, and every node that checks the chain can see that something is wrong. To hide the change, the attacker would have to redo Block 101 *and every block after it*, faster than the rest of the honest network — which is precisely what consensus mechanisms make impractical.

A **Merkle tree** is a data structure that organises all the transactions in a block into a tree of pairwise hashes, ending in a single Merkle root stored in the header (Nakamoto, 2008). Its benefit is efficiency: to prove that one particular transaction is inside a block, you only need a short branch of hashes — a few hundred bytes — instead of the whole block. This makes lightweight verification possible, which matters for mobile users.

> **QUICK NOTE:** A hash is a *fingerprint*, not an encryption. Encryption is meant to be decrypted later by someone with a key; a hash is one-way and is used purely for checking integrity and identity.

4.2.2 The Transaction Lifecycle

Suppose Mei Lin wants to send 0.01 BTC to a supplier. The lifecycle of that transaction in a public blockchain typically runs as follows (Harvey et al., 2021):

1. **Creation.** Mei Lin's wallet software constructs a transaction message: "from my address, to the supplier's address, 0.01 BTC, plus a network fee".
2. **Digital signature.** The wallet signs the transaction with Mei Lin's *private key*, proving authorisation (see Section 4.2.3).
3. **Broadcast.** The signed transaction is sent to the network and spreads from node to node.
4. **Validation.** Nodes independently check the signature, confirm that Mei Lin's address has enough unspent funds, and confirm the funds have not already been spent elsewhere.
5. **Inclusion in a block.** A validator (in Bitcoin, a successful miner) bundles the transaction with others into a candidate block.
6. **Consensus and confirmation.** The network accepts the new block according to its consensus rules; the transaction now has one confirmation.
7. **Settlement finality.** As more blocks are added on top, reversing the transaction becomes progressively harder. In Bitcoin, receivers of large payments conventionally wait for several confirmations.

Compare this with a conventional bank transfer: the bank validates, clears and settles everything internally, and customers simply trust the bank's records. A blockchain distributes those steps across the network, replacing institutional trust with cryptographic proof and economic incentives — what Buterin (2014) called moving from trusted intermediaries to "trustless" execution, meaning users need not trust any single counterparty.

4.2.3 Nodes: Full Nodes, Light Nodes, Mining and Validation

A **node** is any computer running the blockchain's software and participating in its network. Not all nodes do the same work.

- A **full node** downloads and stores the entire blockchain, independently checks every transaction and block against the consensus rules, and relays valid data to other nodes.

Full nodes are the backbone of the network's security: they do not take anyone's word for anything.
- A **light node** (also called a *light client* or *simplified payment verification* client) stores only block headers and uses Merkle proofs to verify specific transactions. Light nodes trade some independence for convenience, which is why wallet apps on mobile phones are usually light clients (Nakamoto, 2008).

Mining and validation describe how new blocks get made. In proof-of-work systems such as Bitcoin, specialised nodes called *miners* compete to solve a computational puzzle — essentially, finding a **nonce** that, when hashed together with the block's contents, produces a hash below a network-set difficulty target. The first miner to find a valid solution broadcasts the block and collects the block reward plus transaction fees. The process is deliberately costly, so that cheating costs more than honesty (Nakamoto, 2008). In proof-of-stake systems, there is no mining race; instead, *validators* are chosen to propose and attest to blocks in proportion to the amount of cryptocurrency they have locked up ("staked") as collateral, and misbehaving validators can lose part of their stake (Buterin, 2014).

> **DID YOU KNOW? (TRIVIA):** Bitcoin's difficulty adjusts automatically every 2,016 blocks so that, no matter how much computing power joins or leaves the network, blocks keep arriving roughly every ten minutes. When mining power surged over the years, the puzzles simply got harder — a self-regulating feature written into the original code.

4.2.4 Public-Key Cryptography and Consensus Mechanisms

Public-key cryptography is the security foundation on which blockchain ownership rests. Each user has a mathematically linked pair of keys: a **private key** (kept secret, like a password that must never be shared) and a **public key** (shared openly, from which a receiving *address* is derived). Data signed with the private key can be verified by anyone using the public key, which proves the owner authorised the transaction without revealing the private key itself (Nakamoto, 2008). Possession of the private key *is* ownership of the coins: lose the key and the funds are unreachable; let someone steal it and the funds are gone. This is why the industry repeats the warning "not your keys, not your coins" — a lesson Malaysian investors should remember when deciding whether to keep assets on an exchange or in self-custody (Arslanian, 2022).

A **consensus mechanism** is the rulebook by which nodes agree on the state of the ledger. The two dominant families are compared in Table 4.2.

Feature	Proof of Work (PoW)	Proof of Stake (PoS)
Who creates blocks	Miners competing with computing power	Validators selected based on staked coins
Resource committed	Electricity and hardware	Capital (locked-up coins)
Energy consumption	Very high by design	Dramatically lower
Dishonesty punished by	Wasted energy and lost rewards	Slashing of the stake
Main examples	Bitcoin	Ethereum (since September 2022)

Table 4.2: Proof of work versus proof of stake.

Proof of work (PoW) secures the ledger by making block production expensive. Its strength is a long, battle-tested security record; its weakness is enormous electricity consumption, because thousands of miners worldwide race to solve the same puzzle and all but one of them waste their effort on every block.

Proof of stake (PoS) replaces the energy race with an economic bond: validators lock up capital and are penalised ("slashed") if they approve invalid transactions. The most important real-world demonstration was **Ethereum's "Merge" in September 2022**, when the second-largest blockchain switched from PoW to PoS, eliminating its mining industry overnight and cutting its energy use to a small fraction of the previous level. The Merge proved that a major live network could change its consensus engine without stopping — a feat sometimes compared to swapping an aircraft's engines mid-flight.

> **BRAINSTORMER:** Proof of work deliberately wastes energy to create security; proof of stake relies on financial penalties instead. Which approach do you think gives stronger *decentralisation* — and does the answer change if a few large companies control most mining hardware, or a few wealthy holders control most staked coins?

In the Malaysian context, consensus design is not an abstract curiosity. Bank Negara Malaysia (BNM) has indicated that a future central bank digital currency would use *permissioned* ledgers rather than open PoW mining (Bank Negara Malaysia [BNM], 2022). That choice reflects exactly the trade-offs in Table 4.2: when participants are known, regulated institutions, there is no need to burn electricity to keep strangers honest.

4.3 Types of Blockchain Networks and Applications

4.3.1 Public, Private and Consortium Networks

Not all blockchains are open to everyone. Networks are usually classified by *who may read the ledger* and *who may validate transactions* (Gupta & Tham, 2019).

Type	Who can read	Who can validate	Typical use
Public (permissionless)	Anyone	Anyone	Cryptocurrencies, open DeFi
Private (permissioned)	One organisation	That organisation	Internal record-keeping, audits
Consortium (permissioned)	A group of organisations	Pre-approved members	Interbank platforms, trade finance

Table 4.3: Types of blockchain networks.

A **public blockchain** such as Bitcoin or Ethereum is open: anyone can download the ledger, create an address and (subject to the consensus rules) participate in validation. Its strengths are openness, neutrality and censorship resistance; its weaknesses are limited throughput, high fees at busy times and energy cost where PoW is used.

A **private blockchain** is run by a single organisation that controls access. It sacrifices decentralisation for speed, privacy and control. Critics note that a private ledger guarded by one

company is arguably just a database with extra cryptography — which is why the decision framework in Section 4.4.4 begins by asking whether multiple mutually distrusting parties are involved.

A **consortium blockchain** (also called a federated or **permissioned blockchain**) is governed by a group of institutions — for example, a panel of banks — that share validation duties under agreed rules. This middle path is the dominant design in finance, because regulated institutions need known counterparties, data confidentiality and clear accountability. Malaysia's digital-asset rules assume exactly this regulated environment: digital asset exchanges must register with the SC as recognised market operators under the *Guidelines on Recognized Markets* (SC, 2024b).

4.3.2 Smart Contracts and Tokenisation

A **smart contract** is a self-executing program stored on a blockchain that automatically carries out the terms of an agreement when predefined conditions are met (Buterin, 2014). The classic analogy is a vending machine: insert the correct payment, and the machine releases the item — no shop assistant needed. In Ethereum's design, smart contracts are code deployed to the blockchain and executed identically by every validating node, so the outcome cannot be quietly altered by either party (Buterin, 2014).

Smart contracts enable escrow without an escrow agent, automatic insurance payouts when a flight is delayed, and the entire field of decentralised finance (DeFi), in which lending, trading and derivatives are performed by code rather than by financial institutions (Harvey et al., 2021). They also carry risks: bugs in the code execute faithfully, and "code is law" offers little comfort when a flawed contract pays out to the wrong address.

Tokenisation is the process of creating a digital token on a blockchain (or other DLT) that represents ownership of, or a claim on, an underlying asset — which may be financial (a bond, a unit in a fund, a deposit) or real-world (property, commodities, carbon credits). Tokenisation promises fractional ownership (you could hold RM1,000 of a RM5 million asset), faster settlement and programmable compliance. Bank Negara Malaysia has taken this seriously: in October 2025 it published a *Discussion Paper on Asset Tokenisation in the Malaysian Financial Sector* to gather industry views on the opportunities and risks of tokenising financial assets in Malaysia (BNM, 2025a). BNM has also signalled work on tokenised deposits and ringgit-denominated stablecoins as part of its 2026 agenda.

4.3.3 Applications: Payments, Trade Finance, Supply Chain, Identity, Tokenisation

Blockchain applications in and around financial services cluster into several families (Pompella & Matousek, 2021; Lee & Shin, 2018).

Payments and cross-border transfers. Public cryptocurrencies allow value to move globally without correspondent banks, and DLT platforms allow regulated institutions to settle obligations with each other more directly. Central bank experiments such as Project Dunbar and Project mBridge — in both of which BNM participates — apply the idea at the wholesale level, aiming to make cross-border settlement faster and cheaper (Bank for International Settlements Innovation Hub [BIS Innovation Hub], 2021).

Trade finance. Traditional trade finance involves exporters, importers, banks, shipping lines and insurers exchanging paper documents, with heavy reconciliation costs. Shared-ledger

platforms let all parties see the same documents and status, reducing disputes and fraud such as double-financing the same shipment. Several international trade-finance networks have been built by bank consortia using permissioned DLT.

Supply chain and halal traceability. Recording each custody transfer of a product on a shared ledger creates an audit trail from farm to shelf. For Malaysia — a global leader in Islamic finance and halal industries — blockchain-based provenance is attractive because it can document halal integrity: who slaughtered, processed, stored and transported a product, and under which certifications. Scholars of Islamic fintech identify traceability and transparency as natural complements to Shariah requirements on authenticity and ethical dealing (Oseni & Ali, 2019).

Digital identity. Blockchains can anchor self-sovereign identity credentials, in which users hold verifiable proofs (for example, of qualifications or licences) that institutions can check without phoning the issuer. In finance, identity systems interact closely with electronic know-your-customer (e-KYC) processes, which BNM permits for non-face-to-face onboarding under its e-KYC policy (BNM, 2024).

Asset tokenisation. As discussed above, tokens can represent securities, sukuk, fund units or property. Malaysia's regulators were early movers: since January 2019, digital currencies and digital tokens have been prescribed as securities under the 2019 Prescription Order, so offers of such instruments fall within securities law (Capital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019).

> **MALAYSIAN SPOTLIGHT:** Malaysia regulates digital assets rather than banning them. The SC's *Guidelines on Digital Assets* — first issued on 28 October 2020 and revised on 19 August 2024 under section 377 of the Capital Markets and Services Act 2007 — govern digital token offerings, which may only be conducted through registered **initial exchange offering (IEO) operators** (SC, 2024a). Trading takes place on SC-registered digital asset exchanges (DAXs) such as Luno Malaysia, SINEGY Technologies, Tokenize Technology and Hata Digital. Notably, the SC's Shariah Advisory Council resolved in 2020 that digital assets are recognised as *mal* (property) from a Shariah perspective, and that trading them on SC-registered DAXs is permissible subject to conditions — an important ruling for the world's leading Islamic finance hub. Remember, though: cryptocurrency is *not* legal tender in Malaysia.
>
> **QUICK NOTE:** "Regulated" does not mean "safe from loss". SC registration means an exchange meets rules on custody, governance and market conduct — it does not guarantee the price of any digital asset, which can still fall sharply, as investors worldwide relearned when the FTX exchange collapsed in November 2022.
>
> **BRAINSTORMER:** If your college issued diploma certificates as blockchain tokens, employers could verify them instantly and forgeries would be nearly impossible. But who should control the "keys" — the college, the student, or a government agency? And what should happen when a graduate wants an old disciplinary record forgotten?

4.4 Benefits, Limitations and Real-World Uses of Blockchain

Balanced judgement requires looking at both sides of the ledger, so to speak. Blockchain is neither magic nor fraud; it is a tool with specific strengths and specific costs.

4.4.1 Benefits

Immutability. Once data is buried under subsequent blocks, altering it is computationally and economically impractical on a well-secured network. This gives blockchains their value as tamper-evident audit trails (Arslanian, 2022).

Transparency and shared truth. All authorised participants see the same ledger, updated simultaneously. In settings where parties currently reconcile separate databases — interbank settlement, trade documentation, supply chains — a single shared source of truth removes disputes and duplicated record-keeping (Gupta & Tham, 2019).

Disintermediation. By letting strangers transact under rules enforced by consensus, blockchains can reduce dependence on certain intermediaries and their fees. Nakamoto's (2008) original motivation was precisely electronic cash "without going through a financial institution". In practice, complete disintermediation is rare; more often, *new* intermediaries (exchanges, custodians, wallet providers) replace old ones, which is why regulators now supervise them (SC, 2024b).

Programmability and resilience. Smart contracts automate conditional payments, while replication across many nodes removes single points of failure — a property relevant to operational-resilience rules for financial institutions such as BNM's Risk Management in Technology (RMiT) requirements (BNM, 2025b).

4.4.2 Limitations and Risks

The scalability trilemma. A widely discussed design constraint holds that a blockchain can optimise only two of three properties — decentralisation, security and scalability — at once (Harvey et al., 2021). Bitcoin is highly decentralised and secure but processes only a handful of transactions per second, whereas conventional payment networks handle thousands. For perspective, Malaysia's own national payment infrastructure is vast: Payments Network Malaysia (PayNet) processed 8.44 billion digital payment transactions in 2025 — a workload far beyond any current public blockchain (Payments Network Malaysia, 2026).

Energy use. PoW networks consume electricity on the scale of entire countries, which sits uneasily with sustainability commitments. Ethereum's Merge in September 2022 showed one escape route — moving to PoS — but Bitcoin remains PoW by design.

Regulatory uncertainty and consumer risk. Rules differ across jurisdictions and continue to evolve. Within Malaysia the framework is clear — digital assets are securities, platforms must be registered — but assets held on *foreign* platforms, and novel products such as DeFi lending, remain largely outside local investor protection (SC, 2024a).

51% attacks. If a single actor gains majority control of a PoW network's mining power (or a PoS network's stake), it can rewrite recent history and double-spend. A **51% attack** is economically irrational against large networks like Bitcoin but has occurred on smaller chains — one reason market capitalisation matters for security.

The oracle problem. Blockchains can verify on-chain data, but they cannot know whether real-world facts are true. A **smart contract** that pays out "if the shipment arrives" must

trust an external data feed — an *oracle* — which reintroduces exactly the trusted third party the design sought to remove. The **oracle problem** is a hard limit on disintermediation for real-world applications (Harvey et al., 2021).

> **QUICK NOTE:** Immutability is a double-edged sword. If you send funds to a wrong address, or a scammer's address, no bank manager can reverse the transaction. This is why Malaysia's anti-scam defences — such as the National Scam Response Centre hotline 997 — focus on the *banking* side of transfers, where reversal is still possible.

4.4.3 Real-World Uses Relevant to Malaysia

Multi-CBDC platforms. Project Dunbar (2021), examined in the case study below, prototyped a shared platform on which multiple central bank digital currencies could be exchanged directly between commercial banks across borders (BIS Innovation Hub, 2021). Its successor in spirit, Project mBridge, reached minimum-viable-product stage in 2024, with BNM participating. BNM's Financial Sector Blueprint 2022–2026 commits to a multi-year, three-phase CBDC exploration — wholesale cross-border first, full-scale wholesale next, and retail consideration last — and BNM has run wholesale digital ringgit proof-of-concept trials, while emphasising that no decision has been made to issue a retail CBDC (BNM, 2022).

Trade-finance networks. Internationally, consortia of banks have operated permissioned DLT platforms for letters of credit, guarantees and supply-chain finance, demonstrating that consortium designs can pass regulatory muster when all validators are known institutions (Pompella & Matousek, 2021).

Halal supply-chain provenance. Pilot projects in the halal economy use shared ledgers to record certification and custody events, supporting Malaysia's ambition to lead the global halal market; transparency of this kind aligns with Shariah values of authenticity and trustworthiness (Oseni & Ali, 2019).

Digital asset markets. Domestically, registered DAXs and IEO operators provide regulated venues for trading and fundraising with digital tokens (SC, 2024a, 2024b) — evidence that, in Malaysia, blockchain finance grows *inside* the regulatory perimeter.

4.4.4 When to Use Blockchain: A Decision Framework

Given the costs, students should resist "blockchain for everything". Ask these questions in order:

1. **Do multiple parties who do not fully trust each other need to write to a shared record?** If one organisation owns the data, a conventional database is cheaper, faster and simpler.
2. **Is there no acceptable trusted central party?** If a regulator, bank or platform operator is available and acceptable, centralisation is usually more efficient.
3. **Do records need strong tamper-evidence and a common history?** If auditability across organisations is the core requirement, DLT adds real value.
4. **Can the needed data be verified on-chain (or via acceptable oracles)?** If not, the oracle problem may defeat the design.
5. **Do throughput, privacy and governance constraints fit a public, consortium or private design?** Match the network type to the trust model (Table 4.3).

Rule of thumb: *blockchain earns its keep when many writers share one ledger, trust is limited, and tamper-evidence matters more than raw speed.* Otherwise, use a database — and spend the savings elsewhere.

Case Study 4.1: Project Dunbar (2021) — A Multi-CBDC Platform with Malaysian Participation

Cross-border payments today typically travel through a chain of correspondent banks, each holding accounts with the next, each taking time and fees, and each operating in limited hours. A payment from Kuala Lumpur to Johannesburg may pass through several intermediaries before it settles. Project Dunbar asked a radical question: what if central banks issued their digital currencies onto a *shared platform*, so that commercial banks could transact directly with each other across borders in central bank money?

The project was led by the BIS Innovation Hub together with Bank Negara Malaysia, the Reserve Bank of Australia, the Monetary Authority of Singapore and the South African Reserve Bank (BIS Innovation Hub, 2021). In 2021 the partners built working prototypes of a multi-CBDC (mCBDC) platform: a common ledger on which each central bank issued its own wholesale CBDC, and on which commercial banks in each jurisdiction could hold and exchange these currencies directly. Two technical prototypes were developed on different enterprise DLT platforms (one built with R3's Corda and another with a Partior/Quorum-based stack), allowing the partners to compare designs.

What it proved. The prototypes demonstrated that a shared mCBDC platform is *technically feasible*: multiple central banks can issue on one platform; commercial banks can transact directly in foreign central bank money; and settlement can be atomic — payment and counter-payment completing together or not at all — reducing settlement risk and potentially cutting the layers of intermediaries in correspondent banking (BIS Innovation Hub, 2021). For Malaysia, participation gave BNM hands-on expertise that feeds directly into the three-phase CBDC exploration announced in the Financial Sector Blueprint 2022–2026 and into later initiatives such as Project mBridge, which reached minimum-viable-product stage in 2024 (BNM, 2022).

Its limits. Dunbar was a prototype, not a production system. It did not resolve the hardest questions, which are political and legal rather than technical: Who governs a platform shared by sovereign nations? Under what conditions may foreign banks hold a country's CBDC? How are disputes, outages and monetary-policy spillovers managed? The partners themselves stressed that the experiment implied *no commitment* to issue any CBDC — and BNM continues to state that no decision has been made on a retail digital ringgit.

Discussion Questions

1. Explain how a shared mCBDC platform could reduce both the *cost* and the *time* of cross-border payments compared with correspondent banking. Which features of DLT are doing the work?
2. Project Dunbar showed the technology works, yet no partner has issued a CBDC as a result. Identify two non-technical barriers and discuss why they may be harder to solve than the engineering.

3. BNM's blueprint sequences CBDC exploration from wholesale cross-border use, to full-scale wholesale, to retail consideration. Why do you think this order was chosen rather than starting with a retail digital ringgit for the public?
4. Suppose Malaysia and Singapore both issued wholesale CBDCs on a common platform. Discuss one opportunity and one risk this would create for Malaysian SMEs that export to Singapore.

Chapter Summary

- **4.1:** Distributed ledger technology is a shared, replicated ledger maintained across many nodes; a blockchain is a *type* of DLT that groups transactions into cryptographically linked blocks (Nakamoto, 2008).
- **4.1:** Three properties define a blockchain: an append-only ledger, cryptographic linking of blocks, and agreement by a consensus mechanism.
- **4.1:** Blockchain history runs from Nakamoto's 2008 white paper and Bitcoin's 2009 launch, through Ethereum's smart contracts (Buterin, 2014), to enterprise DLT and central bank experiments.
- **4.2:** A block contains a header (hash, previous-hash, timestamp, nonce, Merkle root) and a transaction list; previous-hash linkage makes tampering evident.
- **4.2:** Transactions move through creation, signing, broadcast, validation, block inclusion and confirmation; finality grows as blocks accumulate.
- **4.2:** Full nodes store and verify everything; light nodes verify specific transactions with Merkle proofs. Public-key cryptography ties ownership to a secret private key.
- **4.2:** Proof of work secures by expending energy; proof of stake secures by bonding capital. Ethereum's September 2022 Merge moved the second-largest chain from PoW to PoS.
- **4.3:** Networks may be public, private or consortium; financial applications favour permissioned consortium designs with known validators.
- **4.3:** Smart contracts are self-executing blockchain programs; tokenisation represents assets as on-chain tokens, the subject of BNM's 2025 discussion paper.
- **4.3:** Malaysia regulates digital assets as securities under the 2019 Prescription Order, with SC guidelines governing DAXs and IEO operators; crypto is not legal tender.
- **4.4:** Benefits include immutability, transparency and reduced reliance on intermediaries; limitations include the scalability trilemma, PoW energy use, regulatory uncertainty, 51% attacks and the oracle problem.
- **4.4:** Real-world uses relevant to Malaysia include Project Dunbar and mBridge, trade-finance networks, halal supply-chain provenance and regulated digital-asset markets; the decision framework shows when blockchain beats a database.

Key Terms

- **Blockchain**: a type of distributed ledger in which validated transactions are grouped into blocks that are cryptographically linked in chronological order.
- **Distributed ledger technology (DLT)**: a digital record-keeping system in which a ledger is shared, replicated and synchronised across multiple network nodes without a single central authority.

- **Consensus mechanism**: the rules and incentives by which nodes in a distributed network agree on valid transactions and the official state of the ledger.
- **Block**: a bundle of validated transactions together with a header containing metadata such as the hash, previous-hash, timestamp, nonce and Merkle root.
- **Hash**: the fixed-length output of a one-way hash function; a digital fingerprint that uniquely represents any input and changes completely if the input changes.
- **Nonce**: an adjustable number in a block header that proof-of-work miners vary when searching for a hash that meets the network's difficulty target.
- **Merkle tree**: a tree of pairwise hashes over a block's transactions, ending in a single root that enables efficient, lightweight verification of individual transactions.
- **Node**: a computer running a blockchain's software that stores, relays and (depending on type) validates ledger data according to the consensus rules.
- **Full node**: a node that downloads the entire blockchain and independently verifies every transaction and block against the consensus rules.
- **Light node**: a node storing only block headers that uses Merkle proofs to verify specific transactions, suitable for wallets and mobile devices.
- **Public-key cryptography**: an encryption scheme using a linked private key (secret, used to sign) and public key (shared, used to verify), forming the basis of blockchain ownership.
- **Proof of work (PoW)**: a consensus mechanism in which miners compete to solve a computational puzzle, making block production costly and attacks economically impractical.
- **Proof of stake (PoS)**: a consensus mechanism in which validators lock up cryptocurrency as a stake and are penalised for dishonesty, avoiding PoW's energy race.
- **Smart contract**: a self-executing program stored on a blockchain that automatically enforces the terms of an agreement when coded conditions are met.
- **Tokenisation**: the creation of a blockchain-based digital token representing ownership of, or a claim on, a financial or real-world underlying asset.
- **Permissioned blockchain**: a network in which read and/or validation rights are restricted to approved participants, covering private and consortium designs.
- **51% attack**: an attack in which one actor controls the majority of a network's mining power or stake, enabling it to rewrite recent history and double-spend.
- **Oracle problem**: the limitation that blockchains cannot verify off-chain, real-world facts, so smart contracts must rely on trusted external data feeds called oracles.

Test Your Knowledge

Part A: Multiple Choice

Q1. Which statement correctly describes the relationship between blockchain and DLT? A. Blockchain and DLT are two names for exactly the same thing. B. Blockchain is one type of DLT; other distributed ledgers may not use chained blocks. C. DLT is one type of blockchain used only by banks. D. DLT always requires cryptocurrency, whereas blockchain does not.

Q2. The previous-hash field in a block header primarily serves to: A. Encrypt the block's transactions for privacy. B. Compress transactions to save storage space. C. Link blocks so that altering an earlier block breaks the chain's verifiable sequence. D. Reward miners for creating the block.

Q3. The main purpose of a Merkle tree inside a block is to: A. Allow efficient verification that a specific transaction is included, without downloading the whole block. B. Hide the identities

of transaction senders. C. Determine the order in which blocks are produced. D. Store the private keys of network users.

Q4. A friend's phone wallet app wants to confirm that her payment was included in a block without storing the entire blockchain. Which kind of node best suits this need? A. A full node. B. A mining node. C. A light node. D. A central bank node.

Q5. In proof of work, miners compete to: A. Stake the largest amount of cryptocurrency. B. Find a nonce that produces a block hash below the network's difficulty target. C. Obtain the most votes from token holders. D. Be randomly selected by a central administrator.

Q6. Ethereum's "Merge" of September 2022 is significant because the network: A. Launched the first smart contract platform. B. Switched from proof of work to proof of stake, drastically cutting energy use. C. Doubled the block reward to 6.25 ETH. D. Merged with the Bitcoin network.

Q7. A group of ten banks wants a shared ledger for letters of credit, with only the banks allowed to validate and regulators granted read access. Which network type fits best? A. A public permissionless blockchain. B. A private blockchain owned by one bank. C. A consortium (permissioned) blockchain. D. No ledger is needed; e-mail is sufficient.

Q8. Smart contracts, as implemented on Ethereum (Buterin, 2014), are best described as: A. Legally binding paper contracts scanned onto a blockchain. B. Self-executing programs that carry out agreement terms automatically when conditions are met. C. Artificial-intelligence systems that negotiate contracts between humans. D. Contracts that can always be reversed by the platform operator.

Q9. A 51% attack refers to a situation in which: A. 51% of a network's users forget their private keys. B. One actor controls the majority of mining power or stake and can rewrite recent history. C. Transaction fees rise by 51% during congestion. D. Regulators shut down 51% of exchanges.

Q10. Under Malaysian law and regulation: A. Cryptocurrency is legal tender issued by Bank Negara Malaysia. B. Digital currencies and digital tokens are prescribed as securities, and token offerings must go through SC-registered IEO operators. C. Digital asset trading is completely banned. D. Digital assets are regulated by the Ministry of Domestic Trade.

Part B: Short Answer

Q1. Explain what "append-only" means and how cryptographic linking makes a blockchain tamper-evident.

Q2. Distinguish between a full node and a light node, and explain why light nodes matter for the practical adoption of blockchain wallets in a mobile-first country like Malaysia.

Q3. Compare proof of work and proof of stake in terms of (a) the resource committed, (b) energy consumption, and (c) how dishonest behaviour is punished.

Q4. BNM states that any Malaysian CBDC would be a central bank liability, not a crypto-asset, and would use permissioned ledgers. Explain what this means and why a central bank would reject public proof-of-work mining for its digital currency.

Q5. A halal food exporter asks whether to record its supply chain on a blockchain. Apply the five-question decision framework from Section 4.4.4 and state your recommendation with reasons.

Answer Guide

Part A: - **Q1: B** — All blockchains are DLT, but DLT includes non-blockchain structures (Section 4.1.1). - **Q2: C** — Previous-hash linkage creates the tamper-evident chain (Section 4.2.1). - **Q3: A** — Merkle proofs allow light verification of individual transactions (Section 4.2.1). - **Q4: C** — Light nodes store only headers and use Merkle proofs, ideal for mobile wallets (Section 4.2.3). - **Q5: B** — PoW miners search for a nonce yielding a hash below the difficulty target (Section 4.2.3). - **Q6: B** — The Merge switched Ethereum from PoW to PoS in September 2022 (Section 4.2.4). - **Q7: C** — Known institutional validators with restricted access = consortium design (Section 4.3.1). - **Q8: B** — Smart contracts are self-executing on-chain code (Section 4.3.2). - **Q9: B** — Majority control of mining/stake enables history rewriting and double-spending (Section 4.4.2). - **Q10: B** — The 2019 Prescription Order makes digital assets securities; the SC's Guidelines on Digital Assets channel offerings through IEO operators; crypto is not legal tender (Section 4.3.3).

Part B (model answers): - **Q1:** Append-only means entries can be added but past entries are never edited or deleted; corrections are new entries. Because each block stores the previous block's hash, changing old data changes that block's hash and breaks the match with the next block, so every node can detect the tampering. - **Q2:** A full node stores the whole blockchain and verifies everything independently; a light node stores only block headers and relies on Merkle proofs to verify particular transactions. Light nodes make blockchain usable on smartphones with limited storage and data plans, which is essential in a mobile-first market where most users access services through phones. - **Q3:** PoW commits computing power and electricity; PoS commits capital as locked-up stake. PoW consumes very large amounts of energy by design, while PoS needs only enough to run validator software. PoW punishes dishonesty through wasted energy and forfeited rewards; PoS slashes (confiscates part of) the offender's stake. - **Q4:** A CBDC would be a direct claim on Bank Negara Malaysia, like physical ringgit, rather than a privately issued speculative asset. A permissioned ledger restricts validation to approved, accountable institutions, so the central bank retains control over issuance, governance, privacy and performance — and there is no need to spend huge amounts of electricity to keep anonymous strangers honest. - **Q5:** (1) Multiple parties — farmers, processors, certifiers, logistics firms and buyers — do write to a shared record: yes. (2) No single party is trusted by all to keep the master record: yes. (3) Tamper-evident shared history matters for halal integrity: yes. (4) Much of the data (certificates, custody events) must come from the real world, so trustworthy oracles/certifiers are needed: a caveat. (5) A consortium permissioned chain fits the trust model. Recommendation: proceed with a consortium blockchain *if* recognised halal certifiers act as data sources; otherwise strengthen conventional certification first.

Problem-Solving Corner

Problem 1: Following the Chain — A Hash-Linkage Reasoning Exercise

A simplified teaching blockchain uses this toy rule for a block's hash:

> hash = (previous-hash value + data value) mod 65,536, expressed as a hexadecimal number.

The ledger currently contains three blocks recording payments (in RM):

Block	Data value (RM)	Previous-hash stored	Computed hash
1	500	0 (genesis)	?
2	750	hash of Block 1	?
3	250	hash of Block 2	?

(a) Compute the hash of each block. **(b)** An attacker changes Block 2's data from RM750 to RM5,000. Recompute Block 2's hash. Does the previous-hash stored in Block 3 still match? What does the network conclude? **(c)** To hide the change, what else must the attacker do, and why does consensus make this impractical on a real network?

Solution. (a) Block 1: (0 + 500) mod 65,536 = 500 → hex **0x01F4**. Block 2: (500 + 750) mod 65,536 = 1,250 → hex **0x04E2**. Block 3: (1,250 + 250) mod 65,536 = 1,500 → hex **0x05DC**.

(b) Tampered Block 2: (500 + 5,000) mod 65,536 = 5,500 → hex **0x157C**. But Block 3 still stores the old previous-hash value 1,250 (0x04E2). Since 1,250 ≠ 5,500, the chain is broken at that point: any node re-verifying the chain sees the mismatch and rejects the altered history.

(c) To make the tampering internally consistent, the attacker must recompute Block 2 *and* every block after it (Block 3's new hash would become (5,500 + 250) mod 65,536 = 5,750, and so on to the chain tip), and then convince the network to accept this rewritten version. On a real network, consensus rules require each rewritten block to satisfy the current difficulty (PoW) or valid validator signatures (PoS) *while the honest majority keeps extending the original chain*, so the attacker must outpace the honest majority — effectively needing majority control, i.e., a 51% attack. Hash-linkage therefore turns a one-record forgery into an impossibly large computational or economic task.

Problem 2: Counting the Cost — A PoW versus PoS Energy Scenario

Two hypothetical payment networks each process 100 million transactions per year.

- **Chain A (proof of work):** mining consumes 2,000 GWh of electricity per year.
- **Chain B (proof of stake):** validator operations consume 2 GWh per year.
- Electricity tariff: RM0.50 per kWh.

(a) Compute the energy used per transaction on each chain. **(b)** Compute each chain's annual electricity bill. **(c)** Express Chain B's energy saving per transaction as a percentage reduction, and comment on the policy relevance.

Solution. (a) Convert units: 1 GWh = 1,000,000 kWh. Chain A: 2,000 GWh = 2,000,000,000 kWh ÷ 100,000,000 transactions = **20 kWh per transaction** — roughly enough to run a household air-conditioner for a day. Chain B: 2 GWh = 2,000,000 kWh ÷ 100,000,000 transactions = **0.02 kWh per transaction**.

(b) Chain A: 2,000,000,000 kWh × RM0.50 = **RM1.0 billion per year**. Chain B: 2,000,000 kWh × RM0.50 = **RM1.0 million per year**.

(c) Reduction = (20 − 0.02) ÷ 20 × 100% = **99.9%** less energy per transaction. *Comment:* The two chains deliver the same number of transactions, but PoW's security budget is paid in electricity while PoS's is paid in locked capital. This is why Ethereum's Merge mattered environmentally, and why BNM indicates any Malaysian CBDC would run on permissioned ledgers — known validators do not need an energy race to be kept honest. (Figures are illustrative teaching values, not measurements of any real network.)

Problem 3: Blockchain or Database? — A Decision Case

Scenario A. Tenun Sari Sdn. Bhd., a textile SME in Terengganu, wants to digitise its internal stock records (currently a spreadsheet). Only its own three employees enter data. Its accountant recommends "moving to blockchain" at a quoted setup cost of RM120,000 plus RM30,000 per year.

Scenario B. Five halal-industry players — a poultry farm, a slaughterhouse, a processor, a logistics firm and a retailer — each keep separate records. Disputes over certification and delivery events cost the group an estimated RM200,000 per year in reconciliation, audits and rejected shipments. A consortium DLT platform would cost RM250,000 to set up (shared) plus RM100,000 per year (shared equally).

Task: For each scenario, apply the Section 4.4.4 framework and recommend a design, including a simple cost comparison.

Solution. Scenario A. Question 1 fails immediately: a single organisation writes the data, so there is no multi-party trust problem. Questions 2–5 are moot. A conventional database (even a modest cloud accounting package at, say, RM6,000 per year) does the job faster and cheaper. *Comparison:* blockchain = RM120,000 + RM30,000/year vs database ≈ RM6,000/year. Over three years: RM210,000 vs RM18,000 — the blockchain option costs over eleven times more while adding no trust benefit. *Recommendation:* conventional database.

Scenario B. Q1: five mutually distrusting writers — yes. Q2: no single member is accepted by all as record-keeper — yes. Q3: tamper-evident shared history of certification and custody events is the core need — yes. Q4: data originates off-chain, so recognised halal certifiers and IoT/logistics feeds must serve as trusted oracles — a manageable caveat. Q5: known parties with restricted access → consortium permissioned blockchain. *Comparison:* shared cost = RM250,000 setup + RM100,000/year; per party = RM50,000 + RM20,000/year, i.e., RM110,000 per party over three years. Against group losses of RM200,000/year (RM600,000 over three years), the platform costs RM550,000 (group total, three years) and removes most of the RM600,000 loss — roughly break-even at worst, clearly positive if even 20% of disputes remain avoided plus audit savings. *Recommendation:* consortium blockchain, with certifiers contracted as data oracles.

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Chapter 5: Introduction to Algorithmic Trading and Robo-Advisory

Chapter Outline

- 5.1 Overview of Algorithmic Trading and Automated Trading
- 5.2 Trading Strategies and Their Applications
- 5.3 Robo-Advisory and Digital Investment Services
- 5.4 Benefits, Limitations, Risks and Ethical Considerations

Learning Outcomes

After studying this chapter, you should be able to:

1. **Define** algorithmic trading, high-frequency trading and robo-advisory (digital investment management), giving examples of each. *(Bloom's level: Remember)*
2. **Explain** how an algorithmic trading pipeline (signal, risk checks, order routing, execution) and a robo-advisor's investment process (risk profiling, portfolio construction, rebalancing) operate. *(Bloom's level: Understand)*
3. **Apply** common trading strategies — momentum, mean reversion, arbitrage and market making — to simple numerical market situations. *(Bloom's level: Apply)*
4. **Analyse** a basic backtest, distinguishing genuine strategy performance from overfitting, transaction costs and slippage. *(Bloom's level: Analyse)*
5. **Evaluate** the benefits, limitations, risks and ethical concerns of algorithmic trading and robo-advisory for Malaysian investors. *(Bloom's level: Evaluate)*
6. **Design** a suitable robo-advisory investment plan for a given Malaysian investor profile, justifying asset allocation, rebalancing rules and cost considerations. *(Bloom's level: Create)*

Opening Vignette

Aina Sofea, 24, graduated from a polytechnic in Shah Alam last year and now works as a junior executive in Kuala Lumpur, earning RM3,200 a month. After essentials, she can set aside about RM300 each month. Her colleagues trade shares on their phones and talk about

"algos" and "robos". Aina is curious but cautious: she has never bought a stock, and her mother's unit trust charges fees she does not understand.

One evening, Aina downloads a robo-advisory app licensed by the Securities Commission Malaysia. It asks her ten questions about her income, goals and how she would react if her investment fell 20%. Within minutes, the app recommends a diversified portfolio of exchange-traded funds (ETFs) and offers to invest her RM300 automatically every month, rebalancing the portfolio whenever it drifts from its target mix. No branch visit, no RM1,000 minimum, no agent.

Is Aina making a wise choice? How do the algorithms behind trading systems and robo-advisors actually work, what can go wrong, and what protections does Malaysian regulation give her? This chapter answers these questions.

5.1 Overview of Algorithmic Trading and Automated Trading

Walk into the dealing room of a large investment bank today and you will find far fewer humans shouting into telephones than in the movies. Most trading is now done by computers that follow precise, pre-programmed instructions. This section defines algorithmic trading and high-frequency trading, explains the pipeline that turns a trading idea into an executed order, and describes how these activities are permitted and supervised on Bursa Malaysia.

5.1.1 What Is Algorithmic Trading?

Algorithmic trading is defined as the use of computer programs that automatically generate and submit buy or sell orders to an exchange according to a defined set of rules — such as price, quantity, timing or mathematical models — with little or no human intervention at the moment of execution (Liaw, 2021). The "algorithm" is simply a recipe: *if* condition X occurs in the market, *then* send order Y of size Z. A human designs and tests the recipe; the machine applies it tirelessly, consistently and at great speed.

Algorithmic trading is a subset of the broader fintech transformation of financial services, in which technology reshapes how financial products are produced and delivered (Arner et al., 2016). Thakor (2020) notes that automation lowers the marginal cost of trading and portfolio management, which is why both wholesale market trading and retail investment services have been reorganised around software.

It is important to separate two ideas that students often confuse:

- **Automation of decisions** — the algorithm decides *whether* to trade (for example, a trend-following model generates a buy signal).
- **Automation of execution** — the human has already decided what to trade, and the algorithm decides *how* to trade it (for example, slicing a RM50 million institutional order into thousands of small child orders across the day to reduce market impact).

Both forms exist on modern exchanges. A pension fund selling a large block of shares may use an execution algorithm even though its investment decision was entirely human; a quantitative hedge fund lets its models make both the decision and the execution.

> **QUICK NOTE:** Algorithmic trading does not always mean ultra-fast trading. An algorithm that rebalances a portfolio once a month is still algorithmic trading —

what defines it is rule-based automation, not speed.

5.1.2 High-Frequency Trading

High-frequency trading (HFT) is a specialised form of algorithmic trading characterised by extremely short holding periods, very high order volumes, and heavy investment in low-latency technology — fast computers, optimised code and servers located physically close to the exchange's matching engine — to exploit opportunities that may exist for only milliseconds or microseconds (Hill, 2018).

Typical HFT activities include electronic market making (continuously quoting buy and sell prices to earn the spread) and rapid arbitrage between related instruments or venues. HFT firms rarely carry large positions overnight; they aim to end each day nearly "flat". Because profit per trade is often a fraction of a sen, HFT is a volume business: millions of tiny trades must add up.

How significant is automation in global markets? It is a textbook-accepted estimate that algorithmic trading executes the majority of equity-market volume in developed markets such as the United States and Europe (Hill, 2018; Pompella & Matousek, 2021). In other words, most of the time, when a share changes hands in New York or London, at least one side of that trade — and very often both — was initiated by a computer program rather than a human clicking a button.

> **DID YOU KNOW? (TRIVIA):** In developed equity markets, the majority of trading volume is now generated by algorithms rather than humans. The "trader" on the other side of your buy order is statistically more likely to be a server in a data centre than a person.

5.1.3 The Algorithmic Trading Pipeline

Whether simple or sophisticated, almost every trading algorithm follows the same four-stage pipeline. Understanding this pipeline helps you see where humans remain in control and where errors can occur.

Stage 1: Signal generation. The algorithm consumes market data — prices, traded volumes, order-book depth, sometimes news feeds — and applies its rules to produce a **signal**: an instruction to buy, sell or do nothing. Example rule: "If the 3-day simple moving average of the share price rises above the 5-day simple moving average, generate a BUY signal for 1,000 shares." You will build exactly such a signal in the Problem-Solving Corner.

Stage 2: Pre-trade risk checks. Before any order leaves the system, automated risk controls validate it. Typical checks include: Is there enough cash or credit? Does the order exceed maximum position limits? Is the price within a sensible range of the last traded price (a "fat-finger" check)? Is the instrument on a restricted list? Regulators expect these checks to be systematic and documented; in Malaysia, technology risk requirements for financial institutions are set out in Bank Negara Malaysia's Risk Management in Technology (RMiT) policy document (Bank Negara Malaysia [BNM], 2025b).

Stage 3: Order routing. The validated order is sent to a market. If the trader's firm is a member of the exchange, orders flow through the member's systems; institutional clients may instead use **direct market access (DMA)**, an arrangement in which the client transmits orders electronically through the broker's infrastructure directly into the exchange's order book, subject to the broker's automated pre-trade controls. The router may also choose among venues

or slice the order using an execution algorithm such as VWAP (volume-weighted average price), which spreads a large order across the day in proportion to expected trading volume.

Stage 4: Execution and confirmation. The exchange's matching engine pairs the order with opposing orders according to price-time priority. Executed trades ("fills") flow back to the algorithm, which updates positions, computes profit and loss, and may generate the next signal. Post-trade, the positions move into clearing and settlement like any other trade.

Stage	Question answered	Simple example
Signal	Should we trade?	3-day SMA crosses above 5-day SMA → BUY
Risk checks	May we trade?	Order value RM10,300 ≤ RM50,000 limit → pass
Order routing	Where and how?	Send via DMA to Bursa Malaysia; slice into 5 child orders
Execution	What happened?	Filled 1,000 shares at average RM1.03; update position

5.1.4 Algorithmic Trading on Bursa Malaysia

Malaysia's exchange has accommodated automated trading for many years. Bursa Malaysia permits algorithmic trading and direct market access under its rules: approved participating organisations may offer DMA to clients, provided that orders pass through the broker's automated pre-trade risk filters and that the broker retains responsibility for all orders entered under its identity (Bursa Malaysia Securities Berhad, n.d.). Retail adoption of algorithmic tools is growing, helped by brokers offering application programming interfaces (APIs) and by retail platforms that automate recurring investments.

Two practical consequences matter for you as a future finance professional or investor. First, even Malaysian retail investors now interact daily with algorithmic systems: the robo-advisor rebalancing your portfolio, the execution algorithm working a fund manager's order, and the market-making algorithms quoting bid and ask prices on the exchange. Second, regulation follows function, not label — a firm that trades algorithmically on its own account faces market-conduct rules, while a firm that manages your money algorithmically is licensed and supervised as a fund or investment manager, as Section 5.3 explains.

> **MALAYSIAN SPOTLIGHT:** Malaysia's capital market regulator, the Securities Commission Malaysia (SC), introduced the Digital Investment Management (DIM) framework in May 2017 — the first regulatory framework for robo-advisory in the region — while Bursa Malaysia's rules have long accommodated algorithmic and direct market access trading (SC, 2017; Bursa Malaysia Securities Berhad, n.d.). Together, these frameworks mean that automation in Malaysian markets operates inside clear licensing and conduct perimeters rather than in a legal vacuum.

5.2 Trading Strategies and Their Applications

An algorithm is only as good as the strategy it encodes. This section explains five widely used strategy families — momentum, mean reversion, arbitrage, market making and index-fund

rebalancing — each with a simple numeric example, and then examines how strategies are tested before real money is risked.

5.2.1 Momentum and Trend-Following

A **momentum strategy** (or trend-following strategy) buys assets whose prices have been rising and sells assets whose prices have been falling, on the assumption that trends persist for a while before reversing (Gupta & Tham, 2019). The classic implementation is the moving-average crossover: compute a short-window average and a long-window average of past prices; buy when the short average crosses above the long average, and sell when it crosses below.

Numeric example. Suppose shares of a fictional company, DanaTech Bhd, trade at RM1.00, RM1.00, RM1.01, RM1.00, RM0.99, RM1.00 and then RM1.03 on day 7. The 3-day simple moving average (SMA) rises from RM0.9967 to RM1.0067, crossing above the 5-day SMA of RM1.006. A momentum algorithm generates a BUY at RM1.03. The price rallies to RM1.10 and later fades; when the 3-day SMA crosses back below the 5-day SMA at RM1.04, the algorithm sells. Gross profit: RM1.04 − RM1.03 = RM0.01 per share. The example shows both the appeal and the weakness of trend-following: the algorithm captured part of an RM0.10 up-move, but because moving averages are *lagging* indicators, it gave back most of the gain before exiting. You will reproduce this exact backtest in the Problem-Solving Corner.

5.2.2 Mean Reversion

Mean reversion is the opposite bet: prices that move unusually far from their historical average tend to return ("revert") to that average (Gupta & Tham, 2019). A mean-reversion algorithm buys when the price is abnormally low and sells when it is abnormally high.

Numeric example. A fictional real estate investment trust (REIT), AmanahREIT, has traded near RM2.00 for a year, with a typical range of RM1.95–RM2.05. News of a temporary tenant vacancy pushes the price to RM1.80 — 10% below its mean. A mean-reversion algorithm buys 5,000 units. Two weeks later, the price recovers to RM1.98 and the algorithm sells, earning RM0.18 × 5,000 = RM900 before costs. The danger is obvious: if the price fell for a *permanent* reason (the building is structurally unsound), there is no mean to revert to, and the algorithm keeps buying a falling asset. This is why mean-reversion systems always include stop-loss rules.

5.2.3 Statistical and Cross-Market Arbitrage

Arbitrage is the simultaneous purchase and sale of the same or economically equivalent assets in different markets or forms to profit from a price difference, with little or no net risk (Hill, 2018). Pure arbitrage is rare and fleeting; "statistical arbitrage" instead trades baskets of historically related securities whose price relationship has temporarily broken down.

Numeric example (cross-market). An ETF that tracks the FTSE Bursa Malaysia KLCI trades on Bursa Malaysia. At 11:00 a.m., the combined value of its underlying shares — its indicative net asset value (iNAV) — is RM2.50 per unit, yet the ETF's units are changing hands at RM2.46. An arbitrage algorithm buys 100,000 ETF units at RM2.46 and simultaneously sells short an equivalent basket of the underlying shares for RM2.50. When the gap closes, it unwinds both legs, locking in roughly RM0.04 × 100,000 = RM4,000 minus costs. Trades like this are socially useful: they keep ETF prices aligned with their underlying value, which protects ordinary investors such as Aina from overpaying.

Numeric example (statistical). Two Malaysian plantation companies historically move together with a near-constant price ratio of 1.25. The ratio suddenly jumps to 1.40 with no fundamental news. A statistical-arbitrage model shorts the relatively expensive stock and buys the cheap one, profiting when the ratio normalises. Unlike pure arbitrage, this trade can lose money if the historical relationship has genuinely broken — a phenomenon traders call model breakdown.

5.2.4 Market Making and Index-Fund Rebalancing

Market making is the practice of continuously quoting both a buy (bid) price and a sell (ask) price for a security, earning the bid–ask spread while bearing the risk of holding inventory (Hill, 2018). Electronic market-making algorithms are a major HFT activity and an important source of liquidity on exchanges.

Numeric example. A market-making algorithm quotes a fictional stock at RM5.00 bid / RM5.02 ask. A seller hits the bid: the algorithm buys 2,000 shares at RM5.00. Minutes later, a buyer lifts the offer: it sells 2,000 shares at RM5.02. Gross profit is the spread: RM0.02 × 2,000 = RM40. Repeat this thousands of times a day and the income is meaningful — but if the price collapses to RM4.80 while the algorithm holds inventory, one adverse move (RM0.20 × 2,000 = RM400 loss) wipes out ten successful rounds. Inventory risk control is therefore the heart of market-making algorithms.

Index-fund rebalancing is a quieter, calendar-driven strategy. Index funds must match the weights of their benchmark index. When the index provider changes constituents or weights — for example, when a stock enters the KLCI and another exits — every index fund tracking that benchmark must buy and sell accordingly, usually on the same effective date. Algorithms execute these large, predictable trades, and other algorithms try to anticipate them. Rebalancing logic also appears inside robo-advisors, as Section 5.3 shows.

5.2.5 Backtesting, Overfitting, Transaction Costs and Slippage

Before a strategy touches real money, quants **backtest** it: they apply its rules to historical market data and measure the hypothetical performance. **Backtesting** is defined as the simulation of a trading strategy on past data to estimate how it would have performed (Pompella & Matousek, 2021). A good backtest mimics reality: realistic prices, realistic order sizes, and realistic costs.

The great trap of backtesting is **overfitting** — tuning a strategy's parameters so precisely to past data that it memorises history instead of learning a durable pattern. A model with twenty fine-tuned parameters can produce a beautiful upward-sloping historical equity curve that collapses immediately in live trading, because the "pattern" was noise. Defences include keeping strategies simple, testing on data not used in development ("out-of-sample" testing), and demanding an economic reason for the pattern to exist.

Two real-world frictions turn many paper profits into live losses:

- **Transaction costs** — brokerage commissions, clearing fees, stamp duty and, for unit trusts, sales charges. A strategy trading 200 times a year at 0.2% per side pays away 80% of a 10% gross annual return before anything else.
- **Slippage** — the difference between the price at which a trade was expected to execute and the price at which it actually executes. A buy signal generated at a last traded price of RM1.03 may fill at RM1.035 in a fast market; that half-sen difference, repeated over thousands of trades, materially changes performance.

Numeric example. The momentum backtest above earned RM0.04 per share gross. Add a brokerage cost of 0.2% per side across its four transactions (buy RM1.03, sell RM1.04, buy RM1.07, final value RM1.10) and costs total about RM0.0085 per share, cutting net profit to roughly RM0.0315 — a 21% reduction. Backtests that ignore costs flatter every active strategy.

> **QUICK NOTE:** Overfitting is the silent killer of backtests. If a strategy only works with parameters like "buy on the 17th day of months ending in 'r' ", it has memorised the past, not discovered a law of markets.

> **BRAINSTORMER:** Suppose your classmate's backtest shows a 40% annual return over five years of historical KLCI data. What three questions would you ask before believing the result, and what does your suspicion tell you about the difference between historical and live performance?

5.3 Robo-Advisory and Digital Investment Services

Sections 5.1 and 5.2 described algorithms that *trade*. This section describes algorithms that *invest*: robo-advisors, which automate portfolio management for ordinary savers. In Malaysia these services are regulated under the Securities Commission Malaysia's Digital Investment Management framework, introduced in May 2017 — the first such framework in the region (Securities Commission Malaysia [SC], 2017).

5.3.1 What Is Robo-Advisory?

Robo-advisory is defined as the provision of automated, algorithm-driven investment advice and portfolio management services through a digital platform, with minimal human supervision (Chishti & Barberis, 2016). In Malaysian regulatory language, the same activity is called **digital investment management (DIM)**: the SC's framework licenses firms that provide automated portfolio management — including risk profiling, suitability assessment, asset allocation and rebalancing — delivered online with minimal human intervention (SC, 2017).

The label "robo" is misleading if it suggests a science-fiction robot. A robo-advisor is a regulated asset-management company whose advice engine, portfolio construction and operations are implemented in software. The investment logic itself — diversify, match risk to the client, keep costs low, rebalance mechanically — is decades old; what is new is the delivery channel and the cost structure. Fintech scholars describe this as the unbundling and re-bundling of financial services: software strips the advice-and-portfolio function out of the traditional bank branch and re-delivers it through an app (Gomber et al., 2018; Lee & Shin, 2018).

5.3.2 The Malaysian DIM Landscape

Because Malaysia was an early mover, its robo-advisory sector is comparatively well developed. Approximately nine DIM providers are registered with the SC (SC, 2026). Well-known examples include:

Provider	Notes
StashAway	First DIM licence granted by the SC, in 2018
Wahed Invest	Shariah-compliant portfolios
MYTHEO	Operated by GAX MD

Provider	Notes
Raiz	Micro-investing, including round-ups of spare change
Akru	Malaysian home-grown platform
Versa	Digital cash and investment management
Kenanga Digital Investing (KDI)	Backed by Kenanga Investment Bank
BEST Invest	Shariah-compliant, by BIMB Investment

The presence of Shariah-compliant DIMs such as Wahed Invest and BEST Invest reflects Malaysia's position as a global leader in Islamic finance, anchored by the Malaysia International Islamic Financial Centre and the Shariah governance structure under the Central Bank of Malaysia Act 2009 (DinarStandard & Elipses, 2026). For Muslim investors, a Shariah robo-advisor screens portfolios to exclude non-compliant securities (for example, conventional banking, alcohol or gambling stocks) and may use sukuk funds in place of conventional bond funds.

Adoption is meaningful but still small relative to the overall fund industry. More than one million DIM accounts have been opened, and 51% of those accounts are held by investors in their 20s — evidence that robo-advisory is primarily a young Malaysian's gateway to investing (SC, 2026). Yet total DIM assets under management remain below 1% of Malaysia's RM1.07 trillion fund management industry, although they are growing — about 15% in 2023 alone (SC, 2026). Low minimums explain the youth appeal: while conventional unit trusts and adviser-led portfolios often expect lump sums in the thousands of ringgit, several DIMs accept initial investments of RM100 or less, and some have no minimum at all.

> **DID YOU KNOW? (TRIVIA):** More than half (51%) of robo-advisory accounts in Malaysia belong to investors in their 20s. The "typical" Malaysian robo-investor is not a wealthy professional — she is closer to Aina in our opening vignette.

5.3.3 How a Robo-Advisor Works: The Investment Pipeline

Despite different brands, nearly every robo-advisor follows the same three-stage pipeline.

Stage 1: Risk profiling and suitability. Onboarding begins with a questionnaire covering your age, income, savings, investment horizon, experience and — critically — your behavioural tolerance for losses. **Risk profiling** is the process of assessing an investor's ability and willingness to bear investment risk so that a suitable portfolio can be recommended (SC, 2017). Under the DIM framework, the algorithm's suitability assessment is not a marketing formality: the licence obliges the firm to match portfolios to client profiles and to disclose how the algorithm works, its assumptions and its limitations (SC, 2017).

Stage 2: Portfolio construction. The engine maps your risk profile to a target asset allocation built from **Modern Portfolio Theory (MPT)** — the framework, originating with Harry Markowitz, showing that combining assets with less-than-perfect correlation can produce the best expected return for a given level of risk (Markowitz, 1952). In practice, robo-advisors implement MPT portfolios using **exchange-traded funds (ETFs)**: listed, passively managed funds that track an index (such as the KLCI or a global equity index) and trade intraday like a share. A moderate-risk Malaysian portfolio might combine a local equity ETF, a global developed-market equity ETF, a sukuk or bond ETF and a gold ETF. ETFs keep the structure cheap and transparent, because passive index funds carry far lower management fees than actively managed unit trusts.

Stage 3: Automatic rebalancing. Markets move, so allocations drift. **Rebalancing** is the periodic restoration of a portfolio to its target asset allocation by selling what has grown overweight and buying what has fallen underweight. If a 60/40 equity/bond portfolio drifts to 66/34 after a rally, the algorithm sells equities and buys bonds back to 60/40 — systematically "selling high and buying low" without emotion. Rebalancing may be calendar-based (for example, quarterly) or threshold-based (for example, whenever any asset class drifts more than 5 percentage points from target). Dividends and your monthly deposits are also automatically reinvested according to the target weights, which reduces the need for taxable or fee-bearing sales.

5.3.4 Robo-Advisors versus Conventional Channels

How does a DIM compare with the unit trust channel familiar to many Malaysian families? The table below summarises typical structural differences; specific fees vary by provider and fund.

Feature	Robo-advisor (DIM)	Conventional unit trust
Minimum investment	Low (often RM100 or less)	Often RM1,000 or more
Typical annual management/advisory fee	Lower (commonly well under 1%)	Higher (commonly around 1.5%–2%)
Upfront sales charge	Usually none	Frequently up to several percent
Underlying instruments	Passive ETFs	Actively managed funds
Human adviser	Minimal or none	Agent/bank adviser
Rebalancing	Automatic, rule-based	Varies; investor or manager discretion
Regulation	SC — DIM framework (SC, 2017)	SC — unit trust framework

Two balanced observations are in order. First, lower cost is a structural advantage, not a guarantee of better returns; a skilful active manager can still outperform after fees, though most do not over long periods. Second, human advisers add value that algorithms cannot easily replicate — coaching clients through panic, handling complex family finances, and integrating takaful, estate and zakat planning. The channels are complements for many households rather than strict substitutes.

> **QUICK NOTE:** A robo-advisor is not a stock-picking machine. It is a disciplined portfolio manager: its value comes from diversification, low cost, automatic rebalancing and removing emotional decision-making — not from forecasting which share will rise next month.

> **BRAINSTORMER:** If robo-advisors are cheaper, more accessible and more disciplined, why do you think they still manage less than 1% of Malaysia's RM1.07 trillion fund industry? Consider trust, financial literacy, marketing budgets and the role of human agents.

5.4 Benefits, Limitations, Risks and Ethical Considerations

Automation makes markets faster, cheaper and more inclusive — and introduces failure modes that did not exist when humans traded by telephone. This final section weighs both sides,

because a balanced assessment is exactly what regulators, employers and your own money deserve.

5.4.1 Benefits of Automation

Four benefits stand out, especially for young Malaysian investors:

1. **Lower cost.** Software scales: the thousandth client costs a robo-advisor almost nothing extra to serve, so annual fees are typically well below those of adviser-led channels. Over decades, fee differences compound dramatically, as the Problem-Solving Corner demonstrates.
2. **Low minimums and accessibility.** Anyone with a smartphone, a MyKad and a small sum can open a globally diversified portfolio. Electronic know-your-customer (e-KYC) rules enable fully digital onboarding (BNM, 2024), and the youthful DIM account base (51% in their 20s) shows this accessibility is real, not theoretical (SC, 2026).
3. **Discipline and consistency.** Algorithms do not panic-sell in a crash or chase last month's hot sector. Rule-based rebalancing enforces the classic discipline of selling high and buying low, which humans find psychologically difficult.
4. **Market quality.** Algorithmic market making and arbitrage generally narrow bid–ask spreads and keep related prices aligned, reducing trading costs for everyone, including traditional investors (Hill, 2018).

5.4.2 Model Risk and Flash Crashes

Model risk is the risk of loss arising from decisions based on models that are wrong, mis-specified, mis-calibrated or used outside the conditions for which they were built. Every algorithm embeds assumptions — about correlations, volatility, liquidity — and assumptions fail. The 2008 global financial crisis and the fintech re-organisation that followed it taught regulators that model-driven finance requires active governance (Arner et al., 2016).

The most dramatic market-level illustration is the 2010 US Flash Crash. On 6 May 2010, within minutes, the Dow Jones Industrial Average plunged roughly 1,000 points — about 9% — before recovering most of the loss the same day; some shares briefly traded at one cent. An official US investigation found that a large automated sell order in equity index futures, executed aggressively by an algorithm, interacted with HFT activity and evaporating liquidity to cascade into a **flash crash**: an extremely rapid, deep and largely self-reversing fall in prices driven by interacting automated systems (U.S. Commodity Futures Trading Commission & U.S. Securities and Exchange Commission, 2010). The lesson for market design: exchanges now deploy circuit breakers and trading halts, and regulators expect pre-trade risk controls precisely because algorithms can transmit errors at machine speed.

Overfitting (Section 5.2.5) is the micro-level cousin of model risk: a strategy that is perfect on paper and fragile in live markets. For robo-advisory clients, the analogous risk is a suitability model that mis-classifies their risk tolerance because the questionnaire was too short or the questions were poorly designed.

5.4.3 Behavioural and Ethical Concerns

Gamification and herding. App design borrows from gaming — confetti animations, badges, streaks, leaderboards. **Gamification** is the application of game-design elements to non-game

contexts; in investing apps it can encourage over-trading and treat wealth-building as entertainment. When many users follow the same nudges, trending lists or model portfolios, **herding** results: correlated behaviour that can amplify bubbles and crashes. If every momentum algorithm sells on the same signal, who is left to buy?

Algorithmic bias. When systematic and unfair discrimination is embedded in an algorithm's data, design or deployment, the result is **algorithmic bias** — outputs that treat groups of people differently without legitimate justification. A credit or suitability model trained on historical data may quietly encode past discrimination; a risk-profiling engine might steer certain demographic groups toward inappropriate products. The regional benchmark for managing these issues is the Monetary Authority of Singapore's FEAT principles — Fairness, Ethics, Accountability and Transparency — which call on financial institutions to be able to explain and justify algorithmic decisions (Monetary Authority of Singapore, 2018). Malaysia is moving in the same direction: BNM's discussion paper on artificial intelligence in the financial sector examines how AI and machine-learning models should be governed in Malaysian finance (BNM, 2025a).

Suitability and disclosure duties. When advice is automated, who is responsible for bad advice? In Malaysia the answer is clear: the licensed DIM operator. The DIM framework requires that algorithms be designed, tested and monitored by competent persons; that clients receive clear disclosure of how the service works, what it costs and what its limitations are; and that portfolios remain suitable for the client's profile (SC, 2017). A robo-advisor cannot hide behind its code. Clients who believe they have been treated unfairly may escalate complaints to the Ombudsman for Financial Services, Malaysia's independent dispute-resolution body.

5.4.4 A Balanced Assessment

Algorithmic trading and robo-advisory are tools, and tools are judged by use. For institutions, automation delivers speed, consistency and lower costs, but demands investment in model governance, risk controls and cyber-resilience of the kind required by BNM's RMiT framework (BNM, 2025b). For young Malaysians like Aina, a licensed DIM offers a disciplined, low-cost start to investing — provided she understands that market risk is not automated away: her ETF portfolio can and will fall in value during downturns. The ethical frontier lies in design choices — what the app encourages you to do, whose interests the algorithm serves, and whether you can get a straight answer about how it works.

Case Study 5.1: Aina's RM300-a-Month Robo-Advisory Journey

Recall Aina Sofea, 24, earning RM3,200 a month in Kuala Lumpur. She chooses a DIM provider licensed under the SC's framework and begins onboarding.

Risk profiling. The app asks ten questions covering her horizon (long: she invests for goals 10+ years away), income stability (permanent employment, moderate), existing savings (RM4,000 emergency fund), experience (none) and behaviour ("If your portfolio fell 20%, would you sell, hold or buy more?"). Aina answers "hold". The engine classifies her as *moderately aggressive*: young, long horizon, stable income, average risk tolerance. Because she requested it, only Shariah-screened portfolios are shown.

Portfolio construction. The recommended allocation is: 45% global Shariah equity ETF,

25% Malaysian Shariah equity ETF, 20% sukuk fund, 10% gold ETF. Under Modern Portfolio Theory, the mix targets the highest expected return for her risk band by combining assets that do not move perfectly together (Markowitz, 1952). Her first RM300 buys fractional units of all four funds in exact 45/25/20/10 proportions.

Rebalancing. The service uses 5-percentage-point threshold bands plus quarterly checks. After eight months, a global equity rally lifts her global ETF from 45% to 52% of the portfolio. The algorithm directs her next two monthly deposits entirely into sukuk and gold, and sells a small slice of the global ETF, restoring 45/25/20/10 — automatically selling high and buying low.

Fees versus a unit trust. Aina pays an all-in annual advisory fee of about 0.6% with no sales charge. A comparable adviser-sold unit trust would charge an upfront sales charge of several percent plus an annual management fee around 1.8%. On RM300 a month for 10 years at a 5% gross return, the fee gap alone is worth roughly RM2,700 (worked fully in the Problem-Solving Corner).

What she still bears. Market risk, model risk, and the behavioural risk of abandoning her plan in a crash — risks no algorithm can remove (SC, 2017).

Discussion Questions

1. Which features of Aina's profile justify a 70% equity/gold allocation rather than a conservative 40/60 sukuk-heavy mix? Which single change in her circumstances should trigger re-profiling?
2. Explain how threshold-based rebalancing made Aina "sell high and buy low". What behavioural bias would a human investor typically display instead?
3. Aina's friend argues that the conventional unit trust is worth its higher fees because "a human adviser cares about me". Evaluate this claim using the suitability and disclosure duties in the DIM framework (SC, 2017).
4. If the robo-advisor's app introduced achievement badges and a friends' returns leaderboard, which risks from Section 5.4 would increase, and what duties would the provider still owe Aina?

Chapter Summary

- **Algorithmic trading** is the use of computer programs to generate and submit orders according to defined rules, with little or no human intervention at execution (Liaw, 2021); **high-frequency trading** is its ultra-fast, short-horizon subset (Hill, 2018).
- Every trading algorithm follows a four-stage pipeline: **signal generation → pre-trade risk checks → order routing → execution and confirmation**. Bursa Malaysia permits algorithmic and direct market access trading under its rules, with brokers retaining responsibility for orders (Bursa Malaysia Securities Berhad, n.d.).
- Algorithmic systems execute the majority of equity-market volume in developed markets (Hill, 2018).
- **Momentum** buys recent winners; **mean reversion** bets prices return to their average; **arbitrage** exploits price differences for equivalent assets; **market making** earns the bid–ask spread while managing inventory risk; **index-fund rebalancing** trades to match benchmark changes.
- **Backtesting** simulates a strategy on historical data; its chief trap is **overfitting** — memorising noise — and realistic **transaction costs** and **slippage** must be deducted from

paper profits.

- **Robo-advisory** — in Malaysia, **digital investment management (DIM)** — is automated, algorithm-driven investment advice and portfolio management with minimal human supervision, licensed by the SC under a framework introduced in May 2017, the first in the region (SC, 2017).
- About nine DIMs are registered; examples include StashAway (first licensed, 2018), Wahed Invest and BEST Invest (both Shariah-compliant), MYTHEO, Raiz, Akru, Versa and Kenanga Digital Investing (SC, 2026).
- Robo-advisors operate a three-stage pipeline: **risk profiling** for suitability, portfolio construction using **Modern Portfolio Theory** implemented with **ETFs** (Markowitz, 1952), and automatic **rebalancing** to target weights.
- Over one million DIM accounts exist; 51% belong to investors in their 20s; DIM assets remain under 1% of the RM1.07 trillion fund industry but are growing (SC, 2026).
- Benefits include low cost, low minimums, discipline and accessibility; risks include **model risk**, **flash crashes** (the 2010 US Flash Crash is the canonical example), overfitting, **gamification**, herding and **algorithmic bias**.
- Automated advice remains regulated advice: DIM licensees owe suitability and disclosure duties, and regional AI-governance standards such as the MAS FEAT principles set the ethical benchmark (Monetary Authority of Singapore, 2018).

Key Terms

- **Algorithmic trading**: the use of computer programs to generate and submit buy or sell orders automatically according to defined rules, with little or no human intervention at execution.
- **High-frequency trading (HFT)**: a specialised form of algorithmic trading using low-latency technology to trade very large volumes over extremely short holding periods, often milliseconds.
- **Direct market access (DMA)**: an arrangement allowing a broker's client to transmit orders electronically through the broker's infrastructure directly into an exchange's order book, subject to pre-trade controls.
- **Signal**: the output of a trading algorithm's rules — an instruction to buy, sell or hold — generated from market data such as prices and volumes.
- **Momentum strategy**: a trading approach that buys assets whose prices have risen and sells those that have fallen, assuming trends persist; often implemented with moving-average crossovers.
- **Mean reversion**: a trading approach based on the tendency of prices that move abnormally far from their historical average to return toward that average.
- **Arbitrage**: the simultaneous purchase and sale of the same or economically equivalent assets in different markets or forms to profit from a price discrepancy.
- **Market making**: continuously quoting both bid and ask prices for a security to earn the bid–ask spread, while managing the risk of holding inventory.
- **Backtesting**: simulating a trading strategy on historical market data to estimate how it would have performed before risking real capital.
- **Overfitting**: tuning a model so precisely to historical data that it memorises noise rather than learning a durable pattern, causing live performance to collapse.
- **Slippage**: the difference between a trade's expected execution price and its actual execution price, arising from market movement and limited liquidity.
- **Robo-advisory**: automated, algorithm-driven investment advice and portfolio manage-

ment delivered through a digital platform with minimal human supervision.

- **Digital investment management (DIM)**: the Malaysian regulatory term for robo-advisory; a licensed capital-market activity under the SC's framework introduced in May 2017.
- **Risk profiling**: the process of assessing an investor's ability and willingness to bear investment risk so that a suitable portfolio can be recommended.
- **Modern Portfolio Theory (MPT)**: Markowitz's framework showing that combining assets with less-than-perfect correlation can achieve the best expected return for a given level of risk.
- **Exchange-traded fund (ETF)**: a listed, usually passively managed fund that tracks an index and trades intraday on an exchange like an ordinary share.
- **Rebalancing**: restoring a portfolio to its target asset allocation by selling overweight assets and buying underweight ones, on a calendar or threshold basis.
- **Flash crash**: an extremely rapid, deep and largely self-reversing fall in market prices driven by interacting automated trading systems, exemplified by the 2010 US Flash Crash.

Test Your Knowledge

Part A: Multiple Choice

A1. Which statement best defines algorithmic trading? *(Remember)* - A. Any purchase of shares using a mobile phone application - B. The use of computer programs to generate and submit orders automatically according to defined rules - C. Trading that only large investment banks are legally allowed to perform - D. The manual execution of orders recommended by a research analyst

A2. The correct order of an algorithmic trading pipeline is: *(Understand)* - A. Execution → signal → order routing → risk checks - B. Signal → pre-trade risk checks → order routing → execution - C. Order routing → signal → execution → risk checks - D. Risk checks → execution → signal → order routing

A3. A share has averaged RM4.00 for a year but suddenly falls to RM3.40 on temporary bad news. An algorithm buys, expecting recovery toward RM4.00. This strategy is: *(Apply)* - A. Momentum - B. Market making - C. Mean reversion - D. Index rebalancing

A4. An ETF's underlying shares are worth RM2.50 per unit, but the ETF trades at RM2.46. An algorithm buys the ETF and shorts the underlying basket simultaneously. This is an example of: *(Apply)* - A. Gamification - B. Cross-market arbitrage - C. Risk profiling - D. Slippage

A5. In Malaysia, robo-advisors are licensed and supervised by: *(Remember)* - A. Bank Negara Malaysia under the digital banks framework - B. The Securities Commission Malaysia under the Digital Investment Management framework - C. PayNet under the payments framework - D. Bursa Malaysia under its listing requirements

A6. Which of the following is TRUE about Malaysia's DIM sector? *(Understand)* - A. It holds more than half of the RM1.07 trillion fund industry - B. It was first regulated in 2020 - C. Over one million accounts exist, with 51% held by investors in their 20s - D. It is open only to investors depositing RM10,000 or more

A7. A strategy tuned with 25 parameters earns 38% a year on historical data but loses money immediately in live trading. The most likely diagnosis is: *(Analyse)* - A. Slippage alone - B. Overfitting to historical noise - C. Excessive diversification - D. Low brokerage commissions

A8. The 2010 US Flash Crash is best described as: *(Understand)* - A. A multi-year bear market caused by subprime mortgages - B. A slow decline caused by rising interest rates - C. A rapid, deep and largely self-reversing price collapse driven by interacting automated systems - D. A currency crisis originating in Asia

A9. A robo-advisor's engine maps a client's questionnaire answers to a diversified ETF allocation. Which two pipeline stages does this describe? *(Analyse)* - A. Order routing and execution - B. Risk profiling and portfolio construction - C. Market making and arbitrage - D. Backtesting and slippage control

A10. A trading app adds confetti animations, achievement badges and a leaderboard of friends' returns. The MAIN regulatory/ethical concern is that these features may: *(Evaluate)* - A. Increase exchange matching-engine latency - B. Gamify investing and encourage over-trading and herding - C. Violate stamp duty rules - D. Prevent e-KYC onboarding

Part B: Short Answer

B1. Distinguish between "automation of decisions" and "automation of execution" in algorithmic trading, giving one example of each. *(Understand)*

B2. Explain why moving averages are called *lagging* indicators, and how this affected the DanaTech momentum trade in Section 5.2.1. *(Analyse)*

B3. List and briefly explain the three stages of a robo-advisor's investment pipeline. *(Understand)*

B4. "Low fees matter more than most young investors realise." Justify this statement with reference to compounding and the fee comparison in this chapter. *(Evaluate)*

B5. Identify four ethical or conduct risks associated with algorithmic trading and robo-advisory, and state one regulatory response for each relevant to Malaysia or the region. *(Evaluate)*

Answer Guide

Part A. - A1 — **B.** Algorithmic trading is rule-based automated order generation and submission; the channel (phone) and user (bank) are irrelevant. - A2 — **B.** A signal is validated by pre-trade risk checks, then routed to the market, then executed. - A3 — **C.** Betting that an abnormally depressed price reverts to its historical average is mean reversion; momentum (A) would instead sell into the fall. - A4 — **B.** Buying the undervalued ETF while shorting its equivalent underlying basket is cross-market arbitrage. - A5 — **B.** DIM is a capital-market activity regulated by the SC; BNM regulates banks, not robo-advisors. - A6 — **C.** The SC reports over one million DIM accounts, 51% held by investors in their 20s; DIM assets are under 1% of the industry (SC, 2026). - A7 — **B.** Beautiful in-sample results that fail live are the signature of overfitting. - A8 — **C.** On 6 May 2010, US markets fell about 9% within minutes and largely recovered, driven by automated selling and evaporating liquidity. - A9 — **B.** Questionnaire → profile is risk profiling; profile → ETF allocation is portfolio construction. - A10 — **B.** Game-style rewards can stimulate excessive trading and herding, raising suitability and conduct concerns.

Part B (model answers). - **B1.** Automation of decisions means the algorithm decides *whether* to trade — for example, a moving-average crossover generating a buy signal with no human approval. Automation of execution means a human has already decided *what* to trade, and the algorithm decides *how* — for example, a fund manager using a VWAP algorithm to

slice a RM50 million sell order into small child orders across the day to limit market impact. - **B2.** Moving averages are computed from *past* prices, so they only change after prices have already moved — they lag the market. In the DanaTech trade, the algorithm bought at RM1.03 only after the trend had begun and sold at RM1.04 only after the decline was underway, capturing just RM0.01 of an RM0.10 round-trip move because both entry and exit waited for the averages to confirm. - **B3.** (1) *Risk profiling*: a questionnaire assesses the client's horizon, finances, experience and loss tolerance to produce a suitability classification. (2) *Portfolio construction*: the profile is mapped to a target allocation across diversified ETFs using Modern Portfolio Theory. (3) *Automatic rebalancing*: deposits, dividends and trades are managed to restore target weights when markets cause drift, enforcing buy-low/sell-high discipline. - **B4.** Fees compound like returns, but in reverse: each ringgit paid in fees is a ringgit that stops growing for decades. On RM300 a month for 10 years at a 5% gross return, a 1.8% unit-trust fee leaves about RM42,361 while a 0.6% robo-advisor fee leaves about RM45,119 — a gap of roughly RM2,758, or about 7.7% of the RM36,000 contributed, from fees alone. Over longer horizons the gap widens further. - **B5.** Any four, for example: *model risk* — governed through model validation and RMiT-style technology-risk requirements (BNM, 2025b); *flash-crash/systemic risk* — exchange circuit breakers and pre-trade risk controls under Bursa Malaysia's rules; *gamification/herding* — conduct, suitability and disclosure duties under the SC's DIM framework (SC, 2017); *algorithmic bias* — fairness, ethics, accountability and transparency expectations such as the Monetary Authority of Singapore's FEAT principles (2018) and BNM's AI governance work (BNM, 2025a).

Problem-Solving Corner

Problem 1: The Compounding Cost of Fees (Robo-Advisor versus Unit Trust)

Aina invests RM300 at the end of every month for 10 years (120 contributions). Assume her portfolio earns a gross annual return of 5%, compounded monthly. Compare the outcomes under (i) a robo-advisor charging an all-in annual fee of 0.6%, and (ii) a unit trust charging an annual management fee of 1.8% (ignore upfront sales charges for simplicity).

Solution.

Net annual returns: robo-advisor = 5% − 0.6% = 4.4%; unit trust = 5% − 1.8% = 3.2%.

The future value (FV) of a series of monthly contributions is:

FV = PMT × [((1 + r)^n − 1) ÷ r]

where PMT = RM300, n = 120 months, and r = annual net return ÷ 12.

Robo-advisor (net 4.4%; r = 0.003667):

FV = 300 × [((1.003667)^120 − 1) ÷ 0.003667] FV = 300 × [(1.5513 − 1) ÷ 0.003667] FV = 300 × 150.40 ≈ **RM45,119**

Unit trust (net 3.2%; r = 0.002667):

FV = 300 × [((1.002667)^120 − 1) ÷ 0.002667] FV = 300 × [(1.3766 − 1) ÷ 0.002667] FV = 300 × 141.23 ≈ **RM42,361**

Interpretation. Both investors contributed exactly RM36,000. The 1.2-percentage-point annual fee difference cost the unit-trust investor approximately RM2,758 over just 10 years — about 7.7% of total contributions, sacrificed to fees alone. Had the unit trust also levied a

3% upfront sales charge on every contribution, the gap would widen by a further RM1,000 or more. Small percentage fees, compounded over decades, become large ringgit amounts. This is the strongest structural argument for low-cost automated investing — and a fair counter is that fees are only worth paying if the active manager demonstrably adds more than 1.2% a year after costs, which most do not over long periods.

Problem 2: Backtesting a Moving-Average Crossover

The table below shows 18 consecutive daily closing prices (in RM) for fictional DanaTech Bhd shares, together with the 3-day and 5-day simple moving averages (SMA). A trading rule says: **BUY** when the 3-day SMA crosses from below to above the 5-day SMA; **SELL** when it crosses back below. Assume you trade 1,000 shares per signal at that day's close.

Day	Close (RM)	3-day SMA	5-day SMA
1	1.00	—	—
2	1.00	—	—
3	1.01	1.003	—
4	1.00	1.003	—
5	0.99	1.000	1.000
6	1.00	0.997	1.000
7	1.03	1.007	1.006
8	1.06	1.030	1.016
9	1.09	1.060	1.034
10	1.10	1.083	1.056
11	1.08	1.090	1.072
12	1.04	1.073	1.074
13	1.00	1.040	1.062
14	0.98	1.007	1.040
15	1.00	0.993	1.020
16	1.04	1.007	1.012
17	1.07	1.037	1.018
18	1.10	1.070	1.038

Compute: (a) all trading signals; (b) the gross profit of the strategy; (c) the profit net of brokerage of 0.2% per transaction side; (d) compare with buy-and-hold.

Solution.

(a) Signals. A crossover requires comparing each day with the previous day:

- **Day 7 — BUY at RM1.03.** Day 6: 3-day SMA (0.997) < 5-day SMA (1.000). Day 7: 3-day (1.007) > 5-day (1.006). Cross from below → BUY.
- **Day 12 — SELL at RM1.04.** Day 11: 3-day (1.090) > 5-day (1.072). Day 12: 3-day (1.073) < 5-day (1.074). Cross from above → SELL.
- **Day 17 — BUY at RM1.07.** Day 16: 3-day (1.007) < 5-day (1.012). Day 17: 3-day (1.037) > 5-day (1.018). Cross from below → BUY. The position is still open at Day 18, so we value it at the final close of RM1.10 (mark-to-market).

(b) Gross profit.

- Trade 1: $(1.04 - 1.03) \times 1{,}000 = \text{RM}0.01 \times 1{,}000 =$ **RM10 gain**

- Trade 2 (open, marked at RM1.10): (1.10 − 1.07) × 1,000 = RM0.03 × 1,000 = **RM30 unrealised gain**
- Total gross profit = **RM40** on RM1,030 of capital ≈ **3.9%**

(c) Net of costs. Four transaction sides occur: buy RM1,030, sell RM1,040, buy RM1,070, and the closing valuation RM1,100. Brokerage = 0.002 × (1,030 + 1,040 + 1,070 + 1,100) = 0.002 × 4,240 = **RM8.48**. Net profit = 40 − 8.48 = **RM31.52**, about 3.1% — costs consumed 21% of gross profit.

(d) Buy-and-hold comparison. Buying at Day 1 (RM1.00) and holding to Day 18 (RM1.10) earns RM0.10 × 1,000 = **RM100**, or 10%. The crossover strategy underperformed badly here because moving averages are *lagging*: the system bought only after prices had risen and sold only after they had fallen, and the sideways period (Days 12–16) generated a late exit and late re-entry. **Lessons:** (i) backtests reveal weaknesses as well as strengths; (ii) trend systems shine in sustained trends and suffer "whipsaws" in choppy markets; (iii) 18 days is far too short a sample to validate any strategy — a real backtest would use years of data, out-of-sample testing and realistic costs.

Problem 3: Restoring Balance — A Rebalancing Calculation

One year ago, Mei Ling's robo-advisor invested RM12,000 for her in a 70% equity ETF / 30% sukuk fund portfolio: RM8,400 in equities and RM3,600 in sukuk. Strong markets have since lifted the equity holding to RM11,000, while the sukuk holding slipped to RM3,400. Her DIM uses a 5-percentage-point threshold band. (a) What are her current weights? (b) Is rebalancing triggered? (c) If so, what trades restore the 70/30 target?

Solution.

(a) Total value = 11,000 + 3,400 = RM14,400. Current weights: equities = 11,000 ÷ 14,400 = **76.4%**; sukuk = 3,400 ÷ 14,400 = **23.6%**.

(b) The equity weight has drifted 76.4% − 70% = **6.4 percentage points**, which exceeds the 5-point band → **rebalancing is triggered**.

(c) Target amounts: equities = 0.70 × 14,400 = RM10,080; sukuk = 0.30 × 14,400 = RM4,320.

- **Sell RM920 of the equity ETF** (11,000 − 10,080).
- **Buy RM920 of the sukuk fund** (4,320 − 3,400).

Post-trade check: 10,080 ÷ 14,400 = 70.0%; 4,320 ÷ 14,400 = 30.0% ✓. Note the discipline embedded in the rule: Mei Ling *sold* the asset that had risen and *bought* the one that had lagged — precisely the opposite of what emotion suggests. In practice, a robo-advisor first directs new deposits toward the underweight asset to minimise sales and their costs, exactly as Aina's provider did in the case study.

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Chapter 6: Introduction to Cryptocurrencies and Digital Assets

Chapter Outline

- **6.1** Bitcoin: Development, Characteristics and Transactions
- **6.2** Ethereum, Altcoins and Stablecoins
- **6.3** Digital Assets (NFTs, ICOs and STOs)
- **6.4** Accounting, Taxation, Regulation and Investment Risks

Learning Outcomes

By the end of this chapter, you should be able to:

1. **Define** key terms such as blockchain, cryptocurrency, mining, stablecoin and non-fungible token. *(Bloom's level: Remember)*
2. **Explain** how Bitcoin transactions are created, validated and recorded on the blockchain. *(Bloom's level: Understand)*
3. **Apply** percentage, drawdown and mining-reward calculations to realistic ringgit-denominated scenarios. *(Bloom's level: Apply)*
4. **Compare** proof of work with proof of stake, and fiat-backed with algorithmic stablecoins. *(Bloom's level: Analyse)*
5. **Evaluate** the adequacy of Malaysia's regulatory framework for digital assets in protecting retail investors. *(Bloom's level: Evaluate)*
6. **Design** a simple risk-management plan for a Malaysian retail investor buying cryptocurrency on a registered exchange. *(Bloom's level: Create)*

Opening Vignette

Aina Sofia, a 20-year-old accounting diploma student in George Town, Penang, reads about Bitcoin on social media and decides to invest RM500 from her part-time salary. She downloads the app of a digital asset exchange registered with the Securities Commission Malaysia, verifies her identity using her MyKad and a selfie, deposits ringgit through online banking, and buys her first fraction of a Bitcoin. Two weeks later, the price has fallen 18% and she wonders whether she has made a terrible mistake. Her cousin tells her to move her coins to

a “hardware wallet” for safety; her friend says she should have bought a “stablecoin” instead; and a stranger on a messaging app offers her a “guaranteed 10% weekly return” if she transfers her coins to his platform. Aina’s questions — what did I actually buy, who protects me, where should I keep it, and what could go wrong — are exactly the questions this chapter answers.

6.1 Bitcoin: Development, Characteristics and Transactions

6.1.1 The Nakamoto Whitepaper and the Double-Spending Problem

In October 2008, a person or group using the pseudonym Satoshi Nakamoto published a nine-page whitepaper titled *Bitcoin: A Peer-to-Peer Electronic Cash System* (Nakamoto, 2008). The Bitcoin network itself began operating in January 2009. The whitepaper proposed electronic money that could be transferred directly between users “without going through a financial institution” (Nakamoto, 2008). This was a radical idea, because digital money faces a fundamental technical obstacle known as the **double-spending problem**: since digital information can be copied perfectly, what stops a user from spending the same digital coin twice? In conventional finance, banks solve this problem by maintaining central ledgers and debiting accounts; Bitcoin’s innovation was to solve it without any central party.

Bitcoin’s solution rests on the **blockchain**, defined as a distributed, append-only ledger of transactions grouped into blocks, where each block is cryptographically linked to the previous one, and identical copies are maintained by thousands of independent computers (nodes) around the world (Arslanian, 2022). Because every node holds the full history, anyone can verify that a coin has not already been spent. Transactions are confirmed through a competitive process called mining (discussed below), and altering past records would require overpowering most of the network’s computing power — practically impossible on a large network. This combination of public verifiability and economic incentives is what allows strangers to transact without trusting each other or any intermediary (Nakamoto, 2008).

Key characteristics of Bitcoin include:

- **Decentralisation**: no central bank, company or government issues or controls it.
- **Fixed supply**: the protocol caps total issuance at 21 million coins.
- **Borderless and permissionless**: anyone with an internet connection can hold or send bitcoin.
- **Pseudonymity**: users transact through addresses that are not directly tied to real-world identities (see Section 6.1.4).
- **Volatility**: its price is determined purely by supply and demand and can move sharply in either direction.

  QUICK NOTE: “Bitcoin” with a capital B usually refers to the network and protocol; “bitcoin” with a small b refers to the unit of currency. One bitcoin is divisible into 100 million units called satoshis, so you can buy a tiny fraction — you do not need to purchase a whole coin.

6.1.2 The UTXO Model and the Transaction Lifecycle

Unlike a bank account, which stores a single balance, Bitcoin tracks ownership using the **unspent transaction output (UTXO)** model. A UTXO is a discrete, spendable "chunk" of bitcoin recorded on the blockchain — rather like individual notes and coins in a physical wallet. Your wallet balance is simply the sum of all UTXOs that your keys can unlock. When you spend, your wallet selects one or more UTXOs as *inputs*, consumes them entirely, and creates new *outputs*: the payment to the recipient and, usually, a "change" output returning the leftover amount to you (Arslanian, 2022).

Worked example. Suppose Aiman's wallet contains two UTXOs worth 0.003 BTC and 0.004 BTC, and he wants to pay Bella 0.005 BTC for a second-hand laptop. His wallet combines both UTXOs (0.007 BTC) as inputs and creates two outputs: 0.005 BTC to Bella's address and 0.0019 BTC back to himself as change, leaving 0.0001 BTC as the transaction fee paid to the miner. Just as you cannot tear a RM10 note in half to pay RM5, a UTXO must be spent in full, with change returned.

The lifecycle of this transaction illustrates how the whole system works:

1. **Creation and signing.** Aiman's wallet software constructs the transaction and signs it with his private key, proving he owns the inputs without revealing the key itself.
2. **Broadcast.** The signed transaction is broadcast to the peer-to-peer network and enters the *mempool* — a waiting area of unconfirmed transactions held by each node.
3. **Validation.** Nodes check that the signature is valid and that the inputs have not already been spent (the double-spending check).
4. **Block inclusion.** Miners assemble pending transactions into a candidate block and race to solve the proof-of-work puzzle; the winner's block, including Aiman's transaction, is added to the chain.
5. **Confirmation.** Each subsequent block built on top counts as one additional confirmation. Six confirmations is a common convention for considering a payment effectively irreversible.

   DID YOU KNOW? (TRIVIA): The first real-world Bitcoin purchase is widely reported to have taken place in May 2010, when a programmer in Florida paid 10,000 BTC for two pizzas. The date, 22 May, is still celebrated by enthusiasts as "Bitcoin Pizza Day" — a striking illustration of how much perceptions of Bitcoin's value have changed.

6.1.3 Mining, Proof of Work, the 21 Million Cap and Halvings

Mining is the process by which new bitcoins are issued and new blocks of transactions are added to the blockchain. Miners compete to find a valid solution to a computational puzzle, and the winner earns the right to add the next block and claim the block reward (Nakamoto, 2008). The underlying consensus mechanism is called **proof of work (PoW)**, defined as a system in which participants must demonstrate that they have expended real computational effort — and therefore real electricity and hardware costs — before they may add to the shared ledger (Arslanian, 2022). Because cheating is expensive and honest behaviour is rewarded, PoW aligns miners' incentives with the network's security. The difficulty of the puzzle adjusts automatically roughly every two weeks so that, on average, a new block is found about every ten minutes regardless of how many miners join.

New bitcoins enter circulation only through block rewards, on a schedule fixed in the code:

- **21 million cap**: the total number of bitcoins that can ever exist is capped at 21 million, giving the asset a digitally enforced scarcity often compared to gold.
- **Halvings**: approximately every 210,000 blocks (about four years), the block reward is cut in half. The reward began at 50 BTC in 2009, fell to 25 BTC in 2012, 12.5 BTC in 2016, 6.25 BTC in 2020, and — at the **April 2024 halving** — to **3.125 BTC**. Halvings will continue until the reward becomes negligible, at which point miners will be compensated mainly by transaction fees.

This predictable, slowing issuance schedule is central to the argument made by Bitcoin supporters that it is a "hard" asset. Critics counter that scarcity alone does not guarantee value and that PoW mining consumes very large amounts of electricity, an environmental cost that has attracted sustained debate (Scott, 2022).

> **BRAINSTORMER:** Bitcoin's proof of work deliberately makes block production costly in energy terms, while newer systems achieve consensus with little energy use. Is high energy consumption an acceptable price for security and neutrality in a monetary network, or an unjustifiable environmental burden? What evidence would you need to decide?

6.1.4 Wallets, Keys, Addresses and Pseudonymity

To hold and spend bitcoin you do not store "coins" in a file; you control cryptographic keys that unlock UTXOs on the blockchain. Three linked concepts matter:

- A **private key** is a secret, randomly generated number that proves ownership and authorises spending. Whoever knows the private key controls the funds.
- A **public key** is mathematically derived from the private key and can be shared safely.
- An **address** is a shortened, encoded representation derived from the public key — the destination to which others send bitcoin, analogous to a bank account number.

A **wallet** is software or hardware that manages keys and constructs transactions. **Hot wallets** (mobile or desktop apps, exchange accounts) are connected to the internet and convenient but exposed to hacking; **cold wallets** (hardware devices or paper backups kept offline) are safer for larger amounts. The community maxim — "not your keys, not your coins" — captures the risk of leaving assets with a third party, a lesson brutally reinforced by the FTX collapse discussed in Section 6.4.

Bitcoin is **pseudonymous**, not anonymous: transactions occur between addresses that do not carry names, but every transaction is permanently public. Blockchain-analysis firms routinely trace flows, and when an address is ever linked to a real identity (for example, when a user onboards at a regulated exchange), the entire history of that address can become attributable. This matters for both crime-fighting and privacy, and it explains why regulated exchanges — including those in Malaysia — can and do perform identity checks.

> **QUICK NOTE:** A private key cannot be "reset" like a banking password. If you lose the only copy of your private key or recovery phrase, the bitcoin it controls is permanently inaccessible. Estimates commonly cited in the industry suggest millions of bitcoins may already be lost this way.

6.2 Ethereum, Altcoins and Stablecoins

6.2.1 Ethereum and Smart Contracts

Bitcoin proved that decentralised money is possible, but its scripting capabilities are deliberately limited. In late 2013 and early 2014, programmer Vitalik Buterin proposed a more general platform in the whitepaper *Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform* (Buterin, 2014). The Ethereum network launched in 2015. Its native currency, ether (ETH), pays for computation on the platform.

The central innovation is the **smart contract**, defined as a self-executing programme stored on a blockchain that automatically carries out the terms of an agreement when coded conditions are met, without needing an intermediary (Buterin, 2014). A simple analogy is a vending machine: insert the correct payment and the machine automatically dispenses the item — no cashier required. On Ethereum, developers can build decentralised applications (dApps) for lending, trading, gaming, record-keeping and much more, all running on a shared, tamper-resistant world computer.

Every operation on Ethereum consumes **gas**, defined as the unit measuring computational work on the network; users pay gas fees in ether, which compensates validators and prevents spam by making computation costly (Buterin, 2014). When the network is congested, gas fees rise — an important practical limitation of public blockchains.

6.2.2 The Merge: From Proof of Work to Proof of Stake

Ethereum originally used proof of work, but in **September 2022** it completed "The Merge", switching its consensus mechanism to **proof of stake (PoS)**. Under PoS, defined as a consensus mechanism in which validators are selected to propose and confirm blocks based on the amount of cryptocurrency they lock up ("stake") as collateral, with dishonest behaviour punished by destruction of part of that stake (Harvey et al., 2021). Because PoS does not require an energy-intensive computing race, The Merge eliminated the vast majority of Ethereum's electricity consumption overnight — a landmark event in blockchain history. Table 6.1 summarises the contrast.

Table 6.1: Proof of Work vs Proof of Stake

Aspect	Proof of Work (Bitcoin)	Proof of Stake (Ethereum)
Who adds blocks	Miners solving puzzles	Validators staking coins
Cost of attacking	Computing power and electricity	Acquiring and risking a huge stake
Energy use	Very high	Very low
Penalty for cheating	Wasted energy ("slashing" of sunk costs)	Direct destruction of staked funds
Hardware	Specialised mining machines	Ordinary servers

> **DID YOU KNOW? (TRIVIA):** The Merge was executed while the Ethereum network kept running — developers compared it to swapping an aeroplane's engine mid-flight. After the switch, Ethereum no longer needed miners at all, and its energy use became a small fraction of its former level.

6.2.3 Categories of Altcoins

Any cryptocurrency other than bitcoin is commonly called an **altcoin** ("alternative coin"). Thousands exist; most are small, illiquid and speculative. Textbook categories include:

- **Platform coins** (e.g., ether): fuel for smart-contract platforms.
- **Payment coins**: aimed at faster or cheaper transfers than Bitcoin.
- **Utility tokens**: provide access to a specific application's services.
- **Governance tokens**: confer voting rights over a protocol's future development.
- **Meme coins**: created as jokes or community experiments, with value driven almost entirely by social-media sentiment.
- **Stablecoins**: tokens engineered to hold a steady price (Section 6.2.4).

Students should note that the vast majority of altcoins have failed or lost nearly all their value; survival rates are poor, and marketing claims often far exceed technical substance (Arslanian, 2022).

6.2.4 Stablecoins: Designs and the TerraUSD Collapse

Because bitcoin's price swings make it awkward as everyday money, **stablecoins** emerged — tokens designed to maintain a stable value, usually pegged one-to-one to a currency such as the US dollar (Harvey et al., 2021). Three designs dominate:

1. **Fiat-backed stablecoins** (e.g., Tether's USDT, Circle's USDC): the issuer holds reserves — bank deposits, short-term government securities — intended to back every token 1:1. Stability depends on the quality and transparency of those reserves.
2. **Crypto-backed stablecoins** (e.g., DAI): users lock up crypto collateral worth more than the stablecoins issued (over-collateralisation), with automatic liquidation if collateral value falls.
3. **Algorithmic stablecoins**: software expands or contracts token supply to defend the peg, often with a linked, freely floating sister token absorbing volatility — with no hard reserves at all.

The cautionary textbook example is **TerraUSD (UST)**, an algorithmic stablecoin that collapsed in May 2022. UST maintained its US$1 peg through an arbitrage link with its sister token LUNA. When heavy redemptions shook confidence, the mechanism spiralled: UST lost its peg, LUNA hyperinflated into worthlessness, and tens of billions of US dollars in market value evaporated within days. The lesson is fundamental: an algorithmic peg is only confidence in code, and confidence can vanish. Fiat-backed stablecoins are sturdier but still carry reserve, redemption and regulatory risks.

> **BRAINSTORMER:** A stablecoin promises "1 token = US$1, always". What questions would you ask about the issuer before trusting that promise with your savings? How do your questions differ for a fiat-backed versus an algorithmic design?

6.2.5 Decentralised Finance (DeFi): An Overview

Decentralised finance (DeFi) is defined as an ecosystem of financial applications built on public blockchains — chiefly Ethereum — that replicate services such as lending, borrowing, trading and derivatives through smart contracts rather than through banks or brokers (Harvey et al., 2021). Core building blocks include decentralised exchanges that match trades via automated market-maker pools instead of order books; lending protocols that algorithmically

set interest rates; and “yield farming”, where users move funds between protocols chasing returns.

DeFi promises open access (anyone with a wallet can participate), transparency (code and balances are public) and composability (protocols snap together like building blocks). The risks, however, are severe: smart-contract bugs and exploits, oracle failures, extreme leverage, anonymous developers, scams, and an almost complete absence of consumer protection or recourse (Harvey et al., 2021). For Malaysian retail investors, DeFi platforms operate outside the Securities Commission Malaysia’s regulatory perimeter, meaning losses are typically unrecoverable.

> **QUICK NOTE:** “Decentralised” describes an aspiration, not always a fact. Many DeFi projects depend on small founding teams, centralised admin keys or a handful of large token holders. Always check who can actually change the rules or freeze the system.

6.3 Digital Assets (NFTs, ICOs and STOs)

6.3.1 Fungible and Non-Fungible Tokens

A **fungible** asset is one whose units are identical and interchangeable: one ringgit equals any other ringgit, one bitcoin equals any other bitcoin. A **non-fungible token (NFT)** is defined as a unique digital token recorded on a blockchain that represents ownership of, or a claim linked to, a specific digital or physical item and cannot be exchanged one-for-one with another token of the same type (Arslanian, 2022). NFTs are typically minted on smart-contract platforms; the token records provenance (who created and owned it) and transfers automatically with each sale.

Use cases include digital art and collectibles, in-game items, event tickets, membership passes and experiments in certifying ownership of physical assets. A crucial subtlety: buying an NFT generally purchases the *token* and whatever rights its terms grant — it does not automatically transfer copyright in the underlying artwork. NFT markets boomed spectacularly in 2021 and then contracted sharply; many collections lost most of their value, illustrating extreme speculative risk and thin liquidity.

6.3.2 The ICO Boom of 2017–2018 and Its Failures

An **initial coin offering (ICO)** is a fundraising method in which a project sells newly created digital tokens directly to the public, often before any product exists (Arslanian, 2022). ICOs exploded in 2017–2018, when projects worldwide raised billions of US dollars, frequently from retail buyers, on the strength of little more than a whitepaper and a website. The boom ended badly. A large share of projects were abandoned, delivered nothing, or were outright frauds — plagiarised whitepapers, fake teams, “exit scams” in which founders vanished with the funds. Because most ICOs bypassed securities laws, aggrieved investors had almost no legal recourse. Regulators worldwide responded by clarifying that most tokens sold to raise funds are, in substance, securities and fall under existing capital-markets law (Zetzsche et al., 2017). The ICO episode remains the canonical lesson in why disclosure rules and gatekeepers exist.

6.3.3 STOs, IEOs and Tokenised Securities

A **security token offering (STO)** is defined as the regulated issuance of blockchain-based tokens that represent legally recognised securities — for example, shares, bonds or units in an investment fund — subject to securities law, disclosure duties and investor-protection rules (Pompella & Matousek, 2021). Unlike ICO tokens, security tokens deliberately embrace regulation. **Tokenisation** more broadly means representing a real-world asset as a digital token on a blockchain, potentially enabling fractional ownership (a small investor could hold RM1,000 of a commercial building) and faster settlement. Interest in asset tokenisation is growing among regulators globally; Bank Negara Malaysia (BNM) issued a discussion paper on asset tokenisation in the Malaysian financial sector in October 2025 (Bank Negara Malaysia [BNM], 2025).

An **initial exchange offering (IEO)** is a token sale conducted through a digital asset exchange, which performs due diligence and lists the token — a structure that Malaysia chose as its regulated route for token fundraising (Securities Commission Malaysia [SC], 2024).

> **MALAYSIAN SPOTLIGHT:** Malaysia was an early mover in regulating digital assets. The Capital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019, effective 15 January 2019, prescribed digital currencies and digital tokens as *securities* under Malaysian law, bringing them inside the Securities Commission Malaysia's jurisdiction under the Capital Markets and Services Act 2007. The SC's *Guidelines on Digital Assets* — first issued on 28 October 2020 under section 377 of that Act and revised on 19 August 2024 — now govern digital token offerings and the IEO operators that conduct them, while digital asset exchanges are registered as recognised market operators under the *Guidelines on Recognized Markets* (SC, 2024).

6.3.4 Malaysia's Digital Asset Framework: DAXs, Shariah and Legal Status

Malaysia's approach combines permission with guardrails. Key elements:

- **Registered exchanges.** Only digital asset exchanges (DAXs) registered with the SC may operate in Malaysia. The registered list, updated regularly, has included **Luno Malaysia, SINEGY Technologies, Tokenize Technology and Hata Digital**. Registered DAXs must meet requirements on governance, custody, cybersecurity and market conduct as recognised market operators (SC, 2024).
- **Securities status.** Digital currencies and tokens are prescribed as securities under the 2019 Prescription Order, so capital-markets law applies to offerings and trading.
- **Not legal tender.** Cryptocurrency is **not legal tender** in Malaysia: no one is obliged to accept it as payment, and it is not issued or guaranteed by Bank Negara Malaysia. The ringgit remains the only legal tender.
- **Shariah position.** In 2020 the SC's Shariah Advisory Council (SAC) resolved that digital assets may be recognised as **mal** (property) from a Shariah perspective, and that trading them on SC-registered DAXs is permissible, subject to conditions — a landmark ruling that opened the door for Muslim investors to participate within a regulated environment (Oseni & Ali, 2019, provide broader context on fintech in Islamic finance).
- **AML/CFT.** DAXs are reporting institutions under the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (AMLA), and must conduct customer due diligence and report suspicious transactions.

This framework explains why Aina, in our opening vignette, could verify her identity and buy Bitcoin on a regulated platform — and also why nobody in Malaysia is required to accept her bitcoin as payment for teh tarik.

> **QUICK NOTE:** "Regulated" does not mean "risk-free". SC registration imposes conduct and security duties on the exchange; it does not guarantee the price of any digital asset, insure your holdings, or compensate you if prices crash or if you fall for a scam.

6.4 Accounting, Taxation, Regulation and Investment Risks

6.4.1 The IFRS Accounting Gap

How should a company account for cryptocurrency holdings? International Financial Reporting Standards (IFRS) contain **no standard written specifically for cryptoassets**, a gap much discussed in the accounting literature (Arslanian, 2022). Because cryptocurrencies generally fail the definitions of cash (they are not legal tender issued by a government) and of financial assets (holding bitcoin gives no contractual right to receive cash from another party), holdings are commonly fitted, imperfectly, into two existing standards:

- **IAS 38 Intangible Assets**: the default treatment for many corporate holders — the crypto is an intangible asset, typically held at cost less impairment, which means price falls are recognised but subsequent recoveries are not written back up (an asymmetric result many consider misleading).
- **IAS 2 Inventories**: applicable where the entity is a broker-trader holding crypto for sale in the ordinary course of business, which permits measurement at fair value less costs to sell with changes recognised in profit or loss.

The takeaway for accounting students like Aina: classification depends on the holder's business model, measurement choices significantly affect reported profit, and the standard-setters' project pipeline should be watched for a dedicated standard. Professional judgement and clear disclosure are essential meanwhile.

6.4.2 Malaysian Income-Tax Principles

Malaysia has no crypto-specific tax statute, so general principles of the Income Tax Act 1967 apply. The widely accepted general principle — which you should treat as indicative rather than as tax advice — is:

- **Revenue (taxable) treatment** may apply where trading is frequent, organised and business-like: repeated buying and selling, short holding periods, an intention to profit from trading. In such cases gains may constitute business income taxable at the taxpayer's marginal rate.
- **Capital (generally not taxable) treatment**: Malaysia does not levy a general capital gains tax, so an individual's *occasional* disposals of long-held digital assets have generally not been taxed.

The line between the two is a question of fact (the traditional "badges of trade" analysis), outcomes depend on individual circumstances, and practice continues to evolve — anyone with significant activity should obtain professional advice. Businesses accepting crypto as payment must also keep proper records of ringgit values at transaction dates.

6.4.3 AMLA and the FATF Travel Rule

Digital assets attract anti-money-laundering scrutiny because value can move globally, pseudonymously and quickly. Malaysia's principal statute is **AMLA 2001**, under which regulated intermediaries — including registered DAXs — must identify customers, keep records and file suspicious transaction reports (Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001). Internationally, the Financial Action Task Force (FATF) — the global standard-setter for AML/CFT — extended its **travel rule** to virtual assets: the requirement that identifying information about the originator and beneficiary of a transfer must "travel" with the transaction between service providers, mirroring the long-standing rule for wire transfers (Financial Action Task Force [FATF], 2021). For students, the practical consequences are exactly what Aina experienced: mandatory identity verification at onboarding, screening of transfers, and possible requests for the source of funds.

6.4.4 Investment Risks: Volatility, Hacks, Scams and FTX

Digital assets are among the riskiest assets a retail investor can buy. The principal risks:

1. **Volatility.** Bitcoin has repeatedly fallen 50% or more from peaks; smaller tokens fare worse. Prices can gap violently overnight — crypto trades 24/7, so there is no closing bell to halt a rout.
2. **Exchange hacks and failures.** Numerous platforms worldwide have been hacked or have failed. The defining case is **FTX**, once one of the world's largest crypto exchanges, which collapsed in **November 2022** amid revelations that customer funds had been misused by its affiliated trading firm; millions of customers became unsecured creditors in a lengthy bankruptcy. The lessons: an exchange's size and celebrity endorsements are no guarantee; leaving assets on any platform creates counterparty risk; and segregated custody, proof of reserves and strong regulation matter.
3. **Scams.** Fake investment schemes, "pig-butchering" romance scams, phishing of recovery phrases and impersonation of regulators are rampant. In Malaysia, the **National Scam Response Centre (hotline 997)** can rapidly act on scam-linked bank transfers, and PDRM's **SemakMule** portal lets the public check bank account and phone numbers linked to mule accounts — but funds already converted to crypto are very hard to recover.
4. **No safety nets.** Cryptocurrency holdings are not deposits: they carry no deposit insurance, no investor compensation scheme, and — since crypto is not legal tender — no central-bank backing.
5. **Self-custody risk.** Holding your own keys removes exchange counterparty risk but makes *you* the single point of failure; lost keys mean lost funds.

   DID YOU KNOW? (TRIVIA): When FTX collapsed in November 2022, customers worldwide discovered they were treated as unsecured creditors — meaning they queued behind other claimants for whatever the bankruptcy estate could recover, rather than simply withdrawing "their" coins. The episode is now a standard case study in counterparty risk.

   BRAINSTORMER: If a friend said, "Crypto is regulated in Malaysia now, so it must be safe," how would you respond? Distinguish between regulation of the *intermediary* and protection of the *investment*.

Investor-protection takeaways for Malaysian retail investors: use only SC-registered DAXs; invest only what you can afford to lose entirely; beware of any "guaranteed return"; en-

able all available security features; consider moving significant long-term holdings to a well-understood cold wallet; keep records for tax purposes; and treat social-media hype as a warning sign, not research.

Case Study 6.1: Aina Buys Bitcoin on a Registered DAX

Aina Sofia decides to invest RM2,000 in Bitcoin through Luno Malaysia, a digital asset exchange registered with the Securities Commission Malaysia. This case study traces her journey and examines what protection she does — and does not — have.

Onboarding. Aina downloads the app and completes electronic know-your-customer (e-KYC) verification: she photographs her MyKad, takes a selfie for facial matching, and answers questions about her source of funds. This non-face-to-face onboarding is enabled by BNM's e-KYC policy (BNM, 2024) and by AMLA obligations on the exchange as a reporting institution. Her ringgit deposit arrives via online banking. She notes that the exchange is a registered recognised market operator subject to the SC's *Guidelines on Recognized Markets*, which impose requirements on governance, custody of client assets and cybersecurity.

The purchase. With Bitcoin trading at RM400,000 per coin, Aina's RM2,000 buys 0.005 BTC. The app shows her holdings in ringgit terms, updated every second.

Custody decision. Aina faces a choice. She can leave the bitcoin with the exchange (*custodial* holding — convenient, recoverable password, but exposed to exchange failure), or withdraw to her own hardware wallet (*self-custody* — no counterparty risk, but she alone must protect the recovery phrase). Remembering the FTX collapse of November 2022, where customers of a huge global exchange became unsecured creditors, she decides to keep only a small trading amount on the platform and move the rest to a hardware wallet, storing the written recovery phrase in a locked drawer.

What protects her — and what does not. Aina benefits from: SC registration and supervision of the exchange; AMLA safeguards; the Shariah Advisory Council's 2020 resolution giving clarity that trading on registered DAXs is permissible for Muslim investors; and scam-response infrastructure such as the NSRC (997) and SemakMule. She is *not* protected by: deposit insurance, any investor compensation scheme, price guarantees, or legal-tender status. If the market falls 40%, that loss is entirely hers.

Discussion Questions

1. Identify the specific regulatory instruments and institutions that applied at each stage of Aina's onboarding and purchase. What does each one protect?
2. Compare custodial and self-custody holding from Aina's perspective. Which would you choose for RM2,000, and for RM200,000? Justify your answer.
3. Using the FTX collapse as your reference, explain why "the exchange holds my coins, so they are mine" is legally inaccurate. What is the investor's actual position in an insolvency?
4. Aina's friend argues that because the SC regulates the exchange, Bitcoin is "government-approved and safe". Evaluate this claim and write a two-sentence reply Aina could send.

Chapter Summary

- **6.1** Bitcoin was proposed in Satoshi Nakamoto's 2008 whitepaper and launched in January 2009 as peer-to-peer electronic cash without a central intermediary (Nakamoto, 2008).
- **6.1** Bitcoin solves the double-spending problem with a public blockchain secured by proof-of-work mining; ownership is tracked through the UTXO model rather than account balances.
- **6.1** Supply is capped at 21 million coins; the block reward halves roughly every four years, falling to 3.125 BTC in April 2024.
- **6.1** Users control coins via private keys, public keys and addresses; wallets are hot (online) or cold (offline); transactions are pseudonymous but permanently public.
- **6.2** Ethereum (Buterin, 2014; launched 2015) introduced general-purpose smart contracts paid for with gas, and switched from proof of work to proof of stake at The Merge in September 2022.
- **6.2** Stablecoins aim for price stability via fiat backing (USDT, USDC), crypto collateralisation, or algorithms; the 2022 TerraUSD collapse showed algorithmic pegs can fail catastrophically. DeFi rebuilds financial services in smart contracts but carries severe, largely unprotected risks (Harvey et al., 2021).
- **6.3** NFTs are unique, non-fungible tokens representing specific items; the 2017–2018 ICO boom showed the dangers of unregulated token fundraising; STOs and IEOs bring token issuance within securities law.
- **6.3** Malaysia prescribed digital currencies and tokens as securities in 2019; the SC's *Guidelines on Digital Assets* (2020, revised August 2024) govern IEOs; Luno, SINEGY, Tokenize and Hata are registered DAXs; the SC's Shariah Advisory Council resolved in 2020 that digital assets are *mal* and tradable on registered DAXs; crypto is not legal tender.
- **6.4** IFRS has no dedicated crypto standard, so holdings are commonly treated under IAS 38 or IAS 2; in Malaysia, frequent trading may be taxed as business income while occasional individual gains are generally not taxed — always seek professional advice.
- **6.4** Key risks are volatility, exchange failures (FTX, November 2022), scams, the absence of safety nets, and self-custody errors; using SC-registered DAXs and prudent custody are essential mitigations.

Key Terms

- **Bitcoin**: the first decentralised cryptocurrency, proposed by Satoshi Nakamoto in 2008 and launched in 2009, enabling peer-to-peer value transfer without intermediaries.
- **Blockchain**: a distributed, append-only ledger of transaction blocks, cryptographically linked and replicated across many independent computers.
- **Double-spending problem**: the risk that digital money, being copyable, is spent more than once; blockchains solve it without a central ledger-keeper.
- **Unspent transaction output (UTXO)**: a discrete, spendable chunk of cryptocurrency recorded on the blockchain, consumed entirely as an input when spent, with any remainder returned as change.
- **Proof of work**: a consensus mechanism requiring participants to expend computational effort to add blocks, making attacks economically costly.

- **Mining**: the competitive process of validating transactions and adding blocks to a proof-of-work blockchain in return for block rewards and fees.
- **Halving**: the programmed reduction, roughly every four years, of Bitcoin's block reward by half; the April 2024 halving cut it to 3.125 BTC.
- **Private key**: a secret cryptographic number that proves ownership of cryptocurrency and authorises spending; losing it means losing the funds permanently.
- **Public key**: a cryptographic value derived from a private key, used to generate receiving addresses; safe to share.
- **Smart contract**: a self-executing programme on a blockchain that automatically enforces agreed terms when coded conditions are met.
- **Gas**: the unit measuring computational work on Ethereum, paid in ether, which compensates validators and deters spam.
- **Proof of stake**: a consensus mechanism where validators lock up cryptocurrency as collateral to propose and confirm blocks, with dishonesty punished by slashing.
- **Altcoin**: any cryptocurrency other than bitcoin, spanning platform, payment, utility, governance, meme and stablecoin categories.
- **Stablecoin**: a digital token engineered to hold a stable price, typically pegged to a fiat currency, via reserves, over-collateralisation or algorithms.
- **Decentralised finance (DeFi)**: an ecosystem of blockchain-based applications that provide lending, trading and other financial services through smart contracts without intermediaries.
- **Non-fungible token (NFT)**: a unique blockchain token representing a specific item, not interchangeable one-for-one with other tokens.
- **Initial coin offering (ICO)**: a largely unregulated public sale of newly created tokens to raise funds, notorious for failures and fraud during 2017–2018.
- **Security token offering (STO)**: a regulated issuance of blockchain tokens representing legally recognised securities, subject to securities law and disclosure requirements.

Test Your Knowledge

Part A: Multiple Choice

Q1. The double-spending problem refers to: *(Understand)* A. Paying twice the intended transaction fee B. The risk that the same digital coin can be copied and spent more than once C. Two miners finding a block simultaneously D. Sending bitcoin to the wrong address

Q2. Under Bitcoin's UTXO model, when you spend, you must: *(Understand)* A. Spend exactly half of each UTXO B. Consume selected UTXOs entirely and receive any leftover as change C. Convert UTXOs into an account balance first D. Obtain the miner's permission

Q3. After the April 2024 halving, Bitcoin's block reward became: *(Remember)* A. 6.25 BTC B. 3.125 BTC C. 1.5625 BTC D. 12.5 BTC

Q4. The Merge (September 2022) refers to Ethereum's switch from: *(Remember)* A. Proof of stake to proof of work B. Proof of work to proof of stake C. A private blockchain to a public one D. Gas fees to subscription fees

Q5. Gas on Ethereum is best described as: *(Understand)* A. A tax paid to the Malaysian government B. A unit measuring computational work, paid in ether, that compensates validators and deters spam C. The electricity used by Bitcoin miners D. A stablecoin used to pay

transaction fees

Q6. The collapse of TerraUSD in 2022 primarily illustrates: *(Analyse)* A. The strength of fiat-backed reserves B. That algorithmic stablecoin pegs can fail catastrophically when confidence collapses C. That all stablecoins are illegal D. The success of over-collateralisation

Q7. In Malaysia, digital currencies and digital tokens are legally treated as: *(Understand)* A. Legal tender B. Deposits insured by the government C. Securities, under the 2019 Prescription Order D. E-money licensed by Bank Negara Malaysia

Q8. Which of the following is a digital asset exchange registered with the Securities Commission Malaysia? *(Remember)* A. FTX B. Binance C. Luno Malaysia D. TerraForm Labs

Q9. The SC Shariah Advisory Council's 2020 resolution held that digital assets: *(Understand)* A. Are forbidden (haram) in all circumstances B. May be recognised as *mal* (property) and traded on SC-registered DAXs, subject to conditions C. Must be converted to gold before trading D. Are legal tender for Islamic transactions

Q10. A Malaysian investor's bitcoin is held on an exchange that collapses. Which protection applies? *(Evaluate)* A. Full deposit insurance reimbursement B. Automatic compensation from a government fund C. The investor generally ranks as an unsecured creditor, as FTX customers discovered D. Bank Negara Malaysia restores the coins

Part B: Short Answer

Q1. Explain, in your own words, how proof of work helps prevent double-spending on the Bitcoin network.

Q2. Distinguish between a hot wallet and a cold wallet, and state one situation in which each is more appropriate.

Q3. Compare the three main stablecoin designs and explain why the TerraUSD collapse damaged confidence in one of them specifically.

Q4. Outline the main components of Malaysia's regulatory framework for digital assets, naming at least one statute, one set of guidelines and one type of registered operator.

Q5. "Regulated exchanges make crypto a safe investment." Critically evaluate this statement for a Malaysian retail investor.

Answer Guide

Part A: 1. **B** — digital data is copyable, so a mechanism is needed to prevent the same coin being spent twice. 2. **B** — UTXOs are consumed in full as inputs; change is returned as a new output. 3. **B** — the April 2024 halving cut the reward from 6.25 to 3.125 BTC. 4. **B** — The Merge moved Ethereum from proof of work to proof of stake in September 2022. 5. **B** — gas prices computation in ether, aligning incentives and preventing spam. 6. **B** — UST's algorithmic peg collapsed in a confidence death spiral, destroying tens of billions in value. 7. **C** — the 2019 Prescription Order prescribed digital currencies and tokens as securities. 8. **C** — Luno Malaysia is SC-registered; FTX collapsed and Binance has not been registered in Malaysia. 9. **B** — the SAC recognised digital assets as *mal* and permitted trading on registered DAXs with conditions. 10. **C** — no insurance or compensation applies; customers of failed exchanges are unsecured creditors.

Part B (model answers): 1. Proof of work forces miners to spend real computing power and electricity to add blocks. Transactions are grouped into blocks that all nodes verify against the shared history, so a coin already spent is rejected. Rewriting history to double-spend would require out-computing most of the honest network, which is practically and economically infeasible (Nakamoto, 2008). 2. A hot wallet is connected to the internet (app or exchange account): convenient for small, frequent transactions but exposed to hacking and platform failure. A cold wallet keeps keys offline (hardware device or paper backup): safer for larger, long-term holdings but requires careful protection of the recovery phrase. 3. Fiat-backed stablecoins hold reserve assets intended to back tokens 1:1; crypto-backed ones use over-collateralisation with automatic liquidation; algorithmic ones rely on supply-adjusting code and confidence. TerraUSD was algorithmic: when redemptions broke confidence, its peg and sister token collapsed in a death spiral, showing that code without hard reserves cannot guarantee a peg. 4. The 2019 Prescription Order (under the Capital Markets and Services Act 2007) made digital currencies and tokens securities; the SC's *Guidelines on Digital Assets* (2020, revised 2024) govern token offerings and IEO operators; DAXs such as Luno Malaysia are registered as recognised market operators under the *Guidelines on Recognized Markets*; AMLA 2001 imposes AML/CFT duties; and crypto is not legal tender. 5. SC registration regulates the intermediary's conduct, custody and security but does not guarantee prices, insure holdings or compensate losses; crypto is volatile and scams abound, so the investment itself remains high-risk even on a regulated platform.

Problem-Solving Corner

Problem 1: Percentage Gain/Loss and Maximum Drawdown

Aina buys RM5,000 of Bitcoin when the price is RM400,000 per BTC.

(a) How many bitcoins does she own?
(b) The price rises to RM520,000. What is her holding worth, and her percentage gain?
(c) The price then falls to RM312,000. What is her percentage gain/loss relative to her RM5,000 cost?
(d) Compute the **maximum drawdown** — the percentage fall from the peak price to the trough price.

Solution

(a) Units purchased = RM5,000 ÷ RM400,000 = **0.0125 BTC**.

(b) Value at peak = 0.0125 × RM520,000 = **RM6,500**. Percentage gain = (RM6,500 − RM5,000) ÷ RM5,000 × 100% = RM1,500 ÷ RM5,000 × 100% = **+30%**.

(c) Value at trough = 0.0125 × RM312,000 = **RM3,900**. Gain/loss on cost = (RM3,900 − RM5,000) ÷ RM5,000 × 100% = −RM1,100 ÷ RM5,000 × 100% = **−22%** (a loss).

(d) Maximum drawdown = (trough − peak) ÷ peak × 100% = (RM312,000 − RM520,000) ÷ RM520,000 × 100% = −RM208,000 ÷ RM520,000 × 100% = **−40%**.

Insight: Aina lost only 22% on her cost, but an investor who bought at the peak lost 40%. Note also that recovering from a 40% drawdown requires a gain of 66.7% (RM312,000 → RM520,000) — losses and gains are asymmetric, which is why drawdown matters when assessing volatile assets.

Problem 2: Mining Rewards and Halving Arithmetic

Recall: blocks are produced on average every 10 minutes; the block reward was 6.25 BTC in the 2020–2024 era and 3.125 BTC after the April 2024 halving; halvings occur every 210,000 blocks (≈ four years).

(a) How many blocks are produced per day on average, and how many new bitcoins entered circulation per day after April 2024?
(b) How many new bitcoins were created during the entire 2020–2024 era (210,000 blocks at 6.25 BTC each)?
(c) What will the block reward be after the next two halvings (expected around 2028 and 2032)?

Solution

(a) Blocks per day = 24 hours × 60 minutes ÷ 10 minutes = **144 blocks**. New bitcoins per day after April 2024 = 144 × 3.125 = **450 BTC per day** (≈ 164,250 BTC per year, ignoring fees and minor timing variation).

(b) Total issuance, 2020–2024 era = 210,000 blocks × 6.25 BTC = **1,312,500 BTC**.

(c) Reward halves each era: 3.125 ÷ 2 = **1.5625 BTC** (c. 2028); 1.5625 ÷ 2 = **0.78125 BTC** (c. 2032).

Insight: The geometric decline (50 → 25 → 12.5 → 6.25 → 3.125 → …) means most of the 21 million coins were issued early, and issuance asymptotically approaches the cap without ever exceeding it. Miners must therefore rely progressively more on transaction fees — an open question for Bitcoin's long-term security budget.

Problem 3: Stablecoin De-peg Scenario

Encik Rahman converts RM20,000 into a US-dollar stablecoin at par (1 token = US$1), intending to earn yield in a DeFi protocol. At purchase, his token position is worth exactly RM20,000.

(a) The stablecoin de-pegs to US$0.80. What is the ringgit value of his position, and his percentage loss?
(b) The peg partially recovers to US$0.98 and he sells. What is his ringgit value and overall percentage result?
(c) Contrast his likely outcome had he instead held an *algorithmic* stablecoin that failed like TerraUSD in 2022.

Solution

(a) Token value falls to 80% of par: RM20,000 × 0.80 = **RM16,000**; loss = (RM16,000 − RM20,000) ÷ RM20,000 = **−20%** (a RM4,000 loss).

(b) At US$0.98: RM20,000 × 0.98 = **RM19,600**; overall result = −RM400 ÷ RM20,000 = **−2%** (a RM400 loss).

(c) TerraUSD's peg collapsed almost entirely, and its sister token LUNA became worthless; holders recovered only a tiny fraction of their value. Encik Rahman's likely outcome would have been a near-total loss (for example, a fall to US$0.05 would leave RM20,000 × 0.05 = **RM1,000**, a −95% loss). The scenario shows why the *design* of a stablecoin — reserves versus algorithms — is the first thing an investor must examine, and why DeFi yield cannot compensate for the failure of the underlying asset.

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Chapter 7: Artificial Intelligence and Digital Technologies in Finance

Chapter Outline

- 7.1 AI and Machine Learning (ML) in Financial Services
- 7.2 Applications of AI in Banking and Financial Services
- 7.3 Financial Data, Information Systems and Data Analytics
- 7.4 Benefits, Limitations and Ethical Considerations of AI

Learning Outcomes

After studying this chapter, you should be able to:

1. **Define** artificial intelligence, machine learning, deep learning and generative AI, and distinguish between them. *(Bloom's level: Remember)*
2. **Explain** how supervised, unsupervised and reinforcement learning differ, using financial services examples. *(Bloom's level: Understand)*
3. **Apply** the data analytics pipeline (collection, cleaning, modelling, visualisation, decision) to a Malaysian banking scenario. *(Bloom's level: Apply)*
4. **Analyse** how AI is used in credit scoring, fraud detection, AML monitoring, customer service and robo-advisory in Malaysia. *(Bloom's level: Analyse)*
5. **Evaluate** the benefits and ethical risks of AI in finance, including bias, privacy and model risk, with reference to the MAS FEAT principles and Bank Negara Malaysia's 2025 AI discussion paper. *(Bloom's level: Evaluate)*
6. **Interpret** a confusion matrix for a fraud-detection model and design a simple rule-based credit table while identifying its bias risks. *(Bloom's level: Create)*

Opening Vignette

Aina, a 22-year-old graduate in Penang, runs a small online business selling phone accessories. She earns a decent income, but she has never had a credit card or a bank loan. When she applies for RM5,000 of working-capital financing from a conventional bank, she is rejected: she is a "thin-file" customer with no credit history. A few months later, Aina opens

an account with one of Malaysia's new digital banks. The app analyses patterns in her transaction behaviour — regular supplier payments, steady e-wallet inflows from her customers, consistent bill payments — and, months later, offers her a small credit line. Nothing about Aina changed; what changed was *how the lender looked at her data*. Behind that offer sit machine-learning models, data pipelines and governance rules of the kind you will study in this chapter. The same technology that helped Aina can also discriminate, invade privacy and fail in ways humans struggle to explain — which is why Bank Negara Malaysia is paying close attention.

7.1 AI and Machine Learning (ML) in Financial Services

7.1.1 What Are AI, ML, Deep Learning and Generative AI?

Artificial intelligence (AI) is the broad field of computer science concerned with building systems that can perform tasks that normally require human intelligence — such as understanding language, recognising images, making decisions and predicting outcomes (Chishti et al., 2020). In financial services, AI refers to the use of such systems to automate or augment decisions about lending, investing, risk and customer service (Chishti et al., 2020).

Machine learning (ML) is a subset of AI in which computers learn patterns from data and improve their performance on a task *without being explicitly programmed* with fixed rules (Chishti et al., 2020). A traditional computer programme follows instructions written by a programmer ("if income > RM4,000, approve"). An ML model instead studies thousands of past examples and discovers its own patterns ("customers with these 50 characteristics tended to repay").

Deep learning is a further subset of ML that uses multi-layered **neural networks** — computational structures loosely inspired by the neurons in the human brain — to learn very complex patterns from large volumes of data (Chishti et al., 2020). Deep learning powers image recognition (for example, verifying your identity card photo during e-KYC onboarding) and speech recognition.

Generative AI is a class of AI models that can *create* new content — text, images, code or audio — in response to prompts, rather than merely classifying or predicting (Chishti et al., 2020). Large language models (LLMs), such as those behind modern chat assistants, are generative AI. In banking, generative AI is being explored for drafting customer communications, summarising documents and supporting software developers.

The relationships are nested: generative AI ⊂ deep learning ⊂ machine learning ⊂ artificial intelligence.

> **QUICK NOTE:** Not all AI is machine learning, and not all machine learning is deep learning. A rule-based fraud filter ("block any transaction above RM50,000 at 3 a.m.") is a simple form of automation, but it is *not* ML because it does not learn from data.

7.1.2 Three Types of Machine Learning: Supervised, Unsupervised and Reinforcement

ML practitioners group learning approaches into three main families (Chishti et al., 2020):

1. Supervised learning. The model is trained on *labelled* data — historical examples where the correct answer is already known. In finance, labels might be "defaulted" or "repaid" on past loans, or "fraud" or "genuine" on past card transactions. The model learns to map inputs (income, transaction amounts, locations) to the known outputs, then predicts labels for new cases. **Credit scoring** and **fraud detection** are classic supervised-learning tasks.

Finance example: A Malaysian bank trains a model on 200,000 past personal loans, each labelled with whether the borrower defaulted. The model learns that combinations of high debt-service ratios, short employment tenure and certain spending patterns predict default. It then scores new applicants in seconds.

2. Unsupervised learning. The model is given *unlabelled* data and must find structure on its own — for example, grouping (clustering) customers with similar behaviour, or flagging observations that look unusual (anomaly detection). There is no "correct answer" to learn from.

Finance example: An anti-money-laundering (AML) system clusters accounts by transaction patterns and flags a small group whose behaviour is radically different from all others — a possible smurfing network splitting large sums into many small deposits. Nobody labelled these accounts as suspicious beforehand; the model simply noticed they were *different*.

3. Reinforcement learning. The model learns by trial and error, receiving rewards or penalties for its actions, and gradually discovers a strategy that maximises long-run reward. This is how systems learn to play games — and how some trading and portfolio-rebalancing systems learn to act in changing markets.

Finance example: An execution algorithm learns, over many simulated trading episodes, how to break a large share order into smaller pieces across the trading day to minimise market impact — earning a "reward" for each basis point of cost saved.

> **DID YOU KNOW? (TRIVIA):** Reinforcement learning rose to fame when DeepMind's AlphaGo defeated world Go champion Lee Sedol in 2016 — a game with more possible board positions than atoms in the observable universe. The same "reward-maximising agent" idea is now applied to trading and treasury management.

7.1.3 Why Finance Is Data-Rich

Finance is unusually well suited to AI because it is one of the most **data-rich** industries in the world. Every card swipe, e-wallet top-up, FPX transfer, DuitNow QR scan and securities trade creates a digital record. To appreciate the scale: Bank Negara Malaysia reported **18.4 billion e-payment transactions in Malaysia in 2025** — an average of 538 transactions per Malaysian, up 25% from 14.7 billion in 2024 (Bank Negara Malaysia [BNM], 2026). PayNet alone processed 8.44 billion digital payment transactions in 2025 (Payments Network Malaysia [PayNet], 2026). Each of those transactions carries features — amount, time, channel, counterparty, device — that can feed ML models.

Financial data also has properties that suit ML:

- **Volume:** billions of transactions per year provide enough examples to train statistical models.
- **Velocity:** data arrives in real time, enabling real-time decisions (approve/decline a payment in milliseconds).
- **Variety:** structured ledgers, text complaints, voice calls and identity documents.

- **Labels:** finance is blessed with natural labels — loans either default or do not; transactions are later confirmed fraudulent or genuine.

However, data richness cuts both ways: more data means more responsibility for privacy, security and fair use, as Section 7.4 explains.

7.1.4 BigTech and Data-Driven Finance

A defining trend of the past decade is the entry of large technology firms — "BigTech" — into financial services. Zetzsche, Buckley, Arner and Barberis (2017) describe this as a shift "from FinTech to TechFin": whereas FinTech means technology firms serving finance, **TechFin** (or **BigTech finance**) means technology giants such as platform and e-commerce firms using their existing data, users and infrastructure to *offer financial services directly*. Their competitive advantage is **data-driven finance**: combining payments data, e-commerce data, location data and social data to assess customers in ways traditional banks cannot (Zetzsche et al., 2017).

Malaysia illustrates this vividly. **GXBank**, the first of Malaysia's five licensed digital banks to begin operations (on 1 September 2023), is led by Grab — a BigTech platform with years of ride-hailing, food-delivery and e-wallet data — together with the Kuok Group, under BNM's digital-bank licensing framework (BNM, 2020). Similarly, Boost Bank combines the Axiata/Boost digital ecosystem with RHB Banking Group, and Ryt Bank brings together Sea Ltd (parent of Shopee) and YTL Digital Capital. These ownership structures are precisely the data-driven finance model Zetzsche et al. (2017) analysed: platform data plus banking licences.

> **BRAINSTORMER:** Zetzsche et al. (2017) argue that whoever controls data may eventually control finance. If your e-wallet provider knows more about your spending habits than your bank does, who should be allowed to use that data to decide whether you deserve a loan — and under what conditions?

7.2 Applications of AI in Banking and Financial Services

AI is not a single product; it is a toolkit applied across the whole value chain of a financial institution. Bank Negara Malaysia's **Discussion Paper on Artificial Intelligence in the Malaysian Financial Sector** (BNM, 2025) surveys exactly these uses and proposes how they should be governed in Malaysia. This section walks through the major applications, from lending to customer service.

7.2.1 Credit Scoring with Alternative Data

Credit scoring is the use of statistical models to estimate the probability that a borrower will repay a loan. Traditional scoring relies on formal credit bureau histories — past loans, credit cards, repayment records. But what about the millions of Malaysians who have never borrowed formally? World Bank data show that account ownership in Malaysia reached 88.7% of adults in 2024, with the poorest 40% at 82.0% (World Bank, 2025) — yet having an account is not the same as having a credit history. These are **thin-file customers**: people with little or no bureau data.

Alternative data means data *not traditionally used* in credit decisions — for example, mobile-phone top-up patterns, e-wallet transaction flows, utility bill payments, e-commerce sales history, rental payments and even device metadata (with consent). ML models can find predictive signals in this data, allowing lenders to score thin-file customers (Chishti et al., 2020; Zetzsche et al., 2017).

This application is central to the mission of Malaysia's digital banks. BNM licensed five digital banks in 2022 under the *Licensing Framework for Digital Banks* (BNM, 2020), with an asset limit of RM3 billion during a foundational phase of three to five years, precisely so they could focus on **underserved and unserved segments** — micro-SMEs, gig workers and thin-file individuals. BNM's Governor noted in 2025 that deposits from underserved segments are growing and credit products are rolling out gradually (BNM, 2026).

> **MALAYSIAN SPOTLIGHT:** Malaysia's five digital banks — GXBank, Boost Bank, AEON Bank, KAF Digital Bank and Ryt Bank — were licensed to serve the underserved. AEON Bank (launched 26 May 2024) is Malaysia's first Islamic digital bank, and KAF Digital Bank the second, showing that AI-driven inclusion also extends to Shariah-compliant finance (BNM, 2020).

7.2.2 Fraud and Anomaly Detection

Fraud detection uses ML models to score each transaction for suspiciousness in real time, learning from patterns of past confirmed fraud. Unlike static rules, ML models capture subtle combinations — an unusual merchant category, an unfamiliar device, a transaction velocity inconsistent with the customer's history. **Anomaly detection** (often unsupervised) flags behaviour that deviates from a customer's own normal baseline, even if the pattern has never been seen before.

The stakes in Malaysia are high. Scams and mule accounts are a national concern: the **National Scam Response Centre (NSRC, hotline 997)**, established in 2022, coordinates the rapid freezing of scam-linked funds, and the police **SemakMule** portal lets the public check whether a bank account or phone number has been linked to scams. BNM has directed banks to tighten controls — including account "kill-switches", blocking SMS one-time passwords for high-risk transactions, and restricting accounts to one secured device each (BNM, 2025; PayNet, 2026). ML-driven monitoring works alongside these measures, and PayNet operates fraud-intelligence and alert initiatives with banks.

7.2.3 AML Transaction Monitoring

Transaction monitoring for anti-money laundering (AML) means screening customer transactions for patterns consistent with money laundering or terrorism financing, as required under the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (AMLA). Traditional systems use fixed thresholds (for example, flag all cash deposits above a set amount), which criminals easily evade by "smurfing" — splitting sums into many small deposits.

ML improves this in two ways. First, unsupervised models detect *behavioural* anomalies — networks of accounts moving money in circular patterns — rather than fixed thresholds. Second, supervised models trained on past investigated cases reduce **false positives** (legitimate transactions wrongly flagged), which otherwise overwhelm compliance teams. Lower false-positive rates mean investigators spend time on genuinely suspicious activity.

QUICK NOTE: A *false positive* is a legitimate transaction flagged as suspicious (a false alarm); a *false negative* is real fraud that the system misses. Tuning a model always trades one against the other — a theme you will revisit in the Problem-Solving Corner.

7.2.4 Chatbots and Virtual Assistants in Malaysian Banking Apps

Chatbots are AI systems that converse with customers in natural language, answering questions and performing simple transactions. Modern chatbots combine natural-language understanding (a deep-learning task) with scripted workflows. In Malaysia, conversational assistants are embedded in leading banking apps — for example, Maybank's MAE app and CIMB's digital platforms use automated assistants to handle balance enquiries, card blocking and FAQs around the clock, reducing call-centre load. E-wallet providers such as Touch 'n Go eWallet similarly automate first-line customer support.

Generative AI is raising the bar: instead of selecting from pre-written answers, LLM-based assistants can compose fluent, context-aware replies. Banks remain cautious, however, because generative models can *hallucinate* — confidently state false information (see Section 7.4).

7.2.5 Robo-Advisory (Link to Chapter 5)

Robo-advisory — automated, algorithm-driven investment advice and portfolio management with minimal human intervention — is AI applied to wealth management. As covered in Chapter 5, Malaysia regulates robo-advisors under the Securities Commission Malaysia's **Digital Investment Management (DIM)** framework, introduced in May 2017 — the first such framework in the region (Securities Commission Malaysia [SC], 2017). Nine DIM providers are registered, including StashAway (first licensed, 2018), Wahed Invest (Shariah-compliant), MYTHEO and Kenanga Digital Investing. DIM services automate risk profiling, suitability assessment, asset allocation and rebalancing. Notably, 51% of DIM accounts are held by investors in their 20s — young Malaysians are comfortable letting algorithms manage their money (SC, 2026).

7.2.6 Personalised Marketing and Algorithmic Underwriting

Personalised marketing uses ML models (especially clustering and recommendation techniques) to match customers with relevant products — for example, offering a travel-insurance product to a customer whose transaction history shows frequent flight bookings, or timing an offer of micro-financing to an SME's seasonal cash-flow dip. Done well, this improves relevance and reduces marketing cost; done badly, it becomes intrusive or exploitative.

Algorithmic underwriting extends automated decision-making from credit into insurance: models estimate a policyholder's risk from application data, telematics (for motor insurance) or health data, and price the policy accordingly. The same techniques — supervised learning on labelled historical claims — apply. The fairness concerns are also identical: if the training data reflects past discrimination, the model can reproduce it at scale (Section 7.4).

BRAINSTORMER: Your banking app knows you get paid on the 25th of each month and that your balance usually runs low by the 20th. Is it helpful — or predatory — for the app to offer you a small loan on the 19th? Where would you draw the line, and who should draw it: the bank, the customer, or the regulator?

7.2.7 BNM's 2025 Discussion Paper on AI in the Malaysian Financial Sector

In August 2025, BNM issued its **Discussion Paper on Artificial Intelligence in the Malaysian Financial Sector** (BNM, 2025). The paper recognises AI's potential to improve efficiency, risk management and financial inclusion, while identifying risks including opacity ("black-box" models), bias, data privacy, third-party concentration and consumer harm. It signals BNM's expectations that financial institutions adopting AI should maintain strong governance, risk management and accountability, proportionate to the materiality of the AI use case — a board-approved framework, model validation, and clear human accountability. The discussion paper complements BNM's existing technology-risk framework, the **Risk Management in Technology (RMiT)** policy document (BNM, 2025), which sets technology and cyber-resilience requirements for financial institutions. For students, the key message is that Malaysian regulators are neither banning AI nor leaving it unregulated: they are shaping *responsible* adoption.

7.3 Financial Data, Information Systems and Data Analytics

7.3.1 Types of Financial Data

AI models are only as good as the data they consume. Financial institutions work with several broad categories:

Data type	Description	Malaysian finance examples
Structured data	Highly organised data in rows and columns (databases, spreadsheets)	Core banking ledgers, RENTAS settlement records, transaction logs
Unstructured data	Data without a predefined model — text, audio, images, video	Customer complaint emails, recorded call-centre conversations, identity-card scans used in e-KYC
Semi-structured data	Partially organised (e.g., JSON messages between payment systems)	ISO 20022 payment messages, API payloads
Alternative data	Data from outside traditional financial records used to infer creditworthiness or behaviour	E-wallet flows, utility payments, e-commerce sales, mobile top-ups (Section 7.2.1)

Structured data is the historical backbone of banking; **unstructured data** is now the fastest-growing category, and deep learning (especially natural-language processing and computer vision) is what makes it usable — for example, extracting text from an identity document during BNM-compliant electronic know-your-customer (e-KYC) onboarding (BNM, 2024).

7.3.2 Core Banking and Information Systems

Behind every AI model sits a stack of **information systems** — the hardware, software, databases and networks that record and process an institution's operations. The heart of a conventional or Islamic bank is its **core banking system**: the centralised platform that maintains customer accounts, posts transactions, calculates interest or profit, and produces regulatory reports. Around the core sit channel systems (branch, ATM, internet banking such as Maybank2u — Malaysia's first internet banking platform, launched in 2000 — and mobile apps), payment connections (to PayNet's rails such as IBG, FPX, DuitNow and DuitNow QR), risk systems and data warehouses.

A **data warehouse** consolidates data from these source systems into a single, analysis-ready store; increasingly, institutions add a **data lake** for raw unstructured data and use cloud infrastructure for elastic computing power. Two points matter for AI:

1. **Data must flow reliably.** If the core system records a transaction but the fraud model sees it only the next day, real-time fraud scoring is impossible. This is why digital banks built their stacks "cloud-native" from day one.
2. **Lineage must be traceable.** Under RMiT and BNM's AI governance expectations, an institution must be able to show *where* the data feeding a model came from (BNM, 2025).

7.3.3 The Data Analytics Pipeline

Data analytics is the process of examining data to draw conclusions and support decisions. A typical analytics pipeline in a bank has five stages:

1. **Collection** — capturing raw data from core banking, apps, payment rails, e-KYC documents and external sources (e.g., credit bureaus), with customer consent where required.
2. **Cleaning (preparation)** — fixing errors, removing duplicates, standardising formats (e.g., one date format), handling missing values. Practitioners often say this consumes the majority of project time.
3. **Modelling** — applying statistics or ML to the prepared data: building a credit scorecard, training a fraud classifier, clustering customers.
4. **Visualisation** — presenting results through dashboards, charts and alerts so humans can grasp them (e.g., a branch manager's dashboard of weekly DuitNow QR sign-ups).
5. **Decision** — acting on the insight: approve the loan, block the card, launch the campaign — and feeding the outcome back into the data, creating a learning loop.

   QUICK NOTE: "Garbage in, garbage out" (GIGO) is the oldest rule in analytics: a sophisticated ML model trained on dirty or biased data will produce confident, precise, *wrong* answers. Data quality is not a technical detail — it is a governance duty.

7.3.4 Four Levels of Analytics — with Malaysian Banking Examples

Analytics maturity is commonly described in four ascending levels:

Level	Question answered	Technique	Malaysian banking example
Descriptive	What happened?	Aggregation, reports, dashboards	BNM reporting 18.4 billion e-payment transactions in 2025, up 25% year-on-year (BNM, 2026)
Diagnostic	Why did it happen?	Drill-down, correlation analysis	A bank discovers its app log-ins dropped because an update broke fingerprint authentication on one phone model
Predictive	What is likely to happen?	ML classification/regression, forecasting	A digital bank predicts which thin-file applicants are likely to repay; a card issuer predicts which transactions are likely to be fraudulent
Prescriptive	What should we do?	Optimisation, simulation, reinforcement learning	A treasury system recommends the optimal daily liquidity position; a collections system recommends the best channel and time to contact each overdue customer

Descriptive and diagnostic analytics explain the past; **predictive analytics** and **prescriptive analytics** shape the future — and it is these forward-looking levels where AI and ML contribute most, and where governance matters most.

> **DID YOU KNOW? (TRIVIA):** The average e-money transaction in Malaysia was just RM43 in 2025 (up from RM33 in 2024) (BNM, 2026). Small-ticket, high-frequency data like this is exactly what trains anomaly-detection models — your RM6 teh tarik purchase helps define your "normal".

7.3.5 Data Quality and Governance

Data quality has several dimensions: *accuracy* (is the value correct?), *completeness* (are fields missing?), *consistency* (does the same customer appear with different details in different systems?), *timeliness* (is the data current?) and *validity* (does it conform to expected formats and ranges?). **Data governance** is the framework of policies, roles and controls that ensures data is managed as an institutional asset: clear data ownership, access controls, retention rules, quality monitoring and audit trails.

In Malaysia, governance operates within the **Personal Data Protection Act 2010 (Act 709)**

(amended 2024), which regulates how commercial organisations collect, use, disclose and protect personal data, and within BNM's RMiT requirements for technology risk (BNM, 2025). For AI specifically, BNM's 2025 discussion paper emphasises that institutions remain fully accountable for decisions made by their models — which is impossible without knowing the quality and provenance of the data underneath (BNM, 2025).

7.4 Benefits, Limitations and Ethical Considerations of AI

7.4.1 The Benefits: Efficiency, Speed and Inclusion

AI delivers three headline benefits to financial services:

1. **Efficiency and cost reduction.** Automating document checks, customer enquiries and monitoring reduces operating cost. For digital banks operating under a RM3 billion asset limit in their foundational phase (BNM, 2020), lean, automated operations are not optional — they are the business model.
2. **Speed and scale.** ML models score loan applications or screen transactions in milliseconds, at volumes no human team could match — a necessity when PayNet alone processed 8.44 billion digital payment transactions in 2025 (PayNet, 2026).
3. **Financial inclusion.** Alternative-data credit scoring extends lending to thin-file customers whom traditional models exclude entirely (Zetzsche et al., 2017). With 11.3% of Malaysian adults still outside account ownership in 2024 and the poorest 40% at 82.0% ownership (World Bank, 2025), inclusion remains a real policy goal — and AI is one of the tools BNM's digital-bank licensing strategy deploys toward it.

7.4.2 Black-Box Opacity and the Explainability Problem

The most accurate ML models — deep neural networks, gradient-boosted ensembles — are often the least interpretable. The **black-box problem** is that a model may produce a decision (decline this loan, flag this account) without any human being able to say precisely *why* in terms a customer or regulator can understand. **Explainability** (often called explainable AI, or XAI) is the set of techniques and practices that make model decisions interpretable — for example, identifying which features most influenced a particular credit decision.

Why does this matter so much in finance? Because financial decisions affect livelihoods and are regulated. If a bank cannot explain why Aina (from our vignette) was declined, it cannot demonstrate the decision was fair, cannot correct data errors, and cannot satisfy BNM's accountability expectations (BNM, 2025). Many institutions therefore accept slightly lower accuracy in exchange for interpretable models for high-stakes decisions.

7.4.3 Bias and Discrimination: The Apple Card Lesson

ML models learn from historical data — and history contains bias. If past lending systematically disadvantaged a group (by gender, ethnicity, location or occupation), a model trained on that data can learn and *amplify* the pattern, even if the sensitive attribute itself is removed: **proxy variables** (such as postcode or occupation) can stand in for the protected trait.

The textbook illustration is the **2019 Apple Card controversy**. Apple Card, issued by Goldman Sachs in the United States, was accused in November 2019 of offering married women

much lower credit limits than their husbands — even when the couple shared assets and the wife had the better credit score. The complaints, which went viral, triggered a regulatory investigation by the New York State Department of Financial Services. Whether or not deliberate discrimination occurred, the episode showed three things: algorithmic decisions can produce *disparate outcomes* without any human writing a discriminatory rule; customers will publicly challenge black-box decisions; and regulators will hold the *institution*, not the algorithm, responsible.

> **BRAINSTORMER:** Suppose a credit model never sees gender or race, yet consistently offers lower limits to customers from certain postcodes. Has the bank "discriminated"? If a human loan officer did the same thing, would your answer change? Why or why not?

7.4.4 Data Privacy under the PDPA 2010

AI in finance runs on personal data, so it operates inside privacy law. Malaysia's **Personal Data Protection Act 2010 (Act 709)**, amended in 2024, governs the processing of personal data in commercial transactions: it requires, among other things, consent and notice for collection, purpose limitation, reasonable security safeguards, and data-subject rights of access and correction (Personal Data Protection Act 2010). AI creates special tensions:

- **Purpose creep.** Data collected to operate an e-wallet is reused to train a credit model — is that within the original consent?
- **Profiling and inference.** ML can *infer* sensitive attributes (health, religion, political leaning) from innocuous data such as shopping patterns.
- **Retention and security.** The more data hoarded "just in case", the larger the target for breaches — a direct RMiT concern (BNM, 2025).

Responsible institutions adopt **data minimisation** (collect only what is needed), purpose-bound consent, anonymisation where possible, and privacy impact assessments before deploying new models.

7.4.5 Hallucination, Job Displacement and Model Risk

Hallucination. Generative AI models can produce fluent, confident, *fabricated* statements — inventing a policy clause, misstating an interest rate, or citing a non-existent regulation. In a banking chatbot, a hallucinated answer about fees or Shariah compliance could mislead customers and breach fair-treatment expectations. This is why generative AI in customer-facing roles requires strict grounding, human review and escalation paths (BNM, 2025).

Job displacement. Automation of routine tasks — data entry, first-line customer service, document processing, even parts of credit analysis — will reduce demand for some roles while creating others (model validators, data stewards, AI risk officers). The policy challenge is *transition*: reskilling affected workers. As a student entering the industry, the durable skills are judgement, ethics, communication and the ability to supervise AI systems — the things models cannot (yet) do well.

Model risk. Model risk is the risk of financial loss, customer harm or bad decisions arising from models that are wrong, misused or used outside their valid range. Models can *drift* as the world changes (a credit model trained before a recession may misjudge risk during it); they can be *overfitted* to historical quirks; and they can fail under conditions never seen in training data. Sound practice — reflected in BNM's AI discussion paper — requires model inventories, independent validation, back-testing, monitoring and clear ownership (BNM, 2025).

7.4.6 Human-in-the-Loop and Regional Governance Benchmarks

A widely accepted safeguard is the **human-in-the-loop (HITL)** principle: for high-stakes decisions, a human reviews, can override and remains accountable for the AI's recommendation. The degree of human involvement can be calibrated to risk — automatic approval for low-value, reversible decisions; mandatory human review for loan declines, account closures or AML escalations.

Regionally, the benchmark for AI ethics in finance is the Monetary Authority of Singapore's **FEAT principles** (2018): firms using AI and data analytics in Singapore's financial sector should ensure their use is **Fair, Ethical, Accountable and Transparent**. *Fair* means decisions do not systematically disadvantage individuals or groups without justification; *Ethical* means AI use aligns with ethical standards and values; *Accountable* means boards and management answer for AI-driven decisions; *Transparent* means data subjects receive clear explanations where feasible. BNM's 2025 discussion paper develops Malaysia's own expectations in the same spirit, embedding governance, risk management and accountability into AI adoption (BNM, 2025). Together with RMiT and the PDPA, these frameworks sketch the emerging Malaysian rulebook: innovate, but remain explainable, fair and accountable.

Case Study 7.1: Lending to the Invisible — AI and Alternative Data at a Malaysian Digital Bank

When BNM announced five digital-bank licences on 29 April 2022, the explicit policy goal was to reach **unserved and underserved** customers — gig workers, micro-SMEs and thin-file individuals whom conventional credit scoring leaves out (BNM, 2020). GXBank, led by Grab together with the Kuok Group, was the first to begin operations on 1 September 2023, operating within a RM3 billion asset cap during its foundational phase of three to five years.

Consider how a digital bank in GXBank's position can serve a thin-file borrower — think of Aina, the Penang seller from our opening vignette. A conventional scorecard sees nothing: no credit card, no past loan. But a digital bank observes *consented behavioural data* flowing through its own ecosystem: the rhythm of her e-wallet inflows from customers, her prompt utility and supplier payments, the seasonality of her sales, how her balance behaves before and after payday. An ML credit model trained on thousands of similar micro-entrepreneurs — where repayment outcomes are known — can estimate her probability of default with usable accuracy. The result: a small starting credit line, say RM2,000–RM5,000, sized conservatively and allowed to grow as her repayment behaviour generates real credit history. This is the data-driven finance model described by Zetzsche et al. (2017), now operating under a Malaysian banking licence.

The benefits are genuine. Inclusion: customers excluded by bureau-only scoring gain access to formal credit. Speed: decisions arrive in-app in seconds, at marginal cost near zero. Pricing: risk-based pricing can be *lower* for good thin-file risks than informal-moneylender rates. And Malaysia needs this — the poorest 40% of adults had 82.0% account ownership in 2024 (World Bank, 2025), and BNM's Governor noted in 2025 that digital banks' deposits from underserved segments are growing while credit products roll out gradually (BNM, 2026).

But fairness safeguards are essential. First, **consent and purpose limitation** under the PDPA 2010: behavioural data may be used for credit only with informed consent. Second, **bias testing**: the model must be audited for disparate impact — do applicants with identical

repayment capacity receive different outcomes by gender, ethnicity or state? The Apple Card episode shows how quickly opaque disparities become scandals. Third, **explainability and recourse**: a declined applicant should receive the main reasons and a route to human review, consistent with BNM's AI governance expectations and the regional FEAT benchmark. Fourth, **human-in-the-loop** for edge cases and overrides. Fifth, **conservative sizing** — small initial limits protect both the customer (from over-indebtedness) and the bank (from model error while its models are young).

Get the balance right, and alternative-data lending is one of the most socially valuable applications of AI in Malaysian finance. Get it wrong, and it automates exclusion at scale.

Discussion Questions

1. Identify two types of alternative data a Malaysian digital bank could use to score a thin-file micro-entrepreneur, and explain what each might reveal about repayment capacity.
2. Using the FEAT principles (Fair, Ethical, Accountable, Transparent), evaluate the fairness safeguards described in the case. Which principle do you think is hardest to satisfy in practice, and why?
3. Discuss the tension between data minimisation under the PDPA 2010 and the ML desire for "more data is better". How might a bank resolve it?
4. BNM caps digital banks' assets at RM3 billion during the foundational phase. How might this cap act as an indirect consumer-protection measure for AI-driven lending?

Chapter Summary

- **7.1** Artificial intelligence is the broad field of systems performing tasks requiring human intelligence; machine learning (a subset) learns patterns from data without explicit programming; deep learning uses multi-layered neural networks; generative AI creates new content (Chishti et al., 2020).
- **7.1** Supervised learning trains on labelled data (credit default, fraud); unsupervised learning finds structure in unlabelled data (clustering, anomaly detection); reinforcement learning learns by trial and error with rewards (trade execution).
- **7.1** Finance is data-rich — 18.4 billion Malaysian e-payment transactions in 2025 alone (BNM, 2026) — giving ML models volume, velocity, variety and natural labels.
- **7.1** Zetzsche et al. (2017) describe the shift "from FinTech to TechFin": BigTech platforms leverage their data to offer financial services — the model behind Malaysia's platform-linked digital banks such as GXBank and Ryt Bank.
- **7.2** Major AI applications include alternative-data credit scoring for thin-file customers, fraud and anomaly detection, AML transaction monitoring under AMLA 2001, chatbots in banking apps, robo-advisory under the SC's DIM framework (2017), personalised marketing and algorithmic underwriting.
- **7.2** BNM's 2025 Discussion Paper on AI in the Malaysian Financial Sector sets out governance, risk-management and accountability expectations for AI adoption, complementing RMiT.
- **7.3** Financial data spans structured, unstructured, semi-structured and alternative data; core banking systems and data warehouses feed the analytics pipeline: collection → cleaning → modelling → visualisation → decision.
- **7.3** Analytics maturity rises through descriptive, diagnostic, predictive and prescriptive levels; data quality and governance (within PDPA 2010 and RMiT) underpin everything.

- **7.4** AI delivers efficiency, speed and inclusion, but raises black-box opacity, bias and discrimination (the 2019 Apple Card controversy), privacy risks under the PDPA 2010, generative-AI hallucination, job displacement and model risk.
- **7.4** Key safeguards are human-in-the-loop decision-making, explainability, bias testing, and governance anchored in the MAS FEAT principles (2018) and BNM's 2025 AI discussion paper.

Key Terms

- **Artificial intelligence (AI)**: the field of building computer systems that perform tasks normally requiring human intelligence, such as understanding language, recognising images and making decisions.
- **Machine learning (ML)**: a subset of AI in which computers learn patterns from data and improve at a task without being explicitly programmed with fixed rules.
- **Deep learning**: a subset of ML using multi-layered neural networks to learn complex patterns from large volumes of data.
- **Generative AI**: AI models that create new content — text, images, code or audio — in response to prompts.
- **Supervised learning**: ML trained on labelled historical examples to predict outcomes for new cases, such as default or fraud.
- **Unsupervised learning**: ML that finds structure, clusters or anomalies in unlabelled data without known correct answers.
- **Reinforcement learning**: ML in which an agent learns by trial and error, receiving rewards or penalties to discover optimal strategies.
- **Neural network**: a computational structure of interconnected processing layers, loosely inspired by the brain, used in deep learning.
- **Alternative data**: data outside traditional financial records — e.g., e-wallet flows, utility payments, mobile top-ups — used to infer creditworthiness.
- **Credit scoring**: the use of statistical models to estimate the probability that a borrower will repay a loan.
- **Fraud detection**: the use of ML models to score transactions for suspiciousness in real time based on learned fraud patterns.
- **Transaction monitoring**: the screening of customer transactions for patterns consistent with money laundering or terrorism financing, as required under AMLA 2001.
- **Chatbot**: an AI system that converses with customers in natural language to answer queries and perform simple transactions.
- **Robo-advisory**: automated, algorithm-driven investment advice and portfolio management with minimal human intervention; regulated in Malaysia under the SC's DIM framework.
- **Black-box problem**: the inability of humans to understand or explain precisely why a complex ML model produced a particular decision.
- **Human-in-the-loop (HITL)**: a safeguard in which a human reviews, can override and remains accountable for AI recommendations in high-stakes decisions.
- **Model risk**: the risk of loss or customer harm from models that are wrong, misused or applied outside their valid range.
- **FEAT principles**: MAS's 2018 benchmark requiring AI and data analytics in finance to be Fair, Ethical, Accountable and Transparent.

Test Your Knowledge

Part A: Multiple Choice

Q1. Which statement correctly describes the relationship between the fields? A. Machine learning is a subset of deep learning B. Generative AI is separate from machine learning C. Machine learning is a subset of artificial intelligence, and deep learning is a subset of machine learning D. Artificial intelligence is a subset of machine learning

Q2. A bank trains a model on 200,000 past loans, each labelled "defaulted" or "repaid", to score new applicants. This is an example of: A. Unsupervised learning B. Supervised learning C. Reinforcement learning D. Generative AI

Q3. An AML system groups accounts by transaction behaviour and flags a small cluster whose activity differs radically from all others, with no prior labels. This is an example of: A. Unsupervised learning B. Supervised learning C. Reinforcement learning D. A rule-based filter

Q4. "Thin-file customer" refers to: A. A customer with a poor credit score B. A customer with little or no formal credit bureau history C. A customer with a low account balance D. A customer who only uses cash

Q5. Which of the following is the best example of alternative data for credit scoring? A. A borrower's previous bank loan repayment record B. A borrower's credit card statement C. A borrower's e-wallet inflows and utility bill payments D. A borrower's CCRIS record

Q6. The FEAT principles were issued in 2018 by which regulator? A. Bank Negara Malaysia B. Securities Commission Malaysia C. Monetary Authority of Singapore D. Bank for International Settlements

Q7. The 2019 Apple Card controversy is used as a textbook example of: A. Fraud detection failure B. Algorithmic bias and black-box credit decisions C. Generative AI hallucination D. A data breach under privacy law

Q8. "What is likely to happen?" is the question answered by which level of analytics? A. Descriptive B. Diagnostic C. Predictive D. Prescriptive

Q9. A generative AI chatbot confidently tells a customer a false statement about the bank's fees. This risk is known as: A. Model drift B. Hallucination C. Overfitting D. Smurfing

Q10. Malaysia's Personal Data Protection Act is: A. Act 709 of 2010 B. Act 758 of 2013 C. Act 613 of 2001 D. Act 701 of 2009

Part B: Short Answer

Q11. Define machine learning and explain how it differs from traditional rule-based programming, with one finance example of each.

Q12. Explain why finance is considered a "data-rich" industry, citing at least two characteristics of financial data and one Malaysian statistic from this chapter.

Q13. A digital bank wants to use alternative data to lend to gig workers. Describe two benefits and two risks of this approach, and one governance safeguard the bank should adopt.

Q14. Distinguish the four levels of data analytics (descriptive, diagnostic, predictive, prescriptive) and give a Malaysian banking example for any two levels.

Q15. What is the human-in-the-loop principle, and why do regulators such as BNM and MAS emphasise it for high-stakes AI decisions?

Answer Guide

Part A

1. **C** — The fields are nested: generative AI ⊂ deep learning ⊂ ML ⊂ AI.
2. **B** — Training on labelled historical examples is the definition of supervised learning.
3. **A** — Clustering/anomaly detection on unlabelled data is unsupervised learning.
4. **B** — Thin-file customers lack bureau history, regardless of income or balance.
5. **C** — Alternative data comes from outside traditional credit records; A, B and D are traditional sources.
6. **C** — FEAT (Fairness, Ethics, Accountability, Transparency) is the MAS 2018 framework; BNM's own AI discussion paper followed in 2025.
7. **B** — Apple Card illustrated how opaque algorithmic decisions can produce apparently discriminatory outcomes and trigger regulatory investigation.
8. **C** — Predictive analytics forecasts likely future events; prescriptive recommends actions.
9. **B** — Hallucination is fluent but fabricated output from generative AI.
10. **A** — The PDPA 2010 is Act 709 (amended 2024); Act 758 is the FSA 2013, Act 613 is AMLA 2001, Act 701 is the Central Bank of Malaysia Act 2009.

Part B (model answers)

11. **Machine learning** is a subset of AI in which computers learn patterns from data without being explicitly programmed with fixed rules (Chishti et al., 2020). Rule-based programming follows human-written instructions, e.g., "block transactions above RM50,000 at 3 a.m." ML instead infers patterns from examples, e.g., a fraud model trained on thousands of confirmed-fraud and genuine transactions that learns subtle feature combinations no programmer specified.

12. Finance is data-rich because every payment, transfer and trade creates a digital record, giving data with high **volume** (BNM reported 18.4 billion e-payment transactions in Malaysia in 2025), high **velocity** (real-time transaction streams enabling millisecond decisions), **variety** (ledgers, documents, voice) and natural **labels** (loans default or not). Any two characteristics plus the statistic suffice.

13. *Benefits:* inclusion of thin-file customers excluded by bureau scoring; faster, cheaper, more accurately priced credit decisions. *Risks:* bias/discrimination learned from historical or proxy data; privacy intrusion and purpose creep under the PDPA 2010. *Safeguard (any one):* informed consent and purpose limitation; bias testing for disparate impact; explainable reasons for decline with human recourse; human-in-the-loop review; conservative initial limit sizing.

14. *Descriptive* — what happened (BNM reporting 18.4 billion e-payments in 2025); *diagnostic* — why it happened (drilling into an app log-in drop to a faulty update); *predictive* — what is likely to happen (scoring a thin-file applicant's default probability); *prescriptive* — what to do (optimising daily liquidity or collections contact strategy). Correct definitions plus any two valid examples.

15. Human-in-the-loop means a human reviews, can override and remains accountable for AI recommendations in high-stakes decisions. Regulators emphasise it because accountability cannot rest on an opaque algorithm: customers need recourse, boards must an-

swer for decisions (MAS FEAT's "Accountable"; BNM's 2025 AI governance expectations), and human review catches model errors, drift and edge cases.

Problem-Solving Corner

Problem 7.1: Reading a Confusion Matrix for Fraud Detection

A Malaysian card issuer tests an ML fraud-detection model on a sample of **10,000 real transactions**, of which **100 are actually fraudulent** (confirmed later) and 9,900 are genuine. The model's results are:

	Model says FRAUD	Model says GENUINE
Actually fraud (100)	80 (true positives, TP)	20 (false negatives, FN)
Actually genuine (9,900)	90 (false positives, FP)	9,810 (true negatives, TN)

(a) Calculate the model's **accuracy**.
(b) Calculate its **precision** and **recall** for the fraud class.
(c) The bank's compliance manager says: "98.9% accuracy — excellent!" Critique this claim.
(d) If the bank wants to catch more fraud, what trade-off must it accept?

Solution 7.1

(a) Accuracy = (TP + TN) ÷ total = (80 + 9,810) ÷ 10,000 = **98.9%**.

(b) Precision = TP ÷ (TP + FP) = 80 ÷ (80 + 90) = 80 ÷ 170 ≈ **47.1%**. Of everything the model flagged as fraud, less than half was actually fraud. **Recall** = TP ÷ (TP + FN) = 80 ÷ (80 + 20) = 80 ÷ 100 = **80.0%**. The model caught 80 of the 100 real frauds; 20 slipped through.

(c) The accuracy figure is deeply misleading because of **class imbalance**: 99% of transactions are genuine, so a lazy model that calls *everything* genuine scores 99.0% accuracy while catching zero fraud. The meaningful metrics are precision and recall. This model's precision of 47.1% means analysts will investigate 170 flagged transactions to find 80 frauds — 90 customers experience false alarms (blocked cards, friction). Its recall of 80% means one in five frauds is missed. A fraud model must be judged on the precision–recall trade-off, not accuracy.

(d) To raise recall (catch more fraud), the bank lowers the alert threshold — but this flags more legitimate transactions too, *reducing* precision: more false positives, more analyst workload, more customer friction. Raising precision (fewer false alarms) means a stricter threshold, which lowers recall (more missed fraud). There is no free lunch; the optimal point depends on the *costs* — the RM loss per missed fraud versus the cost of investigating each false positive and the customer harm of wrongly blocked transactions. In practice banks combine the model with rules, human review of borderline cases, and controls such as BNM's kill-switch and one-secured-device requirements.

Problem 7.2: Building and Stress-Testing a Credit Decision Table

A campus entrepreneur club designs a simple rule-based scorecard for a micro-loan of RM3,000 to students with small businesses. Rules (apply in order; first match decides):

Rule	Condition	Decision
1	Business operating < 6 months	DECLINE
2	Monthly business revenue < RM1,500	DECLINE
3	Missed any utility/phone bill payment in last 6 months	DECLINE
4	Revenue ≥ RM3,000 AND business ≥ 12 months old	APPROVE RM3,000
5	Otherwise	APPROVE RM1,500 (half loan)

Applicants: **Ali** — 14 months in business, revenue RM3,200/month, no missed bills. **Mei** — 7 months, revenue RM2,800/month, one missed phone bill. **Sara** — 9 months, revenue RM1,800/month, clean record.

(a) Apply the decision table to Ali, Mei and Sara.
(b) Identify two bias or fairness risks in this rule set.
(c) Suggest one improvement, explaining its effect.

Solution 7.2

(a) - **Ali:** passes Rules 1–3; matches Rule 4 (revenue ≥ RM3,000, ≥ 12 months) → **APPROVE RM3,000**. - **Mei:** passes Rules 1–2; Rule 3 catches her missed bill → **DECLINE**. (She never reaches Rule 5, where her revenue would otherwise earn RM1,500.) - **Sara:** passes Rules 1–3; fails Rule 4 (revenue < RM3,000); matches Rule 5 → **APPROVE RM1,500**.

(b) Bias and fairness risks (any two): 1. **Proxy discrimination and history bias:** "missed any bill in 6 months" punishes one-off hardship (e.g., a family emergency) and disproportionately excludes lower-income students — a single RM40 phone-bill slip outweighs strong revenue. Like the Apple Card lesson, an apparently neutral rule can systematically exclude a group. 2. **Arbitrary thresholds (cliff effects):** revenue of RM2,999 versus RM3,000 changes the outcome dramatically although the applicants are nearly identical; hard cut-offs are brittle and easily gamed once known. 3. **Maturity bias:** Rules 1 and 4 favour older businesses, systematically disadvantaging younger (and often younger-aged, newer) entrepreneurs — precisely the thin-file segment the scheme claims to serve. There is no mechanism for context, growth trajectory or human review.

(c) Improvement (any one, with effect): - Replace hard cut-offs with a **points-based scorecard** (points for revenue bands, business age, bill-payment record) so small differences produce small score differences — reducing cliff effects and gaming. - Add **proportionate tolerance** for isolated small-value missed bills with explanation (human review) — reducing exclusion from one-off hardship and introducing a human-in-the-loop element. - Add **bias testing**: track approval rates by gender, ethnicity and state to detect disparate impact, and document reasons for each decline — supporting fairness and accountability in line with FEAT and BNM's AI expectations.

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Chapter 8: Digital Banking Risk, Regulation and AI Ethics

Chapter Outline

- 8.1 Key Risks in Digital Banking and FinTech (Cybersecurity, Fraud, Scams)
- 8.2 Risk Management in Digital Banking Businesses
- 8.3 AI Ethics (Bias, Fairness, Transparency and Accountability)
- 8.4 Data Privacy, Consumer Protection and Regulatory Technology (RegTech)

Learning Outcomes

After studying this chapter, you should be able to:

1. **Define** key terms related to digital banking risk, technology regulation and AI ethics. *(Bloom's level: Remember)*
2. **Explain** the main categories of cyber, fraud and scam risks facing Malaysian digital banking customers and institutions. *(Bloom's level: Understand)*
3. **Apply** a likelihood–impact risk matrix and cost–benefit analysis to evaluate digital banking risks and anti-fraud controls. *(Bloom's level: Apply)*
4. **Analyse** the failure points in a Malaysian scam case, tracing how funds move through mule accounts and where controls should bite. *(Bloom's level: Analyse)*
5. **Evaluate** AI governance frameworks, including the MAS FEAT principles and BNM's policy direction on artificial intelligence. *(Bloom's level: Evaluate)*
6. **Design** a basic consumer-protection and RegTech control set for a hypothetical Malaysian digital bank. *(Bloom's level: Create)*

Opening Vignette

Puan Rohana, 52, runs a busy nasi lemak stall in Kota Bharu. One Tuesday morning, between serving customers, she received a call from a man claiming to be from her bank's "fraud department". He knew her full name and the last four digits of her identity card number. He told her that her account was "under audit" for suspicious activity and that she must transfer RM9,500 to a "safe holding account" to protect her savings. Frightened, and impressed by how professional he sounded, she opened her banking app and made the transfer. Only when she told her daughter that evening did she realise she had been scammed. Her daughter immediately called **997**, the National Scam Response Centre hotline, and part of the money was frozen before it could be withdrawn. Rohana's story raises the questions at the heart of

this chapter: What risks does digital banking create? Where did controls fail? And how do regulation and ethics protect Malaysians like her?

8.1 Key Risks in Digital Banking and FinTech (Cybersecurity, Fraud, Scams)

8.1.1 Digital growth expands the attack surface

Malaysia is now one of the most digitally connected banking markets in the region. Account ownership among adults reached 88.7% in 2024 (World Bank, 2025), and Malaysians made 18.4 billion e-payment transactions in 2025 — an average of 538 transactions per person, and 25% more than the 14.7 billion recorded in 2024 (Bank Negara Malaysia [BNM], 2026). Payments Network Malaysia (PayNet), the BNM-owned operator of the national payment rails, processed 8.44 billion digital payment transactions in 2025 (Payments Network Malaysia [PayNet], 2026). DuitNow QR alone handled roughly 3 billion transactions across about 3 million merchant touchpoints in 2025, double the volume of the previous year (BNM, 2026).

Every one of these transactions is also a potential target. **Cybersecurity** — the protection of computer systems, networks and data from digital attack, theft and damage (Gupta & Tham, 2019) — has therefore become a core banking function rather than a back-office concern. Scholars of the fintech revolution note that the same forces that make digital finance fast, cheap and inclusive — data, connectivity and automation — also create new vulnerabilities that criminals can exploit at scale (Gomber et al., 2018). Trust is the foundation of banking, and technology failures or fraud can destroy that trust faster than almost any other event (Thakor, 2020). Lee and Shin (2018) similarly identify security threats and regulatory uncertainty as among the most serious challenges facing fintech firms.

8.1.2 A taxonomy of cyberattack, fraud and scam risks

Risk professionals classify digital banking threats into recognisable categories so that each can be measured and controlled. The taxonomy below follows standard industry usage (Gupta & Tham, 2019; Hill, 2018).

Risk	Definition	Typical Malaysian example
Cyberattack	A deliberate attempt to compromise the confidentiality, integrity or availability of systems	Ransomware or a distributed denial-of-service (DDoS) attack on a bank's servers
Phishing	Fraudulent messages (e-mail, SMS "smishing", or voice "vishing") that impersonate trusted entities to steal credentials	An SMS claiming to be from a bank, linking to a fake login page
Social engineering	Manipulating people, rather than systems, into revealing information or performing actions	A caller posing as a "bank officer" or "police inspector"

Risk	Definition	Typical Malaysian example
Malware	Malicious software installed on a device to steal data or control it	A fake "security app" (APK file) downloaded outside official app stores
Account takeover (ATO)	A criminal gains control of a genuine customer's account using stolen credentials	Fraudster logs in with a phished password and changes the registered mobile number
Authorised push payment (APP) scam	The victim is deceived into personally authorising a transfer to the criminal	Rohana's transfer to the "safe holding account" in the vignette

Two distinctions matter here. First, in *unauthorised* fraud (such as ATO), transactions happen without the customer's approval; in *authorised* fraud (APP scams), the customer personally approves the payment, which makes reimbursement claims far more complicated. Second, many attacks combine categories: a phishing SMS harvests credentials, malware captures one-time passwords (OTPs), and the result is an account takeover.

> **QUICK NOTE:** The term "scam" is usually reserved for cases where the victim is psychologically manipulated into cooperating, whereas "fraud" may involve no victim participation at all. This is why APP scams defeat many traditional security controls — the genuine customer presses "confirm".

8.1.3 Mule accounts, data breaches and third-party concentration risk

A **mule account** is a bank account used to receive and move the proceeds of crime, often operated by a legitimate account holder who has "rented" or sold access to a syndicate. Recruitment is brazen: social-media advertisements offer students and gig workers RM300–RM800 for the use of their accounts. The account holder may claim innocence, but renting out an account facilitates money laundering — an offence under the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (Act 613) — and typically results in the account being blacklisted. Mule accounts matter because they break the link between the scammer and the money: funds pass through several mules within minutes, making recovery difficult.

A **data breach** is the unauthorised access to, or disclosure of, personal or financial data. Breaches at banks, merchants or third-party vendors leak names, identity card numbers, phone numbers and account details. These leaks arm scammers: when a caller already knows your name and partial IC number, as in Rohana's case, the script sounds convincing. Data breaches therefore sit *upstream* of scams — today's breach becomes next year's vishing call.

Finally, **concentration risk** arises when many institutions depend on a small number of third-party providers, such as the few hyperscale cloud computing platforms that host much of the world's financial infrastructure. An outage or cyber incident at one provider can cascade across many banks at once, turning a firm-level problem into a systemic one. Analysts of data-driven finance warn that the industry's growing reliance on a handful of large technology providers is one of the defining regulatory challenges of the digital era (Zetzsche et al., 2017).

> **DID YOU KNOW? (TRIVIA):** Malaysia's National Scam Response Centre (NSRC) can be reached by dialling just three digits — **997**. The hotline, established in 2022, links victims directly to a centre connecting the police, Bank Negara Malaysia and

the banks, so that scam-linked funds can be frozen within the critical "golden hours" before they are withdrawn.

8.1.4 Malaysia's anti-scam response: NSRC, SemakMule, bank controls and PayNet

Malaysia has built a layered national response to the scam epidemic:

- **National Scam Response Centre (NSRC, hotline 997, established 2022).** A joint initiative linking the Royal Malaysia Police (PDRM), BNM, financial institutions and telecommunications companies. Its key function is the rapid freezing of scam-linked funds once a victim reports.
- **PDRM SemakMule.** A public portal and mobile app that lets anyone check whether a bank account number or phone number has been reported in connection with scams — a practical tool before transferring money to a stranger.
- **BNM-directed bank measures.** BNM has directed banks to tighten fraud controls, including a **kill-switch** (a self-service feature allowing customers to freeze their own accounts and cards immediately if they suspect compromise), blocking SMS-delivered OTPs for high-risk transactions (forcing approval through more secure channels), and restricting each account to **one secured device** (device binding), so that stolen credentials alone cannot be used on a fraudster's phone.
- **PayNet fraud initiatives.** As the operator of the national payment rails, PayNet runs fraud intelligence and alert-sharing initiatives with participating banks, helping the industry spot mule-account patterns and suspicious payment flows across institutions (PayNet, 2026).

The logic of these measures is layered defence: prevention (device binding, OTP changes), detection (fraud intelligence, SemakMule checks by the public) and response (kill-switch, NSRC freezing). No single layer is sufficient on its own.

> **MALAYSIAN SPOTLIGHT:** Payments Network Malaysia (PayNet) is owned by Bank Negara Malaysia and operates the rails behind RENTAS, DuitNow, FPX, JomPAY and MyDebit. In 2025 it processed 8.44 billion digital payment transactions (PayNet, 2026). Because all participating banks connect through PayNet, it is uniquely positioned to run industry-wide fraud intelligence initiatives — spotting, for example, the same mule account receiving scam proceeds from customers of many different banks.

8.2 Risk Management in Digital Banking Businesses

8.2.1 The three lines of defence

How should a digital bank organise itself so that risks like those in Section 8.1 are controlled? The most widely used governance model in financial institutions is the **three lines of defence**, a structure that assigns clear risk responsibilities across the organisation (Hill, 2018):

Line	Who	Responsibility
First line	Business and operational units (product teams, branches, app developers)	Own and manage the risks they create, day to day

Line	Who	Responsibility
Second line	Risk management and compliance functions	Set policies, challenge the first line, monitor risk appetite
Third line	Internal audit	Provide independent assurance to the board that the first two lines work

Above all three lines sits the board of directors, which is ultimately accountable for risk governance. To see the model in action, imagine a Malaysian digital bank launching a new e-wallet feature. The product team (first line) must build fraud controls into the feature. The compliance and risk department (second line) reviews whether the launch meets BNM requirements and the bank's own risk appetite. Months later, internal audit (third line) independently tests whether the controls actually operate as designed. When any line fails — say, the first line ships the feature without transaction limits — the weakness cascades.

8.2.2 Operational risk and operational resilience

Much of what this chapter covers falls under **operational risk**, formally defined as the risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events (Hill, 2018). Cyberattacks, APP scams, cloud outages and data breaches are all operational risk events. Banks manage operational risk through tools such as risk and control self-assessments (RCSA), key risk indicators (KRIs) and internal loss-event databases.

A related but distinct concept is **operational resilience**: the ability of an institution to continue delivering critical operations — such as payments and customer access to funds — even when disruptions occur. Risk management asks, "How do we prevent incidents?" Resilience asks, "When an incident happens anyway, how do we keep serving customers and recover quickly?" For a digital bank with no branches, where the app *is* the bank, resilience is existential: extended downtime erodes customer trust immediately, and in a market where switching costs are low, customers can leave with a few taps.

> **QUICK NOTE:** Do not confuse the two: *operational risk* is a category of risk to be measured and minimised; *operational resilience* is an outcome to be engineered. A bank can have excellent risk assessments and still fail the resilience test if it has no backup systems.

8.2.3 BNM's Risk Management in Technology (RMiT) policy

The centrepiece of Malaysian technology-risk regulation is BNM's **Risk Management in Technology (RMiT)** policy document, first issued in 2020, revised in 2023 and updated in November 2025 (BNM, 2025b). RMiT applies to licensed financial institutions — including the five new digital banks licensed under BNM's 2020 Licensing Framework for Digital Banks, which must meet these requirements even during their foundational phase (BNM, 2020). Its main expectations can be grouped into four areas:

1. **Governance.** The board and senior management are directly accountable for technology risk. Institutions must appoint competent technology-risk and cybersecurity leadership, maintain a board-approved technology risk appetite, and ensure technology decisions receive the same scrutiny as credit or market decisions.

2. **System availability and resilience.** Critical systems — especially customer-facing payment and banking systems — must meet defined availability and recovery standards, supported by capacity planning, redundancy and regular testing.
3. **Cyber controls.** Institutions must maintain layered ("defence-in-depth") security: security operations centres, vulnerability management, penetration testing, secure software development, and customer-side protections such as device binding and fraud surveillance — the very measures described in Section 8.1.4.
4. **Alignment with national critical-infrastructure law.** Financial services are designated critical infrastructure under the **Cyber Security Act 2024 (Act 854)**, which established the National Cyber Security Agency (NACSA) as the sector lead for national cyber security. Banks, as National Critical Information Infrastructure (NCII) entities, must comply with the applicable NCII Code of Practice — covering cyber hygiene standards, incident notification to authorities and periodic audits — alongside RMiT itself.

RMiT is principle-based but demanding: BNM expects institutions to demonstrate not only that controls exist on paper, but that they are tested and effective in practice.

8.2.4 Third-party and cloud risk, incident reporting and business continuity

Modern banks buy much of their technology rather than building it. **Third-party risk management** therefore covers due diligence before engaging vendors, contractual rights to audit, data-protection clauses, subcontracting controls and — critically — exit plans that allow the bank to switch providers without disrupting service. Cloud computing illustrates the trade-off vividly: hyperscale cloud providers offer security and resilience that few individual banks could match in-house, yet industry-wide reliance on a handful of providers creates the concentration risk discussed in Section 8.1.3 (Zetzsche et al., 2017). Under the "shared responsibility" model, the provider secures the cloud platform, but the bank remains accountable to BNM for everything built on it.

Incident reporting is another RMiT expectation: institutions must notify BNM promptly of major technology and cyber incidents, enabling the central bank to assess systemic spillovers and, where necessary, coordinate an industry response. The Cyber Security Act 2024 adds parallel notification duties to NACSA for NCII entities.

Finally, every institution must maintain a **business continuity plan (BCP)**: a documented, tested strategy for maintaining or rapidly recovering critical functions during disruption. Good BCPs define recovery time objectives (RTO — how quickly a service must be restored) and recovery point objectives (RPO — how much data loss is tolerable), and are validated through regular disaster-recovery drills. For a payments system, an RTO might be measured in minutes; for an internal HR system, in days.

> **BRAINSTORMER:** Hyperscale cloud providers offer world-class security, but if most Malaysian banks rely on the same few providers, a single failure could affect the whole system. Should regulators cap how much of the banking industry one cloud provider may host — or would that simply force banks onto weaker, smaller platforms? Who should bear this systemic risk?

8.3 AI Ethics (Bias, Fairness, Transparency and Accountability)

8.3.1 AI in Malaysian finance and where bias comes from

Artificial intelligence (AI) now sits behind credit scoring, fraud detection, customer-service chatbots, anti-money-laundering monitoring and robo-advisory services licensed under the Securities Commission Malaysia's digital investment management framework (Chishti et al., 2020). AI promises faster, cheaper and sometimes more consistent decisions than humans — but it can also discriminate at scale.

Algorithmic bias is a systematic error in an algorithm's outputs that produces unfair outcomes for particular groups of people (Chishti et al., 2020). Its main sources include:

- **Biased historical data.** A model trained on past lending decisions learns yesterday's patterns — including past exclusion of certain postcodes, occupations or informal-sector workers.
- **Unrepresentative samples.** If training data under-represents rural East Malaysians or gig workers, the model scores them poorly simply because it has seen few people like them.
- **Proxy variables.** Even if ethnicity or gender is removed from the data, correlated variables — postcode, device model, transaction timing — can act as hidden proxies, reproducing the excluded attribute's effect.
- **Feedback loops.** A fraud model that flags one neighbourhood heavily generates more investigations there, which produces more confirmed cases there, which "proves" to the model that the neighbourhood is risky — a self-fulfilling cycle.

Fairness can be measured. Common **fairness metrics** include *demographic parity* (approval rates equal across groups), *equal opportunity* (equal true-positive rates — equally qualified applicants from different groups are approved at the same rate), and *predictive parity* (an approval means the same likelihood of repayment regardless of group). A crucial and sobering finding in the field is that these metrics generally *cannot all be satisfied at once*; choosing among them is an ethical and business judgement, not merely a technical one.

8.3.2 The explainability–accuracy trade-off

Explainability is the degree to which a human can understand and articulate how a model reached a particular decision (Chishti et al., 2020). Traditional credit scorecards and small decision trees are highly explainable: one can state exactly why an applicant was declined. Complex models — gradient-boosted ensembles, deep neural networks — often predict defaults or fraud a few percentage points more accurately, but their internal logic is opaque even to their builders: the "black box" problem.

In regulated finance, explainability is not a luxury. Banks owe customers meaningful reasons for adverse decisions; model validators and internal auditors must be able to test a model's logic; and regulators increasingly expect institutions to understand their own models. Techniques exist to open the black box partially — feature-importance rankings, surrogate (simplified stand-in) models and other post-hoc explanation methods — but they offer approximations, not full transparency. Institutions therefore face a genuine trade-off and must decide, use case by use case, how much accuracy they are willing to sacrifice for interpretability. For high-stakes decisions such as loan approvals, many institutions deliberately retain simpler, more explainable models.

BRAINSTORMER: Suppose a digital bank's deep-learning credit model approves 3% more good borrowers than its explainable scorecard, but cannot tell rejected applicants why they failed. Would you deploy it? Who should make that decision — data scientists, the board, or the regulator?

8.3.3 Accountability, human oversight and governance frameworks

Accountability answers the question: when an AI system harms someone, who is responsible? Sound practice assigns clear ownership of AI outcomes to named senior individuals, keeps a *human in the loop* for high-impact decisions (the model recommends; a person decides), and manages models across their lifecycle — inventory, independent validation before deployment, ongoing monitoring for drift and bias, and retirement.

The regional benchmark for AI ethics is the Monetary Authority of Singapore's **FEAT principles** — Fairness, Ethics, Accountability and Transparency — issued in 2018 to guide the use of AI and data analytics in finance (Monetary Authority of Singapore [MAS], 2018). FEAT's core demand is that firms using AI should be able to demonstrate that their models do not systematically disadvantage individuals, that data use is ethical, that responsibility is clearly assigned, and that decisions can be explained to affected customers.

Malaysia is developing its own approach. In August 2025, BNM published a **Discussion Paper on Artificial Intelligence in the Malaysian Financial Sector**, signalling a risk-based, proportionate regulatory stance: stronger governance, risk-management and accountability expectations for AI use cases that carry greater potential harm to consumers and financial stability, with industry feedback invited before final policy (BNM, 2025a).

Data-protection law adds a further layer. Profiling and automated decision-making engage the Personal Data Protection Act 2010 (Act 709): the General Principle requires consent for the processing of personal data, and the Notice and Choice Principle requires individuals to be informed about how their data will be used (Personal Data Protection Act 2010). Unlike the European Union's GDPR, the PDPA contains no express right to refuse purely automated decisions, so in Malaysia the fairness, transparency and oversight expectations of FEAT and BNM's policy development effectively fill that gap for financial services.

DID YOU KNOW? (TRIVIA): Algorithms already manage money for young Malaysians: 51% of robo-advisory (digital investment management) accounts are held by investors in their 20s, across more than one million accounts nationwide (Securities Commission Malaysia [SC], 2026). That makes the fairness and transparency of investment algorithms a question that directly affects your generation.

8.4 Data Privacy, Consumer Protection and Regulatory Technology (RegTech)

8.4.1 The Personal Data Protection Act 2010 and its 2024 amendments

Malaysia's primary data-protection law is the **Personal Data Protection Act 2010 (Act 709)**, which regulates the processing of personal data in *commercial transactions*. Digital banks, e-wallet issuers and fintech firms are "data users" under the Act and must observe its **seven principles** (Personal Data Protection Act 2010):

#	Principle	Core requirement
1	General	Process personal data only with consent and for lawful purposes
2	Notice and Choice	Inform individuals, in writing, about how their data will be used
3	Disclosure	Disclose data only for the purposes notified, to the parties notified
4	Security	Protect data against loss, misuse and unauthorised access
5	Retention	Keep data no longer than necessary
6	Data Integrity	Ensure data is accurate, complete and up to date
7	Access	Allow individuals to access and correct their own data

Amendments passed in 2024 significantly strengthened the Act. Most notably for this chapter, they introduced **personal data breach notification**: a mandatory duty on data users to notify the authorities and affected individuals when a serious breach occurs — closing a gap that previously allowed breaches to go unreported. The amendments also introduced requirements such as the appointment of data protection officers and substantially higher penalties for non-compliance. For digital banks, breach notification connects directly to Section 8.1.3: because leaked data arms future scammers, rapid notification allows customers to be warned and vigilant.

> **QUICK NOTE:** The PDPA applies to *commercial* transactions. It does not cover purely personal or household affairs (such as your private contact list), and the Federal and State governments are largely outside its scope — one reason why financial-sector rules from BNM impose additional privacy duties on banks.

8.4.2 Consumer protection: FSA 2013, the Ombudsman and fair treatment

Beyond data privacy, Malaysian financial consumers are protected by a dedicated conduct framework. The **Financial Services Act 2013 (Act 758)** gives BNM broad powers over the conduct of licensed financial service providers, including requirements of fair dealing and a schedule of prohibited business conduct — such as misleading or deceptive statements and exerting undue pressure on customers (Financial Services Act 2013).

When disputes arise, the **Ombudsman for Financial Services (OFS)** provides consumers with a free, independent alternative to the courts. Customers of member financial institutions can lodge complaints about matters such as unauthorised transactions, unfair charges or poor claims handling; the OFS investigates and its decisions are binding on the institution, though the consumer remains free to reject the outcome and pursue other remedies.

BNM is also consolidating conduct expectations through its **Fair Treatment of Financial**

Consumers policy development (exposure draft issued in 2024), which aims to require institutions to treat customers fairly across the whole product lifecycle — from product design and marketing, through sales and advice, to after-sales service and complaint handling (BNM, 2024b). Together, these instruments shift the compliance question from "did we disclose it in the fine print?" to "did the customer actually receive fair value and understand what they bought?"

8.4.3 RegTech and SupTech: technology for compliance and supervision

Compliance with everything described above — RMiT, PDPA, FSA conduct rules, anti-money-laundering obligations — is expensive when done manually. This has produced two related fields.

RegTech (regulatory technology) is the use of technology by regulated firms to meet regulatory and compliance obligations more efficiently and effectively; the term emerged as post-crisis compliance burdens made manual approaches unsustainable (Arner et al., 2016). In a data-driven financial system, the volume and speed of transactions make technology-assisted compliance a necessity rather than a choice (Zetzsche et al., 2017). Major RegTech applications in Malaysia include:

- **Electronic know-your-customer (e-KYC):** remote identity verification — such as scanning a MyKad and matching a selfie — enabling non-face-to-face account opening under BNM's e-KYC policy document (first issued 2020, revised April 2024) (BNM, 2024a). Every digital bank depends on e-KYC to onboard customers without branches.
- **AML screening:** automated checking of customers and transactions against sanctions and watchlists, supporting obligations under the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (Act 613).
- **Transaction monitoring:** machine-learning systems that flag suspicious patterns — for example, the sudden high-velocity, pass-through transfers characteristic of mule accounts.
- **Regulatory reporting:** automation of the large volume of statistical and compliance reports that institutions must file with BNM, reducing errors and cost.
- **Fraud intelligence sharing:** industry utilities — such as PayNet's fraud intelligence and alert initiatives with banks — that pool signals so a mule identified at one bank is quickly flagged everywhere (PayNet, 2026).

SupTech (supervisory technology) is the mirror image: the use of technology by *regulators themselves* to supervise more effectively — for example, collecting granular data through automated pipelines and applying analytics to detect emerging risks across the industry (Zetzsche et al., 2017). BNM's Financial Sector Blueprint 2022–2026 explicitly identifies SupTech as part of the central bank's modernisation agenda (BNM, 2022), and the Securities Commission Malaysia faces a parallel need as it supervises fast-growing capital-market fintech segments such as P2P financing, equity crowdfunding and digital asset exchanges (SC, 2026).

The direction of travel is clear: as finance becomes more digital, both compliance and supervision must become more technological. For students, this also signals a career opportunity — RegTech and SupTech roles combine finance, law and data skills.

Case Study 8.1: Anatomy of a Malaysian Scam

Encik Faizal, 34, a technician in Shah Alam, received a call one afternoon from someone claiming to be from his bank's fraud department. The caller sounded professional, knew Faizal's full name and the last four digits of his identity card number — details later traced to a leaked customer database from an earlier data breach — and told him that a suspicious RM3,200 transaction had been detected on his account. To "reverse" it, the caller said, Faizal must verify his identity and temporarily move his savings to a "Bank Negara safe account" while the audit ran.

The caller stayed on the line for over an hour, coaching him step by step: add a new payee, transfer RM10,000, then RM8,000, and read out the SMS one-time passwords "for verification". The "safe account" belonged to Aiman, a 21-year-old student who had rented out his bank account to a syndicate for RM500 — a classic mule account. Within twenty minutes of receiving the money, it was split and moved onward through two further accounts.

That evening, Faizal's wife entered the recipient's account number into PDRM's SemakMule portal: it had been reported dozens of times. Faizal called the NSRC at 997, roughly ninety minutes after the first transfer. The centre coordinated with the banks and froze RM4,300 of the second transfer before it was withdrawn; the rest was gone.

Failure points: (1) an earlier data breach armed the scammer with convincing personal details; (2) caller ID was spoofed, so the call looked genuine; (3) psychological pressure — urgency, authority, keeping the victim on the line — defeated his judgement; (4) SMS OTPs could simply be read aloud to the caller; (5) a mule account was easily recruited; and (6) instant payments moved the money faster than any manual review.

How today's controls are designed to break this chain: device binding (one secured device per account) stops a fraudster from logging in himself even with stolen credentials; blocking SMS OTPs for high-risk transactions forces in-app approval with clearer warnings; the kill-switch lets a suspicious customer freeze his own account in seconds; and NSRC 997 coordinates rapid freezing across banks when prevention fails.

Discussion Questions

1. Classify each of the six failure points above as primarily *human*, *process* or *technology* failures. Which single failure, if removed, would most likely have saved Faizal's money? Justify your answer.
2. Walk through the scam step by step and explain exactly where (a) device binding and (b) blocking SMS OTPs for high-risk transactions would have interrupted the chain. Could the scammer adapt? How?
3. Who should bear the RM13,700 loss — Faizal, his bank, or Aiman the mule-account holder? Consider the concepts of customer negligence, bank duty of care and shared responsibility.
4. Mule-account recruiters openly target students with "easy money" offers. Design a one-paragraph awareness message for your campus that would deter students from renting out their accounts.

Chapter Summary

- **8.1:** Digital growth has made Malaysians heavy users of e-payments — 18.4 billion transactions in 2025 (BNM, 2026) — and every transaction is a potential target, making cybersecurity a core banking function (Gupta & Tham, 2019).
- **8.1:** The key risk categories are cyberattacks, phishing, social engineering, malware, account takeover, mule accounts, APP scams, data breaches and third-party concentration risk; APP scams are especially dangerous because the victim personally authorises the payment.
- **8.1:** Malaysia's layered response combines NSRC 997 (established 2022) for rapid freezing, PDRM's SemakMule for public checking, BNM-directed bank controls (kill-switch, blocking SMS OTPs for high-risk transactions, one secured device per account) and PayNet's industry fraud intelligence initiatives.
- **8.2:** The three lines of defence assign risk ownership to business units, oversight to risk and compliance functions, and independent assurance to internal audit, under board accountability.
- **8.2:** Operational risk concerns losses from failed processes, people, systems or external events; operational resilience is the ability to keep critical services running through disruption.
- **8.2:** BNM's RMiT policy (2020; revised 2023; updated November 2025) sets expectations on governance, system availability, cyber controls and third-party risk, reinforced nationally by the Cyber Security Act 2024 and NACSA's NCII Code of Practice (BNM, 2025b).
- **8.3:** Algorithmic bias arises from biased historical data, unrepresentative samples, proxy variables and feedback loops; fairness metrics such as demographic parity and equal opportunity cannot all be satisfied simultaneously.
- **8.3:** There is a real trade-off between model accuracy and explainability; accountability requires named ownership, human oversight of high-impact decisions, and lifecycle model management, guided regionally by MAS's FEAT principles (2018) and in Malaysia by BNM's 2025 AI Discussion Paper.
- **8.4:** The PDPA 2010 imposes seven data-protection principles on commercial data users; 2024 amendments added mandatory breach notification, data protection officers and higher penalties.
- **8.4:** Consumer protection rests on the Financial Services Act 2013, the Ombudsman for Financial Services and BNM's Fair Treatment of Financial Consumers policy development.
- **8.4:** RegTech helps firms comply (e-KYC, AML screening, transaction monitoring, regulatory reporting, fraud intelligence sharing), while SupTech helps BNM and the SC supervise a digital industry (Arner et al., 2016; BNM, 2022).

Key Terms

- **Phishing**: fraudulent messages that impersonate trusted entities to trick recipients into revealing login credentials or personal data.
- **Social engineering**: manipulating people, rather than attacking systems, into divulging information or performing actions that compromise security.
- **Malware**: malicious software, such as fake apps or trojans, installed on a device to steal data or give attackers control.
- **Account takeover (ATO)**: unauthorised access in which a criminal gains control of a

genuine customer's account using stolen credentials.

- **Mule account**: a bank account used to receive and move criminal proceeds, often rented or sold by its legitimate holder to a syndicate.
- **Authorised push payment (APP) scam**: a fraud in which the victim is deceived into personally authorising a transfer to the criminal's account.
- **Data breach**: unauthorised access to or disclosure of personal or financial data, often arming future scams.
- **Concentration risk**: systemic exposure created when many institutions depend on a small number of third-party providers, such as major cloud platforms.
- **Kill-switch**: a self-service bank feature allowing customers to freeze their own accounts and cards immediately upon suspecting compromise.
- **Operational risk**: the risk of loss from inadequate or failed internal processes, people, systems, or external events.
- **Operational resilience**: an institution's ability to continue delivering critical operations and recover quickly during disruption.
- **Three lines of defence**: a governance model assigning risk ownership to business units, oversight to risk and compliance, and independent assurance to internal audit.
- **Business continuity plan (BCP)**: a documented, tested strategy for maintaining or rapidly recovering critical functions during a disruption.
- **Algorithmic bias**: systematic error in an algorithm's outputs that produces unfair outcomes for particular groups of people.
- **Explainability**: the degree to which a human can understand and articulate how a model reached a particular decision.
- **RegTech**: the use of technology by regulated firms to meet regulatory and compliance obligations more efficiently and effectively.
- **SupTech**: the use of technology by regulators and supervisors themselves to monitor and supervise the financial industry more effectively.
- **Personal data breach notification**: the mandatory duty, introduced by the 2024 PDPA amendments, to notify authorities and affected individuals of serious data breaches.

Test Your Knowledge

Part A: Multiple Choice

Q1. A Malaysian who has just realised they transferred money to a scammer should immediately call: - A) 999 - B) 997 - C) 112 - D) 911

Q2. BNM's RMiT policy document stands for: - A) Risk Management in Technology - B) Risk Mitigation in Transactions - C) Regulatory Monitoring of Islamic Technology - D) Remote Management of Information Technology

Q3. The defining feature of an authorised push payment (APP) scam is that: - A) malware transfers funds silently without the customer's knowledge - B) the victim is deceived into personally authorising the transfer - C) a bank employee embezzles customer deposits - D) criminals skim card data at an ATM

Q4. In the three lines of defence model, independent assurance to the board is provided by: - A) business and operational units - B) the risk management and compliance functions - C) internal audit - D) external customers

Q5. A risk is scored likelihood = 4 and impact = 5 on five-point scales, using bands where a score of 15–25 is "Critical". Its classification is: - A) Low - B) Medium - C) High - D) Critical

Q6. A scammer has obtained a customer's password and SMS OTP through phishing and tries to log in from his own phone. Which BNM-directed measure most directly stops him? - A) The kill-switch - B) One secured device per account (device binding) - C) SemakMule - D) e-KYC

Q7. The primary purpose of a mule account in a scam chain is to: - A) store the scammer's identity documents - B) receive and rapidly move stolen funds to obscure the money trail - C) convert ringgit into cryptocurrency - D) generate phishing messages automatically

Q8. The 2024 amendments to the Personal Data Protection Act 2010 introduced, among other things: - A) mandatory personal data breach notification - B) the abolition of consent requirements - C) a ban on all cross-border data transfers - D) the creation of the National Scam Response Centre

Q9. The FEAT principles (Fairness, Ethics, Accountability, Transparency) for AI in finance were issued in 2018 by: - A) Bank Negara Malaysia - B) the Securities Commission Malaysia - C) the Monetary Authority of Singapore - D) Payments Network Malaysia

Q10. A bank's deep-learning model detects fraud slightly more accurately than its old rule-based system, but no one can explain why individual transactions were blocked. This illustrates the trade-off between: - A) accuracy and explainability - B) speed and cost - C) privacy and security - D) liquidity and profitability

Part B: Short Answer

Q1. Distinguish between phishing, malware and account takeover, and explain how the three can combine in a single attack chain against a Malaysian banking customer.

Q2. Using the example of a digital bank launching a new e-wallet feature, describe the role of each of the three lines of defence.

Q3. Explain how device binding, the kill-switch and the NSRC 997 hotline complement one another across the timeline of a scam (before, during and after the loss).

Q4. Identify two sources of algorithmic bias in credit scoring and explain one fairness metric a bank could use to test its model. Why is choosing a fairness metric partly an ethical decision?

Q5. Define RegTech and SupTech, and give one Malaysian example of each.

Answer Guide

Part A

1. **B** — 997 is the National Scam Response Centre hotline, established in 2022 for rapid freezing of scam-linked funds. 999 and 112 are general emergency numbers.
2. **A** — RMiT is BNM's Risk Management in Technology policy (2020; revised 2023; updated November 2025).
3. **B** — In APP scams the victim personally approves the payment, which is what makes prevention and reimbursement so difficult.
4. **C** — Internal audit is the third line; business units own risk (first line) and risk/compliance functions oversee it (second line).
5. **D** — Score = 4 × 5 = 20, which falls in the 15–25 "Critical" band.
6. **B** — Device binding restricts the account to one secured device, so stolen credentials cannot be used on the fraudster's phone.

7. **B** — Mule accounts receive and rapidly pass on scam proceeds, breaking the link between scammer and money.
8. **A** — The 2024 PDPA amendments introduced mandatory breach notification, data protection officers and higher penalties.
9. **C** — MAS issued the FEAT principles in 2018; they are the regional benchmark for AI ethics in finance.
10. **A** — More complex models often gain accuracy at the cost of explainability; the case describes exactly this tension.

Part B (model answers)

1. Phishing uses fraudulent messages impersonating trusted parties to harvest credentials; malware is malicious software installed on the victim's device to steal data or seize control; account takeover is the resulting state in which a criminal controls the genuine customer's account. In a typical chain, a phishing SMS leads the victim to a fake page that captures the password, malware (for example a fake security app) captures OTPs, and together these enable the account takeover.
2. First line: the product and engineering teams own the risks — they must build in transaction limits, fraud monitoring and secure code. Second line: the risk management and compliance functions review the launch against BNM requirements (for example RMiT and e-money rules) and the bank's risk appetite, and can challenge or block it. Third line: internal audit later independently tests whether the controls operate as designed and reports to the board.
3. Before a scam, device binding prevents a fraudster from using stolen credentials on his own device. During the scam, if the customer suspects something is wrong, the kill-switch lets him freeze his own account in seconds, limiting further transfers. After the loss, NSRC 997 coordinates rapid freezing of scam-linked funds across banks while they are still recoverable. The three measures thus cover prevention, self-help response and institutional response.
4. Sources include biased historical data (models learn past exclusion patterns), unrepresentative samples, proxy variables correlated with protected attributes, and feedback loops. A bank could test *equal opportunity*: whether equally creditworthy applicants from different groups are approved at the same rate. Choosing the metric is ethical because different metrics conflict — demographic parity, equal opportunity and predictive parity cannot generally all be satisfied at once, so the choice embeds a value judgement about what "fair" means.
5. RegTech is the use of technology by regulated firms to meet compliance obligations — for example, e-KYC onboarding under BNM's e-KYC policy, or PayNet-facilitated fraud intelligence sharing among banks. SupTech is the use of technology by supervisors themselves — for example, BNM's adoption of supervisory data analytics under the Financial Sector Blueprint 2022–2026, or the SC's technology-enabled supervision of P2P, ECF and digital asset operators.

Problem-Solving Corner

Problem 1: Building a Risk-Assessment Matrix

You are a risk analyst at a Malaysian digital bank. Rate each risk below on a likelihood scale (1 = rare … 5 = almost certain) and an impact scale (1 = negligible … 5 = severe). Your team has agreed the following ratings:

#	Risk event	Likelihood (L)	Impact (I)
1	Large-scale phishing campaign targeting customers	5	3
2	APP scam wave hitting many customers	4	4
3	Data breach exposing customer IC numbers	3	5
4	Biased AI credit-scoring model declines qualified applicants	3	4
5	Outage at the bank's single cloud provider	2	4

Risk score = L × I. Rating bands: 1–4 Low; 5–9 Medium; 10–14 High; 15–25 Critical.

Tasks: (a) Compute the risk score and band for each of the five risks. (b) Rank the risks from highest to lowest priority. (c) For the top two risks, recommend one preventive control and one responsive control each.

Solution to Problem 1

(a) and (b) Computed scores, bands and ranking:

Rank	Risk event	L	I	Score (L × I)	Band
1	APP scam wave	4	4	16	Critical
2=	Phishing campaign	5	3	15	Critical
2=	Data breach	3	5	15	Critical
4	AI credit-scoring bias	3	4	12	High
5	Cloud provider outage	2	4	8	Medium

(c) Controls for the top two risks:

- **APP scam wave (16, Critical):** *Preventive* — device binding plus blocking SMS OTPs for high-risk transactions, so stolen credentials cannot be used and approvals carry clear in-app warnings. *Responsive* — promote the kill-switch and NSRC 997 so losses are frozen quickly; feed confirmed mule accounts into industry fraud-intelligence sharing.
- **Phishing campaign (15, Critical):** *Preventive* — continuous customer education plus SMS sender-ID filtering and takedown of fake websites. *Responsive* — real-time fraud surveillance that detects unusual login patterns, triggering step-up verification or automatic kill-switch prompts to affected customers.

Note that risks 2 and 3 tie at 15; the bank should treat them as joint priorities, especially because phishing is a common *entry point* for APP scams — controlling one reduces the other.

Problem 2: Cost–Benefit Analysis of an Anti-Fraud System

A digital bank is evaluating a new AI-based anti-fraud system with the following figures:

- Initial implementation cost (Year 0): RM2,000,000
- Annual operating cost: RM500,000
- Current annual fraud losses: RM4,800,000
- Expected fraud-loss reduction: 45% per year

Tasks: (a) Compute the annual gross benefit of the system. (b) Compute the annual net benefit (after operating costs). (c) Compute the simple payback period in years and months. (d) Compute the total net benefit over three years and the three-year return on investment (ROI), where ROI = total net benefit ÷ total costs. (e) State two qualitative benefits not captured in these figures.

Solution to Problem 2

(a) Annual gross benefit = 45% × RM4,800,000 = **RM2,160,000** of fraud losses avoided per year.

(b) Annual net benefit = RM2,160,000 − RM500,000 = **RM1,660,000**.

(c) Simple payback period = Initial cost ÷ Annual net benefit = RM2,000,000 ÷ RM1,660,000 ≈ **1.20 years**, i.e. about **14.5 months** (0.20 × 12 ≈ 2.4 months beyond one year).

(d) Three-year position:

- Total net benefit over 3 years = 3 × RM1,660,000 − RM2,000,000 = RM4,980,000 − RM2,000,000 = **RM2,980,000**.
- Total costs over 3 years = RM2,000,000 + (3 × RM500,000) = RM3,500,000.
- ROI = RM2,980,000 ÷ RM3,500,000 ≈ **85.1%**.

The system pays for itself in under 15 months and generates a strong positive three-year return, so on financial grounds the investment is clearly justified.

(e) Qualitative benefits (any two): greater customer trust and retention (customers who feel protected are less likely to leave); improved regulatory standing with BNM, since effective fraud controls support RMiT and fair-treatment expectations; reduced reputational damage from scam headlines; and richer fraud data that improves the bank's models over time.

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Chapter 9: Islamic and Sustainable Digital Finance

Chapter Outline

- 9.1 Islamic Digital Finance and Its Development
- 9.2 FinTech Applications in Islamic Banking and Financial Services
- 9.3 Shariah-Compliant Digital Investment and Financing
- 9.4 Sustainable Finance and Its Role in Digital Finance (Green and Social)

Learning Outcomes

By the end of this chapter, you should be able to:

1. **Define** Islamic digital finance and the core Shariah principles that underpin it. *(Bloom's level: Remember)*
2. **Explain** how Malaysia's legal and regulatory framework supports the development of Islamic digital finance. *(Bloom's level: Understand)*
3. **Apply** Shariah-screening thresholds to assess whether a company's shares are Shariah-compliant. *(Bloom's level: Apply)*
4. **Analyse** how Islamic financial contracts such as murabahah, wakalah, mudarabah and musharakah are digitised in fintech products. *(Bloom's level: Analyse)*
5. **Evaluate** the role of fintech in advancing sustainable, green and social finance in Malaysia, including the risk of greenwashing. *(Bloom's level: Evaluate)*
6. **Design** a simple Shariah-compliant and sustainability-themed digital financing solution for a Malaysian small business. *(Bloom's level: Create)*

Opening Vignette

Farah, a 22-year-old finance diploma student in Kota Kinabalu, wants to invest her first RM500. Her conscience tells her the money must avoid interest (riba) and companies that deal in alcohol or gambling. At the same time, she cares deeply about the environment — Sabah's coral reefs are part of her childhood. On her phone she finds a Shariah-compliant robo-advisor that screens out non-compliant stocks, an Islamic digital bank that opened for business in 2024, and stories about a Sabah solar energy company that issued the world's first green sukuk. Meanwhile, her uncle, who runs a small furniture workshop, just received working-capital financing through a Shariah-compliant peer-to-peer (P2P) platform — no conventional bank branch was involved. Farah realises that finance can be digital, ethical, Islamic and

green at the same time. This chapter explains how Malaysia built the ecosystem that makes her choices possible.

9.1 Islamic Digital Finance and Its Development

Islamic finance is one of the fastest-growing segments of the global financial system, and Malaysia is widely recognised as one of its leaders. When financial technology (FinTech) meets Islamic finance, the result is Islamic digital finance — a field in which Malaysia holds a top-two position among global ecosystems. This section explains the Shariah principles behind Islamic finance, defines Islamic digital finance, and traces the legal and regulatory foundations that Malaysia has built over several decades.

9.1.1 Core Shariah Principles in Finance

Islamic finance refers to financial activities that comply with **Shariah**, the body of Islamic law derived from the Quran and the Sunnah (the traditions of the Prophet Muhammad). Several core principles distinguish it from conventional finance (Oseni & Ali, 2019).

First is the prohibition of **riba**, usually translated as interest or usury. Money is regarded as a medium of exchange, not a commodity to be rented out for a guaranteed return. A conventional loan that charges 8% interest per annum regardless of the borrower's circumstances is therefore not permissible. Instead, returns must be earned through trade, investment or the sharing of risk.

Second is the prohibition of **gharar**, meaning excessive uncertainty or ambiguity in a contract. The object, price, delivery terms and obligations of each party must be clear. Selling something one does not own, or entering a contract whose outcome depends on unknown future events, introduces unacceptable gharar.

Third is the prohibition of **maysir**, which refers to gambling and speculative gain. Transactions that resemble betting — where one party's gain comes purely from another's loss by chance — are forbidden. This principle is why conventional insurance and highly speculative derivatives raise Shariah concerns.

Two positive principles complete the picture. **Risk-sharing** requires parties to a financial arrangement to share profits and losses rather than transferring all risk to one side, as in partnership contracts such as mudarabah and musharakah. **Asset-backing** requires that financial transactions be linked to real, identifiable assets or economic activities, so that finance serves the real economy rather than circulating money for its own sake (Oseni & Ali, 2019).

> **QUICK NOTE:** Islamic finance is not "interest-free conventional finance". A bank may earn a profit, but it must do so through a real transaction — buying and selling an asset, leasing it, or sharing business risk — not by simply charging interest on money lent.

9.1.2 Defining Islamic Digital Finance

Islamic digital finance can be defined as the application of financial technologies — such as mobile platforms, application programming interfaces (APIs), artificial intelligence, big

data analytics and distributed ledger technology — to deliver financial products and services that comply with Shariah principles (Oseni & Ali, 2019). In simpler terms, it is FinTech that passes the Shariah test: the delivery channel is digital, and the underlying contract remains compliant.

The definition has two parts, and both matter. A mobile app that merely sells a conventional interest-based loan to Muslim customers is digital finance, but not Islamic finance. Conversely, a perfectly compliant murabahah (cost-plus sale) contract signed only on paper at a branch is Islamic finance, but not digital. Islamic digital finance exists only when both conditions — digital delivery and Shariah compliance — are satisfied together.

9.1.3 Malaysia's Leadership and Legal Foundations

Malaysia did not become an Islamic finance leader by accident. The country spent more than four decades building a complete ecosystem of institutions, laws and standards. A key institutional anchor is the **Malaysia International Islamic Financial Centre (MIFC)**, established in 2006 to promote Malaysia as an international marketplace for Islamic finance and to connect its industry with global markets.

Three pillars of the legal framework deserve special attention:

1. **Islamic Financial Services Act 2013 (IFSA).** This Act governs Islamic financial institutions in Malaysia and requires them to ensure, at all times, that their aims, operations, business, affairs and activities comply with Shariah. It gives statutory force to Shariah compliance rather than leaving it as a marketing claim (Islamic Financial Services Act 2013).
2. **Central Bank of Malaysia Act 2009 (CBMA).** This Act makes the **Shariah Advisory Council (SAC)** of Bank Negara Malaysia (BNM) the apex authority for the ascertainment of Islamic law for Islamic financial business. Where a court or arbitrator faces a Shariah question concerning Islamic finance, the SAC's rulings prevail (Central Bank of Malaysia Act 2009). This centralisation gives the industry certainty and consistency.
3. **BNM Shariah standards.** BNM issues detailed policy documents on individual contracts — for example the standards on Murabahah, Tawarruq, Wakalah and Bai' al-Sarf — which set out the operational and Shariah requirements that Islamic financial institutions must observe when structuring products on those contracts (Bank Negara Malaysia [BNM], 2018a).

This combination — a statutory compliance duty, an apex Shariah authority and contract-level standards — is frequently cited as the reason global investors trust Malaysian Islamic finance, and it extends naturally into the digital sphere: an Islamic digital bank or a Shariah-compliant P2P platform in Malaysia operates under the same architecture.

> **MALAYSIAN SPOTLIGHT:** The GIFT Report 2025/26 ranks the world's leading Islamic fintech ecosystems as Saudi Arabia and Malaysia at the top, followed by the United Arab Emirates, Indonesia and Bahrain. Malaysia's position rests on the MIFC platform, BNM's Shariah governance framework and a deep talent pool in Islamic finance (DinarStandard & Elipses, 2026).

9.1.4 The Global Islamic Fintech Market

How large is Islamic fintech? According to the GIFT Report 2025/26, the global Islamic fintech market reached an estimated transaction volume of **US$198 billion in 2024/25**, and it is

projected to grow to **US$341 billion by 2029**, a compound annual growth rate of about 11.5% (DinarStandard & Elipses, 2026). Table 9.1 summarises these headline figures.

Table 9.1 Global Islamic fintech at a glance (GIFT Report 2025/26)

Indicator	Figure
Market transaction volume (2024/25)	≈ US$198 billion
Projected volume (2029)	US$341 billion
Compound annual growth rate	≈ 11.5%
Number of Islamic fintech firms worldwide	484
Leading ecosystems	Saudi Arabia, Malaysia, UAE, Indonesia, Bahrain

Source: DinarStandard & Elipses (2026).

Two lessons can be drawn from these numbers. First, Islamic fintech is no longer a niche: a market approaching US$200 billion in annual transactions attracts serious venture capital, regulatory attention and talent. Second, Malaysia's top-two ecosystem ranking means that Malaysian students are studying this subject in one of the world's best natural laboratories — the firms, regulators and products described in this chapter operate in their own backyard.

9.2 FinTech Applications in Islamic Banking and Financial Services

The principles in Section 9.1 come alive in real products. Malaysia now has Islamic digital banks, Shariah-compliant P2P financing and equity crowdfunding (ECF) platforms, digital gold and waqf applications, and automated Shariah-screening tools. This section tours the main application areas.

9.2.1 Islamic Digital Banks

When BNM issued its *Licensing Framework for Digital Banks* in December 2020, it deliberately left room for Islamic models, requiring any digital bank offering Islamic products to comply fully with Shariah requirements (BNM, 2020). Of the five licences announced in April 2022, two went to Islamic digital banks:

- **AEON Bank** (a consortium of AEON Financial Service, AEON Credit Service and MoneyLion) launched on 26 May 2024 as **Malaysia's first Islamic digital bank**.
- **KAF Digital Bank** (led by KAF Investment Bank with partners including Carsome, MoneyMatch, Jirnexu and StoreHub) became the second Islamic digital bank, approved to commence operations on 20 December 2024 (BNM, 2026).

Like conventional digital banks, Islamic digital banks operate without physical branches and onboard customers remotely using electronic know-your-customer (e-KYC) processes under BNM's e-KYC policy (BNM, 2024). During a foundational phase of three to five years, they are subject to an asset limit of RM3 billion, which allows BNM to observe their risk management while they grow (BNM, 2020). The difference lies in the contracts: deposits and financing are structured on Shariah-compliant contracts such as wadiʿah (safe-keeping), qard (benevolent

loan), tawarruq (commodity murabahah) or wakalah (agency), and each bank must maintain a Shariah committee and comply with IFSA's compliance duty.

> **QUICK NOTE:** An Islamic digital bank is still a bank. Customer deposits placed in Shariah-compliant accounts with licensed banks are protected by Perbadanan Insurans Deposit Malaysia (PIDM) up to the applicable limit, just like conventional deposits — the "Islamic" label changes the contract, not the regulatory safety net.

9.2.2 Shariah-Compliant P2P Financing and Equity Crowdfunding

Peer-to-peer (P2P) financing platforms connect businesses seeking funds directly with investors through an online marketplace, regulated in Malaysia by the Securities Commission Malaysia (SC) under the *Guidelines on Recognized Markets* (Securities Commission Malaysia [SC], 2024b). Shariah-compliant P2P platforms do the same thing, but structure each financing note using Islamic contracts — commonly murabahah, tawarruq or wakalah — and maintain Shariah advisors to certify their processes.

The segment is meaningful in size. Of the RM11.32 billion raised cumulatively through Malaysian P2P platforms since 2016, **26% of funds in 2025 were Shariah-compliant** (Securities Commission Malaysia [SC], 2026). **microLEAP** is an example of a P2P operator that offers Shariah-compliant investment notes alongside conventional ones, allowing issuers and investors who require compliance to participate in the same market.

In equity crowdfunding — where investors receive shares rather than notes — the Shariah-compliant share is smaller but present: about **4.1% of ECF funds** raised in 2025 were Shariah-compliant (SC, 2026). **Ethis**, a Malaysia-linked Islamic crowdfunding platform, channels investment into vetted ventures and social impact projects using Islamic structures, demonstrating that crowd-based finance can carry explicit ethical and religious objectives.

> **DID YOU KNOW? (TRIVIA):** Malaysia was the first country in ASEAN to regulate equity crowdfunding (2015) and among the first to regulate P2P financing (2016) — an early-mover advantage that gave Shariah-compliant variants a regulated home from the start (SC, 2026).

9.2.3 Digital Gold, Waqf, Sadaqah and Zakat Platforms

Islamic social finance — zakat (obligatory almsgiving), sadaqah (voluntary charity) and waqf (endowment) — has been transformed by digital channels.

- **HelloGold**, established in 2015, offers Shariah-compliant digital gold savings: users buy small amounts of fully allocated physical gold through an app, with the gold stored in a vault on their behalf. Because each purchase is backed by a real asset, the product applies the asset-backing principle to micro-savings.
- **Global Sadaqah** is a Malaysian platform that digitises waqf and sadaqah giving, vetting campaigns and routing donations digitally to verified beneficiaries, improving transparency over traditional cash collection.
- **Digital zakat and waqf** services are now offered by state Islamic religious councils and Islamic banks through apps and internet banking, so Muslims can calculate and pay zakat, or contribute to waqf funds, in a few taps.

These services matter for financial inclusion. Digital channels lower the minimum contribution, widen the donor base and create auditable trails — features that strengthen trust, which is itself a Shariah value (Oseni & Ali, 2019).

9.2.4 Takaful Technology and Automated Shariah Screening

Takaful is Islamic insurance, in which participants contribute to a common fund used to help members who suffer loss, structured to avoid the gharar and maysir found in conventional insurance. **Takaful tech** applies digital distribution, automated underwriting and claims automation to takaful products — for example, app-based family takaful with online medical declarations, or micro-takaful embedded in e-commerce checkouts. Malaysian takaful operators increasingly partner with fintechs to reach younger, mobile-first customers.

Another behind-the-scenes application is **Shariah screening automation** for equities. To decide whether a listed share is Shariah-compliant, screening methodologies apply qualitative tests (is the core business permissible?) and quantitative financial-ratio tests (for example, how much debt or interest income the company carries). The SC's Shariah Advisory Council publishes its own screening methodology and a regularly updated list of Shariah-compliant securities on Bursa Malaysia. Robo-advisors and brokerage apps then automate these screens: when a portfolio drifts — say a holding becomes non-compliant after a quarterly review — the system flags or rebalances it without human intervention. Automation turns what was once a manual, scholarly exercise into a continuous, real-time control.

> **BRAINSTORMER:** A P2P platform offers both conventional and Shariah-compliant notes on the same website. What governance arrangements would you want in place to prevent a non-compliant note from being wrongly labelled — and who should be accountable if it happens?

9.3 Shariah-Compliant Digital Investment and Financing

Beyond banking and crowdfunding, fintech has opened Shariah-compliant investment to ordinary Malaysians through robo-advisors, digitised Islamic contracts and regulated digital-asset trading. This section examines how these work and the governance challenges they raise.

9.3.1 Shariah Robo-Advisors: Wahed Invest and BEST Invest

The SC introduced its **Digital Investment Management (DIM)** framework in May 2017 — the first robo-advisory licensing framework in the region — covering automated risk profiling, suitability assessment, portfolio allocation and rebalancing with minimal human supervision (SC, 2017). Among the nine registered DIM operators, two are explicitly Shariah-focused:

- **Wahed Invest**, a global Islamic fintech, offers portfolios built from Shariah-screened exchange-traded funds (ETFs), sukuk and gold, with an in-house Shariah review process.
- **BEST Invest**, operated by BIMB Investment (part of the Bank Islam group), provides Shariah-ESG themed portfolios drawing on one of Malaysia's longest Islamic fund-management track records.

The significance for inclusion is considerable. DIM accounts in Malaysia already exceed one million, with 51% held by investors in their 20s — exactly the demographic that expects Shariah-compliant, low-minimum, mobile-first investing (SC, 2026). A student like Farah from our vignette can start with a few hundred ringgit and obtain a diversified, continuously screened portfolio, something that previously required a unit trust agent and a much larger sum.

9.3.2 Digitising Islamic Financial Contracts

Islamic digital products are built from classical contracts, translated into software logic and automated workflows. Table 9.2 summarises the four structures you will meet most often.

Table 9.2 Major Islamic contracts and their digital applications

Contract	Structure	Typical digital application
Murabahah / Tawarruq	Cost-plus sale: the financier buys an asset and sells it to the customer at a disclosed mark-up, payable in instalments; tawarruq uses a commodity as the traded asset	Digital bank personal financing; P2P working-capital notes
Wakalah	Agency: the customer appoints an agent (wakil) to manage funds for a fee	Robo-advisor mandates; takaful operators managing participants' funds
Mudarabah	Profit-sharing partnership: one party provides capital, the other manages; profits are shared by agreement, losses fall on capital	Islamic investment accounts; some crowdfunding structures
Musharakah	Joint partnership: all parties contribute capital and share profits and losses	Home financing (diminishing musharakah); venture-style ECF deals

Source: Contract descriptions follow Oseni and Ali (2019) and BNM Shariah standards (BNM, 2018a).

Digitisation changes how these contracts are executed, not what they are. In a digital tawarruq, for example, the commodity trades that must occur before funds are disbursed are executed automatically through an electronic commodity platform in seconds, with each leg time-stamped for Shariah audit. In a DIM wakalah mandate, the algorithm that allocates your portfolio acts as your agent, and its fee is the wakalah fee. The Shariah question shifts from "is the contract valid?" — settled long ago — to "does the code actually perform the contract's steps in the right order, with the right disclosures?" (Oseni & Ali, 2019).

9.3.3 Digital Assets as Mal: The SAC's 2020 Resolution

Can a Muslim trade Bitcoin or other digital assets? In 2020, the SC's Shariah Advisory Council issued a landmark resolution recognising **digital assets as *mal* (property)** from a Shariah perspective, and ruling that trading them on SC-registered digital asset exchanges is permissible, subject to conditions (SC, 2024a). Because Malaysia had already prescribed digital currencies and tokens as securities under the *Capital Markets and Services (Prescription of Securities) (Digital Currency and Digital Token) Order 2019*, the resolution gave Islamic investors a regulated pathway: compliant trading is possible, but only on registered exchanges, and crypto remains investment property — not legal tender — in Malaysia.

The resolution illustrates a broader point: Shariah scholarship in Malaysia engages with new technology rather than ignoring it. Classifying digital assets as mal places them within the

familiar legal category of property, so established rules on sale, inheritance, zakat and theft can be applied to them.

> **QUICK NOTE:** "Permissible to trade" does not mean "safe investment". The SAC's resolution addresses Shariah status, not price risk. Digital assets remain highly volatile, and investors bear that risk themselves even on registered exchanges.

9.3.4 Challenges: Governing Algorithms and Auditing Smart Contracts

Islamic digital finance inherits all the usual fintech risks — cybersecurity, model risk, data protection — and adds a distinct layer of Shariah-specific challenges (Oseni & Ali, 2019):

1. **Shariah governance of algorithms.** If a robo-advisor's optimisation algorithm selects assets, who certifies that the algorithm's logic — not just its output — is compliant? Shariah committees were designed to review documents and processes, not source code. Institutions increasingly need members or advisors who understand both fiqh muamalat (Islamic commercial jurisprudence) and software.
2. **Standardisation.** Contracts such as tawarruq are accepted in Malaysia but criticised in some other jurisdictions as too close to interest-based lending. A cross-border Islamic fintech must therefore navigate divergent Shariah opinions, raising compliance cost and limiting scalability.
3. **Audit of smart contracts.** If a sukuk is tokenised and profit distributions are executed by self-executing code, an error in the code is an error in the contract. Shariah audit must extend to code review, testing and version control — skills still scarce in the audit profession.
4. **Data and disclosure.** Automated screening depends on timely, accurate company data. If a firm's reported debt level is wrong, the screening output is wrong, and a compliant-labelled portfolio may silently drift out of compliance.

> **BRAINSTORMER:** Suppose a smart contract automatically distributes sukuk profits, but a coding bug overpays investors by 2% one quarter. From a Shariah and legal standpoint, who bears the loss — the issuer, the programmer or the platform — and how would you design the audit process to catch such bugs before deployment?

9.4 Sustainable Finance and Its Role in Digital Finance (Green and Social)

Islamic finance and sustainable finance share a common root: both insist that finance serve real human and social purposes, not merely the multiplication of money. Malaysia has deliberately linked the two agendas, and fintech is the delivery mechanism that makes sustainable finance measurable and scalable.

9.4.1 Defining Sustainable, Green and Social Finance — and ESG

Sustainable finance refers to financial services and investment decisions that integrate environmental, social and governance (ESG) considerations, so that capital supports long-term, inclusive and climate-resilient economic activity (BNM, 2021). Within it:

- **Green finance** funds activities with environmental benefits — renewable energy, energy efficiency, clean transport, pollution control.
- **Social finance** funds activities with social benefits — affordable housing, healthcare, education, financial inclusion and community development.

ESG itself is the analytical lens used by investors: *environmental* criteria (emissions, resource use), *social* criteria (labour practices, community impact, product safety) and *governance* criteria (board conduct, transparency, anti-corruption). A company or instrument is assessed on all three pillars, not on financial return alone (SC, 2019).

Note how naturally this vocabulary fits Islamic finance. The maqasid al-Shariah — the higher objectives of Islamic law, including the protection of life, wealth, intellect and posterity — overlap strongly with ESG goals, which is why Malaysia frames sustainability as an extension of, not an alternative to, Islamic finance (BNM, 2018b).

9.4.2 From VBI to CCPT: Malaysia's Regulatory Architecture

Malaysia's sustainable-finance rules grew in three deliberate steps.

Value-based Intermediation (VBI), 2018. BNM launched VBI together with the Islamic banking industry to re-anchor Islamic finance on its intended outcomes — delivering value to stakeholders, the economy and the community, rather than merely avoiding prohibited elements (BNM, 2018b). VBI encourages Islamic banks to measure and report their impact, for example on entrepreneurship, community empowerment and environmental stewardship, making Islamic banking explicitly outcome-oriented.

Joint Committee on Climate Change (JC3), 2019. Recognising that climate risk crosses institutional boundaries, BNM and the SC jointly established JC3 in 2019, co-chaired by the two regulators, to coordinate the financial industry's response to climate change — from risk management practices to capacity building and disclosure (BNM, 2021).

Climate Change and Principle-based Taxonomy (CCPT), 2021. BNM's CCPT gives financial institutions a common classification language for economic activities: climate-supporting (C1), watchlist (C2) and climate-affected or harmful categories, based on principles rather than a rigid list (BNM, 2021). A bank can now consistently tag its financing portfolio — a solar farm loan as C1, a coal-related exposure as watchlisted — and manage transition risk. Because the taxonomy is principle-based, institutions exercise documented judgement rather than ticking boxes.

9.4.3 Green Sukuk and the SRI Ecosystem

The SC laid the capital-market foundations. Its **Sustainable and Responsible Investment (SRI) framework**, introduced in 2014, allowed funds and sukuk to be designated as SRI products, channelling capital into environmental and social projects (SC, 2014). The **SRI Roadmap (2019)** then set out a five-year strategy to widen the SRI product range, deepen the investor base and build advisory capacity across the Malaysian capital market (SC, 2019).

The flagship result arrived in 2017, when **Tadau Energy** issued the **world's first green sukuk** — an Islamic bond whose proceeds exclusively finance environmentally beneficial projects. The issuance combined two Malaysian strengths in a single instrument: sukuk structuring expertise and an emerging green-finance framework (SC, 2019). A **green sukuk** thus has a dual test to pass: the cash flows must be generated in a Shariah-compliant way, and

the use of proceeds must be environmentally certified — reviewed by both Shariah scholars and environmental assessors. The Case Study below examines Tadau's issuance in detail.

9.4.4 FinTech Roles in Sustainable Finance — and the Greenwashing Risk

Fintech contributes to sustainable finance in at least four ways:

1. **Green crowdfunding and P2P.** Platforms can earmark campaigns for solar installations, energy-efficiency upgrades or social enterprises, letting retail investors fund small green projects that banks overlook. Shariah-compliant platforms can layer both screens at once.
2. **ESG data analytics.** Start-ups aggregate emissions data, satellite imagery, utility records and disclosures into ESG scores, helping banks apply the CCPT classifications and helping investors compare companies on sustainability rather than marketing claims.
3. **Carbon-tracking apps.** Banking and e-wallet apps increasingly estimate the carbon footprint of a user's spending from transaction data, nudging greener consumption — a direct consumer-facing use of payment analytics.
4. **Tokenised green assets.** Fractional, blockchain-recorded ownership of green assets — such as a share of a solar plant's future revenue — can lower minimum investments and improve traceability of how proceeds are used; BNM's 2025 discussion paper on asset tokenisation examines how such instruments could operate within Malaysian regulation (BNM, 2025).

Every one of these channels creates a temptation toward **greenwashing** — the practice of exaggerating or misrepresenting the environmental or social benefits of a financial product to attract capital (BNM, 2021). A "green" fund that holds coal-adjacent stocks, or a "sustainable" bond with proceeds that drift to general corporate use, destroys the trust on which sustainable finance depends. Regulatory responses include taxonomy-based classification (the CCPT), disclosure standards, and independent external review of use-of-proceeds claims. For students, the working rule is simple: trust the verified data trail, not the label. Digital traceability — from tokenised proceeds tracking to ESG analytics — is precisely the tool that makes verification possible, which is why fintech and credible sustainable finance are natural partners.

> **DID YOU KNOW? (TRIVIA):** When Tadau Energy priced the world's first green sukuk in 2017, the instrument had to satisfy two independent review processes at once — Shariah certification of the sukuk structure and environmental certification of the solar project it financed (SC, 2019).

Case Study 9.1: Two Malaysian Firsts — Tadau Energy's Green Sukuk and AEON Bank

Tadau Energy's green sukuk (2017). In 2017, Tadau Energy, a Malaysian renewable energy company, issued the world's first green sukuk — an RM250 million Sustainable and Responsible Investment (SRI) sukuk — to finance a 50-megawatt solar power plant in Kudat, Sabah. The instrument was issued under the SC's SRI sukuk framework introduced in 2014, and it required two layers of assurance: Shariah review confirming the sukuk's structure and

cash flows complied with Islamic principles, and an independent environmental assessment confirming that the proceeds would fund a genuinely green project. The issuance showed that a sukuk could carry both a religious-compliance certificate and a green label without contradiction — the asset-backing of the sukuk matched the physical solar plant, and the social objective of clean energy matched the maqasid of protecting life and posterity. Malaysia thereby created a template that other markets later followed for Islamic green financing (SC, 2014, 2019).

AEON Bank (2024). Seven years later, AEON Bank launched on 26 May 2024 as Malaysia's first Islamic digital bank, formed by AEON Financial Service, AEON Credit Service and MoneyLion (BNM, 2026). Operating entirely through a mobile app under BNM's digital bank framework — including the RM3 billion asset limit during its foundational phase — it onboards customers through e-KYC and structures deposits and financing on Shariah-compliant contracts supervised by its Shariah committee. Its target segments include underserved and unbanked Malaysians, linking Islamic finance to the social-finance goal of inclusion: account ownership among the poorest 40% of Malaysians stood at 82.0% in 2024, leaving a meaningful gap that low-cost digital delivery can help close (World Bank, 2025).

Discussion Questions

1. Tadau Energy's sukuk had to satisfy both Shariah scholars and environmental assessors. What happens if the two reviews conflict — for example, a compliant structure funding a borderline project? Who should have the final say, and why?
2. How does the asset-backing principle make a green sukuk *easier* to design than a purely speculative green investment product?
3. AEON Bank targets underserved customers under a strict RM3 billion asset limit. Evaluate whether this limit protects or hinders financial inclusion during the bank's foundational phase.
4. Propose one way AEON Bank could combine its Islamic identity with a measurable sustainability objective (in the spirit of VBI), and identify the fintech tools it would need.

Chapter Summary

- **Islamic finance** prohibits riba (interest), gharar (excessive uncertainty) and maysir (gambling), and requires risk-sharing and asset-backing so that finance serves the real economy (Section 9.1).
- **Islamic digital finance** is FinTech delivery combined with Shariah-compliant contracts; both conditions must hold together (Oseni & Ali, 2019) (Section 9.1).
- Malaysia's framework rests on MIFC, the Islamic Financial Services Act 2013, and the Central Bank of Malaysia Act 2009, which makes BNM's Shariah Advisory Council the apex authority; BNM issues contract-level Shariah standards (Section 9.1).
- The global Islamic fintech market was about US$198 billion in 2024/25 and is projected to reach US$341 billion by 2029, with 484 firms; Malaysia is a top-two ecosystem (DinarStandard & Elipses, 2026) (Section 9.1).
- AEON Bank (launched 26 May 2024) and KAF Digital Bank are Malaysia's two Islamic digital banks, operating under BNM's digital bank framework with an RM3 billion foundational asset limit (Section 9.2).
- 26% of Malaysian P2P funds in 2025 were Shariah-compliant (microLEAP), and 4.1% of ECF funds (Ethis); HelloGold, Global Sadaqah and digital zakat platforms extend Islamic

social finance online; takaful tech and automated Shariah screening complete the toolkit (Section 9.2).
- Shariah robo-advisors Wahed Invest and BEST Invest operate under the SC's 2017 DIM framework; classical contracts — murabahah/tawarruq, wakalah, mudarabah, musharakah — are executed digitally without changing their legal nature (Section 9.3).
- The SC Shariah Advisory Council's 2020 resolution recognises digital assets as mal, permitting conditional trading on registered exchanges; key challenges are algorithm governance, standardisation and smart-contract audit (Section 9.3).
- Sustainable finance integrates ESG into financial decisions; Malaysia's architecture runs from VBI (2018) through JC3 (2019) to the CCPT (2021) (Section 9.4).
- Tadau Energy issued the world's first green sukuk in 2017 under the SC's SRI framework (2014) and SRI Roadmap (2019); fintech enables green crowdfunding, ESG analytics, carbon tracking and tokenised green assets — while greenwashing remains the central integrity risk (Section 9.4).

Key Terms

- **Riba**: interest or usury; any guaranteed return on money lent, prohibited in Islamic finance because returns must be earned through real trade or investment.
- **Gharar**: excessive uncertainty or ambiguity in a contract's subject matter, price or obligations, which renders the contract invalid under Shariah.
- **Maysir**: gambling or purely speculative gain where one party profits from another's loss by chance; prohibited under Shariah.
- **Islamic digital finance**: the use of financial technologies to deliver products and services that comply with Shariah principles in both contract and operation.
- **Shariah Advisory Council (SAC)**: the apex authority under the Central Bank of Malaysia Act 2009 for ascertaining Islamic law for Islamic financial business in Malaysia.
- **Islamic Financial Services Act 2013 (IFSA)**: the Malaysian statute regulating Islamic financial institutions and imposing a statutory duty of continuous Shariah compliance.
- **Murabahah**: a cost-plus sale contract in which the financier buys an asset and resells it to the customer at a disclosed mark-up, payable in instalments.
- **Tawarruq**: monetisation via commodity murabahah; buying a commodity on deferred terms and selling it for cash to obtain liquidity.
- **Wakalah**: an agency contract in which a principal appoints an agent to act on their behalf, typically for a fee; used in robo-advisory and takaful.
- **Mudarabah**: a profit-sharing partnership where one party supplies capital and the other supplies management; losses fall on the capital provider.
- **Musharakah**: a joint partnership in which all parties contribute capital and share profits and losses by agreed ratios.
- **Takaful**: Islamic insurance based on mutual contributions to a common fund that compensates members who suffer loss, avoiding gharar and maysir.
- **Sukuk**: Shariah-compliant investment certificates representing ownership of underlying assets or ventures, generating returns from those assets rather than interest.
- **Green sukuk**: a sukuk whose proceeds are used exclusively to finance environmentally beneficial projects, passing both Shariah and environmental review.
- **Value-based Intermediation (VBI)**: BNM's 2018 initiative with the Islamic banking industry to deliver measurable value to stakeholders, the economy and the community.
- **Environmental, social and governance (ESG)**: the three-pillar framework for assess-

ing the sustainability and ethical impact of a company or investment.
- **Greenwashing**: exaggerating or misrepresenting the environmental or social benefits of a financial product or activity to attract capital.
- **Digital Investment Management (DIM)**: the SC's 2017 licensing framework for robo-advisors providing automated portfolio management with minimal human supervision.

Test Your Knowledge

Part A: Multiple Choice

1. Which of the following is *not* one of the core prohibitions in Islamic finance discussed in this chapter? A. Riba B. Gharar C. Maysir D. Musharakah

2. Under the Central Bank of Malaysia Act 2009, the apex authority for ascertaining Islamic law for Islamic financial business in Malaysia is: A. The Securities Commission Malaysia B. The Shariah Advisory Council of Bank Negara Malaysia C. The Malaysia International Islamic Financial Centre D. The Ombudsman for Financial Services

3. According to the GIFT Report 2025/26, the global Islamic fintech market is projected to grow from about US$198 billion (2024/25) to approximately _____ by 2029. A. US$250 billion B. US$341 billion C. US$484 billion D. US$198 trillion

4. Malaysia's first Islamic digital bank, launched on 26 May 2024, is: A. KAF Digital Bank B. GXBank C. AEON Bank D. Ryt Bank

5. In 2025, the proportion of Malaysian P2P financing funds that were Shariah-compliant was approximately: A. 4.1% B. 15% C. 26% D. 51%

6. A contract in which a financier buys an asset and resells it to the customer at a disclosed cost-plus price, payable in instalments, is called: A. Wakalah B. Murabahah C. Mudarabah D. Qard

7. The SC Shariah Advisory Council's 2020 resolution held that digital assets are recognised as _____ from a Shariah perspective. A. legal tender B. mal (property) C. riba D. gharar

8. The world's first green sukuk was issued in 2017 by: A. Wahed Invest B. Bank Negara Malaysia C. Tadau Energy D. HelloGold

9. Which BNM policy document (2021) provides financial institutions with a principle-based classification of economic activities by their climate impact? A. Risk Management in Technology (RMiT) B. Value-based Intermediation (VBI) C. Climate Change and Principle-based Taxonomy (CCPT) D. Licensing Framework for Digital Banks

10. A fund advertises itself as "100% green" but quietly holds shares in coal-dependent companies. This practice is best described as: A. Tawarruq B. Greenwashing C. Asset-backing D. Risk-sharing

Part B: Short Answer

1. Explain the difference between riba and maysir, and give one example of each in a modern digital finance context.

2. Why did the Central Bank of Malaysia Act 2009's designation of the Shariah Advisory Council as the apex authority matter for the growth of Islamic fintech in Malaysia?

3. Describe how a wakalah contract is used inside a Shariah robo-advisor such as Wahed Invest, identifying the principal, the agent and the fee.

4. Explain how Value-based Intermediation (2018) connects Islamic finance to sustainable finance in Malaysia.

5. Identify two ways fintech can reduce greenwashing risk in sustainable finance, and explain why each works.

Answer Guide

Part A

1. **D** — Musharakah is a permissible partnership contract, not a prohibition; the prohibitions are riba, gharar and maysir.
2. **B** — The CBMA 2009 makes BNM's SAC the apex authority for Islamic finance Shariah questions.
3. **B** — GIFT 2025/26 projects US$341 billion by 2029 at about 11.5% CAGR.
4. **C** — AEON Bank launched 26 May 2024; KAF Digital Bank was the second Islamic digital bank.
5. **C** — 26% of P2P funds were Shariah-compliant in 2025; 4.1% refers to ECF.
6. **B** — Murabahah is the disclosed cost-plus sale; tawarruq is its commodity-based monetisation variant.
7. **B** — Digital assets were recognised as mal (property), enabling conditional trading on registered DAXs.
8. **C** — Tadau Energy issued the world's first green sukuk in 2017 to finance a solar plant in Sabah.
9. **C** — The CCPT (2021) classifies activities into climate-supporting, watchlist and other categories.
10. **B** — Misrepresenting environmental benefits to attract capital is greenwashing.

Part B (model answers)

1. Riba is a guaranteed return charged on money lent, prohibited because returns must come from real transactions — for example, interest charged on a conventional digital personal loan. Maysir is gain from chance or gambling-like speculation — for example, a binary-options trading app where profit depends purely on guessing short-term price direction. Both transfer gain without genuine economic activity, which is why Shariah forbids them.

2. Centralising Shariah interpretation in one apex council gives legal certainty: fintech firms, banks and investors know that one authoritative ruling applies nationwide, and courts refer Shariah questions to the SAC. This consistency lowers the compliance cost of launching Islamic digital products and builds investor confidence, helping Malaysia's Islamic fintech ecosystem scale.

3. The investor (principal) appoints the robo-advisor as agent (wakil) to manage funds according to an agreed mandate. The algorithm constructs and rebalances a Shariah-screened portfolio, and the robo-advisor charges a wakalah fee for its service. Because the fee is for agency work — not a guaranteed return on capital — the arrangement complies with Shariah.

4. VBI asks Islamic banks to demonstrate real-world outcomes — community empowerment, entrepreneurship, environmental stewardship — rather than merely avoiding prohibited

elements. This outcome orientation aligns Islamic banking with ESG and sustainability objectives, so later instruments such as the CCPT and green sukuk could build on an Islamic-finance industry already measuring impact.

5. First, ESG data analytics let investors and banks verify claims against independent data (emissions, satellite imagery, disclosures) instead of marketing, exposing exaggerated labels. Second, tokenised green assets and digital use-of-proceeds tracking create an auditable trail showing exactly where investors' money went. Both replace trust in labels with verifiable evidence, which is the antidote to greenwashing.

Problem-Solving Corner

Problem 1: Conventional Loan vs Murabahah Cost-Plus Financing

Encik Rahman needs RM50,000 to buy machinery for his workshop, repayable over 3 years in equal monthly instalments. He compares two offers:

- **Offer A (conventional loan):** borrow RM50,000 at a flat interest rate of 8% per annum for 3 years.
- **Offer B (murabahah):** the Islamic bank buys the machinery for RM50,000 and sells it to Encik Rahman at cost plus a profit of 8% per annum on the cost price for 3 years, payable in equal monthly instalments.

(a) Compute the total amount payable and the monthly instalment under each offer. **(b)** Identify two differences between the offers that matter even though the cash flows look identical.

Solution to Problem 1

(a) Offer A — conventional loan.

- Interest = RM50,000 × 8% × 3 years = **RM12,000**
- Total payable = RM50,000 + RM12,000 = **RM62,000**
- Monthly instalment = RM62,000 ÷ 36 = **RM1,722.22** (rounded to sen)

Offer B — murabahah.

- Profit = RM50,000 × 8% × 3 years = **RM12,000**
- Murabahah selling price = RM50,000 + RM12,000 = **RM62,000**
- Monthly instalment = RM62,000 ÷ 36 = **RM1,722.22**

The economic cash flows are identical — a reminder that Shariah compliance is about *how* the return is generated, not about being cheaper.

(b) Key differences.

1. **Ownership and asset risk.** Under murabahah, the bank must genuinely own the machinery before selling it, bearing ownership risk (however briefly) and guaranteeing valid title; under the loan, the bank never owns the asset — it only lends money. This is the asset-backing principle in action.
2. **Nature of the obligation and default treatment.** The RM62,000 in Offer B is a *sale price*, a debt fixed at the moment of sale; it cannot be increased by charging "interest on late interest" (compounding riba). Late-payment charges on Islamic financing are restricted (for example, ta'widh compensation limited to actual loss), whereas a con-

ventional loan can contractually capitalise penalty interest. The legal character of the customer's obligation therefore differs even though the schedule looks the same.

Problem 2: Screening a Portfolio for Shariah Compliance

You are a junior analyst at a Shariah robo-advisor. Using the **illustrative screening thresholds** commonly taught in introductory Islamic finance — a company's total conventional debt must be **below 30%** of its market capitalisation, and its income from non-compliant activities (interest, alcohol, gambling, conventional financial services) must be **below 5%** of total income — screen the three companies below. (Note: these thresholds are teaching illustrations; regulators and index providers, including the SC's Shariah Advisory Council, publish their own detailed benchmarks.)

Company	Core business	Debt / market cap	Non-compliant income / total income
Tekun Bhd	Halal food manufacturing	28%	3%
MegaBuild Bhd	Construction	45%	1%
Serantau Bhd	Logistics	25%	6.5%

(a) Determine which companies pass the screen, showing each test. **(b)** Your robo-advisor currently holds all three. State what action the portfolio algorithm should take.

Solution to Problem 2

(a) Screening results.

- **Tekun Bhd:** qualitative test — halal food manufacturing is a permissible business. Debt ratio 28% < 30% (pass). Non-compliant income 3% < 5% (pass). **Overall: PASSES** — Shariah-compliant.
- **MegaBuild Bhd:** construction is permissible in principle, but debt ratio 45% > 30% (**fails the financial-ratio test**). Non-compliant income 1% < 5% (passes). **Overall: FAILS** — excluded because one failed test is enough.
- **Serantau Bhd:** logistics is permissible; debt ratio 25% < 30% (passes); but non-compliant income 6.5% > 5% (**fails the income test**). **Overall: FAILS** — excluded.

(b) Portfolio action. The algorithm should **divest MegaBuild Bhd and Serantau Bhd**, retaining only Tekun Bhd, and reallocate the proceeds to other compliant securities in the model portfolio. In practice, it would also calculate a **dividend purification** amount: the small non-compliant portion of income earned while holding borderline stocks is donated to charity rather than kept by the investor — an automated feature in Shariah robo-advisors such as Wahed Invest. Finally, screening is continuous: the system re-checks ratios each reporting period, because a pass today can become a fail after the next financial statement.

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Glossary

All terms are defined as they appear in the chapters; the number in brackets is the chapter where the term is first defined.

51% attack [4]: an attack in which one actor controls the majority of a network's mining power or stake, enabling it to rewrite recent history and double-spend.

Account takeover (ATO) [8]: unauthorised access in which a criminal gains control of a genuine customer's account using stolen credentials.

Algorithmic bias [8]: systematic error in an algorithm's outputs that produces unfair outcomes for particular groups of people.

Algorithmic trading [5]: the use of computer programs to generate and submit buy or sell orders automatically according to defined rules, with little or no human intervention at execution.

Altcoin [6]: any cryptocurrency other than bitcoin, spanning platform, payment, utility, governance, meme and stablecoin categories.

Alternative data [3]: non-traditional digital information — such as bank transaction flows, e-payment records and online sales — used to assess creditworthiness when formal records are thin.

Application programming interface (API) [2]: a standardised set of rules allowing one software system to request data or services from another in a controlled, machine-readable way.

Arbitrage [5]: the simultaneous purchase and sale of the same or economically equivalent assets in different markets or forms to profit from a price discrepancy.

Artificial intelligence (AI) [7]: the field of building computer systems that perform tasks normally requiring human intelligence, such as understanding language, recognising images and making decisions.

Authorised push payment (APP) scam [8]: a fraud in which the victim is deceived into personally authorising a transfer to the criminal's account.

Backtesting [5]: simulating a trading strategy on historical market data to estimate how it would have performed before risking real capital.

BigTech [1]: Large technology companies with vast user bases and data advantages that expand into financial services such as payments, lending and banking.

Bitcoin [6]: the first decentralised cryptocurrency, proposed by Satoshi Nakamoto in 2008 and launched in 2009, enabling peer-to-peer value transfer without intermediaries.

Black-box problem [7]: the inability of humans to understand or explain precisely why a complex ML model produced a particular decision.

Block [4]: a bundle of validated transactions together with a header containing metadata such as the hash, previous-hash, timestamp, nonce and Merkle root.

Blockchain [4]: a type of distributed ledger in which validated transactions are grouped into blocks that are cryptographically linked in chronological order.

Business continuity plan (BCP) [8]: a documented, tested strategy for maintaining or rapidly recovering critical functions during a disruption.

Business model [1]: The plan describing how a firm creates, delivers and captures value, including its sources of revenue and cost structure.

Central bank digital currency (CBDC) [2]: a digital form of central-bank money; a direct liability of the central bank, denominated in the national unit of account, in wholesale or retail form.

Chatbot [7]: an AI system that converses with customers in natural language to answer queries and perform simple transactions.

Concentration risk [8]: systemic exposure created when many institutions depend on a small number of third-party providers, such as major cloud platforms.

Consensus mechanism [4]: the rules and incentives by which nodes in a distributed network agree on valid transactions and the official state of the ledger.

Credit scoring [3]: the use of statistical models to estimate the probability that a borrower will repay a debt, typically summarised as a score or risk grade.

Crowdfunding [3]: raising small amounts of money from a large number of people through an online platform to fund a project, cause or business.

Customer due diligence (CDD) [2]: the process by which a financial institution identifies and verifies a customer, understands the purpose of the relationship, and monitors transactions accordingly.

Data breach [8]: unauthorised access to or disclosure of personal or financial data, often arming future scams.

Decentralised finance (DeFi) [6]: an ecosystem of blockchain-based applications that provide lending, trading and other financial services through smart contracts without intermediaries.

Deep learning [7]: a subset of ML using multi-layered neural networks to learn complex patterns from large volumes of data.

Default risk [3]: the risk that a borrower fails to make required payments, causing lenders or investors to lose returns, principal or both.

Digital asset exchange (DAX) [1]: A platform registered with the Securities Commission Malaysia on which digital assets, prescribed as securities under Malaysian law, may be traded.

Digital bank [2]: a licensed bank that delivers all services through digital channels, operating without any physical branch network.

Digital banking [1]: The delivery of licensed banking services wholly or almost wholly through digital or electronic means, with remote onboarding and no reliance on branches.

Digital financing [3]: the provision of financial services, especially credit and capital raising, through digital technologies and platforms rather than traditional branch-based processes.

Digital investment management (DIM) [1]: SC-licensed automated investment services — robo-advisory — covering risk profiling, suitability assessment, asset allocation and rebalancing with minimal human supervision.

Dilution [3]: the reduction in existing shareholders' ownership percentage when new shares are issued to new investors, even though their share count is unchanged.

Direct market access (DMA) [5]: an arrangement allowing a broker's client to transmit orders electronically through the broker's infrastructure directly into an exchange's order book, subject to pre-trade controls.

Distributed ledger technology (DLT) [4]: a digital record-keeping system in which a ledger is shared, replicated and synchronised across multiple network nodes without a single central authority.

Double-spending problem [6]: the risk that digital money, being copyable, is spent more than once; blockchains solve it without a central ledger-keeper.

DuitNow [2]: Malaysia's instant account-to-account transfer service (2018), addressed to proxies such as a mobile or MyKad number instead of a bank account number.

DuitNow QR [2]: Malaysia's national interoperable QR payment standard (2019); one merchant QR code can be scanned by any participating banking app or e-wallet.

E-wallet [1]: A smartphone application that stores e-money and enables payments, transfers and often additional services such as investments or insurance.

Effective annual cost [3]: the total cost of financing, including returns and all fees, expressed as an annual rate on the net funds actually received by the borrower.

Electronic know-your-customer (e-KYC) [1]: Regulator-approved digital processes for verifying a customer's identity remotely, enabling non-face-to-face account opening.

Electronic money (e-money) [1]: Digitally stored monetary value issued under a BNM licence that holders can use to pay merchants or transfer to other users.

Embedded finance [1]: The integration of financial services such as payments, lending or insurance directly into non-financial apps and platforms at the point of need.

Environmental, social and governance (ESG) [9]: the three-pillar framework for assessing the sustainability and ethical impact of a company or investment.

Equity crowdfunding (ECF) [1]: The raising of equity capital by companies from many investors through an online platform registered with the Securities Commission Malaysia.

Exchange-traded fund (ETF) [5]: a listed, usually passively managed fund that tracks an index and trades intraday on an exchange like an ordinary share.

Explainability [8]: the degree to which a human can understand and articulate how a model reached a particular decision.

FEAT principles [7]: MAS's 2018 benchmark requiring AI and data analytics in finance to be Fair, Ethical, Accountable and Transparent.

Financial inclusion [1]: Access of individuals and businesses to useful, affordable financial products and services — payments, savings, credit and insurance — delivered responsibly and sustainably.

Financial technology (FinTech) [1]: Digital innovations and technology-enabled business model innovations in the financial sector; used for both the industry of firms and the underlying technologies.

Flash crash [5]: an extremely rapid, deep and largely self-reversing fall in market prices driven by interacting automated trading systems, exemplified by the 2010 US Flash Crash.

Fraud detection [7]: the use of ML models to score transactions for suspiciousness in real time based on learned fraud patterns.

Freemium [1]: A business model offering basic services free to build a large user base, then charging for premium features or cross-selling other products.

Full node [4]: a node that downloads the entire blockchain and independently verifies every transaction and block against the consensus rules.

Gas [6]: the unit measuring computational work on Ethereum, paid in ether, which compensates validators and deters spam.

Generative AI [7]: AI models that create new content — text, images, code or audio — in response to prompts.

Gharar [9]: excessive uncertainty or ambiguity in a contract's subject matter, price or obligations, which renders the contract invalid under Shariah.

Green sukuk [9]: a sukuk whose proceeds are used exclusively to finance environmentally beneficial projects, passing both Shariah and environmental review.

Greenwashing [9]: exaggerating or misrepresenting the environmental or social benefits of a financial product or activity to attract capital.

Halving [6]: the programmed reduction, roughly every four years, of Bitcoin's block reward by half; the April 2024 halving cut it to 3.125 BTC.

Hash [4]: the fixed-length output of a one-way hash function; a digital fingerprint that uniquely represents any input and changes completely if the input changes.

High-frequency trading (HFT) [5]: a specialised form of algorithmic trading using low-latency technology to trade very large volumes over extremely short holding periods, often milliseconds.

Human-in-the-loop (HITL) [7]: a safeguard in which a human reviews, can override and remains accountable for AI recommendations in high-stakes decisions.

Information asymmetry [3]: a situation in which one party to a transaction holds more or better information than the other, complicating lending and investment decisions.

Initial coin offering (ICO) [6]: a largely unregulated public sale of newly created tokens to raise funds, notorious for failures and fraud during 2017–2018.

Interchange fee [1]: The portion of a card transaction fee paid by the merchant's bank to the cardholder's issuing bank, forming a key revenue source for card issuers.

Investment note [3]: a security issued through a registered P2P platform evidencing the issuer's obligation to repay investors principal plus agreed returns over a fixed tenor.

Islamic digital finance [9]: the use of financial technologies to deliver products and services that comply with Shariah principles in both contract and operation.

Islamic Financial Services Act 2013 (IFSA) [9]: the Malaysian statute regulating Islamic financial institutions and imposing a statutory duty of continuous Shariah compliance.

Issuer [3]: the business that raises funds on a P2P or ECF platform by issuing investment notes or shares, and bears the obligation to repay or to share ownership.

Kill-switch [8]: a self-service bank feature allowing customers to freeze their own accounts and cards immediately upon suspecting compromise.

Light node [4]: a node storing only block headers that uses Merkle proofs to verify specific transactions, suitable for wallets and mobile devices.

Liveness detection [2]: techniques used during e-KYC to confirm a live person is present — such as blinking or head movements — defeating photos, videos and masks.

Machine learning (ML) [7]: a subset of AI in which computers learn patterns from data and improve at a task without being explicitly programmed with fixed rules.

Malware [8]: malicious software, such as fake apps or trojans, installed on a device to steal data or give attackers control.

Market making [5]: continuously quoting both bid and ask prices for a security to earn the bid–ask spread, while managing the risk of holding inventory.

Maysir [9]: gambling or purely speculative gain where one party profits from another's loss by chance; prohibited under Shariah.

Mean reversion [5]: a trading approach based on the tendency of prices that move abnormally far from their historical average to return toward that average.

Merkle tree [4]: a tree of pairwise hashes over a block's transactions, ending in a single root that enables efficient, lightweight verification of individual transactions.

Mining [6]: the competitive process of validating transactions and adding blocks to a proof-of-work blockchain in return for block rewards and fees.

Mobile-first [2]: a design philosophy in which the smartphone application is the primary banking channel and all other channels are built around it.

Model risk [7]: the risk of loss or customer harm from models that are wrong, misused or applied outside their valid range.

Modern Portfolio Theory (MPT) [5]: Markowitz's framework showing that combining assets with less-than-perfect correlation can achieve the best expected return for a given level of risk.

Momentum strategy [5]: a trading approach that buys assets whose prices have risen and sells those that have fallen, assuming trends persist; often implemented with moving-average crossovers.

Mudarabah [9]: a profit-sharing partnership where one party supplies capital and the other supplies management; losses fall on the capital provider.

Mule account [8]: a bank account used to receive and move criminal proceeds, often rented or sold by its legitimate holder to a syndicate.

Murabahah [9]: a cost-plus sale contract in which the financier buys an asset and resells it to the customer at a disclosed mark-up, payable in instalments.

Musharakah [9]: a joint partnership in which all parties contribute capital and share profits and losses by agreed ratios.

Neobank [1]: Global industry term for a digital-only, branchless banking provider; in Malaysia the nearest equivalents are the five fully licensed digital banks.

Neural network [7]: a computational structure of interconnected processing layers, loosely inspired by the brain, used in deep learning.

Node [4]: a computer running a blockchain's software that stores, relays and (depending on type) validates ledger data according to the consensus rules.

Non-fungible token (NFT) [6]: a unique blockchain token representing a specific item, not interchangeable one-for-one with other tokens.

Nonce [4]: an adjustable number in a block header that proof-of-work miners vary when searching for a hash that meets the network's difficulty target.

Online banking [1]: The provision of banking services over the internet as an additional channel of a traditional branch-based bank, such as Maybank2u or CIMB Clicks.

Open banking [2]: the consent-based sharing of customer banking data with authorised third-party providers through secure APIs.

Open finance [2]: the extension of consented, API-based data sharing beyond bank accounts to wider financial data such as investments and insurance.

Operational resilience [8]: an institution's ability to continue delivering critical operations and recover quickly during disruption.

Operational risk [8]: the risk of loss from inadequate or failed internal processes, people, systems, or external events.

Oracle problem [4]: the limitation that blockchains cannot verify off-chain, real-world facts, so smart contracts must rely on trusted external data feeds called oracles.

Origination fee [3]: a charge levied by a platform for processing, assessing and arranging a financing facility, usually a percentage of the amount raised, deducted from the proceeds.

Overfitting [5]: tuning a model so precisely to historical data that it memorises noise rather than learning a durable pattern, causing live performance to collapse.

Peer-to-peer (P2P) financing [1]: The raising of debt funding by businesses from many investors through an online platform registered with the Securities Commission Malaysia.

Permissioned blockchain [4]: a network in which read and/or validation rights are restricted to approved participants, covering private and consortium designs.

Personal data breach notification [8]: the mandatory duty, introduced by the 2024 PDPA amendments, to notify authorities and affected individuals of serious data breaches.

Phishing [8]: fraudulent messages that impersonate trusted entities to trick recipients into revealing login credentials or personal data.

Post-money valuation [3]: a company's value immediately after a funding round, equal to the pre-money valuation plus the amount of new capital raised.

Pre-money valuation [3]: the agreed value of a company immediately before new investment money is added in a funding round.

Private key [6]: a secret cryptographic number that proves ownership of cryptocurrency and authorises spending; losing it means losing the funds permanently.

Proof of stake [6]: a consensus mechanism where validators lock up cryptocurrency as collateral to propose and confirm blocks, with dishonesty punished by slashing.

Proof of stake (PoS) [4]: a consensus mechanism in which validators lock up cryptocurrency as a stake and are penalised for dishonesty, avoiding PoW's energy race.

Proof of work [6]: a consensus mechanism requiring participants to expend computational effort to add blocks, making attacks economically costly.

Proof of work (PoW) [4]: a consensus mechanism in which miners compete to solve a computational puzzle, making block production costly and attacks economically impractical.

Public key [6]: a cryptographic value derived from a private key, used to generate receiving addresses; safe to share.

Public-key cryptography [4]: an encryption scheme using a linked private key (secret, used to sign) and public key (shared, used to verify), forming the basis of blockchain ownership.

Real-time gross settlement (RTGS) [2]: the settlement of high-value interbank payments individually, immediately and with finality, rather than in netted batches.

Rebalancing [5]: restoring a portfolio to its target asset allocation by selling overweight assets and buying underweight ones, on a calendar or threshold basis.

Recognized Market Operator (RMO) [3]: an electronic platform operator registered with the Securities Commission Malaysia to run a market — such as P2P, ECF or digital asset trading — under the Guidelines on Recognized Markets.

RegTech [8]: the use of technology by regulated firms to meet regulatory and compliance obligations more efficiently and effectively.

Regulatory sandbox [1]: A supervised framework allowing firms to test innovative financial services with real customers under relaxed conditions and safeguards before full licensing.

Reinforcement learning [7]: ML in which an agent learns by trial and error, receiving rewards or penalties to discover optimal strategies.

Remittance [2]: a cross-border transfer of funds, typically sent by migrant workers to family members in their home country.

Riba [9]: interest or usury; any guaranteed return on money lent, prohibited in Islamic finance because returns must be earned through real trade or investment.

Risk profiling [5]: the process of assessing an investor's ability and willingness to bear investment risk so that a suitable portfolio can be recommended.

Robo-advisory [5]: automated, algorithm-driven investment advice and portfolio management delivered through a digital platform with minimal human supervision.

Security token offering (STO) [6]: a regulated issuance of blockchain tokens representing legally recognised securities, subject to securities law and disclosure requirements.

Shariah Advisory Council (SAC) [9]: the apex authority under the Central Bank of Malaysia Act 2009 for ascertaining Islamic law for Islamic financial business in Malaysia.

Shariah-compliant [3]: structured in accordance with Islamic law, avoiding prohibited elements such as riba (interest) through contracts like murabahah, tawarruq or wakalah.

Signal [5]: the output of a trading algorithm's rules — an instruction to buy, sell or hold — generated from market data such as prices and volumes.

Slippage [5]: the difference between a trade's expected execution price and its actual execution price, arising from market movement and limited liquidity.

Smart contract [4]: a self-executing program stored on a blockchain that automatically enforces the terms of an agreement when coded conditions are met.

Social engineering [8]: manipulating people, rather than attacking systems, into divulging information or performing actions that compromise security.

Stablecoin [2]: a crypto-asset designed to maintain stable value against a reference asset, usually a fiat currency; fiat-collateralised examples include USDT and USDC.

Sukuk [9]: Shariah-compliant investment certificates representing ownership of underlying assets or ventures, generating returns from those assets rather than interest.

Supervised learning [7]: ML trained on labelled historical examples to predict outcomes for new cases, such as default or fraud.

SupTech [8]: the use of technology by regulators and supervisors themselves to monitor and supervise the financial industry more effectively.

Takaful [9]: Islamic insurance based on mutual contributions to a common fund that compensates members who suffer loss, avoiding gharar and maysir.

Tawarruq [9]: monetisation via commodity murabahah; buying a commodity on deferred terms and selling it for cash to obtain liquidity.

Three lines of defence [8]: a governance model assigning risk ownership to business units, oversight to risk and compliance, and independent assurance to internal audit.

Tokenisation [4]: the creation of a blockchain-based digital token representing ownership of, or a claim on, a financial or real-world underlying asset.

Tokenised deposit [2]: an ordinary commercial-bank deposit represented as a programmable digital token transferable on modern ledgers.

Transaction monitoring [7]: the screening of customer transactions for patterns consistent with money laundering or terrorism financing, as required under AMLA 2001.

Unspent transaction output (UTXO) [6]: a discrete, spendable chunk of cryptocurrency recorded on the blockchain, consumed entirely as an input when spent, with any remainder returned as change.

Unsupervised learning [7]: ML that finds structure, clusters or anomalies in unlabelled data without known correct answers.

Value-based Intermediation (VBI) [9]: BNM's 2018 initiative with the Islamic banking industry to deliver measurable value to stakeholders, the economy and the community.

Virtual card [2]: a payment card credential that exists only in digital form, issued instantly inside an app and usable for online or wallet-based payments.

Wakalah [9]: an agency contract in which a principal appoints an agent to act on their behalf, typically for a fee; used in robo-advisory and takaful.

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